

Pennsylvania Trail Design & Development Principles

Guidelines for Sustainable, Non-motorized Trails





Elk State Forest Equestrian Trail
Photo Credit: Pashek Associates

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Bellefonte Central Rail Trail

Photo Credit: Pashek Associates





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North Country Trail
 Photo Credit: Pashek Associates



Foreword

Every five years, states across the nation must produce a State Comprehensive Outdoor Recreation Plan (SCORP) to remain eligible to receive federal Land and Water Conservation Funds (LWCF). By design, SCORP directs the use of LWCF funds that may come to the state.

The required elements of the plan include a statewide assessment of outdoor recreation needs and supplies, as well as an action plan for future steps to help enhance outdoor recreation. Additional requirements include an update on the status of the state's wetland resources (SCORP, Appendix A), and extensive opportunity for public participation.

One of the recommendations that stemmed from the development of Pennsylvania's Statewide Comprehensive Outdoor Plan (SCORP), by the PA Department of Conservation and Natural Resources (DCNR), partner agencies, and various recreational stakeholders focused on trail planning. SCORP Recommendation 3.2 states:

“standardize trail planning guidelines, information, and funding decision criteria to build a sustainable statewide trail system.”

SCORP further identified Action Step B to accomplish Recommendation 3.2, and tasked the PA State Trails Advisory Committee with,

“adopting statewide guidelines to encourage uniform trail construction, maintenance, and signage; developing a non-motorized trails standards and guidelines publication; and promoting best practices to ensure continued maintenance and future sustainability of trails and related facilities.”

As a leader with a mission to “conserve and sustain Pennsylvania's natural resources for future generations' use and enjoyment,” and with extensive representation on the PA State Trails

Advisory Committee, the DCNR took it upon itself to develop the manual identified by SCORP's Action Step B.

To accomplish such a tremendous undertaking, DCNR assembled an internal working group with representation from the Bureau of Facility, Design, and Construction (FD&C), the Bureau of Forestry (BOF), the Bureau of Recreation and Conservation (BRC), and the Bureau of State Parks (BSP). Building on the past success in developing the Pennsylvania Trail Design Manual for Off Highway Recreational Vehicles, for motorized trails within Pennsylvania, DCNR decided to use this process as a model for the development of this non-motorized manual.

During the initial phase of this manual's development, the BOF - Recreation Section developed an outline for the manual's content and a scope of work with the assistance of FD&C. The Department's internal work group then reviewed and commented on the manual's table of contents and scope of work.

In order to remain responsive and transparent to our recreational stakeholders, we contacted key representatives from SCORP's Technical Advisory Committee (TAC) – PA State Trails Action Committee and the Departments' Recreational Advisory Committee (RAC) for review and comment of the manual prior to submitting it to a vendor to research, design, and edit this manual. The Department selected Pashek Associates, Inc., as the vendor to produce the document.

The Department's internal work group met with Pashek to discuss the project goals and outputs. At that time, both parties recognized that a variety of good references already exist related to trail planning, design, construction, and operations and maintenance, ranging from manuals produced from state agencies, such as Minnesota; to organizations, like the International Mountain Biking Association



(IMBA) and the PA Equine Council (PEC); to Federal Agencies, such as the United States Forest Service.

The National Park Service defines sustainable trails as:

- Supporting current planned and future uses with minimal impact to the natural systems of the area
- Causing negligible soil loss or movement while allowing naturally occurring plant systems to inhabit the area
- Recognizing needed pruning and eventual removal of certain plants over time
- Not adversely affecting the naturally occurring fauna
- Accommodating existing and future uses while only allowing appropriate uses
- Requiring little rerouting and minimal maintenance over extended periods of time

*- National Park Service;
Rocky Mountain Region,
January 1991*

Rather than trying to reinvent the wheel, the Department and Pashek Associates decided the best course of action involved synthesizing the technical material contained in existing manuals, adding real world input from both the vendor's and internal work groups' past experiences with trail development, and incorporating examples of best management practices that have occurred on PA trails, to create this manual.

In an effort to ensure that the needs, wants, and desires of our recreational stakeholders did not get lost during this process, the Department and Pashek reached out to stakeholders at a subsequent TAC meeting for comments on the design chapter relative to their particular form of recreation. At this time, the Department and Pashek also solicited photos of good trails, photos of best management practices, and identification of Pennsylvania trails that meet the unique needs of their particular user group.

Upon completion of each of the individual chapters, members from TAC and RAC had the opportunity to review and comment on the manual in order to remain responsive to our users.

In the end, the content of the manual remains just as important as the process used to develop it.

In this manual, you will find a compilation of best practices and guidelines for the planning, design, construction, and management of your trail. Equally important, it discusses techniques for developing trails that create desirable and enjoyable trail experiences for their users. At the same time, these techniques employ sustainable design elements and construction practices that allow the trail to make use of natural systems so that the trail remains both physically and environmentally sustainable, which in turn leads to minimal maintenance and operational costs, making the trail economically sustainable over the long run.

Every trail project has its own unique set of challenges and opportunities. A trail project needs specific evaluation of the site (which extends to the surrounding area beyond the trails' physical corridor), trail alignment, potential resource impact(s), potential environmental opportunities, and existing physical and environmental constraints, in order to determine the design parameters of the respective trail. These unique challenges and opportunities may require that you stray from or refine the guidelines contained in this manual to take advantage of a particular opportunity, overcome a particular challenge or ensure the health and safety of your trail user.



The safety and well-being of visitors to your trail must always remain your primary concern. This guide does not provide a substitute for the professional expertise needed to make informed decisions regarding design, planning, construction, management, operation, and maintenance specific to your trail.

This comprehensive reference does, however, provide trail advocates, agencies, organizations, local officials, regional officials, and decision-makers with the basic knowledge needed to undertake a trail project with the assistance of professionals.



Trail Volunteers
Lonberger Trail Reroute

Photo Credit: PA DCNR



Dr. J Freeride Trail
North Park, Allegheny County

Photo Credits: Jon Pratt



Planning

The foundation of a sustainable trail ethic requires proper planning, design, layout, and construction and continues with the management, operation, and maintenance of the trails once they are established.

During the planning process you will define a vision, develop goals, and establish implementation strategies to reach your goals. Always ensure your goals and implementation strategies support your project's vision.

For trails, this means understanding the:

- ability of the landscape to accommodate your vision
- capacity of your organization to develop, manage, and operate the trail
- required level of financial and human resources to achieve your vision

A sustainable trail supports responsible outdoor recreation and has a positive impact on the environment, heritage and communities. It also addresses landowner and user needs and concerns.

Good trail design meets the specific needs of the anticipated users in a manner that works with the trail's environment. By design, your trail should be no wider than needed to accommodate its intended users.

Allegrippis Trails at Raystown Lake

32 miles of sustainably planned, designed and constructed single track mountain bike trails.

Photo Credit: Lesile Kehmeier





The Planning Process

The design of a trail system begins with sound planning.

Planning consists of the following steps:

1. Reviewing relevant state, regional, and local planning efforts
2. Selecting a qualified trail designer
3. Engaging the public
4. Identifying the needs, wants, and concerns of the land manager, property owner, and adjacent neighbor(s)
5. Identifying the needs, wants, and desires of trail users
6. Assembling the mapping
7. Analyzing the property to identify opportunities and constraints for developing your trail
8. Determining the type of trail system and level of difficulty
9. Determining the locations of potential trailheads
10. Determining regulatory requirements
11. Identifying environmental and cultural resources
12. Planning for accessibility
13. Establishing the preliminary trail alignment
14. Evaluating potential management structures for management, maintenance, and operations of your trail
15. Planning for management, maintenance, operation, and construction costs
16. Securing funding for trail planning, design, and construction



Boots on the Ground
Allegheny National Forest
Trail Planning & Design

Photo Credit: Pashek Associates

Old logging roads, designed for temporary rather than long-term use, typically make poor trails. They often include sections of fall-line trail (prone to erosion), flat land sections that collect water, and have a wide tread to accommodate logging equipment. When used for recreational trails, old logging roads need frequent maintenance to address erosion and tread degradation. Unsustainable trails consume valuable time and resources.

Logging Road

Photo Credit: Pashek Associates





Step 1: Determining Consistency with State, Regional, County, and Local Planning Efforts

Before planning for a new trail, review state, regional, county, and local planning efforts to determine whether the proposed trail system is consistent with the strategies of other planning efforts. Typical planning efforts that you should review include:

- Pennsylvania State Comprehensive Outdoor Recreation Plan
www.paoutdoorrecplan.com
- Regional and/or County Comprehensive Recreation, Park, and Open Space Plans/Regional and/or County Greenway Plans
www.dcnr.state.pa.us/brc/greenways/
- Local Comprehensive Recreation, Park, and Open Space Plans
- Local Greenway Plans
- Other local municipal plans, e.g. trail feasibility studies, park master plans, corridor studies, waterfront studies

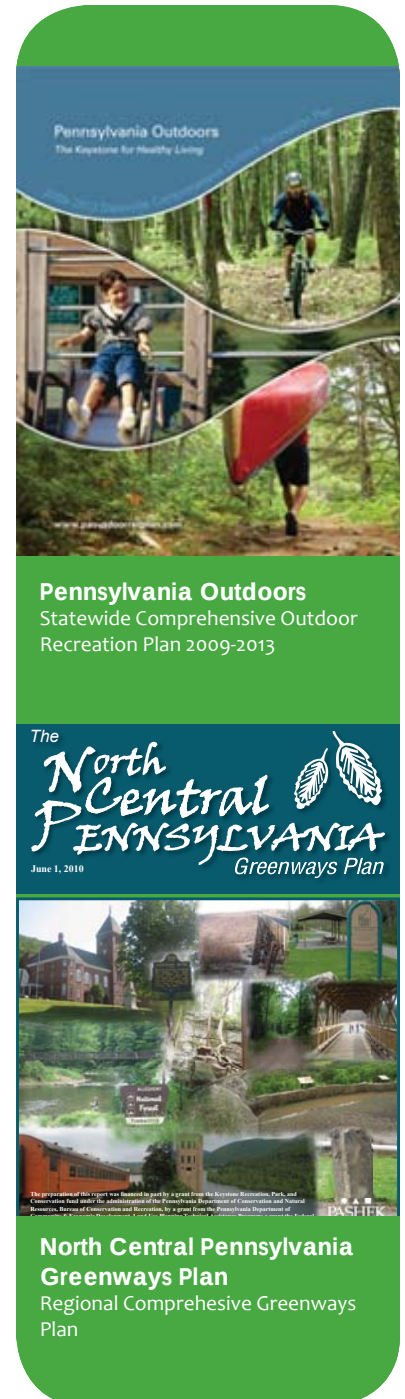
If your proposed trail is not identified in current planning efforts, contact the county and municipal planning agencies having jurisdiction over the project area to determine if the proposed project is consistent with current planning efforts.

Step 2: Selecting a Trail Designer

Whether you appoint someone within your organization to design your trail, or hire a consultant to guide you through the process, it is important to ensure she or he possess the knowledge and experience in the following areas of sustainable trail design:

- Trail planning
- Trail design
- Trail layout and survey
- Trail construction
- Trail bridge design and construction
- Trail signage design and construction
- GIS and CADD capabilities
- Construction document preparation
- Construction observation experience

When you select a trail designer, ask for names and contacts of past clients and conduct reference checks to determine whether he/she will meet the needs of your project. Trail designers are usually landscape architects, engineers, and park, recreation, and forestry professionals with sustainable trail planning and development experience.





You can find DCNR's Consultant Selection Guidelines at:
www.dcnr.state.pa.us/brc/grants/20062007/.../RFPGL_6.pdf

Depending on the resources present and the complexity of addressing them, your trail designer may recommend including other professional disciplines on the design team. These may include surveyors, archeologists, soil scientists, environmental engineers, civil engineers, structural engineers, and architects.

Step 3: Engaging the Public

A critical component of all planning projects is determining the public's needs, interests and concerns related to a proposed undertaking. A well-designed public participation process engages the stakeholders and the public. Involving stakeholders, the public and decision-makers helps diffuse conflict and increase volunteer participation.

Public participation is important because it:

- Provides the stakeholders, leaders and decision makers an opportunity to voice their opinions
- Informs the elected officials of citizen attitudes and needs
- Helps to express broad-based public support for the planning process and planned proposals
- Provides the general public and community leaders with an opportunity to support and be involved with the development of the plan

Public participation techniques typically include some or all of the following:

- Establishing a steering or advisory committee
- Preparing a communication strategy
- Conducting public meetings
- Conducting surveys
- Interviewing key persons
- Conducting focus group meetings

You can find DCNR's Public Participation Guide at:
www.dcnr.state.pa.us/ucmprd2/groups/public/documents/document/dcnr_006106.pdf



Stakeholder Involvement
is critical during all phases of
your trail project.

Photo Credits: Pashek Associates



Steering Committee

Establish a steering or advisory committee to assist your trail designer. The committee also helps you maintain important connections with your community. The committee should include trail user groups, adjacent property owners, local businesses, community leaders, service clubs, cultural and historical groups, recreational groups, and citizens interested in your project.

The size of the committee will depend on the size of the project, but a general rule of thumb is to limit participation to 15 persons or less. You can invite additional people to specific meetings to address a particular area of concern.

Public Meetings

Types of public meetings include informational, data gathering, brainstorming, and work sessions. Before the first public announcement of your meeting, notify your elected officials of the scope and intended schedule for the project, so that they may become well-informed and responsive to their constituents once you make a public announcement.

Traditionally the local media has been the primary avenue to get the word out to the public and to advertise these meetings. However, with the growing popularity of email and social networking, techniques such as email blasts and blog posts can be helpful in encouraging participation in public meetings.

Questionnaire

Developing a survey questionnaire provides planners with an option for gathering input from local citizens. Some people are reluctant to attend or speak out at public meetings. Sometimes you can reach these individuals by giving them an opportunity to complete and return a questionnaire or simply by providing a place for them to mail comments.

Web-based questionnaires are an efficient means to collect data and tabulate responses automatically. You can use email blasts to advertise the survey and encourage participation.

Key Person Interviews

A key person interview typically consists of questioning one individual or groups of individuals with common interests or areas of expertise. Conducting key person interviews can validate and expand upon information gathered by the trail designer, steering or advisory committee, or obtained through other public participation means.

Consensus

Consensus is a decision that takes into account all of the legitimate concerns raised by participants in the process.

Consensus is not unanimity.

Sometimes voluntary agreement of all participants proves impossible to achieve, and a majority decision must be taken.



North Country Trail Hikers
Kellettsville Campground,
Allegheny National Forest



Shenango Trail Hike
Mercer County

Photo Credits: Pashek Associates

Prepare a list of questions or discussion points ahead of time. Additionally, if you intend to request detailed information, inform the interviewee as early as possible so they may gather information and provide it to you during their interview.

A successful public participation process results in understanding the issues and concerns of those who may oppose your project. A successful planning process responds with a plan that adequately addresses the issues and concerns of both those that support and oppose the project. Therefore, identifying concerns and issues early in the public participation process remains crucial to future implementation of your project.

Step 4: Identifying the Needs, Wants, and Concerns of the Land Manager, Property Owner, and Adjacent Neighbors

Understanding the needs, wants, and concerns of the land manager, property owner, and adjacent land owner(s) is just as important as interpreting the needs, wants and desires of your trail's users. Often members of these groups do not use trails. Therefore, their priorities for the land will differ and they will often have legitimate concerns that the designer can address. They may also have concerns based on perceptions, rather than facts. It remains important to identify these concerns early in the planning process. By incorporating an educational component into the planning process you can alleviate most fears.

By listening to land managers, property owners, and adjacent property owners, your trail designer gains an understanding of their views and perspectives. With this information, the designer interprets and evaluates how to address those views and perspectives through the planning and design process. Your designer must have a willingness to listen, to field criticism, and allow angry individuals to vent. However, it is better to have this occur early in the process so you can adjust and respond, rather than later when it can derail the project.



Step 5: Identifying the Needs, Wants, and Desires of Users

When planning, you must gain an understanding of the needs, wants, and desires of potential users to your trail. Users have different goals, desires, and expectations as they travel a trail. If the design of the trail does not meet the standards of its users, the design has failed.

An important step involves defining objectives for creating your trail by answering the following questions:

- What is the purpose of the trail?
- Who will visit the trail?
- How often will they visit the trail?
- What level of trail difficulty will meet the needs of the users?
- Where will the users access the trail?
- What destinations will users want to access from the trail?
- What potential does the trail have to promote tourism and economic development in the area?
- Who will manage the trail?
- What will it cost to maintain the trail?
- What is the liability of developing a trail?

Throughout the planning process, your designer will incorporate and balance users' needs, wants, and desires with the opportunities and constraints presented by the land's natural and cultural resources.

Step 6: Assembling the Mapping

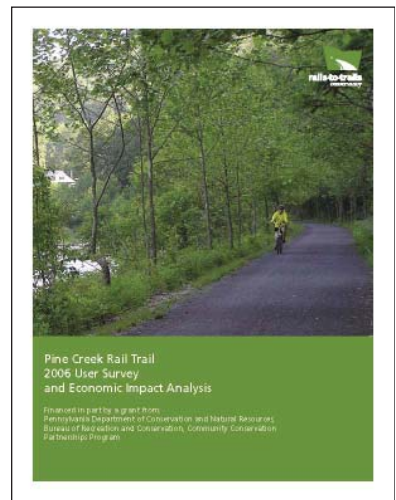
Planning a trail system begins with identifying the location of the proposed trail system and developing a base map of the area.

USGS 7.5 Minute Quadrangles

7-1/2 minute quadrangles, 1-inch = 2,000-feet with 20 foot contour intervals, produced by the United States Geographical Survey (USGS) serve as the most common and readily available mapping. You can download them electronically from the Pennsylvania Spatial Data Access (PASDA), www.pasda.psu.edu. Typically, USGS maps provide a good resource for initial planning and design, but do not provide a sufficient level of detail for final trail design.

PAMAP Program

The PAMAP Program is nearing the completion of an unprecedented project to collect and process high resolution LiDAR elevation data for the entire state. Those familiar with geographic information system (GIS) software can transform LiDAR, an acronym for Light Detection and Ranging, data to



**Pine Creek Rail Trail
User Survey**
Rails to Trails Conservancy

Map Sources

- United States Geological Survey
- PA Map Program
- Penn Pilot
- County Planning Departments
- Municipal Planning Departments



produce various elevation data products including point based digital terrain models (DTM), grid-based digital elevation models (DEM), and contours. The PAMAP Program has processed the LiDAR data using Digital Elevation Models (DEM), two-foot contours, and breaklines (e.g. cultural features such as the edge of a road).

As of this writing, the above LiDAR data products are available for download from PASDA for 54 of Pennsylvania's 67 counties. You can obtain PAMAP LiDAR and derived products at: www.dcnr.state.pa.us/topogeo/pamap/elevation.aspx

Your trail designer can evaluate whether they can use these base maps for design and construction purposes. LiDAR data does not contain property boundaries and parcel information so you need to obtain this information from other sources.

Ortho Photography

Ortho photography is geographically and geometrically corrected aerial photography. A digital form of ortho photographs is frequently used as a base map using GIS or CADD software. Ortho photos are particularly useful in identifying the location of features and destinations. Planners overlay other layers of electronic data over the ortho photos using GIS or CADD. The use of these additional layers enables trail designers to use establish the preliminary trail layout. You can obtain digital ortho photography from county GIS departments. Digital ortho quarter quadrangles (DOQQs) are available for download from PASDA at: www.pasda.psu.edu. A DOQQ covers one quarter of the area contained on a USGS 7-1/2 minute quadrangle.

Historic Photographs

Your trail designer may find historically sequenced aerial photographs extremely useful. They can use historical photographs to see how land use has changed over time. With this information the designer can identify potential problem sites or areas of potential interest to the user, such as historical sites. Aerial photographs covering the past 50 years or more may be available from your county conservation district, and downloaded from Penn Pilot, www.pennpilot.psu.edu. Historical photographs on Penn Pilot date from 1937 to 1942 and 1967 to 1972.

Geographic Information Systems (GIS)

In simple terms, GIS is a tool to combine graphic images and databases of information. GIS is used to analyze data and display the results as spatially oriented visual images. Data sets are graphic representations, with links to all of the data in that particular set. For example, a data set for property may graphically portray property boundary lines and include a link to a database that contains property information, such as: tax parcel number, landowner name, address, acreage, assessed value, etc. GIS software allows your designer to overlay many data sets to create a map of features, interests,



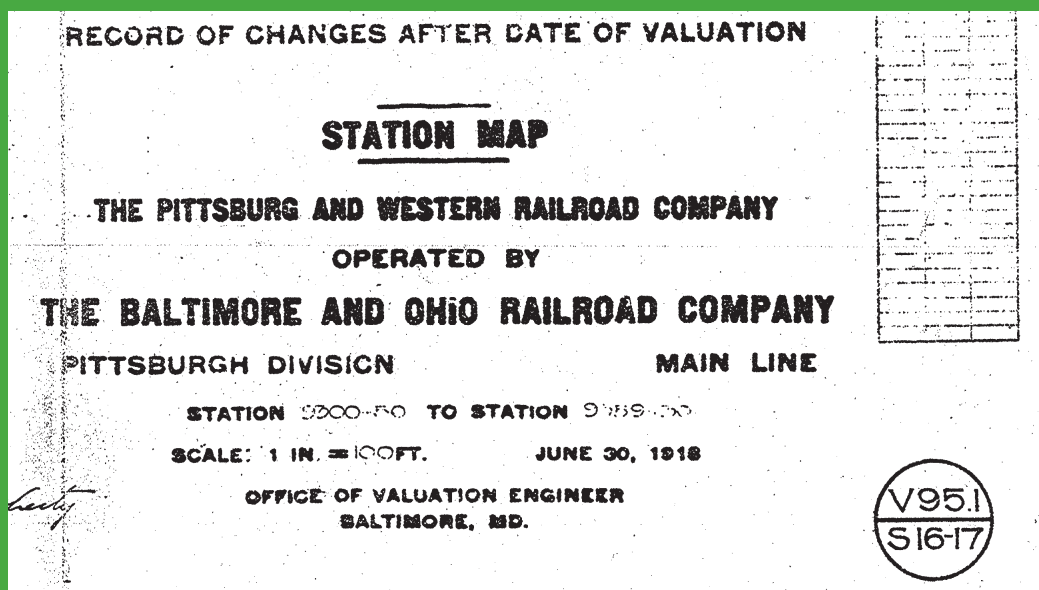
or common data sets for analysis. Your county GIS, planning, 911, and/or assessment offices may have data sets available for your use. You can also download a significant amount of GIS data from PASDA, www.pasda.psu.edu.

All of the information displayed is using real world coordinates and can be displayed to scale. You can quickly measure distances and calculate areas utilizing GIS. You can geo-reference the position of specific point locations acquired with Global Positioning Systems (GPS) to the GIS data.

Data sets of interest to trail designers include: streets and roads, trails, property boundaries, contour lines, streams and lakes, wetlands, floodplains, soils, geology, natural and cultural resources, and sensitive areas. Further, most counties have completed greenway and trail plans. These greenway planning efforts have created data sets that identify sensitive natural resources, as well as proposed trail corridors. Your designer should determine if this data is available for your project area at the beginning of the planning process, and acquire the data if it exists.

Railroad Valuation Maps

Railroad valuation maps can be a good source of information when planning for a rail trail. During World War I the U.S. railroads became grid-locked with war related traffic to such a degree that they literally ground to a halt. The federal government stepped in and “nationalized” the entire system. Among the things they did included requiring the railroad companies to inventory



Railroad Valuation Map
Pittsburg and Western Railroad Company, Station Map
June 30, 1918

Photo Credit: Pashek Associates



everything they owned. This inventory became known as *valuation maps*. Some valuation maps have information on how the land was obtained. Most indicate the width of the rail corridor and identify all structures, bridges, culverts, etc., associated with the corridor at the time.

The railroad valuation maps are warehoused in the National Archives II in College Park, Maryland. For more information on accessing these archives see: www.archives.gov/dc-metro/college-park.

The National Archives maintains a database of its archives in its Archival Research Catalog. You can conduct key word searches in this catalog by visiting: www.archives.gov/research/arc.

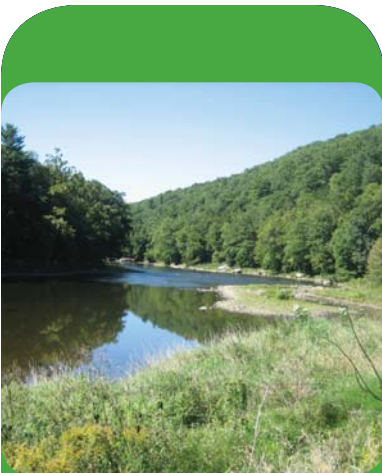
Base Information

Trail planning starts with a good base map of the project area and typically includes:

- boundary survey, locating the metes and bounds of the property, right-of-way, and easements;
- topography survey, identifying the change in elevation throughout the property and including physical features such as streams, wetlands, rock outcroppings, trees and stands of vegetation, buildings, roads, fences, etc.

The scale of the base map can vary. You and your trail designer should mutually agree on the scale of the base map. Ideally, your base map of the trail study area will include the following information:

- Trail name
- Name of municipality, landowner, and trail manager
- Scale, graphic scale, north arrow, date, and legend
- Acreage of site
- Site boundaries with bearings and distances
- Site zoning and zoning of surrounding properties
- Boundary lines of adjacent property parcels where they intersect with the planning site
- Existing uses and ownership of surrounding property
- General location and type of easements on the site
- Deed restrictions on the site
- Topography represented with contours
- Existing structures and facilities including utilities and storm water facilities and rights-of-way
- Circulation patterns (existing access roads/service drives/parking/trails/walks/ramps/paths and bridges)
- Natural and man-made barriers



Streamside Riparian Buffer
Northwest Pennsylvania
Greenways Plan

Photo Credit: Pashek Associates



- Water features (streams/rivers/ponds/lakes)
- Wetlands immediately adjacent to site
- Flood plain - 100 year flood level
- Drainage structures (culverts/catch basins/inlets/ditches/under drains)
- Site control structures (fences/dikes/walls)
- Soils and limiting subsurface conditions
- Boundaries of existing riparian buffers
- Other site features that may impact the use and development of the site

Step 7: Identifying and Documenting Opportunities and Constraints for Developing Your Trail(s)

Once the base map is complete, your designer will then add information to the drawings to help identify opportunities and constraints of the project area. This step typically includes a desktop analysis followed by a site visit to the trail corridor.

This data should include:

- Topographic features such as high points, low points, rock outcroppings, features of topographic and/or geologic interest
- Physical, environmental, historic, and/or aesthetic points of interest
- Views
- Surrounding land uses
- Wind direction
- Environmental issues
- Soil types and drainage capabilities
- Water quality classifications of watersheds
- Riparian (streamside) buffer areas
- Wildlife areas
- Noise/Odors
- Zoning
- Deed restrictions
- Easements that limit use
- Steep slopes
- Abandoned mine features and/or current drilling areas
- Evidence of abandoned mine drainage
- Potential environmental hazards
- Observations on compatibility with adjacent land uses
- Locations and recommended buffers around rare, threatened, and/or endangered species
- Potential vehicular conflicts, note posted speed, estimate traffic volume, and evaluate sight distances
- Locations of buildings and structures including observations of age and historical integrity



- Identification of erosion and sedimentation problems
- Preliminary observations on conditions of structures including bridges, canals, culverts, and/or tunnels
- Locations of infrastructure and utilities noting constraints to potential trail development
- Vehicular intersections and access points
- Potential off-site connections to main streets, adjacent trails, other environmental, cultural, and/or historic resources
- Proximity to motorized trails

A thorough understanding of these elements allows your trail designer to maximize the potential of the trail system while respecting and avoiding sensitive resources that your trail could negatively impact.

Your trail designer will also inventory, locate and analyze any existing non-motorized trails in the area of the proposed system. Document this inventory and analysis on a Trail Assessment Form (refer to Chapter 5 - Management). This analysis will determine whether you can incorporate existing trails into the proposed trail system. For planning purposes, locate existing trails with a handheld GPS unit, accurate to three meters. Then integrate this data into GIS.



Recording Features with GPS
Allegheny National Forest
Equestrian Trails Project

Photo Credit: Pashek Associates

While collecting GPS points along a trail, note basic information about the trail such as:

- Grade of the tread
- Cross slope of the tread
- Tread width
- Tread surface character
- Tread condition
- Road/Railroad crossings
- Location of trailhead and trail access points
- Designated trail use
- Designated level of difficulty
- Location and condition of drainage structures
- Location and condition of signs
- Location and extent of erosion and other negative impacts to the trail
- Location where trail had negative impacts on sensitive resources
- Features trail intersects
- Sustainability of the trail design and layout

This information can also be collected utilizing android or iphone GPS applications and displayed not only on GIS mapping but also in Google Earth.

With this information the trail designer can determine whether you can incorporate the existing trails, in whole or in part, into the proposed trail system; provided the trail meets the designated Trail Management Objectives



(refer to Chapter 4 - Management). Consider closing and obliterating unsustainable segments of the trail and/or segments that you cannot easily maintain. You can incorporate sustainable portions of an existing trail into the new trail system if desired.

Step 8: Determining the Type of Trail System and Level of Difficulty

Your trail designer will synthesize all of the input, mapping and data collected for the proposed trail corridor to determine the appropriate trail system layout(s) and level of difficulty(ies), for your trail.

Trail System Layouts

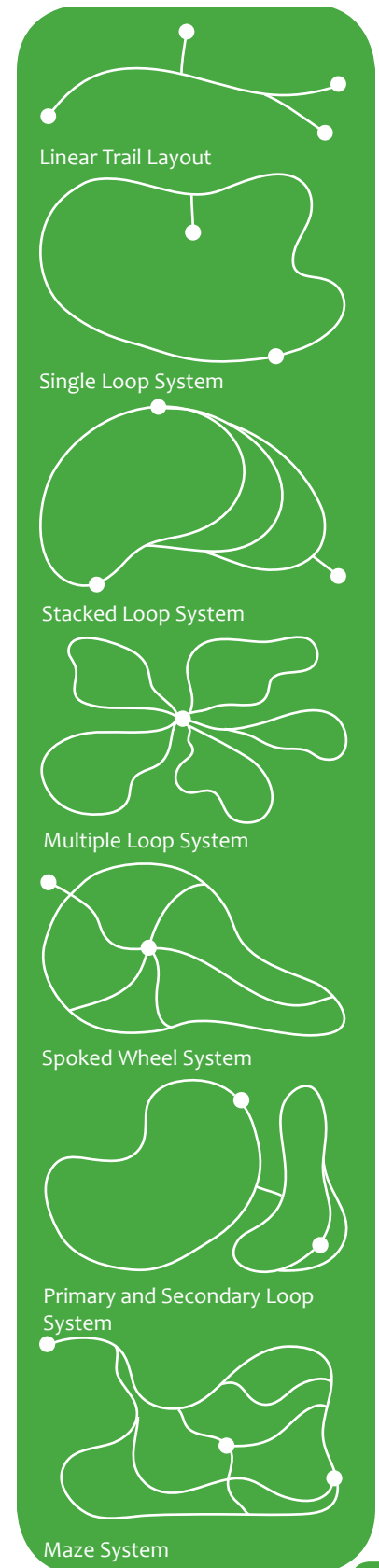
When establishing the preliminary trail layout the first step is to determine the type of layout that best achieves your goals and vision for your trail or trail system. Secondly, it is important for you to determine the level of trail difficulty that you want to provide for your trail users.

Trail designers typically lay out trail systems in one of the following ways:

- **Linear Trail Layout (Point-to-Point)** - The simplest trail layout has a point of origin and a destination. It connects two points or links two trails. Use this layout when points of interest or destinations along a single narrow corridor or to connect other trail systems where the terrain or land ownership restricts the trail to a single narrow corridor. This system may consist of several point-to-point trails that intersect.

Multiple trail access points and trailheads may exist along the corridor. The primary disadvantage to this layout is that the visitor to the trail must return to the point of origin or shuttle vehicles between trailheads. In addition, the trail must have a single level of difficulty over its entire length. Because of their nature, rail trails are normally linear.

- **Single Loop System** - Trail designers use a single loop system most often for a single, long-distance all-day trail, and for a shorter duration trails around lakes, reservoirs, wetlands, and other topographic features. Multiple trailheads may be located along the loop. The disadvantage of a single loop is that the design limits users' choices and the trail must have a single level of difficulty over its entire length.
- **Stacked Loop System** - With a stacked loop system the designer can provide multiple levels of difficulty, with each stacked loop becoming more difficult. This system generally has only one trailhead. A stacked loop system can provide users the opportunity to choose the length and





Locating Existing Trails in Your Area

While conducting your inventory determine if there are trails in your area and whether there is potential to connect to those trails, as part of your project, in the future. To locate existing trails near you visit:

www.explorepatrails.com

duration of their trail experience. This system can allow a party of trail users with varying skill levels to enjoy an outing together.

- **Multiple Loop System** - The multiple loop system typically has a single trailhead, with several loops radiating out from one point of origin. Each loop may have differing level of difficulty, provide a different experience, or provide access to a destination. This system provides the trail designer with the capability to meet user expectations for a variety of users.
- **Spoked Wheel System** - A spoked wheel system consists of an outer loop connected by linear trails to a center trailhead, destination, or center loop. Multiple trailheads along the outer loop may also be established. This system can provide varying levels of difficulty and experiences on each of the spokes.
- **Primary and Secondary Loop System** - This system has a primary loop, with multiple secondary loops. The secondary loops may offer a different level of difficulty, provide a different experience, or provide access to a destination. This system provides the designer with the most flexibility to meet expectations for a variety of users.
- **Maze System** - The maze layout incorporates a system of loops and linear trails that provide many alternative routes. A maze system can provide a variety of experiences, however, it usually accommodates a single level of difficulty. Orienteering events can use this trail system, where participants must visit checkpoints in a particular sequence. Users can become disoriented or temporarily lost in this system. A maze trail layout should be well-mapped and well-marked.



Trail Difficulty Rating System

The second step in establishing the preliminary trail layout is to plan the level of difficulty for your trail(s).

Users of your trail desire to know what to expect before arriving at your trail. CNR, in their publication “Guidelines for Marking Recreational Trails”, has adopted the National Trail Difficulty Rating System, www.dcnr.state.pa.us/ucmprd2/groups/public/documents/document/dcnr_007076.pdf.

Typically, trail designers should strive to design the majority of trails at the “More Difficult” level because this represents a majority of the trail users. Since most trail managers and local tourism officials often have an objective to attract more uninitiated recreationists, the next highest proportion of trails should be rated “easiest.” Trail users desiring a “Most Difficult” rating represent the smallest constituency in a typical trail system.

The National Trail Difficulty Rating System evaluates measurable criteria to determine the trail rating. Criteria examined in this rating system includes tread width, tread surface, trail grade, and natural obstacles. Other conditions can affect difficulty, including exposure, steep drop-offs, and remoteness.

Trails are rated with respect to other trails in the region. The basis for the National Trail Difficulty Rating System revolves around technical challenge, not physical exertion. The three classifications of trail difficulty include:

1. Easiest
2. More Difficult
3. Most Difficult

- **Easiest** - Suitable for beginning trail users and those who do not have the skill or desire to use “more difficult” trails. These trails have a low level of risk for the user and consequently offer less variety than those of greater difficulty. The layout of these routes are appropriate for novice to advanced users and generally follow obvious, well-marked routes. Grades are gentle and few obstacles should be encountered. They require little skill and entail little physical challenge. The tread should be smooth, level, and wide, with generous clearing of trees, limbs, and other vegetation above and to each side of the trail to permit easy passage. Elevation gain or loss is minimal. Most often stream crossings have bridges at this level of difficulty.
- **More Difficult** - Designed to meet the expectations of the majority of trail users. These trails require skills beyond that of a novice and at times will challenge the average trail user. These routes are appropriate for intermediate to advanced users. Terrain may be steeper, trails narrower,

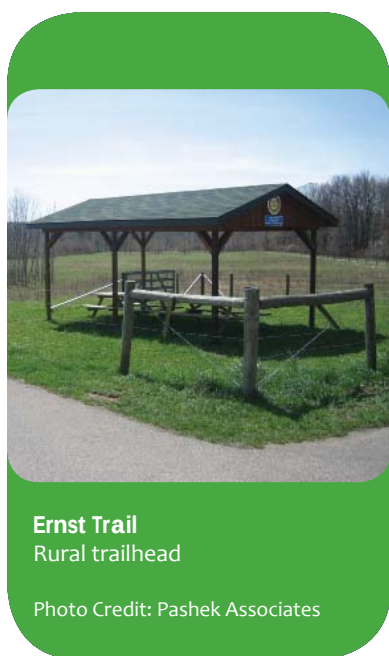


Level of Difficulty Symbols
National Trail Difficulty Rating System



and some obstacles may be encountered. They require a minimal skill level and provide a minimal physical challenge. The tread surface can contain roots and embedded rocks. Clearing of trees, limbs, and other vegetation above and to each side of the trail may result in occasional contact by trail users. Elevation gain or loss is moderate. Most often streams are crossed with fords.

- **Most Difficult** - Designed for trail users with advanced skill, who are seeking a higher risk level. Only advanced to expert users should attempt these routes. The design incorporates steep terrain, and routes with few marks. Trail users should have considerable skill in their chosen activity, as well as knowledge of navigation and survival before attempting these trails. They require a high degree of skill and provide a definite physical challenge. Trail users will seldom encounter a graded tread except on steep side slopes, for safety and prevention of soil erosion. Minimal clearing of trees, limbs, and other vegetation hampers the progress of the user. Usually elevation gain or loss is severe. Streams crossings generally consist of fords and they can sometimes present a challenge.



Ernst Trail
Rural trailhead

Photo Credit: Pashek Associates

Step 9: Determining the Location(s) of Potential Trailheads

Another step in the planning process is determining potential locations for trailheads. Since most trail users drive to the trail for day and weekend trips, emphasize trailhead planning during the earliest phases of planning. Take into account the same concerns you do for trail design: recreation, safety, convenience, and minimal environmental impact.

Not every road crossing should have a trailhead. Avoid or limit trailhead development at dangerous road crossings. Selective development of trailheads regulate the volume and location of users along the trail. Where the trail would deteriorate under heavy traffic, discourage overuse by limiting parking or by developing trailheads only along other nearby, more stable sections of trail.

Remember that trailheads, more than any other trail features, influence the relationship between the trail community and local residents, towns, and counties. People living near the trail judge trail users by what they see at trailheads. Local residents will quickly become impatient with overnight parking in inappropriate locations, persistent highway-safety concerns on narrow or curved roadways, noise and littering, late-night arrivals, and invasions of their privacy from hikers requesting food, water, and the use of phones and facilities.



To be thorough during trailhead planning consider the following¹¹:

- Study existing use patterns: Document use levels along 5 to 15 mile sections of trail by counting parked cars on average and peak weekends. Look at the whole trail system. Remember, you do not need to develop a trailhead at all road crossings. Trailheads at several key road crossings may be all that is required to meet your trail's needs.
- Check with local residents: Local police, municipal or county officials, PennDOT, and local residents can help you avoid or correct potential problem spots. They will appreciate your concern and interest.
- Look for good spots: Inquire or field-check locations that might accommodate parking without additional construction. Wide-shouldered roads, picnic areas, state waysides, sections of old roadbed, and old gravel pits often provide ample space for parking.
- Evaluate motor vehicle access safety: Seek advice from highway safety engineers and local residents to evaluate the safety of access to the roadway, pedestrian crossings, and the local impact of any proposed lot.
- Adhere to standards: Use the standards followed by most highway departments for parking access. On blind hills and curves, parking is unacceptable. On all roadways, clear lines of sight must extend down the roadway, usually 200 to 500 feet, depending on the speed limit.
- Avoid overused areas: Plan to keep parking space to a minimum if an area appears fragile or threatened by overuse.
- Consider landforms: Choose well-drained sites above the 100-year floodplain, without steep banks, on slopes of less than five percent grades; otherwise, the development of sites may be costly, impractical, or require excavation.
- Determine necessary permitting requirements: You may need to acquire a driveway permit to access public roads.
- Discourage vandals: Make policing for vandalism easy by locating parking lots where they are clearly visible from the main road.

Trailhead facilities may include a permanent shelter with restrooms, modular/portable restrooms, picnic areas and shelters, utility or storage buildings, information kiosks, parking, drinking fountains, seating, bike racks, and waste receptacles.

A trail may cross several jurisdictions so it is important for the planner to determine in which of Pennsylvania's 2,562 municipalities the proposed building is located.



Kinzua Valley Trailhead
McKean County

Photo Credit: Pashek Associates



Step 10: Determining Regulatory Requirements

During the planning process it is important for you to identify the regulatory requirements your trail and trail related facilities must meet.

PennDOT Highway Occupancy Permits and Highway Occupancy Agreements

Trail or trailhead access through PennDOT rights-of-way or location of extended utilities, such as sewer, water, and electric, may require a Highway Occupancy Permit (HOP) and or Highway Occupancy Agreement (HOA).

Obtain instructions for HOP applications, and sample HOAs from your regional PennDOT office. Trail designers must identify this need early in the process, so there is sufficient time to coordinate with the PennDOT District Permit Section in order to acquire the necessary permit and/or agreement.

PennDOT Memorandums of Understanding in Lieu of Highway Occupancy Agreements

The Highway Occupancy Agreement process establishes a relationship between PennDOT and trail owners/sponsors when a recreational or transportation trail crosses a State highway. Since this process does not differentiate between types of trails or crossings, PennDOT has decided that these unique areas should be subject to procedures other than the HOA process. This is reflected in the Memorandum of Understanding (MOU) between PennDOT and either a trail sponsor, or DCNR, depending on who is responsible for the respective trail. There are five specific PennDOT Memorandums of Understanding/Agreements for the construction of current and future trails. They include:

- Memorandum of Understanding Between DCNR and PennDOT Regarding Footpaths on DCNR Lands Crossing State Highways
- Amendment to the Memorandum of Understanding Between DCNR and PennDOT Regarding State Forest Hiking Trail Crossings State Highways
- Master Agreement Between Trail Sponsors and PennDOT Regarding Footpaths on Lands Crossing State Highways
- Memorandum of Understanding Between DCNR and PennDOT Regarding Shared Use Paths on DCNR Lands Crossing State Highways
- Master Agreement Between Trail Sponsors and PennDOT Regarding Shared Use Paths on Lands Crossing State Highways

INSERT PennDOT WEB LINK



Water Tank Hollow Vista
State Route 44, Potter County

Photo Credit: Pashek Associates



Uniform Construction Code

Trail related structures may fall under the jurisdiction of the Commonwealth of Pennsylvania's Bureau of Occupational and Industrial Safety who administers and enforces the Uniform Construction Code, Fire and Panic Law, Universal Accessibility Law, Energy Conservation Law, General Safety Law, Boiler Law, and Elevator Law. The Uniform Construction Code applies to construction, alteration, repair, and occupancy of all buildings in the Commonwealth.

The ICC International Building Code is updated triennially and the current version is the 2009 IBC.

Over 90% of Pennsylvania's municipalities have elected to administer and enforce the UCC locally, using their own employees or certified third party agencies. The state enforces the UCC for the remaining municipalities. Inspections required during construction will include:

- Foundation inspection
- Plumbing, mechanical, and electrical inspection
- Frame and masonry inspection
- Wallboard inspection
- Final inspection

If a municipality has "opted out," the Pennsylvania Department of Labor and Industry is responsible for all commercial code enforcement in that municipality.

Local Codes

Trail and trail related development may also be required to meet regulatory requirements of local municipalities. Each of Pennsylvania's 67 counties has established a planning commission to oversee the development of land in the county. In addition, some municipalities have created their own planning commissions to regulate development under their jurisdiction. The planning commission ensures that development in certain areas is consistent with a comprehensive plan which indicated the type(s) of development permitted.

A land development review process facilitates the permit process, requiring developers to obtain approval for land development plans. These plans must show all aspects of a project, including but not limited to: proposed building locations, road/trail alignments, and stormwater management facilities. Additionally, municipal authorities usually ask for evidence indicating that you have obtained all required permits. When designing trails or trailhead facilities you may need to prepare, submit, and obtain approval of a land development plan. Contact your county and/or local planning commission and inquire about any requirements to determine if you need such a plan.



In addition to land development approval, you may need to obtain a zoning waiver or special exception. If the zoning classification does not include trails or trail-related facilities as a permitted use, you may need to apply for a zoning waiver or special exception in order to build a trail or related facility on the property. Similarities exist between this process and the land development process; however, the zoning board grants approval rather than the planning commission.

Sewage Facilities

If the design of the trail includes restroom or other sanitary facilities, there are a few things to consider regarding permitting. If the facilities will connect to a public sewerage system, you must complete a sewage planning module or a sewage planning module waiver. If the facilities are in a remote location and will be self-sufficient, then you will need either a holding tank permit for a holding tank structure or you will need to complete a sewage planning module for a septic tank, sand mound structure, or other approved system. When considering the type of sanitary facilities to install, first contact the sewage enforcement officer (SEO) responsible for the municipality in which the facility will be located. If there is no SEO for the area, contact your local county or regional office of DEP for guidance.



Sweet Smelling Restroom Technology
Lyman Run State Park

Photo Credit: Pashek Associates

Step 11: Identifying Environmental and Cultural Resources

When planning for a sustainable trail, you must identify sensitive environmental and cultural resources and determine whether your proposed trail will have any negative impacts on the identified resources. If negative impacts are possible, then you should determine whether the potential negative impacts can be avoided or appropriately mitigated.

Pennsylvania Natural Heritage Program (PNHP)

To determine whether rare, threatened, or endangered species are located within the vicinity of your project area conduct an online search of the Pennsylvania Natural Heritage Program (PNHP) database.

PNHP partners include: Pennsylvania's Department of Conservation and Natural Resources (DCNR), Pennsylvania Fish and Boat Commission, Pennsylvania Game Commission, U.S. Fish and Wildlife, and the Western Pennsylvania Conservancy.

The PNHP maintains an inventory of areas throughout the state that are home to endangered and/or rare species of plants and/or animals. The identified areas are located and organized within a database, referred to as the Pennsylvania Natural Diversity Inventory (PNDI), www.naturalheritage.state.pa.us.



The online search compares the proposed project location to the PNDI database and identifies any conflicts that might occur between the proposed project and existing natural features, vegetation, wildlife, and habitats. A report then features these results with instructions in instances where potential conflicts exist. Potential conflicts with identified species must be resolved with the appropriate agency.

A PNDI database search must occur when acquiring any PA DEP permit. Trail designers typically need to conduct a PNDI search for trail projects.

Large projects that cannot be drawn utilizing the online Environmental Review Tool interactive map and must be submitted by mail to all of the Jurisdictional Agencies in Pennsylvania for review.

“Large Projects” consist of:

- Projects that exceed 15,000 feet (2.84 miles) East/West x 15,000 (2.84 miles) feet North/South (5,165 acres)
- Township wide, County wide or Statewide Projects. Examples: Act 537 Sewage Plans, Wind Farms, or Roadway Improvements exceeding map limits above.

Many counties have completed County Natural Areas Inventory (CNAI) studies that also identify rare, threatened, or endangered species of plants and animals and other critical natural areas within the county. Contact your county planning office to see if a CNAI is available. If one is available, review it to determine if your proposed project poses any potential conflicts to avoid or remediate during the planning of a trail.

DCNR Bureau of Forestry Environmental Review Policy

Before initiating any project on state forest lands that has the potential to disrupt or alter the environment, the impacted districts must complete an environmental review. DCNR Bureau of Forestry’s Environmental Review Guidelines can be obtained at: <http://www.dcnr.state.pa.us/forestry/ROW/Process/SFER/State%20Forest%20Environmental%20Review%20Guidelines.pdf>

The DCNR Bureau of Forestry environmental review consists of an assessment of the project’s likely adverse and beneficial impacts on each factor. Where the Bureau of Forestry predicts a an adverse impact, the review must contain an explanation of the corrective measures that will be taken, or justification if no corrective action will take place.

Watershed and Water Resources Management

Avoid stream corridors, wetlands, floodplains, and lower portions of north-facing slopes. Placement of trails within these areas can interfere with stream transport of sediment and debris, groundwater movement, and floodplain dynamics.

Environmental Review Agencies

- Department of Conservation and Natural Resources
Bureau of Forestry
Ecological Services
Section
400 Market St.
PO Box 8552
Harrisburg, PA 17105-8552
- Pennsylvania Fish and Boat Commission
Natural Diversity Section
Division of Environmental Services
450 Robinson Lane
Bellefonte, PA 16823-7437
- Pennsylvania Game Commission
Bureau of Wildlife
Habitat Management
Division of Environmental Planning and Habitat Protection
2001 Elmerton Avenue
Harrisburg, PA 17110-9797
- United States Fish and Wildlife Service
Endangered Species
Section
315 South Allen Street
Suite 322
State College, PA 16801



Streambed

Avoid flowing stream crossings

Photo Credit: Pashek Associates

Lower portions of north-facing slopes tend to remain wet because groundwater moving downhill comes to the surface in these areas. Furthermore, the northerly aspect generally receives less direct sunlight than south, east, or west-facing slopes, allowing these areas to remain wetter.

Fish Habitat

Locate trails outside the riparian (streamside) corridor to protect stream banks from erosion, conserve riparian shade, and allow recruitment of large woody debris and sediment to the stream. This habitat helps maintain cooler temperatures during warm weather to protect cool-water fish like salmon and trout.

The roots of riparian vegetation help to anchor soil and stabilize banks. Major disruptions to roots can result in sediment delivery exceeding the natural level of suspended sediment. This may lower water quality and contaminate fish spawning beds. If unchecked, erosion can result in an increase of stream width, sediment in the water, and higher water temperatures.

Social trails and trail crossings located close to streams or wetlands lead to trail compaction and usually loss of vegetation. This can result in an increase of erosion and delivery of sediment to nearby water bodies.

Cursory Wetland Review

The Commonwealth of Pennsylvania, in 25 PA Code Chapter 105, describes wetlands as:

“Those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support a prevalence of vegetation typically adapted for life in saturated soil conditions...”

This definition describes marshes, bogs, swamps, wet meadows, and shallow ponds. Wetlands play a very important role in the Pennsylvania ecosystem. Wetlands help control flooding, help purify water supplies, and offer habitat for many of Pennsylvania’s rare, threatened, or endangered species.

A site must display all of the following three characteristics to be considered as a wetland:

1. The presence of hydric soils (wet or spongy soils);
2. The presence of hydrophytic plants (water-loving plants); and
3. The presence of standing water during the growing season.

A majority of Pennsylvania’s wetlands occur inland and are not subject to tidal influence; however, some regions of Pennsylvania, including areas around Erie and Philadelphia, do support areas of coastal wetlands.



During the planning process for any new trail or trail facility, conduct a desktop analysis of the National Wetland Inventory (NWI) to map and indicate soils on-site. Obtain NWI maps through county conservation districts or the PASDA website. Additionally, you can obtain soils information through the USDA National Resources Conservation Service (NRCS) website.

Although NWI maps indicate current wetlands and site conditions, they do not identify many potential wetlands. A qualified soil or wetland scientist should conduct a field investigation to determine the presence of wetlands on a site if its soils appear hydric. Determine if a site is in or near an area identified on a National Wetland Inventory (NWI) map.

If you find wetlands present in the trail corridor, the regulatory agency must delineate and review boundaries around these areas. Once identified, the Army Corps of Engineers in conjunction with the Pennsylvania Department of Environmental Protection establish regulatory authority over these locations.

Stormwater Management

Do not underestimate the significance of water on the trail. High volume and high velocities of concentrated water flow on a trail results in significant erosion. Erosion also has a negative impact on water quality and adjacent habitats, as well as significantly increases the maintenance needs of the trail.

A smaller footprint limits the amount of water a tread receives and reduces the time it takes to direct the water from it. A sustainable trail follows the contour of the land in lieu of going against the grain of the contour, provides positive drainage away from the trail tread, avoids negative impacts to adjacent sensitive resources, and uses control points and features to keep users on the trail and to define the trail experience.

Avoid Steep Trail Pitches: To prevent water from concentrating on steep trails, avoid creating short trail segments steeper than 10 percent. If water concentrates itself on a trail, erosion problems can occur at the site where the water runs off the trail. Further, you will find it difficult to control drainage on steep earthen trails and is expensive to repair.

Encourage Infiltration: Before selecting a trail's design and materials, evaluate the trail's location and width, as well as the range and level of user groups. Then select designs and materials that facilitate infiltration and prevent stormwater runoff.

Avoid Aligning Trails with the Slope: To avoid water running down the surface of the trail, align trails perpendicular to the slope.

Water Crossings: Perpendicular routes should descend to water crossing on both sides to prevent the stream from flowing down the trail. Consider armoring the trail in areas that dip down to avoid sediment and



trail runoff from entering the watercourse.

Avoid Long, Sustained Grades: Do not construct trails with long, sustained grades that can concentrate runoff on their surface. By installing rolling dips and grade breaks, runoff will direct itself off the trail and grade breaks will provide areas for users to rest.

Avoid Flat Ground and Steep Cross-Slopes: Construct sustainable trails on sideslopes with cross-slopes between 5 and 25 percent, to maximize drainage and decrease trail construction and maintenance efforts. Trails on flat ground do not drain well. However, trails with steep slopes require larger excavations and run the risk of sloughing. Increased erosion and degradation can increase maintenance costs and lead to users forming bypass trails.

Avoid Discharging Trail Runoff: Concentrated runoff from trails can cause damage to fill slopes and unprotected soils adjacent to the trail. Carefully select trail locations to slow runoff velocity so sediments can settle out. Armor fill slopes where you intend to discharge runoff onto them or convey runoff off-site where it can be infiltrated.

Trails can change the timing, quantity, and quality of the natural hydrological system by delivering both sediments and runoff directly to streams, wetlands, and riparian resources.

Avoid Removing Trees and Shrubs at Stream Crossings: Use existing roads and bridges to avoid removing trees and shrubs at crossings and to avoid new stream corridor disturbances. Consider routes that can handle drainage without affecting nearby water resources. Some measures to avoid these impacts include trail narrowing, encouraging filtration, providing frequent drainage, and spreading crushed aggregate to enhance drainage.

Avoid Stacking Switchbacks and Climbing Turns: Carefully locate climbing turns and switchbacks to prevent user cut-throughs and short cuts. Offset them from one another to take advantage of natural benches, slope breaks, and natural screening. To discourage shortcuts, you may increase grades and use rocks, trees, or log barriers while constructing the turn.

Pennsylvania Department of Environmental Protection Permits
Pennsylvania's Department of Environment Protection (DEP) requires permits for many construction activities that may negatively affect the environment and/or its citizens. Some of the permits that DEP may require when designing a trail or trail facilities include:



- **Chapter 102 - Erosion and Sediment Control and Stormwater Management Permits** required for anyone proposing to discharge storm water from construction activities disturbing one acre or more of land.
- **Chapter 105 Waterways and Wetlands Management Permits** required for persons planning to construct, operate, maintain, enlarge, or abandon a water obstruction or encroachment (such as a bridge or channel change) in a watercourse, its 100 year floodway or any lake, pond, reservoir, or wetland. Waters of the Commonwealth of Pennsylvania, including streams and wetlands, springs, vernal pools, and seeps are subject to PA DEP regulations.

The permits described herein are a sampling of the number of permits that DEP could require. Look for information on these and other potential permit requirements at www.depweb.pa.us, keyword: Permit.

Chapter 102 - Erosion and Sediment Control and Stormwater Management Requirements

PA DEP's Chapter 102 - Erosion and Sediment Control and Stormwater Management regulates earth disturbance activities. Chapter 102 requires that persons proposing or conducting earth disturbance develop, implement, and maintain Erosion and Sediment Control (E&SC) Best Management Practices (BMPs) to decrease erosion and the potential for pollution to water resources. In addition, under the Federal Clean Water Act, projects that have 1 acre or more of earth disturbance over the life of the project may require an National Pollutant Discharge Elimination System (NPDES) Permit for Stormwater Discharges Associated with Construction Activities.

Chapter 102 regulations require all earth disturbance activities exceeding 5000 square feet to have an adequate erosion and sediment control (E&SC) plan designed for the site. The plan must show the locations and details of E&SC BMPs that protect against accelerated erosion. Examples of E&SC BMPs include compost filter logs, mulch, diversion channels, sediment traps and basins, and the planting of grasses or similar vegetation. The plan must show the site, location of the BMPs, and timing and sequence of their installation for maximum erosion control.

The county conservation district can offer guidance to anyone developing a plan. Once completed, submit the plan to the conservation district for review. For earth disturbance activities less than 5,000 square feet, you typically do not need a written plan but the appropriate BMPs must be installed.

National Pollutant Discharge Elimination System Permit

Projects disturbing one or more acres must acquire authorization through a NPDES Permit for Stormwater Discharges Associated with Construction Activities before beginning any earth disturbance activity.



Major components of a NPDES Permit application include:

- Erosion and Sediment Control Plan
- Pennsylvania Natural Heritage Program Clearance
- Post Construction Stormwater Management Plan
- Thermal Impact Analysis
- Antidegradation Analysis

Cultural Resource Geographic Information Systems

Cultural Resource Geographic Information Systems (CRGIS) is a map-based inventory of the historic and archaeological sites and surveys stored in PHMC's Bureau for Historic Preservation.

Currently there are 21,473 archaeological sites and 128,985 historic properties on record. Access to paper records is free and open to the public by appointment at the BHP office in Harrisburg.

CRGIS is a means of accessing some of this data without a trip to Harrisburg.

Web access to all of the historic resource data is open to the public. Access to archaeological site locations and detailed site information is restricted and password protected and will be granted to qualified individuals on a need to know basis. Two levels of restricted access are available: one for planning professionals and one for archaeological professionals.

To access the CRGIS visit:
crgis.state.pa.us

Look for information on these and other potential permit requirements at www.depweb.state.pa.us, keyword: Permits. Additional departmental permits, approvals and coordination could include Water Obstruction and Encroachments Permit, Chapter 105 general permits, and Brownfield Remediation. Failure to apply for and obtain all necessary permits and approvals may result in the delay of construction.

Chapter 105 Waterways and Wetlands Management Requirements

PA DEP's Chapter 105 – Waterways and Wetlands Management regulates work in and around streams, wetlands, and other water bodies with a focus on avoiding or minimizing impacts on water quality, flooding frequency, and riparian habitats. Under a Delegation Agreement, the local Conservation District administers DEP General Permits for various types of work in waterways, including wetlands and stream crossings, streambank stabilization, boat ramp and dock installation, intake and outfall structures, and utility line crossings. District staff inspects permitted sites, investigates complaints, and provides regulatory and permitting assistance.

Cultural Resources Management

Under the Pennsylvania History Code and the National Historic Preservation Act, the Pennsylvania Historical and Museum Commission (PHMC) serves as the State Historic Preservation Office. The Commission's Bureau for Historic Preservation administers all official state historic preservation programs and activities.

Pennsylvania Historical and
Museum Commission
Bureau of Historic Preservation
400 North Street, Second Floor
Harrisburg, PA 17120-0093

One role of PHMC involves advising and aiding state and federal agencies in complying with legislation including the National Historic Preservation Act and the PA History Code, which protect historic properties. To meet this goal, the Bureau has established the Environmental Review process, through which it reviews projects for their potential to affect historic properties.



When planning a construction project that may disturb a historic structure or area, you must contact the PHMC to determine if the structure or area has significant historical value. If a site has historic significance the PHMC may require additional research, possibly including archeological surveys of the area.

Any construction on land within the Commonwealth requires a PHMC review. PHMC can also provide assistance at the request of the planner or designer. When planning trails or trail facilities it is prudent to contact PHMC during preliminary design. This allows them to locate any nearby cultural or historical resources and eliminate the need to redesign the project if a resource is found later.

Start by completing and submitting a cultural resource notification. PHMC has developed the Request to Initiate Consultation in Compliance with the *State History Code and Section 106 of the National Historic Preservation Act form*, www.portal.state.pa.us/portal/server.pt/community/historic_preservation/3741/forms_and_guidance/418107.

Document all construction activities involving structural and/or ground disturbance on the form and submit it to PHMC for review.

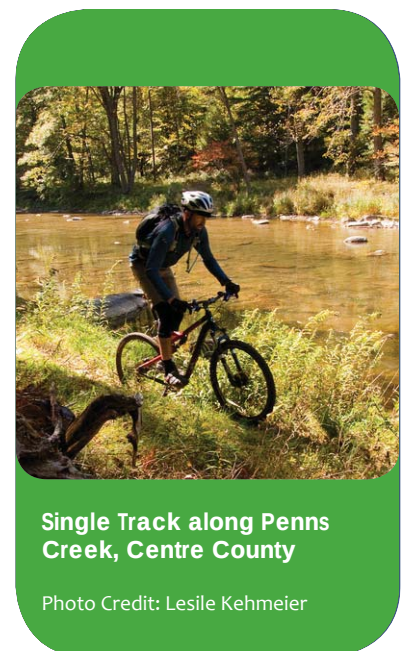
The following basic project information is required on the form:

- **Map Location:** Include a 7.5 USGS Map showing the project boundary and the Area of Potential Effect (APE). The APE should include indirect effects, such as visual and audible impacts. Federal projects must provide an explanation of how the APE was determined.
- **Photos:** Include photos of all buildings or structures in the APE. For properties over 50 years submit a Historic Resource Survey Form with the initial request.
- **Project Description Narrative:** Provide a detailed project description describing the project, any ground disturbance, any previous land use, and age of all affected buildings in the project area. Attach a site map showing the location of all buildings in the project area.

You will typically receive a written response from PHMC within 30 days.

PHMC's response may take one of the following forms, depending on the circumstances:

- A determination of no resources and/or no effect for the project, indicating you have completed consultation with the SHPO office.
- Notification that historic structures and/or archaeological resources



Single Track along Penns Creek, Centre County

Photo Credit: Lesile Kehmeier



Typical Water Obstruction & Encroachment General Permits for Trail Projects

- **PAG-2 NPDES**
GP: Stormwater Discharges Associated with Construction Activities
- **GP-03: Rehabilitation, Bank Protection & Gravel Bar Removal**
- **GP-04: Intake & Outfall Structures**
- **GP-05: Utility Line Stream Crossings**
- **GP-06: Agricultural Crossings & Ramps**
- **GP-07: Minor Road Crossings**
- **GP-08: Temporary Road Crossings**
- **GP-11: Maintenance, Testing, Repair, Rehabilitation, or Replacement of Water Obstructions and Encroachments**

exist within the project area and that project area needs a survey or other investigation completed to evaluate the National Register eligibility of these resources.

- Notification that, although no known archaeological sites exist within the project area, the physical setting of the project, combined with comparative information from similar settings nearby, suggests a high probability that archaeological sites exist within the project area and that the project area will require a survey to locate these sites.
- Notification that, although no known historic buildings, structures, districts, or objects exist within the project area, USGS quadrangle maps show buildings present. Therefore, a survey of the project area is required to identify potential historic buildings, structures, districts, and object.
- Determination that the project will have no adverse effect on known or newly identified historic and/or archaeological resources and that, unless plans change, consultation is complete.
- Determination that the project will have an adverse effect on known or newly identified historic and/or archaeological resources and the project will require further consultation.
- A notice that PHMC received too little information to perform a review, in which case the PHMC will attach an itemized checklist of types of information with their response.

Step 12: Planning for Accessibility

In 1981, the World Health Organization published the International Classification of Impairment, Disability, and Handicap (ICIDH) to create a standardized level of communication and terminology on an international level. Within this document, the ICIDH defined three health-related restrictions of an individual's level of function: 1) impairment, 2) disability, and 3) handicap. Impairment is a function of the individual, while a disability or handicap occurs because of limitations imposed on the individual by the community in which he or she lives. Thus, disability and handicap are functions of the environment rather than the individual.

Functions such as agility, balance, flexibility, endurance, hearing, and problem solving help determine an individual's ability to safely access a facility. Persons with activity limitations, resulting from impairment, may have trouble when performing various actions. It is important for designers to consider these users' needs, as well as to consider how they will meet demands in the future to promote active lifestyles.

The term accessible refers to a facility that persons of all abilities can approach, enter, and use.



During the planning phase it is important to incorporate the philosophy of universal design in the planning process. Specific design requirements for accessible trails is detailed in Chapter 2 - Design Needs for Specific Non-Motorized Trail Users.

Designers must integrate universal design strategies into the design process whenever possible. Newly constructed trails should address these issues during the planning process to eliminate changes in a design during and after construction.

Every aspect of a trail experience should consider including accessible built facilities, amenities, trailheads, and trail corridors to ensure a design's versatility. Moreover, a trail's environment and benefits should be enjoyable and appeal to those with and without disabilities.

Step 13: Establishing the Preliminary Trail Alignment

Once your trail designer has a firm understanding of your vision and goals, the locations of sensitive resources, the opportunities and constraints presented by the land, the desires of the users, and the concerns of land managers, property owners, and adjacent neighbors, you can begin to establish the preliminary alignment for the trail system. Take into consideration the sustainable trail requirements, detailed in Chapter 2, and the specific trail-user design criteria, detailed in Chapter 3, to achieve the desired level of difficulty.

With this information your designer can establish the preliminary trail layout by reviewing the inventory and analysis mapping, and laying out the trail segments following the sustainable trail design criteria referenced in these guidelines.

Sustainable Trail Planning Guidelines

The design of sustainable trails begins during the planning phase. Sustainable trails follow the contour of the surrounding landscape, gradually climb, and direct water away from the trail. Regardless of trail type, your trail can be sustainably designed by following the International Mountain Bicycling Association's (IMBA) five essential elements of sustainable trails⁹:

1. **The Half Rule:** A trail's grade should not exceed the half grade of the hillside or sideslope that the trail traverses. If grade does exceed half of the sideslope, consider it a fall line trail that will be susceptible to erosion.
2. **The Ten Percent Average Grade Guideline:** Generally, a trail with an average grade of 10 percent or less is most sustainable. This does not mean you need to keep all grades less than 10 percent. Many sections of a trail will have short steep sections greater than 10 percent, and some unique situations will allow average grades of more than 10 percent.



York County Heritage Trail
Providing for accessibility with a stable and firm trail tread.

Photo Credit: Carl Knoch



3. **Maximum Sustainable Grade Trails:** Maximum sustainable grade equals the steepest section of trail that is more than ten feet in length. When designing a trail, it is essential to determine early in the process the maximum grades the trail will be able to sustain given local conditions. Variables that impact the maximum sustainable grade include:
 - Soil Type
 - Rock
 - Annual Rainfall Amount
 - Grade Reversals
 - Type of Users
 - Number of Users
 - Planned Level of Difficulty
4. **Grade Reversals:** A grade reversal occurs at spot where a climbing trail levels out and then changes direction, dropping subtly for about 10 to 50 linear feet before rising again. This change in grade forces water to exit the trail at the low point before it can gain volume, velocity, and erosive power. Other names for grade reversals include dips, grade breaks, drainage dips, or rolling dips.
5. **Outslope:** As the trail contours across a hillside, the downhill or outer edge of the trail tread should tilt slightly down and away from the high side. This encourages water to flow across and off the trail.

When you are designing sustainable trails, consider the level of trail development required based upon the location, use, and other factors. Some trails will consist of a natural surface, while others will consist of more developed surfaces.

Common Trail Building Mistakes

Make every effort to avoid the ten most common trail building mistakes¹⁰ as summarized by the International Mountain Bicycling Association (IMBA) to ensure for the design and maintenance of safe, sustainable trails.

Not Getting Land Manager Approval: An important step to complete before starting trail work is to obtain approval from the landowner or manager. Failure to secure permission, usually ends up being the single biggest cause of trail closures and may result in further complications for users, builders, and designers.

Following the Fall Line: Fall line trails usually follow the shortest route down a hill, following the same path water flows. Natural and user-created erosion strips the trail of soil, exposes roots, creates gullies, and scars the environment. To build trails that last, use the Half Rule: trail



grade, or steepness, should not exceed half the grade of the hillside; and the 10 Percent Rule: Keep the overall trail grade 10 percent or less.

Guessing the Grade: Always use a clinometer to confirm the grade when laying out a trail, because no amount of trail improvement can fix a trail built at an unsustainable grade. This accurate, relatively inexpensive tool measures the grade, or steepness, of a trail in degrees or percent for designers and builders to construct a trail route correctly.

Going Against the Flow: All trail builders should ensure a trail has smooth transitions and good flow. This ensures an enjoyable trail experience. By designing smooth turns that users can safely navigate at a consistent speed, designers can limit user conflicts and decrease safety hazards with minimal trail degradation.

Not Constructing the Tread on a Full Bench: The only instances where you should not construct a fully bench cut trail is:

- (a) When the sideslope is so steep—80 percent or greater—that the backslope exceeds 6 feet in height, or;
- (b) When a trail design forces trail builders to build close to the downhill side of a large tree.

In both cases, build a crib wall to support a partial bench, and, as in all trails, the tread should maintain a 5 to 7 percent outslope.

Steep Climbing Turns: Avoid fall line turns, they result in extensive trail erosion. Trail designers should build climbing turns on sideslopes at grades no steeper than 7 to 10 percent to ensure their endurance.

Using Poor Materials: Using poor quality materials when building trail structures reduces their safety and longevity. Doing so leaves trail designers and builders vulnerable to user and trail safety concerns.

Opening a Trail Too Soon: Do not open newly constructed trails until they are ready and free of all safety hazards.

Log Lined Trails: A properly constructed trail should not have logs lining its edges. In fact, lining a trail with logs or rocks traps water on the tread and increases erosion.

Ignoring Old Wounds: Always reclaim eroded areas with check dams—natural obstacles like logs or rocks that divert the flow of water and soil—and reclaim all closed trails with transplanted native vegetation that conceals the old corridor.



Vegetation and Wildlife Habitat Management

Equally important to sustainable design of the trail, is locating the trail in a sustainable manner. Wildlife species function in areas specific to their size, needs, season, and quality of habitat. Within these areas animals most often feed, breed, and winter throughout their lifespan. As a result, human activities can have a negative impact within these sensitive habitat areas.

These impacts affect some species more than others do. For example, some species prosper in the presence of humans while others prefer habitats in seclusion. Especially for bird species, human disturbance may cause shortened feeding times, lowered reproductive success, a decline in parental care, and nest abandonment.

Since most human recreation areas are located in riparian corridors and near streams, conflict between humans and these natural habitats occurs. Most terrestrial species of wildlife depend on riparian corridors for feeding, breeding, moving, and dispersing. Consider the following guidelines when planning for trails located in natural areas:

Minimize Trail Construction: Initial human disturbances sometimes have more impact on wildlife than continuing disturbance. As a result, avoid locating trails in areas that contain high quality habitat, sensitive natural resources, and a variety of species. Consider locating trails in locations with lower habitat quality where you can concentrate use.

Use Existing Disturbance Corridors: Align trails along existing disturbance corridors such as existing rail lines, old roads, utility maintenance corridors, and construction routes. By locating trails in previously disturbed areas, you conserve sensitive resources from additional disturbance and long-term environmental impacts.

Locate Trails at Habitat Edges: Align trails in locations with extensive tension zones of exotic vegetation. Replace these plants with native vegetation to restore the plant community and provide better transition zones between the trail and intact habitat zone.

Avoid Sensitive Habitat Areas: Diverse species and sensitive ecological areas typically occur in areas usually associated with large habitat patches. For this reason, avoid locating trails in areas that may fragment large, intact species.

Maintain Connectivity: Diversity of water resources, access, and seasonal availability help determine the quality of wildlife habitats. Avoid constructing water crossings in areas with high connectivity to prevent wildlife barriers from occurring.



Avoid Small Areas of High-Quality Connector Habitat: Avoid small habitat patches, especially those that contain sensitive or rare species, when planning a trail. Not all habitats need to interconnect to thrive; but in some instances, connection to nearby habitats is essential.

Avoid Habitat for Threatened, Endangered, and Sensitive Species: Avoid placing trails in habitats of threatened, endangered, or sensitive species. Consult a biologist in the preliminary planning stages to determine a precise trail location.

Once the designer has prepared the preliminary layout, she/he will flag the trail in the field and refine the design as necessary to achieve the final layout of a sustainable trail system. Chapter 2 describes this process in greater detail.

Step 14: Evaluating Potential Management Structures for Management, Maintenance, and Operations of Your Trail

There are many successful models for managing, maintaining and operating a trail. They include:

- Multi-county Authority - as owner/operator
- Multi-county Authority - as owner/ with contract operator(s)
- Regional Trail Corporation – 501c3 formed as per Intergovernmental Cooperation Agreement between corporation and original trail owners or sponsors or municipalities or counties
- Multi-municipality Authority – as owner/operator
- Private Community Board of Grassroots Volunteers – with no government involvement
- Private Non-Profit Supporting Charity
- Utility-Type Entity – Fee Based
- Greenway Partnership - as owner/with contract operators
- Greenway Partnership - as owner/with contract local trail chapters as operators
- Greenway Partnership - as supporting organization with local trail chapters as operators
- Greenway Partnership - as contract operator with counties as owners of segments within their jurisdiction
- County Recreation Authorities – each owns and operates/maintains their segment
- Private Company – with contracts with each county to design, develop, maintain, and operate the trail
- Long-Term Lease for Trail(s)
- Municipal Authority

DCNR Publications

DCNR has published resources on multi-municipal cooperation and trail maintenance and operation. They include:

- *Multi-Municipal Cooperation for Recreation and Parks*
- *Rail-Trail Maintenance & Operation*

These publications serve as valuable resources during the trail planning process.

To access these publications visit:

www.dcnr.state.pa.us/brc/publications



- Municipality Owned/Operated
- Regional Trail Commission – formed as per Intergovernmental Cooperation Agreement between commission and original trail owners or sponsors or municipalities or counties
- Regional Trail Land Trust
- Council of Governments - as owner with local trail sponsors as contract operators
- County – as owner of trail easements or right-of-way and DCNR will maintain trail (unique to northern portion of Pine Creek Rail Trail)



Sensitive Habitat Near Trail
Kinzua Valley Trail, McKean
County

Photo Credit: Pashek Associates

Each management structure has its own unique set of strengths and weaknesses depending on the capacity of the participating partners. Potential management structures and the capacity of the participating partners must be evaluated on a case by case basis to determine which management structure would best meet your trail's needs.

Factors to consider when evaluating each structure and the capacity of potential partners includes:

- Project management
- Planning and design
- Acquisition of land or right-of-way or easement
- Property management
- Construction funding
- Other Resources - In-Kind and Donated materials, equipment, labor, etc.
- Maintenance or maintenance agreements
- Insurance or insurability/risk management
- Organization and financial administration
- Stability of operating funding
- Public Relations
- Geographic Area of Jurisdiction and geographic relationship to project area

If an organization is going to be created outside of existing management entities, you need to be aware of the enabling legislation and its requirements regarding the creation of such entities. Typical enabling legislation in Pennsylvania includes:

Intergovernmental Cooperation

The PA Intergovernmental Cooperation Law (Act 177 of 1972, as amended) establishes the requirements for creating an intergovernmental agency to create an intergovernmental trails organization.

If an intergovernmental trails organization is created under provisions of this Act, the Intergovernmental Cooperation Agreement must be formally approved by enactment of an ordinance by each participating governing



body. The County Code, Third Class Cities Code or Borough Code or Second Class Township Code, etc. also define specific procedures and public notice requirements for creating such an organization. The general powers and duties of such an organization must be listed in the ordinance to define the commitments as mutually agreed by the participating entities.

The required features of an intergovernmental cooperation agreement established according to the terms of Act 177 include the following:

1. The conditions of the agreement.
2. The duration of the agreement.
3. The purpose and objectives of the agreement, including the powers and scope of authority delegated in the agreement.
4. The manner and extent of financing the agreement.
5. The organizational structure necessary to implement the agreement.
6. The manner in which property, real or personal shall be acquired, licensed, or disposed of.
7. That the entity created under this section shall be empowered to enter into contracts for policies of group insurance and employee benefits, including social security, for its employees.

Additional information on creating an intergovernmental organization can be found in the Governor's Center for Local Government Publication *Intergovernmental Cooperation Handbook* which can be found at: www.newpa.com/webfm_send/1545

Municipal Authorities

The PA Municipal Authorities Act of 1945, as revised, pertains if a trails authority model is chosen. The Act establishes certain procedures for creating an authority, and its obligations and responsibilities with respect to ownership of facilities and the Authority's financial structure. An authority must be formally authorized by ordinance enacted by the governing body of all participating counties or municipalities. As new municipal entities desire to join a joint authority, a majority of existing municipal entities must agree to expand their jurisdiction. The general and specific powers and duties of such an authority must be listed in the ordinance.

Additional information on creating an intergovernmental organization can be found in the Governor's Center for Local Government Publication *Municipal Authorities in Pennsylvania* which can be found at: www.newpa.com/webfm_send/1936.



Non-Profit Organizations

Guidance on establishing a non-profit organization can be obtained from the Pennsylvania Attorney General's publication Charitable Non-Profit Organization Handbook, www.attorneygeneral.gov/uploadedFiles/Consumers/nonprofitbooklet.pdf, and from Pennsylvania Associations of Non-Profit Organizations publication titled *Pennsylvania Nonprofit Handbook*.

Registration forms and other information about registering a charitable organization can be obtained at: www.dos.state.pa.us/charity/index.

Step 15: Planning for Management, Maintenance, Operation and Construction Costs

During the planning stage it is important to plan for management, maintenance, operation, and construction costs. The trail organization must be confident of its ability to securing funding, resources, and volunteer/in-kind support before advancing into the design of the trail.

Before projecting your costs you must determine who will build and maintain your trail. You can build and maintain trails in many ways. Nonprofit trail clubs (with professional and volunteer workers), municipalities, landscape companies, and construction firms all construct trails. For many projects on both public and private land, volunteers and conservation corps are responsible for building trails.

Many projects use both professional and volunteer labor for trail construction and maintenance. It is important to determine who will build the trail in order to include the appropriate labor costs into the project estimate. Obtain estimates from nonprofit trail clubs, contractors, and conservation corps before completing your project budget. These final numbers may vary widely depending on the labor costs and any hidden costs like food, travel expenses, and safety equipment.

Management, Maintenance and Operation Costs

Management, maintenance and operation costs typically include:

- Land acquisition or purchase of easements
- Office space
- Salaries for paid staff
- Liability and other insurance policies
- Printing and mailing expenses
- Telephone expenses
- Fuel expenses
- Continuing education expenses
- Large equipment expenses



- Small equipment expenses
- Consumable expenses
- Maintenance materials and supplies
- Utility Expenses

Your organization should project anticipated expenses and likely revenue by developing a five year management, maintenance and operations budget. This can be accomplished by interviewing staff of similar trail organizations who have trails of similar scope and size.

Potential expenses and revenues can also be estimated using resources available in the Rails to Trails Conservancy trail Building Toolbox. These resources are located at: www.railstotrails.org/ourWork/trailBuilding/toolbox/index.html.

Construction Costs

Trail managers and designers often use one or more of the following methods to project trail construction costs:

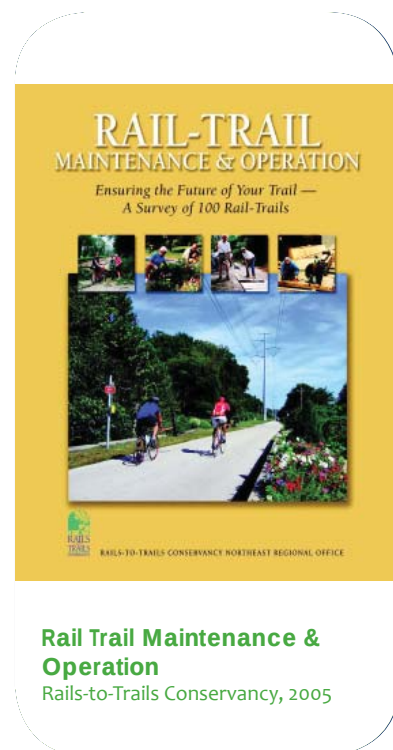
1. Unit costs
2. Crew weeks of labor
3. Per foot or mile costs

With unit costs, a predetermined cost exists for each feature associated with the construction of the trail. You simply multiply a tally of features by their unit cost to determine the total project cost. When using the unit cost method a completed trail work log serves as an important tool for the designer.

Often trail designers estimate construction costs by using crew weeks of labor to determine the amount of time it will take to complete a project. Experienced trail leaders can estimate how much work a crew can complete in a week and divide the project into a number of crew weeks. Usually, one estimates project material costs and labor costs separately, then combines them to estimate the overall cost. Both volunteer and professional trail crews use the crew week method to determine trail project costs. As with other methods, the person estimating the project costs should have a solid background in trail work and first-hand knowledge of the work a crew can perform in a given time period.

Trail Feasibility Studies & Master Plans

Potential funders want to see an overall plan for the proposed trail system. Completing a trail feasibility study for your trail is a way to demonstrate you have developed a plan to manage, construct, maintain and operate your trail. The trail feasibility process is typically designed to address all of the planning steps described in this chapter.



Volunteer Trail Crew
Lonberger Trail, Centre County

Photo Credit: PA DCNR



Parker Dam State Park
Boardwalk over sensitive habitat

Photo Credit: Pashek Associates

A trail feasibility study should document the process, analyze the feasibility, recommend specific improvements, identify the overall costs, provide recommendations for phasing the construction of the trail system, and provide detailed implementation strategies to guide your organization in development of the trail.

PA DCNR's Community Conservation Partnership Program is a competitive grant program that can provide a source of funding to complete a trail feasibility study or master plan. The cost to complete a trail feasibility study or master plan varies. This cost depends on the length of the trail, complexity of land ownership, number of features, structures, and crossings, potential for environmental hazards, etc.

If your organization has interest in exploring the potential to complete a feasibility study or master plan for your trail project, contact your regional DCNR adviser who can advise you on the process, www.dcnr.state.pa.us/brc/grants/Regional_Map.pdf.

To view a sampling of DCNR funded trail feasibility studies visit: www.dcnr.state.pa.us/brc/elibrary/resourcesta/sampleplans/index.htm.

Step 16: **Securing Funding for Trail Planning, Design and Construction**

Where will the money come from to support your proposed trail? Trail projects often receive funding from various federal, state, local governments, and private sources to pay for project costs. Other sources of funding such as donations, trail user fees and taxes offer creative ways to fund a trail project.

The trail sponsor is responsible for seeking funding sources, soliciting volunteer labor, and applying for grants. When considering whether or not to apply for federal or state grants, speak with grant administrators and past recipients to ensure your project and application will meet their needs. Be particularly aware of application deadlines, award dates, and the reporting process to prepare for and continue a project during the waiting period. Always have a backup plan and make allowances for moving forward with any project.

Raising funds for new projects, especially where strong partnerships are developed, remains easier than raising funds for trail maintenance projects. Establish strong community support and solid relationships with donors before seeking funding. You can phase construction over time by dividing the project into smaller components.



Typical sources of funding for trail development includes:

- Allocations from municipal government
- Grants from non-profit foundations
- In-kind and donated services in the form of materials, labor, equipment, operators
- State and Federal grant programs

In Pennsylvania the primary source of State and Federal grant opportunities are administered by the Pennsylvania Department of Conservation and Natural Resources through their Community Conservation Partnership Program. For information on these grant opportunities visit: www.dcnr.state.pa.us/brc/grants/index.aspx.





Design Needs for Specific Non-Motorized Trail User Groups

In order to design a trail it is important to understand the type of trail you are designing and the intended users of the trail.

This chapter details the characteristics of each non-motorized trail type, their respective users, and the corresponding design requirements, as they vary between the different uses. This includes:

- Hiking Trails
- Mountain Biking Trails
- Shared Use Paths and Rail-Trails
- Rails-with-Trails
- Equestrian Trails
- Cross Country Skiing Trails
- Snowshoeing and Winter Hiking Trails

Further, all trails, trail amenities, and trail related facilities should be universally designed to accommodate all users, without and with disabilities, in accordance with [*Accessibility Guidelines for Outdoor Developed Areas*](#). The design requirements for accessible trails are described in the final section of this Chapter.

Your ultimate goal during the design phase involves preparing plans and specifications to communicate all aspects of the design to those responsible for building trails, amenities, and support facilities in a sustainable manner.



Tiadaghton Hike

Photo Credit: PA DCNR



Trail Design

Design and construction of trails is a complex combination of skills and should be accomplished by experts. Experience in trail design, construction, and management is essential for implementing projects that involve poor soils, complex topography, high levels of use (especially when stock animals are involved), and extensive improvements, such as surfacing or structures. Experience is also essential for the design of multiple-use trail corridors that meet standards to allow for the safe use of a trail.

In addition to consulting experts in trail design and construction, it is important to seek advice from experts in resource disciplines. Two of the most common problems of backcountry trails, deterioration through overuse of popular trails and the development of undesired routes at popular destinations, can be avoided by drawing on personnel or outside experts with trail design and management experience and by following commonly accepted standards of trail design after thorough field study.

Observing proposed or existing routes through several seasons, including winter, will assist the planning team in determining the fitness of new corridors for trail development, as well as the level of improvement or rerouting required to achieve sustainability for rebuilt trails.

There are a variety of factors necessary for a sustainable, low-impact trail. By carefully fitting the trail profile to the local topography, erosion will be minimized, thus increasing the durability and sustainability of the natural surfaces.

National Park Service Natural Resource Management, Reference Manual # 77, 2006.



User Characteristics and Design Requirements for Non-Motorized Trails

Hiking Trails

The development of hiking trails provides naturally surfaced routes for pedestrian use. These trails offer hikers and joggers the opportunity to experience and interact with nature with little disturbance from other trail users.

General Characteristics of Hikers

The following information defines the general preferences and motivations of hiking trail users and the design requirements to accommodate use.

User Information

Whether the intended users are walkers, hunters, hikers, overnight backpackers, or destination hikers, hiking trails accommodate users seeking multiple experiences. Motivations for using natural trails vary widely, ranging from physical challenge to experiencing nature.

A large percentage of hikers enjoy escaping from motorized activity and value experiencing nature. Maintaining a natural setting with wooded, rolling terrain that features wildlife-viewing opportunities remains especially important to users of this type of trail. To accommodate overnight backpackers, do not forget to incorporate some amenities like camping areas, access to water, and composting toilets.

Since trail difficulty aids users in determining which trail they will select, include signage containing trail difficulty ratings at trail access points. To help guide trail users include maps route guides, and general information about trail features in publications and signage along each trail. Trail length preferences vary widely based on individual skills and preferences. Beginners prefer short loops of 2-4 miles and day hikers desire trails about 5-9 miles. Plan for routes varying in length from 5-100 miles, with camping areas placed at 5-10 mile intervals to accommodate overnight backpackers.

Destination Hiker Characteristics

Trail Use Pattern

- Seeks out trails for a desired experience (such as solitude), whether near home or some travel distance
- Prefers looped systems over out-and-back trails to vary the experience
- Will seek out trails of varying difficulty
- Likes to stop along the trail to rest, observe, and socialize if hiking in a group
- Expects trail to be of varying difficulty consistent with landscape characteristics



West Rim Trail, Lycoming & Tioga Counties

Photo Credit: PA DCNR



Recreation Setting Preferences

- Large percentage seeks escape from motorized activity, and value experiencing nature
- Natural settings remain important to all users, with wooded, rolling terrain and wildlife viewing opportunities commonly preferred
- Trail difficulty often serves as an important determinant in trail selection, with a desire for a wide range of challenges
- Access to the trail offers a major predictor of use levels
- Length preferences vary widely with skills and preference, with beginners liking shorter loops of 2–4 miles and day hikers preferring 5–9 miles
- Minimum preferred width is 18’

Overnight Backpacker Characteristics

Overnight backpackers have many of the same preferences as a destination hiker, only with a few nuances associated with overnight stays. Other preferences include:

- Camping areas at intervals of 5–10 miles are desired, with average daily hiking distance up to 10 miles
- Access to water, especially at camps
- A desire and need for composting/pit toilets at designated camp areas to minimize environmental impacts
- Outing length varies from 5–100 miles, with 25–35 miles being a common distance

Hiking General Speeds and Distances

Hunter Walker

- 1-2 miles per hour
- 2-4 miles

General Hiker/Backpacker

- 2-3 miles per hour
- 4-10 miles

Fitness Hiker/Backpacker

- 3-4 miles per hour
- 6-15 miles

Trail Jogger

- 6-7 miles per hour
- 3-15 miles

Hiking Trail Classifications

The design of natural hiking trails should accommodate trail users on a local, county, regional, and state level. Designers can further subdivide these trails to accommodate general hikers, walkers, hunters, and overnight backpackers.

General Hiking Trail: General hiking trails often have a natural surface tread. You can find this type of trail in parks or greenways where adequate open space exists to form a trail loop. These trails provide users various opportunities to observe wildlife and interact with nature.

Nature Interpretive Trail: Nature interpretive trails, similar to general hiking trails, allow users to interact with, enjoy, and learn about nature and wildlife. You can typically find trails of this kind within designated conservation areas and arboretums.

Walker/Hunter Trail: Walker/hunter trails often take advantage of old logging access trails and roads. These routes typically have a non-motorized designation. Authorized forest management activities often occur near them.



Hiking Trail Layout Configurations

Hiking trail design normally responds to the landscape's topography and highlights a sequence of naturally occurring events that enhances the trail users' experience. In parks and natural areas the design often integrates a looped trail system at a specific site, while larger landscape settings normally incorporate a linear trail design. Designers should carefully employ design elements in the trail that fit with the existing landscape on all levels to decrease ecological degradation, maximize views, and embrace natural features along the route.

Difficulty Rating System

Difficulty Standards for Hiking Trails, as defined by the Pennsylvania DCNR:

Trail Type	Easiest (Interpretive)	More Difficult	Most Difficult
Clearing Height	8-10'	8'	8'
Clearing Width**	4'	3-4'	3'
Treadway Width***	1.5-2'+	1-1 ½'	1-2'
Treadway Slope*	Less than: 5%	Less than: 12%	Less than: 18%
	Maximum: 20% up to 100'	Maximum: 30% up to 300'	Maximum: 30+% up to 500'
Treadway Cross Slope	0-3%	0-5%	0-8%
Turning Radius	NA	NA	NA
Sight Distance	NA	NA	NA
Surface Materials	Uniform, firm and stable surface. Smooth tread with no obstacles. Pavement may be appropriate in highly developed settings.	Native surface with some imported material. Sidehill trail is constructed. Generally clear of obstacles, steps to 10".	Native surface with constructed sidehill trail. Obstacles, roots, rocks, and steps to 24".

* Upper limit of treadway slope and distance depends on soil type, amount of rock, vegetation type, and other conditions affecting trail surface stability.

** Curve alignment to avoid cutting large trees.

*** Increase tread width six inches on switchbacks or where side slopes exceed 60%.



Golden Eagle Trail Hike

Photo Credit: PA DCNR



Figure 2-1: Two-Way Hiking Trail

Hiking Trail Guidelines and Considerations

Technical Provisions

Trail Grade

- No more than 30% of the total trail length may exceed a trail grade of 8.33%.
- Trail grade of up to 5% can occur for any distance.
- Trail grade of up to 8.33% can occur for up to 200 feet if resting intervals are provided at distances no greater than 200 feet apart.
- Trail grade of up to 10% can occur for up to 30 feet if resting intervals are located at distances no greater than 30 feet apart.
- Trail grade of up to 12.5% can occur for up to 10 feet if resting intervals are located at distances no greater than 10 feet apart.
- At drain dips, a trail grade of 14% can occur for up to 5 feet where the cross slope does not exceed 5%.

Cross Slope

- The cross slope shall not exceed 5%.
- At drain dips, a cross slope of up to 10% can occur at the bottom of the dip where the clear tread width equals at least 42 inches.

Resting Intervals

- Where the trail grade exceeds 5%, provide resting intervals. Resting intervals shall be at least 60 inches long, be at least as wide as the widest portion of the trail segment leading to the resting intervals, and have a slope not exceeding 5% in any direction.



Exception: This provision does not apply where resting intervals cannot be constructed because one or more conditions for departure exist.

Surface

- The trail tread surface shall be both firm and stable.

Exception: This provision does not apply where a firm and stable surface cannot be provided because one or more conditions for departure exist.

Clear Tread Width

- The clear tread width of the trail shall equal at least 36 inches.

Exception 1: You may reduce the clear tread width to no less than 32 inches where one or more conditions for departure exist.

Exception 2: This provision does not apply when you can not provide at least 32 inches clear tread width because one or more conditions for departure exist.

Passing Spaces

- Where the clear tread width of the trail is less than 60 inches, provide passing spaces at intervals of no more than 1000 feet. Passing spaces shall be at least 60 inches by 60 inches in size, or an intersection of two walking surfaces that provide a T-shaped space complying with §304.3.2 of the Architectural Barriers Act Accessibility Standards (ABAAS), where the arms and stem of the T-shaped space extend at least 48 inches beyond the intersection. The cross slope of passing spaces shall not exceed 5% in any direction.

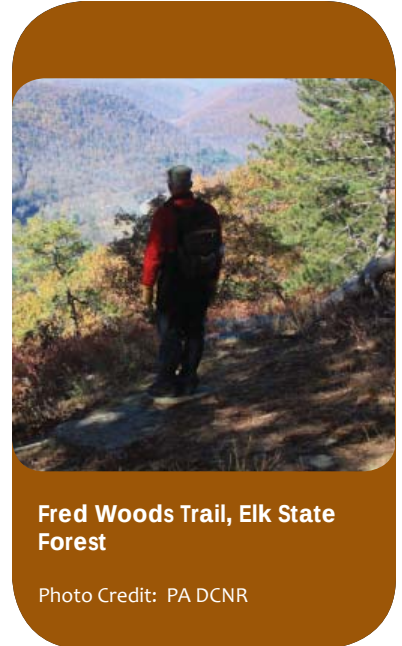
Exception: This provision does not apply where you can not provide passing spaces because one or more conditions for departure exist.

Tread Obstacles

- Where tread obstacles exist, they shall not exceed a height of 2 inches.

Exception 1: Tread obstacles with a maximum height of 3 inches are permitted where trail grade and cross slope are 5% or less.

Exception 2: Where one or more conditions for departure exists that prevent you from meeting exception 1.





Protruding Obstacles

- Protruding objects on trails shall comply with §307 of the ABAAS and shall have at least 80 inches of headroom.

Exception 1: Where a reduction in the vertical clearance of a trail is reduced to less than 80 inches because one or more conditions for departure exist, you must provide a barrier to warn individuals who are blind or visually impaired.

Exception 2: Where one or more conditions for departure exception 1 cannot be met because less than 80 inches of headroom or installation of a warning barrier is precluded.

Openings

- Openings in trail tread surfaces shall be small enough to prevent passage of a 1/2-inch diameter sphere. Elongated openings shall be placed so that the long dimension is perpendicular or diagonal to the dominant direction of travel.

Exception 1: Elongated openings are permitted to be parallel to the dominant direction of travel where the opening does not allow passage of a 1/4-inch diameter sphere.

Exception 2: Perpendicular or diagonal opening are permitted to the dominant direction of travel as long as the size does not allow passage of a 3/4-inch diameter sphere where one or more conditions for departure exist.

Exception 3: Where you cannot meet exception 1 or exception 2 because one or more conditions for departure exist.

Edge Protection

- When you provide edge protection, the edge protection shall have a height of at least 3 inches.

Signs

- If an individual must obtain materials from or manipulated on a sign or kiosk, the sign or kiosk shall be designed to meet the reach ranges in §308 of the Architectural Barriers Act Accessibility Standard. You must post signs at the trailhead of new or altered trails and trail, as well as at the trailhead of trails that have been evaluated for accessibility. In addition to the standard information including the name and length of the trail, these signs shall include the typical and maximum trail grade, typical and maximum cross slope, typical and minimum tread width, surface type and firmness, and obstacles. These signs also should state that the posted information reflects the condition of the trail following construction or assessment. The signs



should include the dates of these procedures.

- When providing more extensive trail information (e.g., an aerial map of the trail and related facilities), identify the location of specific trail features and obstacles that do not comply with the Forest Service Trail Accessibility Guidelines technical provisions and include a profile of the trail grade.

Sequence of Events

Creating a sequence of events through trail design remains especially important when maximizing the hiker's trail experience. This includes the attempt to decrease a user's visibility from another section of trail. Varying land character coupled with numerous access points, overlooks, and camping opportunities are necessary to offset the out-and-back aspect of looped or linear trails and provide specific points of destination along each route.

Ecological Buffers

Maintaining adequate buffers between trails and sensitive ecological systems, as well as limiting the impact on transitional zones between plant communities remains an integral component of natural trail design. If designed effectively and responsibly, natural trails should blend seamlessly into the existing landscape, with minimal impact on an area's natural systems.

Complementary Publications

In order to plan hiking trails, it is important to have an understanding of various types of hiking and those users' needs, as well as hiking trail planning, design, construction, management, and maintenance. The following publications are valuable resources that may be consulted:

- *Standard Specifications for the Construction and Maintenance of Trails*, U.S. Forest Service
- *National Trail Drawings and Specifications*, U.S. Forest Service
- *Trail Construction and Maintenance Notebook*, U.S. Forest Service
- *Federal Trail Data Standards*, U.S. Forest Service
- *Trail Assessment and Condition Survey (TRACS) System*, U.S. Forest Service
- *Trail Bridge Catalog*, U.S. Forest Service
- U.S. Forest Service *Trails Accessibility Guidelines*, U.S. Forest Service
- *Trail Training DVD Services*, U.S. Forest Service
- *Complete Guide to Trail Building and Maintenance, 4th Edition*, Appalachian Mountain Club
- *Appalachian Trail Design, Construction, and Maintenance*, William Birchard
- *Appalachian Trail Club Presidents Handbook*, 2002, Marty (Martha) Lawthers, Kevin Peterson, Katharine Wroth, and Others
- *Backcountry Sanitation Manual*, Appalachian Trail Conference

Hiking Trails Best Practices

- Fred Woods Trail, Cameron County
- Timberdoodle Flats Wildlife Interpretive Trail, McKean County
- Tracy\Johnnycake Trailhead, McKean and Warren Counties (multiple layouts includes shared use trails)
- Minister Creek Recreation Area, Warren and Forest Counties.

Hiking Trail Amenities Best Practices

- Hearts Content, Warren County (interpretive trail, latrine, parking, access to linear ski trail system; other layouts)
- Pine Tree Trail, Cameron County (latrine, campground, interpretive trail, kiosk); Lashure Trail, Forest County
- Quehanna Hiking Trail, Elk, Clearfield, and Cameron Counties (kiosks\parking areas)
- Tracy Ridge Recreation Area (campground, latrine, shelter, camping area, also utilized by Shared Use Trails); Minister Creek Recreation Area, Warren and Forest Counties



You can obtain additional information by contacting the Keystone Trails Association, a volunteer-directed federation of membership organizations and individuals dedicated to providing, preserving, protecting, and promoting recreational hiking trails and hiking opportunities in Pennsylvania. Further, they represent and advocate for the interests and concerns of the Pennsylvania hiking community.

- Keystone Trails Association
101 N. Front Street, 3rd Floor
Harrisburg, PA 17101
(717) 238-7017
www.kta-hike.org



Mountain Biking Trails

The development of mountain biking trails provides a sustainable approach to trail design that encourages more people to explore the outdoors. As a goal the designer should attempt to attract bicyclists seeking a challenging experience that many shared use paths lack and users desire.

General Characteristics for Mountain Bicyclists

The following information defines the preferences and motivations of mountain bikers, as well as common trail building mistakes and alternatives designers should take into consideration when planning mountain bike trails.

User Information

Mountain bikers, ranging from beginner to expert, enjoy the excitement and outdoor activity of the sport. Since mountain bikes typically have shock absorbers and wider tires, bikers have the ability to go off-road and take on more challenging courses than traditional biking allows. Prime motivators of mountain bikers include obtaining exercise, experiencing natural settings, and testing riders' technical skills.

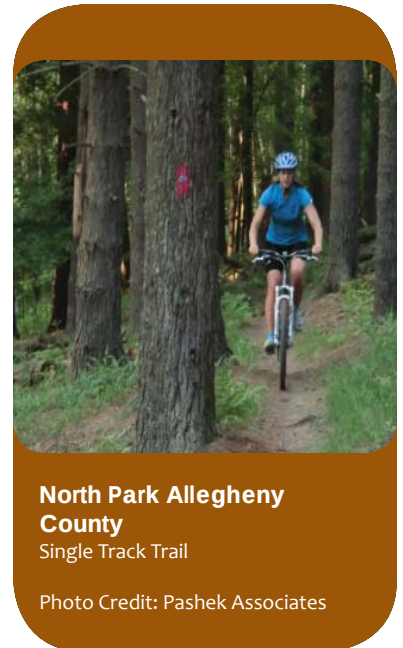
Dedicated mountain bikers seek out trail systems sometimes away from home and travel to them for day or overnight trips. These trails, specifically designed for mountain biking not only provide a good workout, but also present the opportunity for riders to test their skills and immerse themselves in nature.

Mountain bike trails commonly provide 2- 3 hour riding opportunities that consist of approximately 20-25 miles of contiguous trail. Beginner trails typically have a wider tread and bypass routes at technical challenges like boulder or log obstacles. The most technically challenging mountain biking trails require balance, fitness, and strength. In addition to challenging courses with varying terrain, riders appreciate amenities like water spigots located at trailheads to clean bikes after their rides.

Rider Type Considerations

Mountain biking represents one of the fastest growing recreation experiences within America's national forests. Those who share an interest in this sport enjoy various types of riding experiences that appeal to a range of skill levels. Riders ranging from novice to advanced can take part in many types of biking, including:

Recreational Biking: Recreational bike riding offers an enjoy experience for riders of all levels and abilities. Whether young or old, riders can enjoy exercise while being immersed in nature. These riders seek interesting trail routes that involve little to marginal physical effort.



**North Park Allegheny
County**

Single Track Trail

Photo Credit: Pashek Associates



Local Access and Utilitarian Biking: Utilitarian bicycle riders ride for the fun and fitness benefits of the sport. These riders rely on ease of accessibility to local trail loops and desire courses ranging in both length and technical requirement.

Long Distance Biking: Long distance biking requires determination, endurance, and hard work. Bicyclists dedicated to this sport will often seek out new trails to experience on day or overnight trips.

Downhill Biking: A downhill trail allows riders to go as fast as possible while finding the best lines for the smoothest ride. Advanced riders seek out faster lines, vertical drops, and jumps that require a higher level of reaction, strength, and agility.

Pump Track: This cross country style of mountain biking allows riders to attack berms, navigate turns, attempt jumps, and improve their technical ability. Pump track biking offers high levels of fun and effort with lower levels of fear, risk, and even speed.

Technical Challenge or “Free-Rider” Biking: This style of mountain biking celebrates the challenges of technical riding, encouraging riders to soar off jumps, balance on log rides, and teeter on high-rise stunts. The fact that speed and on-edge elements do not mix well with equestrians, hikers, and even other mountain bikers requires separate trails for these activities.

Mountain Bike General Speeds and Distances

Recreational Trail Riding

- 6-15 miles per hour
- 10-15 miles

Recreational Technical Riding in a Park Setting

- 3-12 miles per hour
- 3-12 miles

Advanced Recreational Trail Riding

- 10-20 miles per hour
- 15-50 miles

Advanced Recreational Technical Riding in a Park Setting

- 6-18 miles per hour
- 6-20 miles

Mountain Biker User Characteristics

Trail Use Pattern

- Seeks and travels to trails away from home as a day or overnight trip
- Do not confuse mountain bikers with people who own mountain bikes but do not use them on mountain bike trails
- Commonly desire 2- 3 hour riding opportunities, 20–25 miles of contiguous trail (although fewer miles are acceptable in challenging terrain)

Recreation Setting Preferences

- Best trails have a natural, challenging character and immerse the rider in nature while providing a good workout and opportunity to test skills
- Use a combination of roads, logging roads, and trails as available, safe, and convenient
- In urban/suburban areas, highly prefer developed mountain bike trails offering looped configurations with varying levels of challenge



Mountain Bike Trail Classifications

Mountain Biking Trail: Bikers normally find mountain biking trails in local, county, regional, and state parks or forests where adequate open space exists to form loop trail systems. These naturally surfaced trails offer casual, dual-track trails, single track, and technical single tracks to accommodate riders' needs and abilities. In addition, mountain bike trails should incorporate sustainable design techniques to respond to a site's specific setting and geographic location.

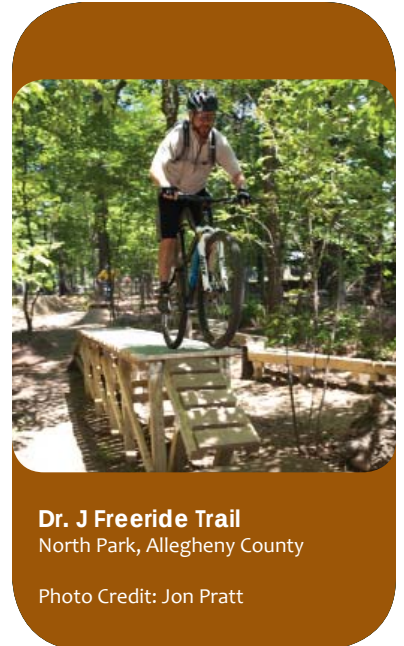
Technical Challenge/"Free-Ride" Course: A technical challenge or "free-ride" bike course consists of numerous dirt jumps, elevated boardwalks, and drops. Technical challenge areas should include a wide variety of challenges, from easy to difficult, to provide for skill progression. Typically you will find these courses located in the middle of a trail system with technical challenges placed to direct competent riders to the course. Best practices recommend surrounding these courses with natural barriers consisting of deadfall to ensure safety and seclusion from other trail users.

Downhill Course: Downhill trails cater to single-use, one-way routes with technical trail elements designed for the sport of downhill biking. Jumps and vertical drop-offs represent key features that designers should include in downhill trails. Designers should separate downhill courses from trails designed for other users to avoid conflict. Furthermore, these trails normally need more tread maintenance and drainage structures than other trails due to their steep nature.

Pump Track: A pump track requires a continuous loop of rollers and berms mountain bikers can ride without pedaling. Pump courses offer steeply banked turns and small roller jumps that riders can generate speed by "pumping" their bikes with their arms and legs, while alternately pulling up and pushing down on their handlebars. The small footprint of these courses allows managers to integrate them into a larger trail system. "Pump" bicyclists can gain an excellent workout, develop skills, and improve fitness while navigating this type of trail.

Mountain Bike Trail Layout Configurations

Mountain bike trail design should provide users with one or two-way trails to accommodate challenging features that appeal to riders. These trails respond to the landscape and highlight interesting natural features throughout. Trails must maintain a consistency or rhythm to create an appealing trail layout. The easiest of trails should maintain a relatively gentle flow with predictable curves, while the most challenging may have tight curves with demanding technical requirements. The design of transitions between sections, which can vary in their degree of openness, should allow riders to predict the upcoming transition and maintain control of their bicycles.





Mountain Bike Trail Guidelines, Level of Difficulty and Other Considerations

Essential Elements of Sustainable Mountain Biking Trails

Take the level of development needed into consideration and base the trail design on a trail's location, use, and other factors. Some trails may have a natural surface tread, while others may consist of more developed surfaces.

Mountain bike trail designs should follow the International Mountain Bicycling Association's (IMBA) five essential elements of sustainable trails. These include:

1. **The Half Rule:** A trail's grade should not exceed the half grade of the hillside or sideslope that the trail traverses. If grade does exceed half of the sideslope, these section are considered a fall line trail which are susceptible to erosion.
2. **The Ten Percent Average Grade Guideline:** Generally, a trail with an average grade of 10 percent or less offers the most sustainable design. This does not mean that you must keep all grades less than 10 percent. Many sections of trail will have short, steep sections greater than 10 percent and some unique situations will allow average grades greater than 10 percent.
3. **Maximum Sustainable Grade Trails:** The maximum sustainable grade represents steepest section of sustainable trail more than 10 feet in length. When designing a trail you should determine the maximum grade a trail can sustain given local conditions. The maximum sustainable grade should be determined early in the design process. Variables that impact the maximum sustainable grade include:
 - Soil Type
 - Rock
 - Annual Rainfall Amount
 - Grade Reversals
 - Type of Users
 - Number of Users
 - Level of Difficulty
4. **Grade Reversals:** Grade reversals, also known as a grade breaks, drainage dips, or rolling dips consist of locations where a climbing trail levels out and then changes direction, dropping subtly for about 10-50 linear feet before rising again. These changes in grade forces water to exit the trail at the low point before it can gain volume, velocity, and erosive power.
5. **Outslope:** As a trail follows the contours across a hillside, the downhill, or outer edge of a trail, a trail's tread should tilt slightly down and away from the high side. This slight tilt encourages water to sheet flow across and off the trail, minimizing erosion and tread damage.



IMBA Difficulty Standards for Mountain Biking Trails:

Aspect	Easiest (White Circle)	Easy (Green Circle)	More Difficult (Blue Square)	Very Difficult (Black Diamond)	Extremely Difficult (Dbl. Black Diamond)
Trail Width	72" or more	36" or more	24" or more	12" or more	6" or more
Tread Surface	Hardened or surfaced	Firm and stable	Mostly stable with some variability	Widely variable	Widely variable and unpredictable
Vertical Clearance	10'	10'	10'	10'	10'
Horizontal Clearance	2'	2'	1'	6'	6'
Average Grade	Less than 5%	5% or less	10% or less	15% or less	20% or less
Maximum Grade	Maximum 10%	Maximum 15%	Maximum 15% or greater	Maximum 15% or greater	Maximum 15% or greater
Natural Obstacles and Technical Features	None	Unavoidable obstacles 2" tall or less' Avoidable obstacles may be present Unavoidable bridges 36" or wider	Unavoidable obstacles 8" tall or less Avoidable obstacles may be present Unavoidable bridges 24" or wider Technical trail feature 24" high or less, width of deck is greater than ½ the height	Unavoidable obstacles 15" tall or less Avoidable obstacles may be present, including loose rocks Unavoidable bridges 24" or wider Technical trail feature 48" high or less, width of deck is greater than ½ the height Short sections may exceed criteria	Unavoidable obstacles 15" tall or less Avoidable obstacles may be present, including loose rocks Unavoidable bridges 24" or narrower Technical trail feature 48" high or greater, width of deck is un- predictable Many sections may exceed criteria





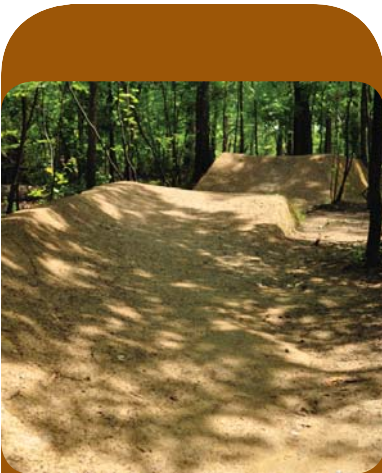
Figure 2-2: Two-Way Mountain Biking Trail

Trail Treads

The construction of a trail's surface must accommodate the current and anticipated level of wear and tear from its users. Materials used may provide varying tread conditions that determine the level of stability and difficulty of a mountain biking trail. For example, a trail that contains a paved or surfaced material that remains firm and stable and then transitions to a trail that possesses a loose condition will provide a level of excitement and unpredictably throughout a designated course.

Natural Obstacles

Riders consider the technical design of a mountain bike trail important. Therefore, the success of a trail design hinges on incorporating technical elements. By incorporating a sequence of events through the use of gateways and existing natural features, designers provide riders with destinations along a trail that increase the trail's interest and challenge. By incorporating natural features and technical challenges throughout a trail system designers distinguish mountain bike trails from shared use natural trails. Some of these features include natural drop-offs, rock gardens, rock slabs, rock chokes, and fallen trees or limbs.



Mountain Bike Skills Park
North Park, Allegheny County

Photo Credit: Pashek Associates



Constructed Technical Design Features

A mix of technical features, both natural and man-made, consistent with each trail rating is important to holding the interest of riders. If the design meets user expectations, riders are more likely to stay on the trail rather than creating new routes. Man-made technical features that improve interest and increase technicality without limiting space requirements include narrow bridges, ladder bridges, anchored bridges, boulders, boardwalks, and stabilized curves. These various elements provide alternate routes, discourage riders from creating new routes, and increase the longevity of each trail.

Complementary Publications

In order to design sustainable mountain bike trails, you must have an understanding of various classifications of mountain biking trails, their riders' needs, as well as the planning, design, construction, management, operations, and maintenance of mountain bike trails. Two International Mountain Biking Association publications provide guidelines for addressing each of these aspects. These publications are:

1. **Trail Solutions: IMBA's Guide to Building Sweet Singletrack**
2. **Managing Mountain Biking: IMBA's Guide to Providing Great Riding**

These guidelines use excerpts from these publications because they provide detailed background and guidance on mountain biking trail issues.

The International Mountain Bicycling Association - Trail Building Resources, www.imba.com/resources/trail_building/index.html, provides a wealth of information about:

- Advocacy and Organizing
- Bike Parks and Freeriding
- Grants and Fundraising
- Kids and Mountain Biking
- Land Access and Protection
- Liability and Insurance
- Logos and Graphics
- Maps and Signs
- Mountain Bike Research
- National Mountain Bike Patrol
- Resource Conservation
- Risk Management
- Trail Building and Design

Further information may be obtained by contacting the International Mountain Biking Association:

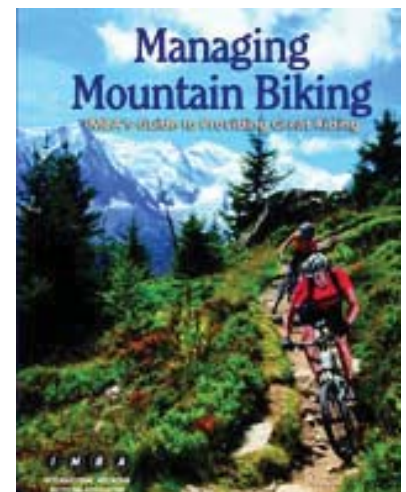
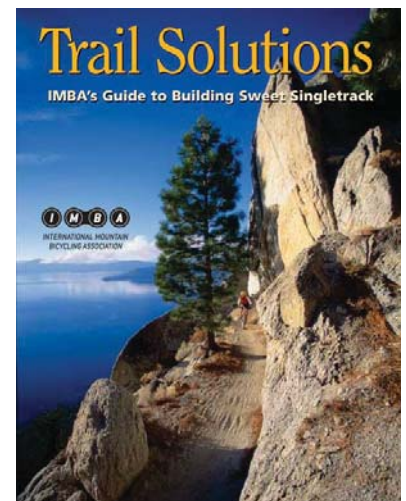
- International Mountain Biking Association
P.O. Box 7578
Boulder, CO 80306
(303) 545-9011
www.imba.com

Single Track Mountain Bike Trails Best Practices

- Allegrippis Trail, Raystown Lake, Huntingdon County
- Eagleton Mine Camp Trail, Clinton County

Technical Challenge/Free Ride Courses Best Practices

- Dr. J Trail, North Park, Allegheny County





Shared Use Paths and Rail Trails

A shared use path is a facility within its own right-of-way, which is separate from the vehicular right-of-way. As its name suggests many different types of users may use a shared use path. Users generally include walkers, joggers, bicyclists, and in-line skaters and sometimes equestrians. Similarly, rails-with-trails often use the same standards and guidelines for their development as shared use paths.

The trail community recognizes the American Association of State Highway Transportation Officials (AASHTO) publication titled “Guide for the Planning, Design, and Operation of Bicycle Facilities” as the authority for shared use path guidelines. AASHTO last published this document in 1999 and it is currently undergoing revision. The guidelines contained herein come from the February 2010 Draft. For purposes of these guidelines they provide general guidance for planning. During the design phases of a shared use path, consult the AASHTO “Guide for the Planning, Design, and Operation of Bicycle Facilities” to ensure that your design follows the guidelines specific to your particular scenario.

General Characteristics of Shared Use Paths

The following information defines shared use path’s various users and how each trails’ design can accommodate its use.

User Information

Shared use paths serve virtually every type of user group, whether users want to walk a dog, push a stroller, jog, bicycle, in-line skate, or even horseback ride. The hard surfacing of this type of path provides a high level of accessibility to users of all abilities. For this reason, shared use paths can accommodate the desires of various users, while minimizing ecological impacts and the amount of trails constructed.

Shared use paved trails function at local, county, regional, and state levels. The amount of use, character, width, length, location, and the type of user(s) accommodated serves as a major distinction between the various levels. Moreover, the extensiveness of a shared use trail network depends on the level of service, rate of use, and geographical characteristics.

The amount of value the public places on a shared use path also serves as a determining factor in the development of shared use trails at all levels. These values include convenience, safety, fitness, recreation, as well as transportation. Planners and designers should consider these values heavily in the planning and design of any trail system.



Altoona Shared Use Path
Blair County



Montour Trail
Allegheny & Washington Counties



East Branch Trail
Crawford County



User Type Characteristics

Shared use paths offer versatility and can accommodate a wide range of user groups. These paths able users to pursue various activities and may offer connections to local and regional trail networks.

Pedestrian (Walker/Jogger)

Pedestrians utilize shared use trail networks to walk and run for recreational and fitness value. Walkers and joggers of all types use trails year-round for social interaction, scenic beauty, and health benefits.

Bicyclist

Bicyclists of all types utilize shared use trails for convenient, safe travel between parks, rest areas, and other local destinations. Whether a bicyclist uses the path for family, recreational, fitness, or transportation purposes, these trail networks provide invaluable economic and health benefits.

In-line Skater

Recreational, fitness, and commuter in-line skaters value smooth, wide paths that allow for easy to navigation. These users' primarily motivations revolve around getting exercise, being outdoors, enjoying skating, and ease of transportation.

Wheelchair User

Disabled persons and wheelchair users value smooth, level, wide paths that are easily navigated. Shared use trail networks allow users to be outdoors, experience nature, and socialize.

Equestrian

Shared use paths allow equestrians to take part in social activity and ride on a local scale. These trails should allow for single-file and staggered riding alongside other types of trail users.

Walker, Jogger, In-Line Skater/Recreational and Fitness Walker/Jogger Characteristics

Trail Use Pattern

- Will use the same trails daily or several times per week if paths offer convenience and ease of access (most live within 3 miles of the trail they are use)
- Recreational user wants trails that provide social interaction, scenic beauty, or both
- Will use sidewalks to get to a trail system in urban and suburban settings
- Will use trails year-round, although spring, summer, and fall are the most popular seasons

Walker, Jogger, and In-Line Skater General Speeds and Distances

Strolling Walker

- 1-2 miles per hour
- 2-3 miles

Casual Walker

- 2-3 miles per hour
- 3-6 miles

Fitness Walker

- 3-4 miles per hour
- 6-9 miles

Fitness Jogger

- 6-7 miles per hour
- 3-15 miles

Casual Inline Skater

- 5-10 miles per hour
- 6-10 miles

Recreational In-line Skater

- 10-15 miles per hour
- 10-20 miles

Fitness In-Line Skater

General

- 15-20 miles per hour
- 10-25 miles

Elite/ Commuting In-Line Skater

- 15+ miles per hour
- 20-30 miles



Recreation Setting Preferences

- Recreational user finds sense of place, natural setting, scenery, and being away from traffic important (less so with fitness user)
- Prefers looped configurations in all settings, with 2-4 miles suitable for beginners and 5-9 miles for fitness walkers
- Has a strong desire for safety and security, lack of safety and security serve as a major reason for underutilized trails.

Recreational In-line Skater Characteristics

Trail Use Pattern

- Seeks out nearby trails for daily use, but will travel to a specific trail on weekends
- Prefers loop system, with 10-15 miles minimum (will use out and back if no other choice is available)

Recreation Setting Preferences

- Values smooth, wide trails; rough trails prove especially troublesome for beginners
- Seeks trails that do not receive heavy use
- Does not prefer technically difficult trails with sharp turns, too many steep hills, or poor stopping conditions
- Does well on trails designed similar to bike trails, especially when they are 10 feet wide or wider
- Routine sweeping of the trail is important

Fitness In-Line Skater Characteristics

Trail Use Pattern

- Prefers routes that offer challenges and have enough distance to get in a good workout (10- 25 miles)
- May go out daily or several times per week and will routinely use the same trails close to home
- Prefers loop system

Recreation Setting Preferences

- Values smooth, wide trails; rough trails prove especially troublesome for beginners
- Primarily uses a series of streets, roads, and trails to create a long enough route
- Does not desire technically difficult trails with sharp turns, too many steep hills, or poor stopping conditions
- Has facility needs similar to those of bicyclists

Commuting In-Line Skater Characteristics

Trail Use Pattern

- Uses skating as a form of transportation
- Uses trails where available, but will also use streets and roads
- Needs traffic enforcement, security, skate friendly routes to and from



- work sites
- Other preferences are similar to those of transportation bicyclists

Family Bicyclist Characteristics

Trail Use Pattern

- Prefers bike trails and quiet streets (to avoid heavy traffic), with preference for trails if conveniently located
- Most activity happens close to home, but will also use trails extensively on vacation
- Rides in family groups, often including small children

Recreation Setting Preferences

- Controlled, traffic-free access to trails is the most important consideration
- Quality of the riding experience is of primary importance, with length being secondary (20 miles maximum)
- Needs good information for planning trips and access to support facilities (rest areas, parking lots, water sources) and prefers restrooms to portable toilets
- Connections to parks and playgrounds are important
- Prefers scenic areas but not challenging terrain, especially when children come along

Recreational Bicyclist Characteristics

Trail Use Pattern

- Seeks out and travels to trails and bicycle-friendly areas away from home, either as a day or overnight trip
- Prefer trails, but will use low volume roads that offer safety, and convenience

Recreation Setting Preferences

- Repeat users do not desire trails shorter than 10 miles; 20 miles is the desired minimum
- Prefer looped configurations of varying lengths over out and back systems
- Sense of place and an interesting experience are important, with riders seeking places that possess scenic quality and interesting natural or (if in urban setting) built forms

Fitness Bicyclist Characteristics

Trail Use Pattern

- Will use a combination of roads and trails that are long and/or challenging experiences offering a good workout
- Prefers trails of sufficient length (20 or more miles) and allow for faster speeds with minimal user conflicts
- Will routinely use the same routes for challenges and timing, often daily

Bicyclist General Speeds and Distances

Family Bicyclist

- 6-10 miles per hour
- 5-10 miles of distance

Recreational Bicyclist

- 10-15 miles per hour
- 10-20 miles

Fitness Bicyclist

- 15-20 miles per hour
- 20-40 miles

Transportation Bicyclist

- 20+ miles per hour
- 40+ miles



- Frequently extends the season by riding earlier in the spring and later in the fall than recreational riders

Recreation Setting Preferences

- Trails need to offer varying difficulty and lengths; prefers interconnected loops
- Not primarily motivated by experiencing a natural setting, but will select this type of trail if other needs are met

Transportation Bicyclist Characteristics

Trail Use Pattern

- Not dependent on trails, but will use them if convenient, safe, and direct

Recreation Setting Preferences

- Lack of a safe system of roads (with bike lanes or routes) is a major barrier
- Prefers low volume roads with low percentage of truck traffic
- Trail design remains critical, with the most important criteria being the ability to go fast with good sight lines and direct routes

Shared Use Paths Guidelines and Considerations

Shared Use Path Widths

The appropriate pavement width for a shared use path is dependent on the context, volume, and mix of users. The minimum width for a two-directional shared use path is 10 feet. However, trends vary in width from 10-14 feet depending on a trail's level of use and variety of users.

In very rare circumstances, designers may use a reduced width of 8 feet in the the following conditions:

- Expectation of low bicycle traffic, even on peak days or during peak hours
- Expectation that pedestrian use of the path is only occasional
- Horizontal and vertical alignments provide safe and frequent passing opportunities
- Regular maintenance traffic will not occur on the path

In addition, designers may specify a path width of 8 feet for a short distance when physical constraints such as environmental features, bridge abutments, utility structures, or fences exist. Designers should consider the addition of warning signs that indicate when a pathway narrows, per the Manual on Uniform Traffic Control Devices (MUTCD), at these locations.

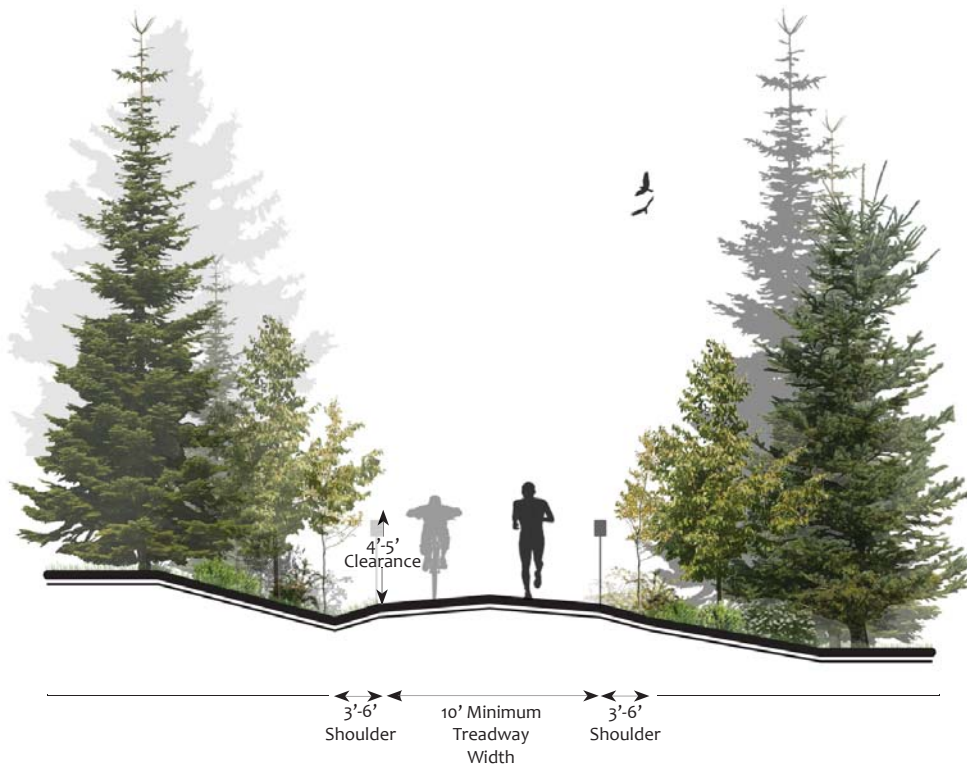


Figure 2-3: Two-Way Shared Use Path

Shared use paths require a wider tread when paths are frequently used by both pedestrians and other users. AASHTO recommends wider pathways, typically 11-14 feet, in locations that serve high user volumes. Eleven-foot wide paths enable a bicyclist to pass another user going the same direction, while another user is approaching from the opposite direction at the same time. AASHTO recommends that designers should incorporate wider paths in the following situations:

- Where significant use by in-line skaters, adult tricycles, or other users that need more operating width, occurs
- Where larger maintenance vehicles use the path
- On steep grades to provide an added passing area
- Through curves to provide more operating space

Shared Use Path Shoulders

Ideally, the design should maintain a graded shoulder at least 3 to 5 feet wide with a maximum cross-slope of 6:1 on each side of a pathway. At a minimum, the design should provide a 2 foot graded shoulder that has a maximum slope of 6:1 to provide clearance from lateral obstructions such as shrubs, large rocks, bridge abutments, and poles.

Where the design introduces features such as bicycle railings or fences with



Butler Freeport Trail
Butler County

Photo Credit:
Butler Freeport Community Trail



appropriate flaring end treatments, a lesser clearance (not less than 1 foot) is acceptable. Mark barrier or rail ends that stray from the 2 foot clear area with object markers and warning signs.

Where a path is adjacent to parallel water hazards or downward slopes are 3:1 or steeper, consider a wider separation. A 5 foot separation from the edge of the pavement to the top of a slope is desirable. Depending on the height of the embankment and condition at the bottom, the shoulder design may require a physical barrier such as dense shrubbery, railing, or fencing.

Safety Rails

Where a recovery area (i.e., distance between the edge of a path pavement and the top of a slope) is less than 5 feet, AASHTO recommends physical barriers or rails be constructed in following situations:

- Slopes 1:1 or steeper, with a drop of 1 foot or greater
- Slopes 2:1 or steeper, with a drop of 4 feet or greater
- Slopes 3:1 or steeper, with a drop of 6 feet (1.8 m) or greater
- Slopes 3:1 or steeper, adjacent to a parallel water hazard or other obvious hazard

Protective railings, fences, or barriers on either side of a shared use path should be a minimum of 42 inches. In some locations consider a 48 inch high railing to prevent bicyclists from falling over the railing in the event of a crash. This includes bridges or bridge approaches where high-speed and/or steep-angle (25 degrees or greater) impacts between a bicyclist and a railing may occur. For example, at a curve at the end of a long, descending grade where the curve radius is less than that appropriate for the design speed or anticipated speed.

On elevated structures, the openings between horizontal or vertical members on railings should be small enough that a 6 inch sphere cannot pass through them in the lower 27 inches. For the portion of railing higher than 27 inches, openings may be spaced so that an 8 inch sphere cannot pass through them. This prevents children from falling through the openings. Where a bicyclist's handlebar may encounter a railing or barrier, a smooth, wide rub-rail may be installed between 36 inches to 44 inches in height, reducing the likelihood that the railing will contact or catch a bicyclist's handlebar.

Railings that protect users from slopes or where the design uses railing to discourage path users from venturing onto a roadway or neighboring property can typically have relatively large openings. A typical design includes two to four horizontal elements with vertical elements spaced widely, but frequently enough to provide the necessary structural support. Where the path-side hazard is a high vertical drop or a body of water, use engineering judgment to determine whether to use a railing suitable for bridges.



In addition to railings, you may use other materials to separate paths from adjacent areas and hazardous conditions, or to discourage pathway users from venturing onto adjacent properties. Berms or vegetation can serve this function.

A 10 foot vertical clearance to obstructions is desirable. Design should not permit fixed objects to protrude within the vertical or horizontal clearance of a shared use path. Eight feet is the recommended minimum vertical clearance for constrained areas. In some situations, the design may require a vertical clearance greater than 10 feet to provide access for maintenance and emergency vehicles.

Grade

Keep grades on shared use paths, in independent corridors, to a minimum, especially on long inclines. Grades greater than 5 percent are undesirable because the ascents prove difficult for many users, and the descents cause some users to exceed speeds which they can navigate competently. In addition, shared use paths generally remain open to pedestrians, therefore the grades on paths must also follow ADA guidelines.

Limit the grades on paths in independent rights-of-way as follows:

- 5% maximum for any distance
- 8.3% maximum for up to 200 feet
- 10% maximum for up to 30 feet
- 12.5% for up to 10 feet

Additionally, no more than 30 percent of the total path length should have a grade exceeding 8.3 percent. Where grades exceed 5 percent, the design should incorporate a resting interval at the end of any segment of maximum length as described above. A resting interval must be at least 5 feet long, be as wide as the path, and have a maximum slope not exceeding 5 percent. Provide smooth, gradual transitions between sloped segments and resting intervals.

Options to mitigate excessive grades on shared use pathways include the following:

- Use higher design speeds for horizontal and vertical curvature, stopping sight distance, and other geometric features.
- When using a longer grade, consider an added 4-6 feet of width to permit slower bicyclists to dismount and walk uphill, and to provide more maneuvering space for fast downhill bicyclists.
- Install the hill warning sign for bicyclists and an advisory speed plaque, if appropriate, per the MUTCD.
- Provide signing that alerts path users to the maximum percent of grade as shown in the MUTCD.



York County Heritage Trail
York County

Photo Credit: Carl Knoch



- Exceed minimum horizontal clearances, recovery area, or protective railings.
- If other designs prove impracticable, use a series of short switchbacks to traverse the grade. If using this method AASHTO recommends widening the path an extra 4 to 6 feet of path to provide maneuvering space.
- Provide resting intervals with flatter grades to permit users to safely stop periodically to rest.

Grades steeper than 3 percent may prove impractical for shared use paths with crushed stone or other unpaved surfaces for both bicycle handling and drainage erosion reasons. Typically, avoid grades less than 0.5 percent, because they do not efficiently convey surface drainage. When building paths on flat terrain, increase the proposed grades to provide a gradual rolling vertical profile. This design conveys surface drainage to outlet locations.

Drainage

Design trails to shed water from the trail as soon as possible as discussed earlier in Chapter 1 – Planning. Typically, a minimum cross slope of one percent provides adequate drainage for a shared use path. A cross slope design is preferred over a crowned design due to ease of construction and simplification of the drainage design. A smooth, even surface prevents the ponding of water on the tread during the winter months. On compacted stone treads, pay particular attention drainage design details to avoid erosion of the tread and adjacent surfaces.

Motorized Vehicle Barriers and In-Line Gates

The unauthorized use of paths by motor vehicles is a problem on many paths. Typically, this is a greater issue on paths that extend through independent corridors as the path may not be visible from adjacent roads and properties.

The AASHTO “Guide for the Planning, Design, and Operation of Bicycle Facilities” discourages the use of bollards and barriers unless a known history of use by unauthorized motor vehicles exists. Bollards, fences, or other similar devices and barriers create permanent fixed object hazards to path users. Bicyclists and other path users often strike bollards on paths, resulting in serious injury.

Splitting the entryway into two sections, by using low landscape features, is the preferred design method to restrict entry to motor vehicles. Each section should be half the nominal path width; for example, split a 10 foot path into two 5 foot sections. Emergency vehicles can still enter if necessary by straddling the landscaping; however, a more appropriate method may entail defining emergency vehicle access through secured access drives. Delineate the approach to the split with solid line pavement markings to guide the path user around the split.



Ernst Trail
Crawford County

Photo Credit: Pashek Associates



Where you can justify the need for bollards or other vertical barriers in the pathway, despite the hazard posed to cyclists, take measures to ensure the bollards or barriers are as safe as possible:

- Mark bollards with a reflective material on both sides and appropriate object markers, per the Section 9B. 26 of MUTCD.
- Bollards should permit passage, without dismounting, for adult tricycles, bicycles towing trailers, and tandem bicycles. Bollards should not restrict access for people with disabilities. Accommodate all legally permitted users; failure to do so increases the likelihood that the bollards will pose a hazard.
- Bollard placement should provide adequate sight distance to allow users to adjust their speed to avoid hitting them.
- Bollards should be a minimum height of 40 inches and minimum diameter of 4 inches. Some jurisdictions recommend taller bollards that users can see from above to reinforce their visibility.
- AASHTO recommends striping an envelope around the approach to the post to guide path users around the object.
- One strategy involves using flexible delineators, which may reduce unauthorized vehicle access without causing injuries commonly incurred by rigid bollards.
- Only install bollards in locations where vehicles cannot easily bypass the bollard. The preferred method is to use of one bollard in the center of a path. When using more than one bollard, space them 6 feet on center. AASHTO does not recommend using two bollards because they direct opposing path users towards the middle creating conflict and the possibility of a head-on collision. Wider spacing allows motor vehicles to enter the path, while narrower spacing may prevent adult tricycles, wheelchair users, and bicycles with trailers to enter the path.
- Set bollards back from the roadway edge a minimum of 30 feet. Bollards set back from the intersection allow path users to navigate the potential hazard before approaching the roadway.
- Install hardware in the ground to hold a bollard or post flush with the surface to avoid creating a safety hazard.
- Lockable, removable (or reclining) bollards allow entrance by authorized vehicles.

Intersections

The way intersections are designed between trails and roads significantly impacts the users' comfort and safety. Intersection designs should address cross-traffic movements, as well as trail users entering and exiting the path. Since conflicts may arise at these junctions, it is important to design intersections in a simple manner, in order to maintain orderly movement between trail traffic and other traffic.

The principles that apply to general safety at crossings, regardless of where a trail crosses a roadway, also apply to trail intersection design. There are a



wide range of design features that improve pedestrian and bicyclist safety at such intersections. This guide provides a general overview of crossing measures recommended by AASHTO in their Guide for the *Planning, Design, and Operation of Bicycle Facilities*.

Trail crossings come in many configurations with many variables, including the number of lanes crossed, divided or undivided roadways, speed and volume of traffic, as well as traffic controls. Since each intersection is unique, designers must exercise judgment to determine an appropriate intersection treatment.

Due to the mixed nature of trail traffic, remember to consider the speed variability of each mode of travel and its resulting effect on design values, when considering design solutions for trail and roadway intersections. Use the fastest vehicle for determining approach speeds, as these users are the most likely to surprise cross traffic at the intersection.

Providing generalized recommendations for the treatment of intersections to accommodate pedestrian and bicycle facilities is difficult. This difficulty stems from the degree of variability in vehicular traffic volumes, pedestrian and/or bicycle traffic volumes, crossing widths, and existing traffic controls. However, some general considerations for the design process include:

Intersection Design Considerations

- Provide positive guidance to pedestrians, bicyclists, and motorists to ensure full awareness at intersections.
- Minimize conflicts and channelize intersections to separate moving conflicts.
- Unavoidable conflicts should occur at right angles.
- Optimize sight triangles, ensure stopping, intersection crossing, and decision sign distances.
- Provide adequate staging and refuge areas for pedestrians and bicyclists.
- Avoid obstacles and highlight unavoidable obstacles.
- At signalized intersections, minimize trail user delay by minimizing traffic signal cycle time.
- Provide adequate signal crossing time for pedestrians.
- Provide easily accessible tactile/audible push buttons.
- Design to assist the user in looking into the direction of the potential hazard.

Traffic Control Features

Adding additional signage to assign the right of ways to pedestrian and bicycle users provides a simple measure of traffic control, especially on rural roads. On more developed roadways, municipalities must ensure the meet the needs of bicyclists and pedestrians in the initial design process.



Typically, the use of traffic control signals are a last resort if you cannot incorporate pedestrian and bicycle users into a controlled intersection environment. Some current traffic control features include activated traffic control signals and pedestrian countdown signals. Only consider these controls after completing a detailed engineering study for an intersection.

Intersection Treatments

Curb Ramps and Aprons

The opening of a trail at a roadway should be at least the same width as the trail itself. If provided, a curb ramp should also be the full width of the trail. The approach should provide a smooth and accessible transition between the trail and the roadway, with a 5 foot radius or flare at appropriate turns. On unpaved trails the design of a trail road intersection should include paved aprons that extend a minimum of 20 feet from paved road surfaces.

Widening Paths at Intersections

For locations where queuing at an intersection results in crowding at the roadway edge, consider widening the path approach. This can increase the crossing capacity and help reduce conflicts at path entrances.

Chicanes

Designers can use chicanes, or horizontal curvatures, to reduce trail users' approach speeds at intersections where sight distance is limited or where users must stop and yield. End chicanes far enough in advance of the intersections to allow the user to focus on the curves of the pathway and then the approaching intersection. Design chicanes for speeds less than 8 mph with a solid centerline to reduce users from cutting corners.

Restricting Motor Vehicle Traffic

Unauthorized access by motor vehicles on pedestrian-only trails, especially those that extend through independent corridors, poses a major issue. The MUTCD permits the R5-3 NO MOTOR VEHICLES sign to be used to reinforce the rules. You can also use bollards or similar barriers to restrict motor vehicle access, but determined individuals who too often use the path illegally usually find ways around the physical barrier.

Crossing Islands

Raised medians significantly lower pedestrian crash rates at multi-lane crossings. Crossing islands particularly benefit trail-roadway intersections with high volumes of speeds, multiple lanes, or excessive roadway width. In addition, crossing islands benefit children, the elderly, and those with disabilities. Design crossing islands to be large enough to accommodate multiple trail users including groups of pedestrians or bicyclists, wheelchairs, and equestrians. Design crossing islands in accordance with the Public Rights-of-Way Accessibility Guidelines (PROWAG).



Additional Crossing Considerations

Transition Zones

It is important to integrate a trail into the existing system of sidewalks and bicycle facilities when the path crosses or terminates at an existing roadway. The terminus should transition smoothly into a safe merging or diverging situation. Use appropriate signage to warn and direct both trail users and motorists regarding these transitions. Each crossing serves as an access point whose design should facilitate movements of path users either entering or exiting the path from the road.

Traffic Calming for Intersections

Traffic calming measures may affect crossing locations positively when the speed of approaching traffic is a concern. Appropriate calming measures include crossing islands, speed cushions, curb extensions, chicanes, raised intersections or crosswalks, and curb reduction at corners. These improvements prepare motorists to yield to path users, reducing the frequency and severity of collision.

Shared Use Paths through Interchanges

Provide separation and continuity where a shared use path travels parallel to a roadway intersection. The design should not require users to exit the path, ride on roads or sidewalks through the intersection, and then resume riding on the path. The designer may need to incorporate grade-separated crossings to enable trail users to conveniently and safely navigate through these exchanges.



NACTO



Urban Bikeway Design Guide

National Association of Transportation Officials

The MUTCD regulates the use and design of all traffic control devices. Trail designers may use signs, pavement markings, and signals in compliance with MUTCD standards to guide and regulate trail user traffic on roads and paths.

Urban Bikeway Design

The National Association of City Transportation Officials (NACTO) *Urban Bikeway Design Guide* (part of the Cities for Cycling initiative) provides guidance on solutions that can help create complete streets that are safe and enjoyable for bicyclists.

The NACTO Urban Bikeway Design Guide stems from the experience gained from the best cycling cities in the world. The designs were developed by cities for cities, since unique urban streets require innovative solutions. Most of these treatments are not directly referenced in the current versions of the AASHTO *Guide to Bikeway Facilities* or the *Manual on Uniform Traffic Control Devices* (MUTCD), although many of the elements are found within these documents.



While this guide does not focus on addressing urban bikeway design, trail designers need to be aware of the NACTO *Urban Bikeway Design Guide*. Those planning trail connections through main street and downtown areas will find it useful. You can download this guide from <http://nacto.org/cities-for-cycling/design-guide/>.

Complementary Publications

In order to plan shared use paths and rail trails, it is important to have an understanding of various types of users, their needs, as well as the planning, design, construction, management, operations, and maintenance needs for shared use paths. The Rails to Trails Conservancy's on-line Trail Building Toolbox, www.railstotrails.org/ourWork/trailBuilding/toolbox/index.html, provides a wealth of information on each of these topics.

In addition, the AASHTO Guide for the Planning, Design, and Operation of Bicycle Facilities establishes the design guidelines recognized by PennDOT.

You can obtain additional information by contacting the Northeast Regional office of the Rails to Trails Conservancy at:

- Rails-to-Trails Conservancy
21221 Ward Court, NW, 5th Floor
Washington, DC 20037
202-331-9696
www.railstotrails.org

Shared Use Trail & Rail Trail Best Practices

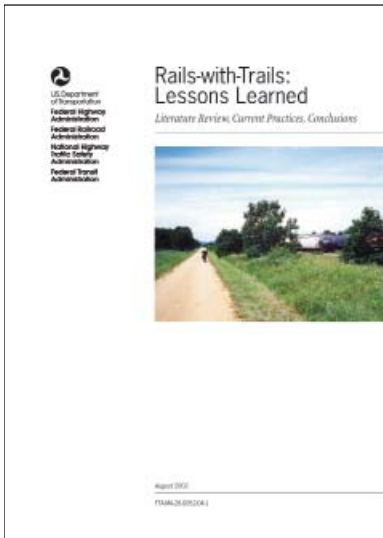
- Great Allegheny Passage, Allegheny, Westmoreland, Fayette, Somerset Counties
- Pine Creek Trail, Lycoming and Tioga Counties
- Butler Freeport Trail, Butler and Allegheny Counties

Shared Use Trail and Rail Trail Trailhead Best Practices

- Pine Creek Rail Trail (various locations, various amenities), Lycoming and Tioga Counties
- Little Toby Rail Trail; Elk and Jefferson Counties
- Lowlands Trail (viewing blinds, parking, kiosk); Cameron and Potter County
- Youghiogheny Trail, Cedar Creek Park, Westmoreland County



Rails-with-Trails



Rails-with-Trails: Lessons Learned
U.S. Department of Transportation

Rails-with-Trails (RWT) are pedestrian pathways constructed on or adjacent to active railroad rights-of-way. Do not confuse Rails-with-Trails with Rails-Trails, which are paths constructed on or along abandoned railroad rights-of-way.

This section uses information obtained from the U.S. Department of Transportation's 2002 publication *Rails-with-Trails: Lessons Learned*, available from www.fhwa.dot.gov/environment/recreational_trails/publications/rwt/index.cfm.

General Considerations for Rails-with-Trails

Currently there are no national guidelines or standards that address rail-with-trail facility design. As a result, designers should use standards associated with shared use paths, pedestrian and railroad facilities, roadway crossings, or safety guidelines of railroad rights-of-way to guide the design and maintenance of these trails. Rails-with-trails must accommodate the operational needs of railroads and trail users, without compromising safety or function for either use. Furthermore, designers should base rail-with-trail designs on specific site conditions, needs of the railroad owner, completion of a feasibility study, as well as other regulatory requirements.

Setback Distances

'Setback' refers to the distance between the edge of a rail-with-trail and the centerline of the closest active railroad track. On the other hand, 'separation' refers to the treatment of the space between a rail-with-trail and the closest active railroad track, fence, vegetation buffer, or ditch.

To determine the minimum setback for rails-with-trails consider train speed and frequency, separation techniques, track curvature, engineering judgment, applicable state standards, maintenance needs, historical problems, and topography.

In 2002, the U.S. Department of Transportation published *Rails-with-Trails: Lessons Learned* that discusses appropriate RWT setback considerations. This information comes from the analysis of existing rails with trail setbacks including:

- 27% are setback 21' to 50'
- 20% are setback 12' to 20'
- 13% are setback 8' to 12'
- 13% are setback 2' to 7'
- 12% are setback 51' to 90'
- 10% are setback 90' to 100'
- 2% have unknown setbacks



The range of trail setbacks on existing rails-with-trails varies from less than 7 feet to as great as 100 feet, with an average of a 33 foot setback from the centerline of the nearest track. For curved tracks, typical minimum setback is 8 1/2 feet to 9 1/2 feet. A comparison of varying setback distances for existing trails to both train frequency and speed reveal little correlation. Over half of the existing rails-with-trails have 25 feet or less setback along railways. Trail managers for these well-established trails report few problems, however, train engineers have indicated that they observe daily trespassing and problems in areas with little setback and no physical separation.

Currently, no guidelines exist that contain recommended setback requirements for rails-with-trails, nor a method of determining these needs. Most often, trail planners use AASHTO's *Guide for the Development of Bicycle Facilities* for guidance. Others use their State's Public Utilities Commission's minimum setback or clearance standards for adjacent walkways to guide trail design.

The guidelines contained in *Rails-with-Trails: Lessons Learned* represent minimum setbacks based on the physical size of railroad cars and railroad companies often recommend them along their rail lines and at their public grade crossings.

As noted, some railroads and states have established their own standards. The Alaska Railroad Corporation's rule of thumb for setbacks along main tracks equals one railcar length, or 60 to 70 feet. In contrast, the Maine Department of Transportation allows for trail setbacks that range from a minimum of 18 feet from track centerline, to 12.5 feet in constrained circumstances. The, the BNSF Railroad's policy on RWTs establishes the following setbacks:

Line Type	Train Speed	Minimum Required Setback
Branch	25 mph or less	30' from centerline
Mainline	25 mph to 50 mph	50' from centerline
Mainline	50 mph to 70 mph	between 50' and 100' from centerline
Mainline	70 mph to 90 mph	100' from centerline

* Where train speeds are greater than 90 miles per hour, trails are not acceptable.

** No trails less than 30 feet from centerline of track for any reason.

Other concerns in establishing setbacks include flying/loose debris associated with rail cars and retaining access for rail line maintenance. Trains blow up debris from the railbed, including rocks and other objects deliberately placed on the rails by trespassers. This debris can fall off trains, hang off rail cars, and hit a passerby who travels too close to the tracks. Railroad companies also need access to tracks for routine and emergency maintenance. While



most railroad companies have the ability to maintain tracks from the tracks themselves, it often is more cost effective and less disruptive to access the tracks from maintenance vehicles operating alongside the tracks.

Factors in Determining Rail with Trail Setback Requirements

- Type, speed, and frequency of trains in the corridor
- Separation technique
- Topography
- Sight distance
- Maintenance needs
- Historical problems

Based on this analysis, the authors of *Rails-with-Trails: Lessons Learned* conclude that due to the lack of consensus on acceptable setback distances, the designer, in consultation with local rail operator, must determine the appropriate setback distance on a case-by-case basis. Designers should incorporate an analysis of technical factors.

Corridor Ownership

Corridor ownership may be another determining factor of a trail's setback. Trails proposed for privately-owned property should comply with the railroad's own standards. Planners need to be aware that the risk of injury is likely if a train derails, even for slow-moving trains. Discussions about liability need to factor this into consideration.

Clearance and Setback Width

In many cases, designer can achieve adequate setbacks of 25 feet or greater along most of a corridor. However, certain constrained areas will not allow for desired setbacks. In those circumstances designers should consider additional barrier devices or rights-of-way to ensure the safety of users. In the case of high speed freight or transit lines, designers should locate rails-with-trails as far from the tracks as possible. If the designer cannot achieve adequate setbacks and separation the trail should not be built.

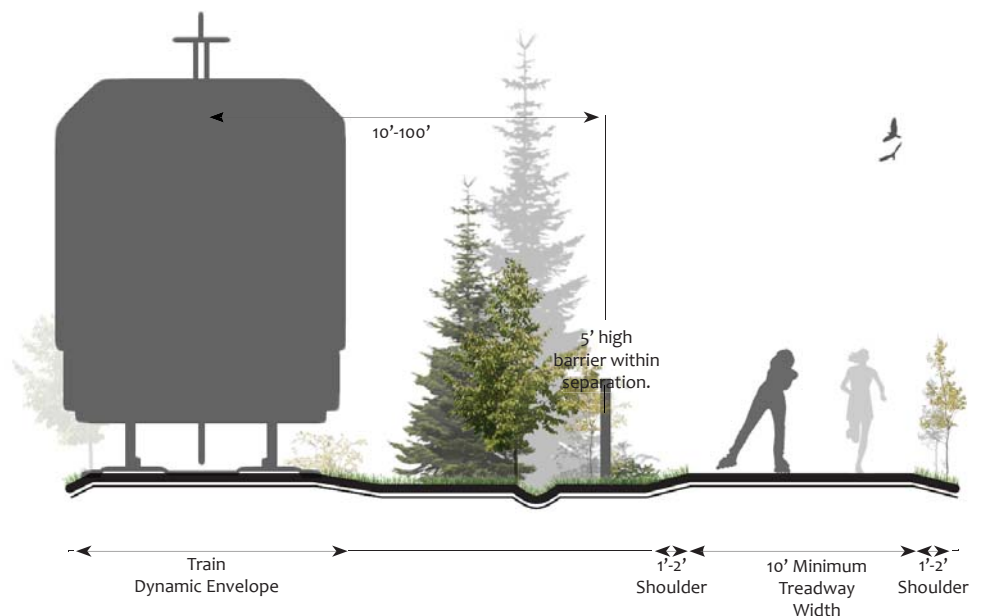


Figure 2-4: Two-Way Rail-with-Trail

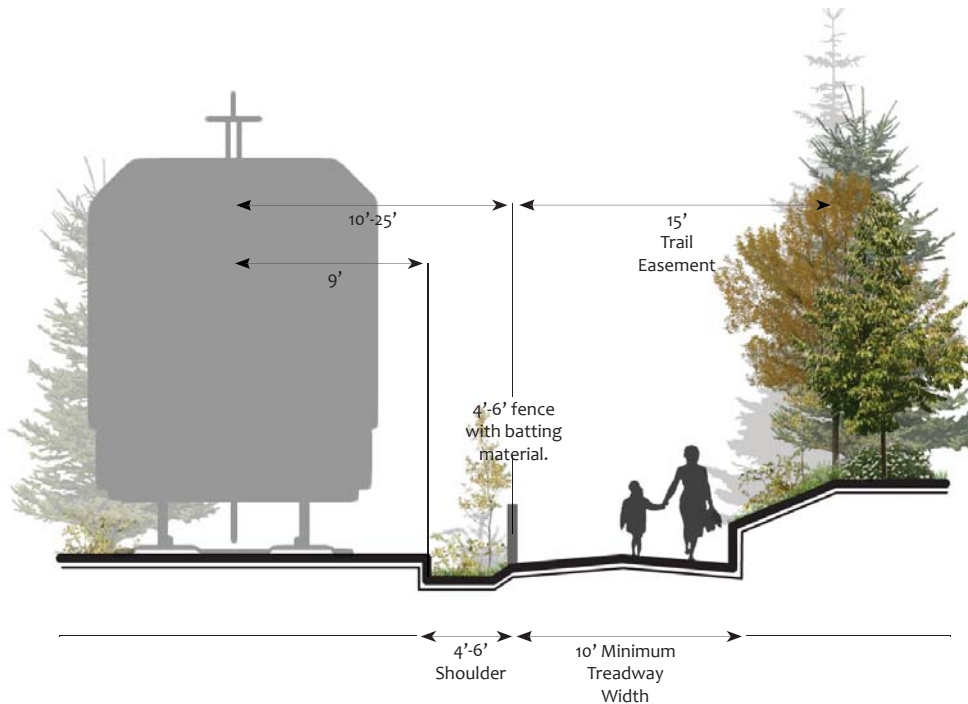


Figure 2-5: Minimum Rail-with-Trail Setback-Constrained Sections
(Depending on Situation)

At an absolute minimum, trail users must be kept outside the dynamic envelope of the track. The ‘dynamic envelope’ is the space required for the train to operate effectively and safely. According to the Manual on Uniform Traffic Control Devices (MUTCD), the dynamic envelope is “the clearance required for the train and its cargo overhang due to any combination of loading, lateral motion, or suspension failure.” Relatively narrow setback distances of 10 feet to 25 feet may be acceptable in certain situations, such as in constrained areas, along relatively low speed and frequency lines, and in areas with a history of trespassing where a trail might help alleviate a current problem.

Pedestrian Barriers and Separation

Over 70 percent of existing rails-with-trails use fencing and other barriers such as vegetation for separation from adjacent active railroads. Barriers include fencing, vegetation, vertical grade, and drainage ditches. Fencing styles vary considerably, from chain link to wire, wrought iron, vinyl, steel picket, and wooden rail, with fencing height ranging from 3 feet to 6 feet, although typical height is 3 to 4 feet.

Most railroad companies require rails-with-trails to provide fencing about 6 feet in height. Fencing may not be needed where a significant deterrent to trespass exists such as a water body, severe grade differential, or dense vegetation. Other barrier types such as vegetation, ditches, or berms often provide separation where a rail-with-trail is located further than 25 feet from the edge of a trail to the centerline of a track, or where the vertical separation



is greater than 10 feet. In constrained areas, an acceptable design may allow for a narrower setback distance with a combination of separation techniques.

When on railroad property, designers must follow the railroad company or agency fencing requirements. In all circumstances designers should coordinate with the railroad to determine appropriate fencing for a site. Typically the trail manager will be responsible for barrier installation and maintenance.

Vertical Separation

Vertical or grade separation achieves many of the same benefits as horizontal separation, and occurs in conjunction with rails-with-trails along cut and fill locations. For example, a trail may be located 20 feet or more below a railroad track on a steep-fill section. In a case such as this, the setback becomes less important than the amount of vertical separation that effectively addresses the negative elements of debris and wind. In cases with vertical separation of greater than 10 feet, the danger from falling objects may increase. A fence or barrier at the top of the slope may help prevent injuries on the trail below.

Vegetation

Vegetation can serve as both a visual and physical barrier between a train track and a trail. The density and species of plants determine how effective a barrier will be in deterring potential trespassers. A dense thicket can be just as effective as a fence, in some cases, keeping trail users off the tracks. Planted barriers generally take a few years to become effective. Therefore designers should recommend installing temporary barriers until plantings matured. Designers should keep in mind that neither vegetation nor fencing must not block the trail user's view of an approaching train at highway-rail crossings.

Fences and Walls

Fences and walls are the most common type of physical barrier used in rail-with-trail corridors. Most railroads require or request fencing, for which the trail management agency will be responsible. The height and type of material used as barriers determines their effectiveness in discouraging trespassing and the resulting impact on required setback distance. For example, a tall wall or fence constructed with materials that are difficult to climb should deter all but the most determined trespasser.

From a trail manager's perspective fencing is a mixed blessing, helping to deter trespassers within these areas yet posing ongoing expense. Installing and maintaining fencing is expensive. Improperly maintained fencing poses a higher liability risk than no fencing at all.

The visual quality of fencing materials can have an impact on illegal activities along rails-with-trails. Particularly for an urban trail in an area with crime



problems, an important design feature may involve maintaining visual access to the trail corridor from adjacent land uses, so that portions of the trail do not become isolated from public view. Fence design in these instances should not block visual access to the trail corridor.

Railroad Track Crossings

Railroad track crossings, or the point at which trails cross active tracks, are areas of greatest concern for railroads, trail planners, and trail users. The FRA, State DOTs, and railroad owners have spent years in an attempt to reduce the number of grade crossings to improve public safety and increase the efficiency of service. Rail-with-trail designs should not create new at-grade crossings wherever possible. Alternative options such as underpass or overpass crossings, as well as modifying existing highway-rail crossings, prove expensive and should only be installed in limited circumstances. These locations include intersections blocked by trains for long periods and extremely dangerous conditions caused by high-speed trains, limited distances, or other conditions.

Some government agencies and railroad companies have adopted policies for new at-grade crossings. In these cases, grade-separated crossings or utilizing existing crossings may be the only options. When designing a trail with trail, designers should carefully evaluate all proposed crossings, with consideration given to:

- Train frequency and speed
- Location of the crossing
- Specific geometrics of the site (angle of the crossing, approach grades, sight distance)
- Crossing surface
- Nighttime illumination
- Types of warning devices (passive or active)

The Railroad Company, State DOT, or Public Utility Commission will need to approve any new crossings, the design of which must comply with the MUTCD. The Railroad-Highway Grade Crossing Handbook (FHWA, 1986) and U.S. DOT Highway-Rail Grade Crossings Technical Working Group (TWG) document, Guidance on Traffic Control Devices at Highway-Rail Grade Crossings (FHWA, 2002) all contain relevant information.

Planned trails have higher quality at-grade crossings with a full complement of automatic gates, warning alarms, and signage. More than half of the existing rails-with-trails in the U.S. include some sort of track crossing, mostly at-grade. Existing at-grade crossings typically have some sort of passive warning device such as a railroad crossing sign. The Lehigh River Gorge Trail is a good example.



Location of the Crossing

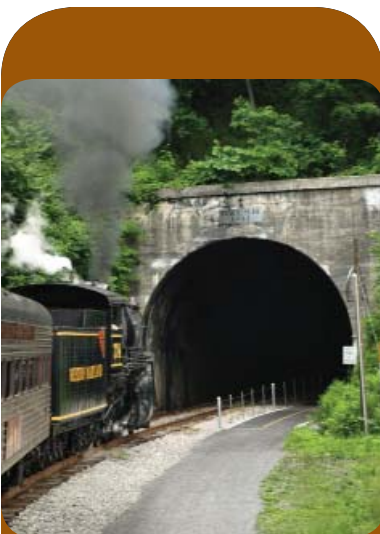
Strategically locate rail-trail grade crossings to channelize trail users to safer crossings, which ultimately reduces illegal trespassing. Do not locate crossings where trains regularly stop, nor at railroad curves. Crossings in these areas are extremely dangerous because of poor sight lines and the opportunity for unauthorized crossing. Where new at-grade crossings are not permitted, the design should channelize users to cross the tracks at roadway locations or develop a grade-separated crossing.

Sight Distance

Adequate sight distance is necessary feature at rail-trail intersections that do not have active warning devices, such as automatic gates or flashing lights. These provide trail users enough time to detect the presence of an approaching train, in order to stop or clear an intersection before the train arrives.

To establish safe movement of trail users across railroad tracks include these three elements:

1. **Advance notice of the crossing:** The first element concerns stopping sight distance, a common consideration in highway intersection design. The stopping sight distance is that distance needed for a trail user to see an approaching train or grade crossing warning devices at the crossing, recognize them, determine what they need to do, and then come to a safe stop at a point 15 feet clear of the nearest rail, if necessary. Pavement markings in advance of the crossing usually mark this point. Measure sight distance along the trail, basing it on a trail user traveling at a given speed, and coming to a safe stop as discussed above.
2. **Traffic control device comprehension:** The second element involves the recognition of the grade crossing warning devices by the approaching user. Remind trail users of the meaning of all traffic control devices in use at grade crossings, such as the fact they should treat a crossbuck sign as a yield sign, or when flashing, flashing lights without gates, must be treated as a stop sign.
3. **Ability to see an approaching train:** The third element concerns the trail user's ability to see an approaching train in order to decide whether it is safe to cross. This involves two different kinds of sight distance considerations for safe movement across the crossing. This third element involves the sight distance available in advance of the crossing, as well as the sight distance present at the crossing itself.



Great Allegheny Passage
Brush Tunnel

Photo Credit:
Allegheny Trail Alliance

Approach sight distance, or corner sight distance, involves a clear sight line in both directions up and down the tracks. This sight line should allow a trail user to determine in advance of the crossing if it is safe to proceed across



the tracks without having to come to a stop. These sight triangles depend on both train speed and trail user speed, as determined by the *Railroad-Highway Grade Crossing Handbook* (FHWA, 2007), http://safety.fhwa.dot.gov/xings/com_roaduser/07010/index.htm.

Often topography, vegetation, or structures obstruct sight triangles. If the design cannot establish clear sight triangles, then place warning signs or reduced speed limit signs in advance of the crossing. As another option you can place stop or yield signs along the trail at the crossing.

Clearing sight distance, which applies to all crossings without automatic gates, involves a clear sight line in both directions present at the crossing itself. A trail user stopped 15 feet short of the nearest rail must see far enough down the track in both directions to determine they can cross the tracks, to a point 15 feet past the far rail, before the arrival of a train. Designers must also consider that at crossings without gates with multiple tracks the presence of a train on one track can the view of a second train approaching on an adjacent track.

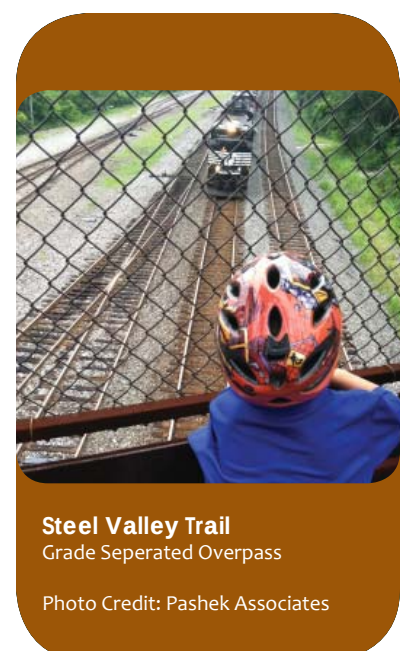
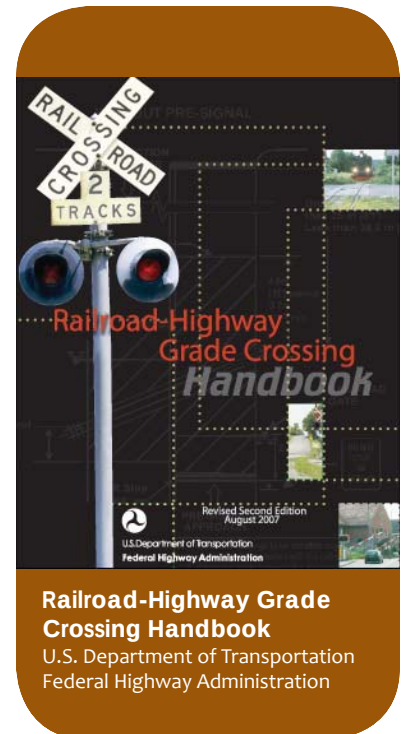
Approach Grades and Angle

AASHTO and ADA specify grade requirements for shared use paths. These requirements permit trail grades over 5 percent for short distances and they do not recommend grades over 5 percent for crossing approaches. In general, keep the trail approach at the same elevation as the track. Steep grades on either side of the track can cause bicyclists to lose control, may block sight lines, and may distract trail users from crossing conditions.

The angle of a crossing represents another critical issue, particularly for bicyclists and those with disabilities. The AASHTO Bike Guide makes the following statement with respect to the crossing angle of a bikeway at a railroad track:

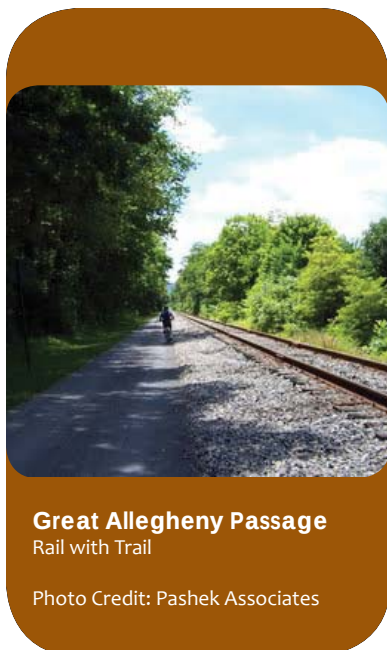
“The risk of a fall is kept to a minimum where the roadway or shared use path crosses the tracks at 90°. If the skew angle is less than 45°, special attention should be given to the bikeway alignment to improve the angle of approach, preferably to 60° or greater, so bicyclists can avoid catching their wheels in the flange and losing their balance.”

Efforts to create a right-angle crossing at a severe skew can have unintended consequences, as the reversing curves required for a right-angle approach can create other problems for bicyclists. It is often best to widen the roadway, shoulder, or bike lane to allow bicyclists to choose the path that suits their needs the best. On extremely skewed crossings (30° or less), it may be impracticable to widen the shoulders enough to allow for 90° crossing; widening to allow 60° crossing or better is often enough. It may also be helpful to post a warning sign at these locations.”



Steel Valley Trail
Grade Separated Overpass

Photo Credit: Pashek Associates



Great Allegheny Passage
Rail with Trail

Photo Credit: Pashek Associates

Crossing Surface

The smoothness of a crossing surface has a profound effect on trail users. For example, uneven surfaces or sudden bumps can cause bicycle riders to lose control and crash. For pedestrians, it is important for trails to have a smooth surface and a design that meets ADA Accessibility Guidelines. Moreover, accessible trail design should include tactile warning strips before at-grade track crossings.

The AASHTO Guide for the Planning, Design, and Operation of Bicycle Facilities notes *“The four most common materials used at railroad crossings are concrete, rubber, asphalt, and timber. Concrete performs best, even under wet conditions, as it provides the smoothest ride. Rubber crossings are quite rideable when new, but they are slippery when wet and degrade over time. Asphalt is smooth when first laid, but can heave over time and must be maintained to prevent a buildup next to the tracks. Timber wears down rapidly and is slippery when wet.”*

Nighttime Illumination

Most rails-with-trails experience nighttime use. Therefore, it is important for designers to consider lighting rail-trail crossings. Recommended lighting levels and appropriate location for rail-grade crossings can be found in Illuminating Engineering Society’s publication *American National Standard Practice for Roadway Lighting*. Shield lighting from the locomotive engineer’s view for safety.

Advanced Warning Devices at Rail-Trail Crossings

A variety of warning devices are available for rail-trail crossings. In addition to the MUTCD standard devices, the use of innovative treatments encourages cautious bicyclist and pedestrian behavior. Since no single standard system of traffic control devices has been proven universally applicable for all highway-rail grade crossings, an engineering study, involving both the railroad company and highway agency, should determine the appropriate system.

Two categories of advanced warning devices for rail-trail crossings exist. Signs or pavement markings serve as passive warning devices to alert trail users of rail-trail crossings. These warnings direct trail users to proceed with caution and look for approaching trains. ‘Active warning devices’ advise trail users of the presence or approach of a train at railroad crossings. These devices consist of flashing lights, bells, and automatic gates triggered by the presence of an approaching train.

Warning Signs

The Manual on Uniform Traffic Control Devices (MUTCD) contains a number of warning signs that can be used to indicate the configuration of nearby crossings and warn users of special conditions. The MUTCD applies to all signs, considered traffic control devices, whether on roads or on shared use paths. The MUTCD provides specifications for sign shapes, colors,



dimensions, legends, borders, and illumination or retroreflectivity. However, these standards do not apply to signs that do not function as traffic control devices, such as “No Trespassing” signs and informational kiosks often are required along railroad tracks.

Some state and local governments, railroad companies, and trail managers have used a number of non-compliant signs at rail-trail crossings. Some state or local roadway and trail design guidelines have adopted some of these supplemental signs to provide more information than available on MUTCD compliant signs. Section 1A.10 of the MUTCD specifies procedures for requesting changes or permission to experiment with traffic control signs and devices.

In the case of paved trails, the MUTCD also requires pavement markings at railroad crossings. At a minimum, the marking should include an “X,” the letters “RR,” and a stop bar. For unpaved trails, consider paving approaches to rail-trail crossings to provide a smooth crossing and appropriate pavement markings. If this is not possible, trail designers may need to consider installing additional warning devices.

Tunnels

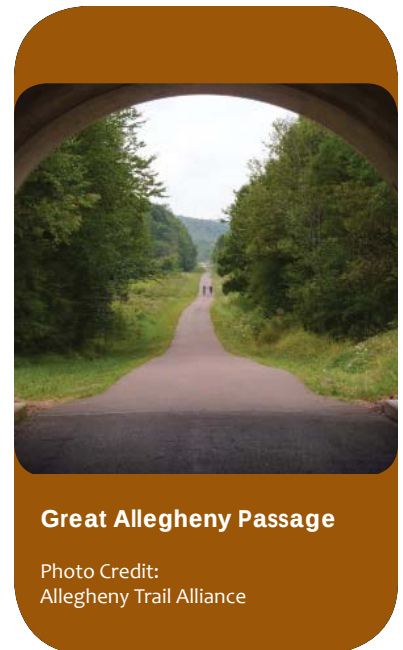
Where single track tunnels exist on a railroad corridor, rails-with-trails generally are not feasible. One known case of a shared rail-with-trail and single track tunnel is along the York County Heritage Trail. Trail users must wait when a train is passing through the tunnel. Usually, tunnels are constructed where the topography dictates the need for going through terrain, meaning that a rail-with-trail would have a difficult time traversing over or around the obstacle to avoid a tunnel.

Trestles and Bridges

The presence of trestles and bridges pose major constraints within the overall feasibility of a rail-with-trail project. Virtually all railroad corridors contain bridges or culverts. In some cases, longer trestles and bridges over roadways, highways, and rivers may exist. However, most no railroad structures are not designed to accommodate pedestrians and therefore are a safety hazard to pedestrians and bicyclists.

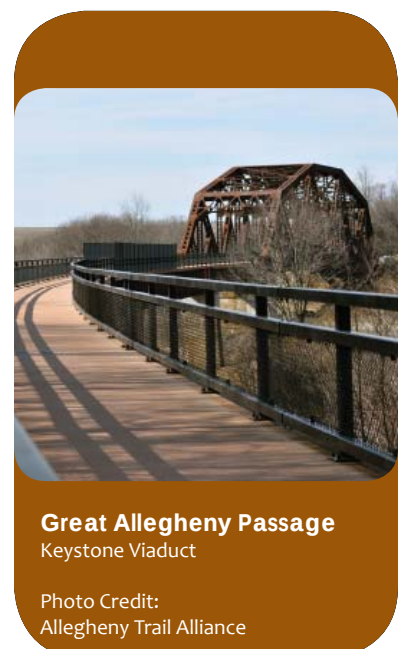
Simple prefabricated bridges constructed over small streams and culverts are not expensive items, but they may affect the feasibility of a project from an environmental standpoint. A new bridge over a highway or on a long trestle may have enormous costs and sometimes may be the single greatest expense of a project.

Rails-with-trails bridges constructed over existing roadways and rail lines must be designed to meet accessibility provisions. This often means the design will include a series of ramps to provide accessibility to the structure.



Great Allegheny Passage

Photo Credit:
Allegheny Trail Alliance



Great Allegheny Passage

Keystone Viaduct

Photo Credit:
Allegheny Trail Alliance



Engineers can design solutions to virtually any challenge. Prototype solutions for bridges along rails-with-trails corridors include :

- Using an existing structure: In rare cases, rails-with-trails have been constructed on existing railroad structures. Another involves designing structure attached in some fashion to an existing trestle or bridge. While method may involve less expense, the designer may need to make structural improvements to the existing bridge and assume maintenance and liability protection for the new combined structure.
- Constructing a new structure: This option offers a simple solution to construct a new structure, rather than trying to use an existing railroad structure. This option will likely be expensive and could have negative environmental impacts if it requires construction in a riparian or other habitat. If constructed over a State highway, it may need time-consuming permit approvals and should adhere to strict design standards.
- Consideration of rail-with-trails operational aspects: After construction of a rail-with-trail, trail maintenance and operations often decrease impacts on railroad companies and offer a safe and pleasant trail user experience.

Invite representatives from railroad operation, track, and signal departments to participate in technical discussions and to provide advice when analyzing the feasibility analysis phase of a rail-with-trail. Rails-with-trails advocates should consider the maintenance and access needs of the railroad operator in the alignment and design of the rail-with-trail. They should provide adequate room for railroad access and operations outside the rail-with-trail and fenced area wherever possible. In areas with narrower than a 25 foot setback, the trail will likely serve as a shared maintenance road. In all cases, the railroad should be provided adequate room and means for access to and maintenance of its tracks and other facilities. The feasibility study and easement/license agreement also should identify the designs and costs of any improvements that would become the responsibility of the rails-with-trails agency.

Considerations for Rails with Trails along Steam Locomotive Lines

Special consideration is warranted trails proposed adjacent to rail lines used by steam locomotives. Depending on operations and the steam locomotive, it may be necessary to blow condensation out of the steam cylinders while the locomotive is standing or moving. The outlets for this escaping steam and moisture are less than 12 inches above the ground. Steam generally shoots out perpendicular to the locomotive and may startle nearby trail users. Also, the reciprocating motion of valves and drive rods require additional lateral clearance for safety reasons. Thus, a feasibility study for



RWTs proposed alongside steam locomotives should take into consideration safety measures and added trail setbacks.

Complementary Publications

In order to plan trails with rails, it is important to have an understanding of various issues related to planning, designing, operating and maintaining trails with rails.

The Rails to Trails Conservancy's on-line Trail Building Toolbox, www.railstotrails.org/ourWork/trailBuilding/toolbox/index.html, provides a wealth of information on the following topics:

Corridor Research

- Active vs. Abandoned Corridors
- Environmental Contaminants

Railbanking

- History of Railbanking
- Railbanking: What and Why?
- How to Railbank
- Example Railbanking Agreement

Acquisition

- Corridor Valuation
- Acquisition Methods
- Salvaging Material
- Financing/Funding

Outreach

- The Railroad
- Opposition & Neighbors
- Public Agencies and Officials
- Supporters
- Making the Case for Rail-Trails

Plan, Design, Build

- Tunnels
- Accessibility
- Crossings
- Design for User Type
- Surface
- Rail-with-Trail
- Gardens
- Public Art
- Bridges

Management & Maintenance

- Liability/Trail Insurance
- Financing/Funding
- Police and Safety
- User Surveys

Rail-with-Trails Best Practices

- Great Allegheny Passage
- York County Heritage Trail
- Little Toby Creek Trail

The U.S. Department of Transportation published *Rails-with-Trails: Lessons Learned* documenting best practices in establishing rails-with-trails and provides recommendations for the planning and design of rails-with-trails. The AASHTO *Guide for the Planning, Design, and Operation of Bicycle Facilities* provides recommendations related to the design of trails with rails. For additional information contact:

- Rails-to-Trails Conservancy
21221 Ward Court, NW, 5th Floor
Washington, DC 20037
202-331-9696
www.railstotrails.org



Equestrian Trails

When planning trail opportunities, give consideration to equestrians. Equestrian trails attract horseback riders seeking a safe, contiguous trail experience in a natural setting. When designing trails for equestrian use strive to meet the goal of providing accessible and pleasurable trail riding experiences with few or no environmental impacts.

General Considerations for Equestrian Trails

The following information defines the common equestrian trail user, the nature of a horse, and design elements to accommodate equestrians.

User Information

Horseback riders may desire to have an individual experience seeking solitude in nature with their horse, or they may desire a social experience riding in groups. Trail riding occurs at a walking pace or slow canter in a couple hours time. This slow pace allows riders to travel safely in groups, but may increase to elevate a trail's difficulty for both the rider and animal.

Horseback riders do not need a wide or highly developed trail; however, they do prefer variety. Water crossings, fallen logs, grade climbs and descents, open areas, and woods create interesting dynamics throughout a trail system. At trailheads where camping is permitted, high lines or roofed tie stalls are preferred and are more desirable than corrals. Horses can easily kick down corrals, corrals take up a lot of space, and require horses to be familiar with each other.

Dedicated equestrians travel to local and regional areas to ride designated trails or a network of varying length looped trails. Local trail riders' trips in the immediate area usually range from 7 to 10 miles per day. Dissimilarly, destination trail riders often travel long distances to publicized trail networks and ride about 10 to 15 miles per day and 25-30 miles on an average weekend trip.

Understanding Horses

The average horse used for recreational riding or driving weighs approximately 800 - 1400 pounds and travels 3 to 5 mph at a walk or slow gait. As herd animals horses feel more secure in groups because there is safety in numbers.

A major concern of equestrians is safety. The safety of horses and their riders depends on minimizing opportunities for horses to be frightened and providing an acceptable trail tread. Good sight lines, clearing width and clearing height of 10 to 12 feet are desirable whether managing the trail as a shared use trail or single use trails.

Understanding Horses

From a horse's point of view, fishing rods look suspiciously like buggy whips. The ticking of bicycle gears may sound like an electric fence charger at home. Boisterous dogs look like wolves and persons with high backpacks or carrying canoes look like large animals. Horses that are not usually scared or spooked by cars, tractors, or ATVs in their home setting may have an entirely different attitude in an unfamiliar area. Although wildlife usually may not concern seasoned trail horses, anything novel such as a grouse flying overhead or a deer dashing across the trail is always something of which to be aware.

**From the U.S. Forest Service
Equestrian Design Guidebook
for Trails, Trailheads, and
Campgrounds**



Local and Destination Equestrian Rider Profile

Trail Use Pattern

- Destination trail riders travel to trails and public land areas to ride designated trails or a network of trails through the forest; local trail riders use trails in the immediate area where they keep or board their horses
- Destination riders ride 10–15 miles per day, 25–30 miles on an average weekend trip; local riders average 7–10 miles per day
- Prefer looped configurations with varying conditions and mileage
- Local riders need direct access to trails from boarding areas
- Riders like to remain self contained, with special trailers used for hauling horses and to house riders at night
- Often travel long distances to trail systems that provide many miles of trail

Recreation Setting Preferences

- Do not need a wide or highly developed trail
- Single-file trails make horses easier to handle and need less maintenance
- Need water nearby for stock
- Variety in a trail is desirable, including water crossings, hill climbs and descents, open areas and woods
- Bridges need to be about 6' to 8' wide and clear zone above the trail has to be at least 9' high but prefer 10' to 12' high
- Large, open flat field with good drainage is best for parking, gravel lots are also acceptable
- High lines or hitch rails are preferred over corrals for day use
- High lines 7' from the ground or roofed tie stalls are recommended for overnight camping

Equestrian Trail Classifications

You will usually find equestrian trails in state, regional, and county trail systems. These trails are typically classified by their managed use, as either accommodating equestrian use, shared use, or carriages.

Equestrian Trail: Equestrian trails typically have natural surface. They are located in natural settings offering scenic beauty, wildlife observation opportunities, and adequate open space for looped trail systems.

Carriage Trail: Carriage trails are routes that have the ability to accommodate both horseback riders and carriage drivers. When considered in a design, a minimum trail width of 8 feet is necessary to accommodate carriages.

Shared Use Path: A shared use path is a path physically separated from motorized vehicular traffic by an open space or barrier and either within the highway right-of-way or within an independent right-of-way.

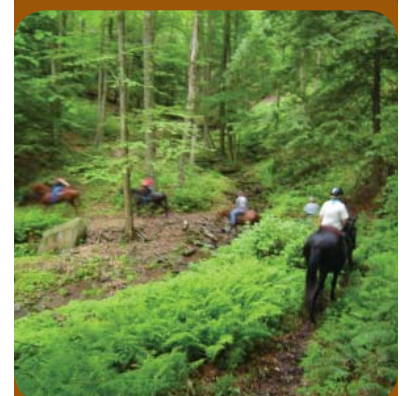
Equestrian General Speeds and Distances

Walking

- 3-5 miles per hour
- 7-10 miles

Walking/ Trotting

- 3-9 miles per hour
- 10-15 miles



PA Grand Canyon

Photo Credit: PA Equine Council



Pedestrians, bicyclists, skaters, wheelchair users, joggers, equestrians, and other non-motorized users share this type of path.

Rule of Thumb for Ride Distances

- One hour loop
3-5 miles
- Two hour loop
6-10 miles
- Three hour loop
8-15 miles

Equestrian Trail Layout Configurations

Plan equestrian trail systems based upon a designed riding experience. This layout should highlight scenic qualities of an area, as well as limit impact on nearby ecological systems. In a normal walking gait, horses travel about 3 to 5 miles per hour. Therefore, it is important to plan trails based on the amount of time needed to complete each ride. Looped trail systems offer a desirable experience for riders as trails of varying lengths are interconnected, giving riders the option of shorter or longer trail rides.

Equestrian Trail Guidelines, Level of Difficulty and Other Considerations

PA DCNR Standards for Equestrian Trails

Trail Type	Easiest (Interpretive)	More Difficult	Most Difficult
Clearing Height	10'	8'	8'
Clearing Width*	8'	6-8'	3-6'
Treadway Width**	2'	2'	18"
Treadway Slope***	Less than: 10%	Less than: 10%	Less than: 15%
	Maximum: 25% up to 300'	Maximum: 25% up to 300'	Maximum: 30% up to 500'
Treadway Cross Slope	0-2%	0-5%	0-10%
Turning Radius	Horses: 4-6' Shared non-motorized use: 6-8' No blind turns		
Sight Distance	Two-way traffic: 50-100' Motorized road crossings: 100-200'		
Surface Materials	Surfacing as needed for stability. Native surface with some imported material. Sidehill trail is constructed. Smooth tread with few obstacles.	Native surface with constructed sidehill trails. Occasional roots and rocks to 6".	Native with limited grading. Roots, rocks, and logs to 12".

** Increase tread width 1 foot on switchbacks.

*** Upper limit of treadway grade and distance depends on soil type, amount of rock, vegetation type, and other conditions affecting trail surface stability.

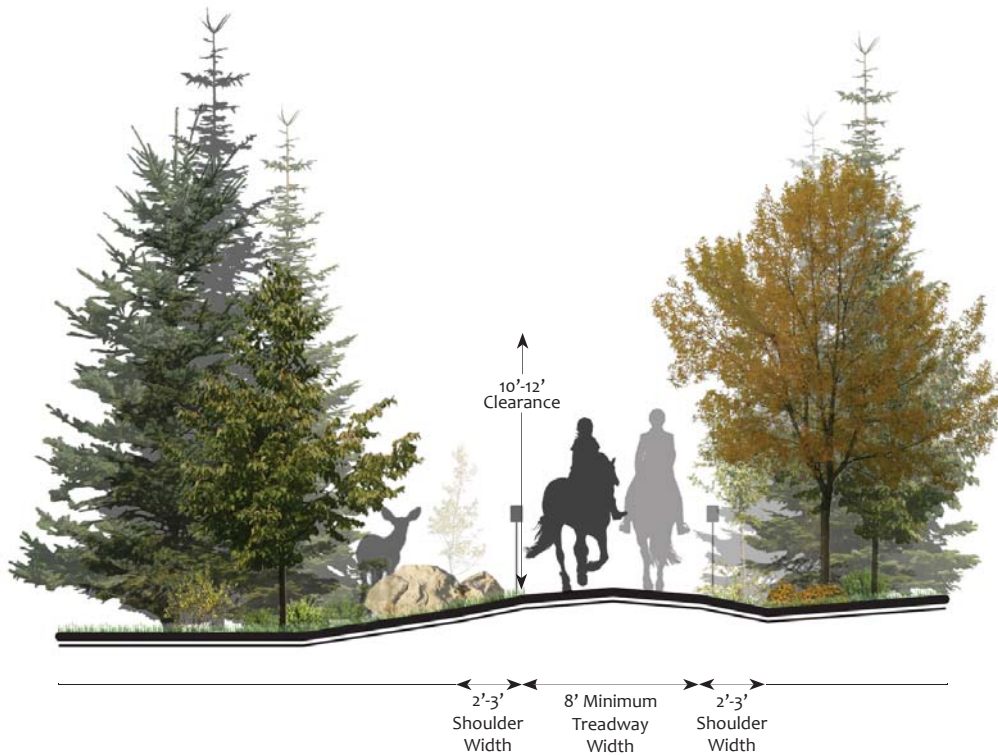


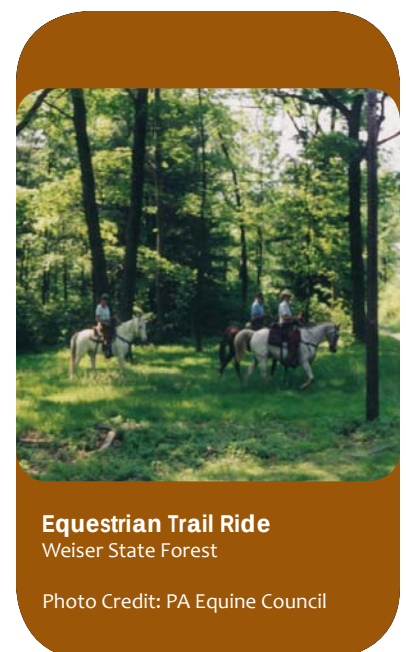
Figure 2-6: Two-Way Equestrian Trail

Equestrian Trail Treads

Design the treadway to accommodate the weight and motion of animals and riders. The choice of varying tread surface treatments affects the speed at which horses and mules can travel. Fine aggregate provides good traction and provides a surface conducive to safe cantering. However, hard surfaces offer poor traction and limit travel to a walk. Remove tread obstacles such as tree roots, holes, dead or leaning trees, and projecting objects for safety reasons. Natural soils can serve as the tread material provided they meet the requirements described in this Chapter.

For durable, all-season urban equestrian trails, plan at least a 2 to 4 foot wide tread surface to accommodate single-file use. Where existing soils cannot adequately support equestrian use, construct a treadway using geotextile, base rock, compacted aggregate, and compacted cushion layer of rounded stone. Horses do not like to walk across angular stone, it is uncomfortable on their hooves. Therefore, avoid using angular stone on the tread surface. Horses tend to favor the outside edges of side hill trail treads.

The trail designer should recommend proper placement and compaction of fill material at trail edges for equestrian safety. This is of special concern where trails are constructed on existing fill where the quality of the fill material and the construction methods are unknown. Fill consisting of large





stones does not accommodate barefoot trail horses, whose numbers are increasing with the natural horsemanship movement. Build retaining walls for newly constructed side hill trails to stabilize older, deteriorating trails.

Equestrian Trail Amenities

On-Trail Amenities

On-trail amenities encourage equestrian use and enhance a trail user's experience. The design of a loop trail system is preferable to linear trails, allowing the connection of multiple routes. By providing these connections, as well as maintaining natural components of the landscape, the design can feature compelling trail sequences and can provide access to water sources throughout the system. Designers should incorporate a hardened, gentle slope leading to trailside streams or ponds to allow horses to access natural source of waters and to prevent erosion.

Trailhead Amenities

Designers should locate trailheads at the beginning of a trail system and include broad range of amenities for trail users. Amenities to consider include: parking, restrooms, picnic areas, signs, and maps. Equestrian trailheads should also include specific amenities to accommodate equestrian use including: potable water, accessible mounting platforms, ordinary mounting blocks (stones or stumps), bunkers for manure disposal, and parking to accommodate haul vehicles and their trailers. Designers should clearly separate equestrian parking from other trail parking by designing parking lanes approximately 30 feet wide by 80 feet long to accommodate large truck and trailer combinations, as well as to allow for the loading and unloading of animals. Riders also desire high lines or hitching posts, covered horse tie stalls for overnight camping, and dedicated equestrian access to the shared use trails or paths.

Shared Use Paths Accommodating Equestrian Use

Heavy equestrian use of a shared use path with a compacted stone surface, or a natural surface trail, may result in the displacement of the trail's surface, and will require a higher level of maintenance to correct. In areas with low horse populations, the amount of equestrian use typically remains limited and result in less tread displacement. In corridors that provide sufficient width, a dual-tread trail can separate equestrians from other trail users. See the section on Shared Use Paths earlier in this Chapter.

Manure remains a concern of non-equestrian users. When visiting a stable, horse track, horse show, county fair, or rodeo, visitors expect the atmosphere surrounding these activities. In particular, horse manure is an accepted nuisance. Although visitors walking or biking on trails that allow equestrian use should expect these conditions, tolerance sometimes wanes. Recognizing this concern, a trailhead can address this issue by providing a length of dedicated equestrian trail connecting to a shared use path. Horses



Moshannon State Forest
Quehanna Equine Camp

Photo Credit: PA Equine Council



tend to relieve themselves at the start of their ride. Therefore, providing a separate trail for the first 500 to 2,500 feet can reduce the non-equestrian users' encounters with horse manure.

Trail managers can minimize conflicts by conveying the message that a trail intends to accommodate equine use from the beginning of the planning process. Equestrian-specific amenities by their very nature establish expectations. Trail users who see these amenities at a trailhead and along the trail corridor often accept horses more readily. This is because they realize the trail supports equine activities. Emphasize equestrian use of a shared use path with signage at trail access points and along the trail to remind users of expectations and etiquette. This understanding should work both ways though on shared use paths. Encourage equestrian users to clear the path of horse manure when possible, just as hikers are expected to clean-up after their animals.

Accessibility on Equestrian Trails

In addition to meeting typical ADA requirements documented earlier, the addition of mounting blocks and ramps at trailheads provides accessibility to those who may not otherwise be able to mount a horse. Mounting blocks and ramps serve a broad range of riders.

Installing mounting blocks or ramps in areas where riders normally dismount and mount can increase usage of trails, trailheads, or campgrounds. Many riders have difficulty getting on and off a horse or mule, especially young children, small or older riders. Many riders in this situation search out large rocks, stumps, or mounds to give themselves a boost. Such objects can be unstable or slippery. Therefore, designers should provide a mounting blocks or ramps instead.

Riders of all abilities and ages can use mounting blocks. A mounting block resembles a short staircase that ends in midair. The rider climbs the stairs to reach the saddled animal standing at the elevated end. Mounting blocks may be made of wood, steel, concrete, plastic, fiberglass, or a combination of these materials. Structures that are more permanent, for example those made of concrete or steel, are most suitable at trailheads or campgrounds that have easy access for construction equipment. Permanent structures also discourage theft. Structures made from wood, fiberglass, or plastic are easier to transport, install, and place along trails.

To meet accessibility requirements, the treads on mounting blocks must be at least 11 inches deep and 36 inches wide. Risers should be uniform and measure between 4 and 7 inches high. Mounting blocks commonly have an overall height of 16 to 28 inches. The need for handrails remains a source of debate. While handrails keep users from falling off platforms, they may catch the animal, rider, assistants, or equipment.



Equestrian Trail Best Practices

- Tiadaughton State Forest
- Bald Eagle State Forest
- Tuscarora State Forest
- Weiser State Forest
- Pine Grove Furnace State Park
- Forbes State Forest
- Valley Forge State Forest
- French Creek State Park
- Jacobsburg Environmental Center, Berks County
- Bucks County Horse Park

Equestrian Trailheads and Camping Area Best Practices

- Showaker Trailhead, Tuscarora State Forest, Perry County
- Thunder Mountain Trailhead, Elk State Forest, Cameron County
- Dyer CCC Camp, Susquehannock State Forest, Potter County
- Kettle Creek State Park, Clinton County
- Quehanna Shared Use Trail (Reservoir Road Camping Area)
- Moshannon State Forest, Clearfield and Cameron Counties
- Keystone Mountain Country Shared Use Trail (12 Mile and Dyer Farm Equestrian Trailheads) Tioga and Lycoming Counties

Riders usually mount from the left side of the animal, passing their right leg over the horse's back. Handrails on the right-hand side of the stair will interfere with the rider's leg movement. This makes a compelling case for leaving handrails off mounting blocks, or for installing handrails that stop before the top step. To meet the ADA/ABAAG requirements, handrails must have extensions, also called returns, at the top and bottom. In this case it is not appropriate to have handrail returns extending into the animal's space.

Complementary Publications

In order to plan equestrian trails, it is important to have an understanding of horses, their riders, and their needs. In 2007, the U.S. Forest Service published the *Equestrian Design Guidebook for Trails, Trailheads, and Campgrounds*, www.fhwa.dot.gov/environment/recreational_trails/publications/fs_publications/07232816/index.cfm. Excerpts from this publication are used herein as it provides detailed background and guidance on equestrian trail planning issues.

Contact the Pennsylvania Equine Council's Trail Stewardship Program for additional information:

- Pennsylvania Equine Council
Post Office Box 62
Huntington Mills, PA 18622-0062
(888) 304-0281
info@pennsylvaniaequinecouncil.com



Cross-Country Skiing Trails

Cross-country skiing routes provide naturally surfaced trails used during the winter season. The design and grooming of these trails provide for a variety of trail users including traditional or classic, skate, and backcountry skiers. Traditional or classic cross-country skiers use a kick or gliding motion, while skate skiers use a unique skating motion to move forward along a trail.

General Considerations for Cross-Country Skiing Trails

The following information defines the general preferences and motivations of cross-country skiers, and provides suggestions for the most effective design of these routes.

User Information

Cross-country skiing is attractive to people of all skill levels and abilities. This winter sport appeals to recreational, fitness, event, and backcountry skiers who enjoy skiing for its range of benefits. Although motivations for skiing differ for each individual, all users enjoy trails groomed trails whose design meets their specific needs.

Trail networks that offer convenience and diverse opportunities for all types of use prove attractive to dedicated skiers. Recreational and fitness skiers prefer a combination of traditional and skate-ski style trails that can accommodate varying skills and preferences. Event skiers prefer hilly, lighted trails that allow skilled users to participate in organized events and train throughout the year.

Cross-country ski trails generally serve as a part of a looped trail system which provides varying conditions and difficulty levels. While recreational skiers do not depend on the technical difficulty of a trail, highly skilled skiers seek out well-groomed trails that offer a mixture of difficulty and length. Trailheads and trail systems alike should offer amenities such as restrooms, warming areas, and drinking water to accommodate users.

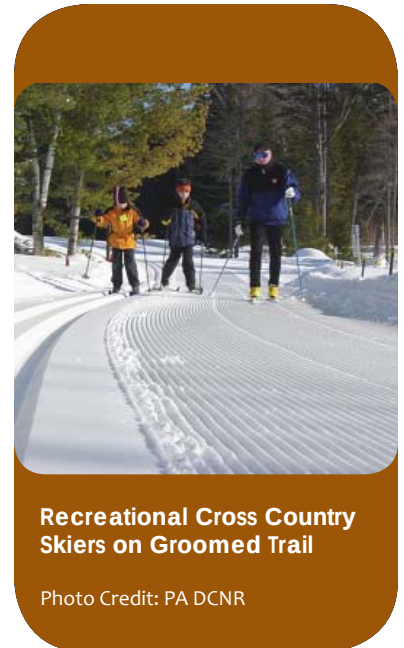
Recreational/Family Skier User Characteristics

Trail Use Pattern

- Seeks out and travels to designated, groomed trails
- Often skis as a family, but you will commonly encounter couples and individual skiers
- Prefers looped configurations with varying conditions

Recreation Setting Preferences

- Prefers larger natural settings
- Attracted to convenience and diverse activity opportunities in the area to accommodate all family members
- Strong preference for well-groomed trails offering a mixture of



Recreational Cross Country Skiers on Groomed Trail

Photo Credit: PA DCNR



difficulty and length, with places for children to practice

- Generally does not want all skate skiing or too many fast skiers on trails
- Prefers a combination of traditional/classic and skate-ski styles to accommodate varying skills and preferences
- Prefers that skate skiers and traditional/classic skiers be on different trails (not side-by-side) at least once in a while in order to have their own space
- Drinking water, restrooms, and warming areas at trailheads remain important
- May do multi-day trips and stay at local lodging facilities or resorts

Fitness Skier User Profile

Trail Use Pattern

- May go daily or several times per week; many use local trails routinely
- Primarily skis on local trails in a park or on a golf course that offers trails providing enough length and challenge

Recreation Setting Preferences

- Needs trail of varying difficulty and length, with looped systems preferred for training
- Must have well-groomed trails; groomed trails are a significant factor in trail selection
- Prefers a natural setting, but having ample trail distance is most important
- Prefers a combination of traditional/classic and skate ski styles to accommodate varying skills and preference; many participate in both kinds of skiing
- Needs and highly supports lighted trails to enable training in the evening during the work week
- Drinking water, restrooms, and warming areas at trailheads remain important

Racer/Event Skier User Profile

Trail Use Pattern

- Uses trails as part of an organized event or competition
- Often falls into the fitness skier category on a day-to-day basis and commonly trains on local trails

Recreation Setting Preferences

- Prefers hilly terrain for good skiing and avoiding boredom during longer events
- Needs support facilities for rest, staging, and comfort
- Needs same trail facilities as fitness skiers for training, including lighted trails
- 10-120 km loops are good for events, as are long linear courses of up to 51 km

Cross-Country Ski General Speeds and Distances

Backcountry Skier

- 1-6 kilometers per hour
- 4-15 kilometers

Recreational/Family Skier

- 3-10 kilometers per hour
- 4-10 kilometers

Fitness Skier

- 5-15 kilometers per hour
- 8-25 kilometers

Racer/Event Skier

- 10-20 kilometers per hour
- 5-50 kilometers



Backcountry Skier User Profile

Trail Use Pattern

- Prefers natural areas with extensive opportunity for skiing
- Does not need groomed trails
- May camp during winter as part of an outing

Recreation Setting Preferences

- Beginners need marked trails with maps and some level of basic maintenance
- Traditional/classic style is predominant
- Prefers looped system, but linear trails offer an acceptable alternative if shuttle service is available
- Length of trail can be less than 5 miles for a day outing and up to 50 miles for a winter camping trip
- Prefers remote settings free of motorized activity

Cross Country Skiing Classifications

Cross-country skiing trails accommodate trail users on a local, county, regional, and state level during the winter months. Local trail systems typically do not support skiing because of their need for specialized grooming. All trails accommodate traditional and skate-style skiers on groomed trails that vary in form. Backcountry skiers typically follow routes with little to no formal maintenance.

Traditional (Classic) Style – one track set/one direction:

Traditional cross-country trails are usually located in park settings rather than in county, regional, and state parks. Total width is 6 to 8 feet.

Traditional (Classic) Style – two track set/one or two directions:

Two-way traditional style routes are the most common type of groomed trails in state parks. These trails receive routine maintenance, especially after snowfall over a few inches. Total width is 8 to 10 feet.

Skate Style – single width/one direction: These trails are typically located in county, regional, or state parks as connector trails from one loop to the next. Trails of this type are common in high use areas where users prefer the separation of skiing styles. Total width is 8 to 10 feet.

Skate style – double width/one or two directions: This configuration is typically found in county, regional, and state parks. Trails of this type are common in high use areas where users prefer the separation of skiing styles. However, uses are often combined because of the increased mileage needed to accommodate separate uses. Total width is 14 to 16 feet.



Ungroomed Cross-Country Trail

Photo Credit: PA DCNR



Combination Traditional and Skate Style – one direction: One-directional use trails are typical in county, regional, and state parks and most popular among skiers. This type of trail helps avoid confusion and conflict, keeps overall tread width narrow, and accommodates high levels of use and a diversity of skier types. Total width is 12 to 14 feet.

Combination Traditional and Skate Style – two direction: This type of trail serves as a linear connector between loops or at trailheads with two-directional use. Total width is 16 to 20 feet.

Cross-Country Skiing Trail Layout Configurations : Cross-country ski trails should offer 5-10 mile looped trail systems of varying terrain consistent with the trail difficulty ratings shown in the following table. Internal connector trails and cutoffs allow different trail lengths and permit easy return access for skiers. Users prefer multiple, short loops ranging from ½ to 3 miles in length, to one long loop.

In addition to looped configurations, linear or point-to-point cross-country ski trails commonly occur with greenways and trail corridors located in developed areas. This approach allows nearby trail users to conveniently access cross-country skiing areas as a major advantage.

PA DCNR Cross-Country Trail Guidelines, Level of Difficulty, and Other Considerations

Trail Type	Easiest (Interpretive)	More Difficult	Most Difficult
Clearing Height	10-12'	10'	8-10'
Clearing Width	8-24" outside of treadway	12-18" outside of treadway	12" outside of treadway
Treadway Width	One-way: 2-4' Two-way: 5-6'	1 ½-4'	1-2' Typically not designed for two-way travel
Treadway Grade	Less than 8% Maximum: 15% up to 150'	Less than 10% Maximum: 20% up to 150'	Less than 15% Maximum: 20% up to 200'
Treadway Cross Slope	0-4%	0-4%	4-8%
Turning Radius	50-100' Gentle turns on downhill slopes. Avoid sharp turns. Never locate a turn at the base of a downhill run.	50-100' Incorporate more turns in trail layout. Avoid sharp turns. Never locate a turn at the base of a downhill run.	50-100' Incorporate more turns in trail layout. Never locate a turn at the base of a downhill run.
Sight Distance	50' on downhill runs, streams, and road crossings.	50' on downhill runs, streams, and road crossings.	50' on downhill runs, streams, and road crossings.
Surface Materials	Consistently smooth treadway. No rocks, roots, dips, bumps, or obstructions. Can be groomed or ungroomed.	Generally smooth treadway. Dips, bumps, or ruts to 8" are uncommon. Can be groomed or ungroomed.	Dips, bumps, or ruts to 12" are common. Occasional surface obstacles. No grooming.



Grades

[Above table does not match recommendations below – both are DCNR publications]

The following guidelines serve as maximum standards for smooth, wide, straight sections of trail with good outruns. Grades should be reduced for less desirable conditions.

A. Beginner Trails (Easiest)

1. 8% maximum for sustained grades more than 300' in length.
2. 10% maximum for grades less than 300' in length.
3. 75 feet maximum vertical difference in any continuous grade.

B. Intermediate Trails (More Difficult)

1. 12% maximum for sustained grades more than 300' in length.
2. 20% maximum for grades less than 300' in length.
3. 150 feet maximum vertical difference in any continuous grade.

C. Experienced Trails (Most Difficult)

1. 15% maximum for sustained grades more than 300' in length.
2. Unlimited gradient for short sections (100' or less).
3. 300 feet maximum vertical difference in any continuous grade.

Tread Tread Widths

[Above table does not match recommendations below – both are DCNR publications]

Light use: 5- 6 feet (one-way traffic)

Heavy use: 7- 8 feet (two-way traffic)

Uphill climbs: 10-12 feet



Figure 2-7: Two-Way Cross-Country Skiing Trail

Groomed cross-country skiing trails typically accommodate two distinct styles, traditional and skating style. Each type of skiing has specific trail width and grooming needs to accommodate individual users. In traditional or classic style cross-country skiing, skiers use a kick and gliding motion to move forward within a set track. A grooming machine or the lead skier, who who “breaks-in” a trail, sets these tracks. Skate skiers also have a unique style, using a skating motion to move forward along a groomed and textured trail route. Skate ski trails do not have tracks on their surface and are usually machine groomed.

Many multi-use trails that accommodate pedestrians, bicyclists, and equestrians during warmer months provide ideal opportunities for cross-country skiing throughout winter months. A minimum of 6 inches of snow on a trail offers excellent skiing opportunities with minimal to no damage of the trail or ski equipment. If the trail sees other winter use, cross-country skiers will often ski off to the side to avoid having their tracks trampled.

Trail Grades, Curves, and Sight Distances

Grade variations enhance the skier’s experience, consistent with the desired difficulty level of a trail. Generally, 1/3 of a given trail provide uphill skiing, 1/3 should provide downhill skiing, and 1/3 should have a rolling grade.

Designers should consider downhill runs as an important feature in the design of cross-country ski trails. The longer and steeper the run, the straighter and



longer the run-out area must be at the bottom of a hill. This allows skiers to dissipate speed to regain a sense of control. Where limited space exists, a rise in grade can offset a loss of run-out distance. Within clearance zones and on two-direction trails, the trail should be wide enough to separate various user types, as well as allow skiers to safely fall and slide several feet.

Since most skiers are not experts, avoid incorporating sharp bends at the base of hills. Long, gentle curves are the preferred design on downhill ski trails; however, in some situations where the design of a trail cannot avoid a curve at the end of a downhill, the design should incorporate a radius of 100 feet or more. This also holds true for all other curves along the trail. In these situations, locate warning signs at least 100 feet before the beginning of the slope.

Forward sight distances are critical on steep downhill runs, and where trails cross waterways, motorized roadways, and other potential hazards. In these cases, incorporate level approaches with forward sight distances of at least 50 feet. On sharp curves and downhill sections, designers need to incorporate a sight distance of at least 100 feet.

Trail Alignment and Preparation

Trail Alignment

The tread of a cross-country skiing trail refers to the underlying trail beneath the compacted and groomed snow. Proper off-season evaluation of trail alignments, tread surface preparation, and trail maintenance must occur to ensure the quality of a trail. Trails should be located in locations where snow will remain the longest and offer the most stable skiing. Important consideration in determining the location of a cross country skiing trail include: sun and wind intensity, solar orientation, location, and overall trail tread. By designing with these factors in mind, trail designers can locate trails geographically to maximize their use throughout the season.

Cross-country ski trails are usually well suited for hardwood and coniferous forests. Hardwood forests offer an excellent setting for these trails because the sun's intensity is lower and the air temperature remains colder than in wide-open flat areas. Incorporating changes in topography offers designers another effective strategy to reduce the extent of sunlight on the trail. Use this strategy along the base of north-facing slopes where the sun is less intense. Also, avoid locating skiing trails at the base of south facing slopes where the sun is most intense.

Before permanently establishing a trail, the designer should conduct field tests to test a trail's alignment, snow displacement patterns, and seasonal wind effects. Even relatively minor shifts in the location of a skiing trail can make a dramatic difference in the impact natural elements will have on it.



Avoid wetland areas and water bodies when aligning cross-country ski trails to minimize ecological impacts, surface quality issues, and safety issues. Do not place ski trails on lakes for the given safety concerns.

Trail Cross Grades

The optimal skiing trail cross-section has a consistent, even grade with a 0 to 2 percent cross-slope. Avoid abrupt changes in grade along a trail to make the route more enjoyable and easier to maintain. The following graphics illustrate optimal, as well as less desirable conditions to consider when designing cross-country skiing trails.

Tread Surface Conditions

The tread surface of a trail is important to consider for overall preparation and grooming maintenance. A level, smooth trail with a short grass cover of 3 to 4 inches across the entire width of the trail offers ideal conditions. This type of tread prevents off-season foot traffic and erosion from creating an uneven tread surface and holds snow better than bare ground or pavement.

Routine mowing maintains the tread surface, as well as reduces the need for brush trimming and controls the growth of woody plants. In protected or environmentally sensitive areas, employ a two-step approach to mowing as the preferred method. The first mowing should occur in late September to cut grass to the desired length and the second mowing should occur a few weeks later to mulch debris left from the first pass. Remove logs, rocks, and other woody debris from the trail shortly before the season begins.

If grass cannot be maintained, woodchips offer the next best alternative. Other surfaces such as bare ground and aggregate surfacing have certain limitations. Asphalt and concrete are the least desirable surfaces since asphalt absorbs more sun energy, loses snow earlier, and is hard on ski equipment when snow cover is thin.

Tread drainage and erosion are important design considerations evaluating overall trail quality and the dual use of ski trails. To prevent erosion the design must stabilize the trail with ground cover during the off-season.

Cross-Country Ski Trail Amenities

Lighted Cross-Country Ski Trails

Lighted ski trails, usually designed as 5 kilometer loops, should incorporate a variety of terrain and challenge into a trail's design to be of interest to skiers. By lighting a previously unlit trail, skiers will have the opportunity to use a trail's facilities for longer periods throughout the day. While it may seem reasonable to light a loop of the recommended length, anything more or less should take into account the added costs for development and maintenance, as well as provide justification through user counts and satisfaction surveys.



Pole-mounted fixtures are commonly spaced every 100 feet with lighted bollards approximately every 80 feet along a trail route. Shroud fixtures to limit light pollution and to eliminate distraction from the experience of skiing at night. The efficient use of transformers, power sources, and wiring should determine the lighting of a specific trail. Consider burying wires, as they provide a better overall aesthetic and ease maintenance efforts.

Trailhead Facilities and Signage

Currently there are no set standards for cross country skiing trailheads or for cross-country ski trails. In larger park settings, the main trailhead is commonly located adjacent to a visitor center. On the local level trailheads usually consist of simple amenities such as a plowed parking area, trail shelter, warming area, portable restroom, and self-registration station.

The signage for cross-country ski trails should be generally consistent with the signage recommendations contained in Chapter 3. It is important to include maps, trail distance, level of difficulty, and warning signs on all cross-country ski trails. Locate trail orientation signs at all trailheads and intersections, with distance markers and warning signs placed along a trail's route. These signs give skiers the information necessary for them to select the proper trail for their needs and avoid potential hazards.

Shared and Summer Uses

Designate cross-country skiing trails for skiing only during the winter months if you want to prevent non-skiers from displacing groomed trails and tracks. Designating these routes for only cross-country skiing minimizes safety concerns and accommodates skiers intolerant of others being on the trail.

Carefully consider the use of trail routes throughout the summer months. Using a common tread for summer and winter trail uses creates a challenge to prevent erosion, maintain groundcover, and maintain the quality of the trail. For these reasons, some agencies keep the two separate by closing off the winter ski trails during summer. Others take a modified approach that allows the use of some trails throughout the year, while other trails remain open for summer or winter use only.

Best Practice Examples

The following represent the use of best practices in the development of cross-country ski trails and amenities. You may find it useful to research and or visit these locations before planning, designing, and constructing your trail.

Cross-Country Ski Trails

- Parker Dam State Park: in northern Clearfield County, grooms and sets track on five miles of ski trails, provides cross-country skiing programming and classes, provides equipment, and guides learners on the trails.



- Bald Eagle: Some park trails and open areas are suitable for cross-country skiing. About seven miles of ungroomed trails are available with proper snow conditions.
- Black Moshannon: All trails are open to cross-country skiers. Trails labeled below with cross-country skiing are recommended.
- Chapman: The 4.4 miles of ski trails connect with numerous trails in the adjoining national forest and state game land. There is a warming hut.
- Clear Creek: A three-mile cross-country ski trail uses portions of Truby, Sawmill, Clear Creek and Ox Shoe trails to make a loop.
- Cook Forest: There are three designated cross-country ski trails.
- Kettle Creek: Cross-country skiing is permitted throughout the park.
- Little Pine: The five-mile Lake Shore Trail follows level terrain and parallels the lake and headwaters and then returns to the starting location. Motor vehicles and snowmobiles are prohibited in this area.
- Ole Bull: Cross-country skiing and snowshoes are allowed throughout the park.
- S. B. Elliott: Ski on more than four miles of easy to moderate roads and trails.
- Sizerville: Park trails provide access to many miles of trails on adjacent state forest land. The average yearly snowfall is 60-70 inches. The park provides parking and restrooms.

Amenities

- Pine Creek Rail Trail, Tioga and Tiadaghton State Forest
- Chapman State Park, Warren County



Complementary Publications

In order to plan cross-country skiing trails, it is important to have an understanding of the various types of skiing, their users' needs, and cross-country ski trail planning, design, construction, management, operations, and maintenance. Excerpts from this publication are used herein as these publications provide detailed background and guidance on cross-country ski trail uses.

- *Best Practices for Recreation: Cross-Country Skiing Model and Signature Sites in the Pennsylvania Wilds*, PA Wilds Recreation Team, 2007
- *Outdoor Recreation Programming Strategy for Cross-Country Skiing in the Pennsylvania Wilds*, PA DCNR

Snowshoeing and Winter Hiking Trails

The development of naturally surfaced trails for winter use allows routes to accommodate the sport of snowshoeing and winter hiking. The amount of snow, as well as the level of grooming on a trail is depends on the type of user group that uses the trail throughout the winter season.

General Considerations for Snowshoeing Trails

The following information defines the general preferences and motivations of both snowshoers and winter hikers, as well as how the design of each type of route can best accommodate its use.

User Information

Snowshoers and winter hikers enjoy being outdoors during the winter season. Winter hiking trails, used for alternative uses during the summer months, can accommodate trail users seeking multiple experiences. Motivations for using natural trails vary widely, ranging from physical challenge to experiencing nature.

A large percentage of natural trail users seek to escape from motorized activity and value experiencing nature. Winter hikers usually seek groomed trails of varying difficulty that provide observation points for users to rest, observe, and socialize. Snowshoers like trails that vary in skill and difficulty levels whether they are groomed or not.

Given the limited demand, designers can typically accommodate both uses on the same trail. With deeper snow, snowshoers tend to frequent the trail most often. Conversely, hikers tend to use trails when the snow is less than a foot deep on unpacked trails. Since both user groups prefer a trail with varying snow conditions, it usually allows users to enjoy the trail without disturbing others.



Recreational Snowshoers

Photo Credit: PA DCNR

Snowshoe and Winter Hike General Speeds and Distances

Snowshoer/Winter Hiker

- 2-4 miles per hour
- 3-5 miles



Snowshoer User Profile

Trail Use Pattern

- Uses trails whether groomed or ungroomed, depending on personal preference and type of snowshoe
- Frequently leaves the established trail
- May walk along groomed ski trails

Recreation Setting Preferences

- Needs unplowed, ungroomed surfaces, although some prefer to follow a groomed trail
- Prefers natural areas
- Snowshoers interested in exercise seek a trail with hills and adequate length

Winter Hiker

Trail Use Pattern

- Seeks out trails for a desired experience, whether near home or some travel distance
- Prefers looped systems over out-and-back trails to vary the experience
- Prefers groomed or plowed trails for ease of walking
- Will seek out trails of varying difficulty
- Likes to stop along the trail to rest, observe, and socialize if hiking in a group
- Expects trail to be of varying difficulty consistent with landscape characteristics

Recreation Setting Preferences

- Large percentage seeks escape from motorized activity, and value experiencing nature
- Natural setting is important to all, with wooded, rolling terrain with wildlife viewing opportunities commonly preferred
- Trail difficulty is an important determinant in trail selection
- Access to the trail serves as a major predictor of use levels
- Length preferences vary widely with skills and individual user preference
- Minimum preferred width should be 18"

Snowshoeing Trail Layout Configurations

Although snowshoeing and winter hiking have grown in popularity over the years, the demand for specialized trails remains small. As a result, the layouts for snowshoeing and winter hiking trails generally remain consistent with cross-country skiing trails. A looped trail system of 3 to 5 miles offers the most desirable design for this type of trail layout in most park settings.



If snowshoers have permission to use cross-country ski trails, trail managers commonly limit access to specific areas of a trail route. Trail blazes often define a route in order to guide trail users and keep them in a specific area and trail managers may relocate the blazes throughout the season.

Snowshoeing Trail Guidelines and Considerations

Slope Considerations for Snowshoeing Trails:

Aspect	Easy	Intermediate	Expert/Advanced
Corridor Clearance (Horizontal)	6-8'	6-8'	6-8'
Corridor Clearance (Vertical)	10'	10'	10'
Average Trail Grade	4%-10%	6%-12%	>12% (most challenging loops)
Maximum Hill Grade	10%-12%	12%-18%	>18%, with 40% max. for short distance

The design and grooming standards for snowshoeing and winter hiking trails are generally consistent with those defined for cross-country ski trails. To accommodate users for both trail types, trail grades should be consistent with easy or intermediate cross-country ski trail difficulty levels. Determine trail widths by the type and level of use the trail will receive. To accommodate one or two-way snowshoeing and/or winter hiking, the tread should be 6 to 8 feet wide.

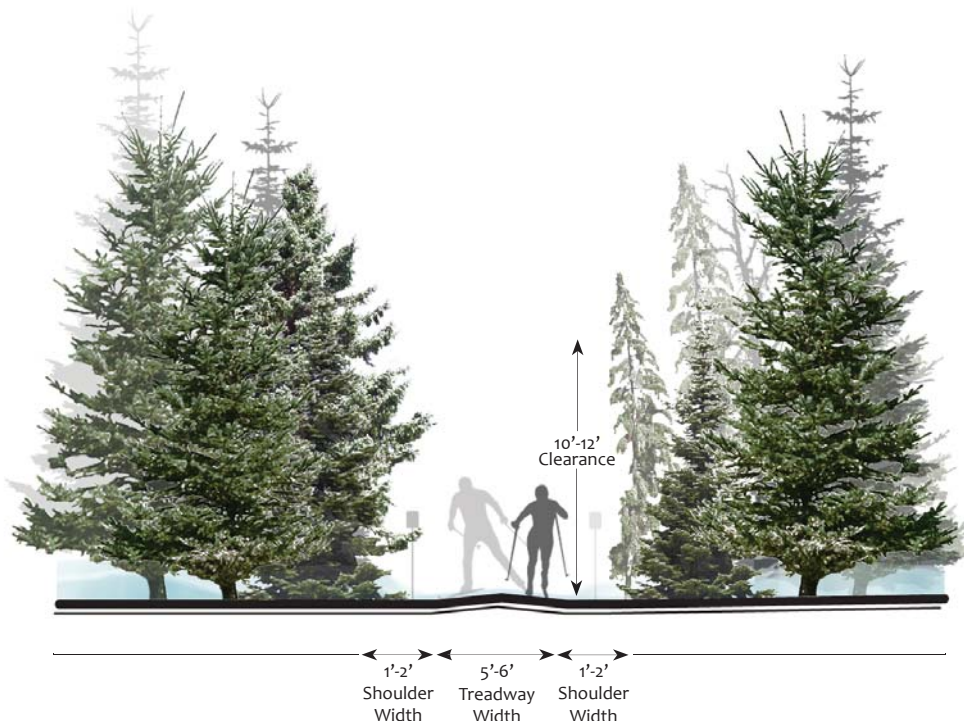


Figure 2-8: Two-Way Cross-Country Skiing Trail



Trail Treads

A trail's tread depends on its seasonal use, type and level of use, and its level of maintenance. Snowshoers and winter hikers commonly follow groomed trails similar to cross-country skiing trails. Maintain packed trails for hiking and snowshoeing once a week or after a significant snowfall. This rate is far less than cross-country ski trails since snowshoes and foot traffic easily accommodate loose or uncompacted snow on a trail. Most often, these trails are located on existing summer-use natural surface trail corridors.

While hikers usually prefer a groomed trail system, snowshoers are more tolerant of ungroomed and uneven surfaces. For ungroomed snowshoeing trails, trail managers typically use tree markers and trail blazes to mark a the route. For cross-country travel, maps and directional signs typically highlight specific areas designated for snowshoeing.

Difficulty Rating System for Snowshoeing Trails

PA DCNR Difficulty Standards for Snowshoeing Trails

Aspect	Easy	Intermediate	Expert/Advanced
Length	1 ½-3 miles	3-10 miles	3-20 miles
Average Trail Grade	4-10%	6-12%	>12% (most challenging loops)
Maximum Hill Grade	10-12%	12-18%	>18%, with 40% max. practical
Character of Trail	Wide trails with ample run-out on hills, nice rolling terrain with easy grade changes.	Introduction to steeper, longer, and more frequent hill climbs, but with ample run-out on hills still important; steepest hills are relatively short; intermediate trails should be combined with easy trails to provide user with diversity and opportunity to work on various skills and endurance	More frequent, steeper, and longer hills with less recovery time in between; run-out area on hills is more constricted, but still safe for skill level; expert trails should be combined with intermediate and easy trails to provide user with diversity and opportunity to work on various skills and endurance; upper end hill grades should be shorter than 50 yards



Accessible Trails

As noted in Chapter 1, Designers must integrate universal design strategies into the design process whenever possible. Newly constructed trails should address these issues during the planning process to eliminate changes in a design during and after construction.

Every aspect of a trail experience should consider including accessible built facilities, amenities, trailheads, and trail corridors to ensure a design's versatility. Moreover, a trail's environment and benefits should be enjoyable and appeal to those with and without disabilities.

Federal regulatory committees or federal legislation have defined four types of accessible routes:

1. Access Routes
2. Outdoor Recreation Access Routes
3. Beach Access Routes
4. Accessible Trails

Access Routes

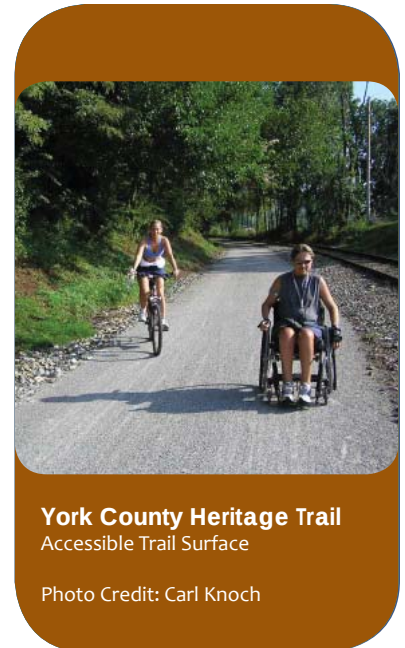
ADA Accessibility Guidelines for Buildings and Facilities (ADAAG) defines Access Routes along walkways. Walkways are exterior pathways (including sidewalks) with a prepared surface intended for pedestrian use. Accessible routes along walks are governed by the Department of Justice regulations titled the [2010 ADA Standards for Accessible Design](http://www.ada.gov/regs2010/2010ADAStandards/2010ADAstandards.htm), www.ada.gov/regs2010/2010ADAStandards/2010ADAstandards.htm.

In the trail environment accessible routes are typically associated with building environments such as comfort stations, nature centers, and the like.

ADA Part II requires at least one accessible route within the site from accessible parking spaces and accessible passenger loading zones; public streets and sidewalks; and public transportation stops to the accessible building or facility entrance they serve.

Accessible routes must meet the following requirements:

- **Floor or Ground Surface:** Floor or ground surfaces shall be stable, firm, and slip resistant. Openings in floor or ground surfaces shall not allow passage of a sphere more than 1/2 inch diameter.
- **Vertical Changes in Elevation:** Changes in level of 1/4 inch high maximum shall be permitted to be vertical.
Beveled: Changes in level between 1/4 inch high minimum and 1/2 inch high maximum shall be beveled with a slope not steeper than 1:2.
Ramps: Changes in level greater than 1/2 inch high shall be ramped.



York County Heritage Trail
Accessible Trail Surface

Photo Credit: Carl Knoch



- **Protrusion Limits:** Objects with leading edges more than 27 inches and not more than 80 inches above the finish floor or ground shall protrude 4 inches maximum horizontally into the circulation path.
- **Vertical Clearance:** Vertical clearance shall be 80 inches high minimum.
- **Slope:** The running slope of walking surfaces shall not be steeper than 5 percent. The cross slope of walking surfaces shall not be steeper than 2 percent.
- **Clear Width:** The clear width of walking surfaces shall be 36 inches minimum.
- **Clear Width at Turn:** Where the accessible route makes a 180 degree turn around an element which is less than 48 inches wide, clear width shall be 42 inches minimum approaching the turn, 48 inches minimum at the turn and 42 inches minimum leaving the turn.
- **Passing Spaces:** An accessible route with a clear width less than 60 inches shall provide passing spaces at intervals of maximum. Passing spaces shall be either: a space 60 inches minimum by 60 inches minimum; or, an intersection of two walking surfaces providing a T-shaped space complying with 304.3.2 where the base and arms of the T-shaped space extend 48 inches minimum beyond the intersection.

Design requirements for accessible walks contain more rigorous requirements than do the requirements for outdoor recreation access routes, trails, and beach access routes.

Outdoor Recreation Access Routes

In 2009 the Federal Access Board released the [Draft Final Accessibility Guidelines for Outdoor Developed Areas](http://www.access-board.gov/outdoor/draft-final.htm) (AGODA), www.access-board.gov/outdoor/draft-final.htm. This document establishes accessible design requirements for outdoor recreation access routes and accessible trails.

AGODA applies to Federal land management agencies, including the U.S. Forest Service, National Park Service, Fish and Wildlife Service, Bureau of Land Management, Bureau of Reclamation, and Army Corps of Engineers. These guidelines also apply to the following non-federal entities that construct or alter facilities on Federal lands on behalf of the Federal government:

- Private entities that construct or alter camping, picnic, or beach facilities on Federal lands pursuant to a concession contract or other arrangement with a Federal agency under which the Federal agency reviews or approves the design of the facility and has a property interest in the facility;
- State or local government entities that construct or alter camping, picnic, or beach facilities on Federal lands pursuant to an agreement with a Federal agency under which the Federal agency reviews or



approves the design of the facility and has a property interest in the facility; and

- Non-profit organizations and state or local government entities that enter into partnerships with a Federal agency to construct or alter trails or viewing areas on Federal lands

Many state agencies and municipalities also view this as the standard for accessible trails within their respective jurisdictions.

Outdoor Recreation Access Routes (ORARs), paths connecting to accessible elements in a picnic area, campground, or trailhead should comply with these recommendations. Table X compares the accessible design requirements of the ADAAG Access Routes and the AGODA Outdoor Recreation Access Routes requirements.



Comparison of Technical Provisions for Access Routes and Outdoor Recreation Access Routes

Aspect	Access Route (ASAAG)	Outdoor Recreation Access Route (ORAR)
Surface	Stable, Firm, Slip resistant	Firm and Stable
Maximum Running Slope	1:12	1: 20 (for any distance) 1: 12 (for max 50 ft) 1: 10 (for max 30 ft)
Maximum Cross Slope	1:50	1:33 <i>Exception: 1: 20 (for drainage purposes)</i>
Minimum Clear Tread Width	36 inches 32 inches (for no more than 24 inches)	36 inches <i>Exception: 32 inches when * applies</i>
Edge Protection	Where provided, minimum of 2 inches.	Where provided, minimum of 3 inches.
Tread Obstacles	(Changes in Level) 1/4 inch (no beveled edge) 1/4 - 1/2 inch must have a beveled edge with a max slope of 1: 2. Over 1/2 inch= ramp.	1 inch high max <i>Exception: 2 inches high max (where beveled with a slope no greater than 1: 2 and where * applies.)</i>
Passing Space	Every 200 feet where clear tread width is less than 60 inches, a minimum 60 X 60 inch space, or a t-shaped intersection of two walks or corridors with arms and stem extending min of 48 inches.	Every 200 feet where clear tread width is less than 60 inches, a minimum 60 X 60 inch space, or a t-shaped intersection of two walking surfaces with arms and stem extending min of 48 inches. <i>Exception: every 300 feet where * applies.</i>
Resting Intervals	(Landings) 60 inch min length, min width as wide as the ramp run leading to it, if change in direction occurs, must have 60 X 60 inch space.	60 inches minimum length, width at least as wide as the widest portion of the trail segment leading to the resting interval and a max slope of 1: 33 <i>Exception: a max slope of 1: 20 is allowed for drainage purposes.</i>

*Refer to the discussion on Conditions for Departure.



Accessible Trails

In addition to defining and establishing accessible design requirements for Outdoor Recreation Access Routes, the AGODA also defines and establishes design requirements for Accessible Trails. At the time of this writing this is the most recent standard for designing and constructing accessible pedestrian trails.

AGODA defines a Trail as follows:

A pedestrian route developed primarily for outdoor recreational purposes. A pedestrian route developed primarily to connect elements, spaces, or facilities within a site is not a trail.

Table X summarizes the accessible design requirements of the AGODA for accessible trails.

Technical Provisions for Trails

Aspect	Trail
Surface	Firm and Stable Exception*
Maximum Running Slope	1: 20 (for any distance) 1: 12 (for max 200 ft) 1: 10 (for max 30 ft) 1: 8 (for max 10 ft) Exception: 1: 7 (for 5 ft max for open drainage structures) Exception*
Maximum Cross Slope	1:20 Exception: 1: 10 (at the bottom of an open drain where clear tread width is a min of 42 inches)
Minimum Clear Tread Width	36 inches for any distance Exception: 32 inches when * applies.
Edge Protection	Where provided, 3 inches minimum height
Tread Obstacles	2 inches high max Exception: 3 inches max (where running and cross slopes are 1: 20 or less) Exception*
Passing Space	Every 200 feet where clear tread width is less than 60 inches, a minimum 60 X 60 inch space, or a t-shaped intersection of two walking surfaces with arms and stem extending min of 48 inches. Exception: every 300 feet where * applies.
Resting Intervals	60 inches min length, width at least as wide as the widest portion of the trail segment leading to the resting interval and a maximum slope of 1: 20. Exception*

* Refer to the discussion on Conditions for Departure.



Accessible Route to Picnic Shelter

Photo Credit: Pashek Associates



Conditions for Departure and Exceptions from the Draft Final Accessibility Guidelines for Outdoor Developed Areas

Compliance will not always result in accessible facilities accessible to all persons with disabilities. Therefore the guidelines recognize this by providing four conditions for departures from the required technical provisions. The Access Board permits departures from certain technical provisions where at least one of four conditions is present for trails, picnic and camping facilities, and beaches.

The four conditions that permit departures from specific technical provisions include:

1. **Where compliance would cause substantial harm to cultural, historic, religious, or significant natural features or characteristics.**

Example: A significant natural feature such as a large rock, outcrop, tree, or water feature may interfere with trail construction or be altered to the extent that the trail could not be made accessible. This includes areas protected under Federal or State laws, such as areas with threatened or endangered species or designated wetlands that could be threatened or destroyed by full compliance with the technical provisions. It also includes areas where compliance would, directly or indirectly, harm natural habitat or vegetation.

2. **Where compliance would substantially alter the nature of the setting, the purpose of the facility, or portion of the facility.**

Examples: A trail intended to offer a rugged experience such as a cross-country training trail with a steep grade or a challenge course with abrupt and severe changes in level. If these types of trails are constructed to comply with the technical provisions for accessible trails, they cannot provide the intended and desired level of challenge and difficulty for users.

Trails that traverse over boulders and rocky outcrops are another example. The purpose of such a trail is to provide people with the opportunity to climb the rocks. To remove the obstacles along the way or reroute the trail around the rocks would destroy the purpose of the trail. The “nature of the setting” may also be compromised by actions such as widening for the construction of imported surfaces on a remote trail or removing ground vegetation in meadows or alpine areas.



3. Where compliance would require construction methods or materials that are prohibited by Federal, State, or local regulations or statutes.

Example: Federally designated and some State designated Wilderness Areas prohibit use of mechanized equipment. Imported materials may be prohibited in order to maintain the integrity of the natural ecosystem. For traditional, historic, or other reasons, many trails are built using only the native soil for surfacing and may not be firm or stable. Some constructed water crossings, which would be needed to provide accessibility, are not permitted under certain laws or regulations.

4. Where compliance would not be feasible due to terrain or prevailing construction practices.

Example: Complying with the technical provisions, particularly running slope, in areas of steep terrain may need extensive cuts or fills that would be difficult to construct and maintain, or cause drainage and erosion problems. Also, in order to construct a trail on some steep slopes the trail may become significantly longer, causing a much greater impact on the environment. Certain soils are highly susceptible to erosion. Other soils expand and contract with water content. If compliance requires techniques that conflict with natural drainage or existing soil, the trail would be difficult if not impossible to maintain.

This condition may also apply where construction methods for particularly difficult terrain or an obstacle would need the use of equipment other than that typically used throughout the length of the trail. One example is requiring the use of a bulldozer to remove a rock outcropping when hand tools are otherwise used.

These conditions for departure do not provide an overall exemption of the entire trail. When the condition for departure no longer exists, the technical provisions are applicable.

These conditions for departures are consistent with the conditions for departure specified in ADAAG and the ADA as well. Example: It may be impracticable in new construction to follow ADAAG where soil and terrain pose obstacles that cannot be remedied. Compliance with the provision for a firm and stable surface might conflict with the prevailing construction practices by requiring the use of a new surfacing material. If the prevailing construction practices do not include importing new surface material and the natural surface material is not firm and stable, the trail may not comply with that specific provision.



The term not feasible specifies what is “reasonably do-able.” It does not refer to the feasibility or possibility of full compliance with the technical provisions. For example, it may be feasible to provide a trail with a 1:20 slope or less up a 1,500 foot tall mountain using heavy construction equipment, but the trail would be at least 5.8 miles long, rather than 2 miles long under a traditional backcountry layout. Further, although feasible, the longer route may cause inappropriate environmental and visual impacts. The intent of this conditional departure is to recognize that the effort and resources needed to comply would be disproportionately high relative to the level of access created. Although technically feasible, the effort and resources needed are not “reasonable.” Therefore, this can be classified as an acceptable condition for departure.

Further, trail construction practices vary greatly, from the use of volunteer labor and hand tools, to professional construction with mechanized equipment. For alterations to an existing trail, the prevailing construction practices are defined as the methods typically used for construction or maintenance of the trail. For new trails, the land manager determines the construction practices employed on each trail. However, this choice of construction practice is primarily determined by the available resources (e.g. machinery, skilled operators, finances) and the environmental conditions (e.g., soil type and depth, vegetation, natural slope). The intent of this conditional departure is to ensure that compliance with the technical provisions does not require the use of construction practices that are beyond the skills and resources of the trail building organization. It is not intended to automatically exempt an organization from the technical provisions simply because of a particular construction practice, (e.g. the use of hand tools or to suggest that hand tools should be used to avoid compliance) when more expedient methods and resources are available.

An accessible trail meets the technical provisions included within the guidelines. A trail is also accessible where one of the exceptions within the technical provisions addresses a specific condition. This is limited to certain exceptions and does not include those that allow for departure from the entire provision.



To comply with AGODA when specifying new signs at trailheads, on newly constructed or altered trails, designers must include trail signs that indicate the following information related to each trail:

1. Length of the trail or trail segment
2. Surface type
3. Typical and minimum tread width
4. Typical and maximum running slope and
5. Typical and maximum cross slope

These signs allow users to determine whether a particular trail is accessible to them at any given time given their abilities.





Design Requirements for Sustainable Trails

Once you have completed the initial planning, and before proceeding with construction, you need to establish the details of the design. The design process builds upon the work completed during the planning phase and results in defining the specific locations, details, and specifications that will guide the construction of the trail. Further, you must consider requirements for designing sustainable trails, trail amenities, and support facilities. These requirements apply to all trail types. When designing trails, you must also consider and address the requirements of all applicable permits as discussed in Chapter 1 - Planning.

With an understanding of the site gained during the planning process, your designer refines the trail alignment based on the trail's design requirements, users' needs, and sustainable design practices. The design requirements for your trail are based on the uses you desire to accommodate.

Your ultimate goal during the design phase is to develop plans and specifications that will communicate all aspects of the design to those responsible for building trails, trail amenities, and support facilities in a sustainable manner.



Dr, J Skills Park, Allegheny County
Hardened tread

Photo Credit: Jon Pratt



Allegrippis Trails
Climbing Turn

Photo Credit: Leslie Kehmeier





Design Requirements for Sustainable Trails

To be successful, a trail must be designed to be physically, ecologically, and economically sustainable. This includes:

- Physical Sustainability**
 Designing trails to retain their structure and form over years of use and under forces of humans and nature is a key factor in sustainability. Trail use promotes change, so trails must be designed in anticipation of change to ensure that they remain physically stable with appropriate maintenance and management.
- Ecological Sustainability**
 Minimizing the ecological impacts of trails, and protecting sensitive natural and cultural resources is fundamental in sustainable trail design and development.
- Economic Sustainability**
 For any trail, the implementing agency or advocacy group must have the capacity to economically support it over its life cycle. Developing and committing to a long-term maintenance strategy is a critical aspect of a successful trail program.

How Does this Translate on the Ground?

While there are many factors that can influence the sustainability of trails, when you get to actually putting them or managing trails on the ground, they should achieve the following objectives.

Connect Positive, and Avoid Negative, Control Points

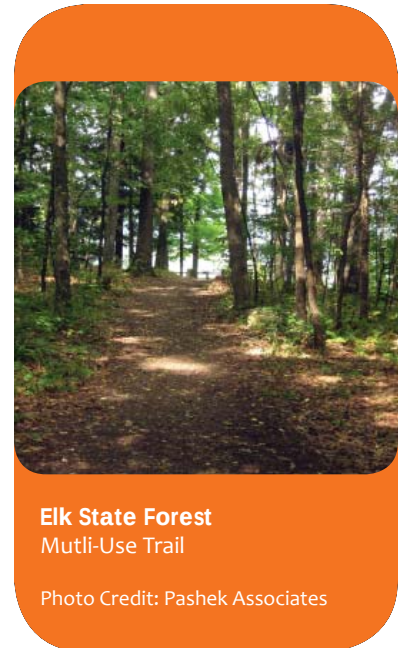
Sustainable trails lead users to desired destinations such as water features, historic sites, vistas, interesting landforms and user facilities; while avoiding wet areas, steep slopes, critical habitats, and other culturally or environmentally sensitive areas.

Keep Water Off the Trail

Erosion is the number one problem for sustainable trails. It damages trails, is expensive to repair and diminishes the user experiences. Water is the primary erosive force. Trails that collect water or channel water will be both environmentally and economically un-sustainable.

Follow Natural Contours

Trails lie on the land in three ways – along a fall-line (in the direction of the slope), on flat ground, or along the contour (perpendicular to the slope). Of these types of trails, only the contour trail on the side-slope easily sheds water and is thus sustainable.



Elk State Forest
Mutli-Use Trail

Photo Credit: Pashek Associates



Hillside Before Trail
Vegetation keeps water sheeting slowly downhill.



Keep Users on the Trail

When users leave the trail tread, they widen it, create braided trails, and create social trails. These can cause environmental damage and raise maintenance costs. Users leave the trail when it becomes eroded or wet, or when the trail does not meet their needs or expectations.

Meet Desired User Experiences

Sustainable trails and trail systems must meet different users' needs and expectations. If they do not, users may abandon the trails and/or create their own, less sustainable trails.

Ultimately, a sustainable trail design will most often be a trail that connects desired control points by roughly contouring along the sides of slopes.

Upon establishing your Trail Management Objectives (see Chapter 4 – Management) and referring to the user characteristics and design requirements for the respective trail type (as detailed in Chapter 2 - Design Requirements for Specific Trail Users), you are ready to begin the design of your trail.

Important considerations in sustainable trail design include:

- Trail Corridor
- Tread Design
- Tread Drainage
- Changes in Trail Grade
- Tread Reinforcement & Trail Structures
- Trailhead Design
- Signage and Markings
- Trail Gates and Barriers
- Native Landscaping

Each of these considerations are described in this Chapter.

Trail Corridor

The trail corridor is the area that your trail will pass through. The trail corridor includes the trail's tread and the area above, the area below, and to the sides of the tread. Trail standards typically define the edges of the trail corridor as the clearing limits. Vegetation is trimmed back, and obstacles, such as boulders, fallen trees, and branches, are removed from the trail corridor to make it possible to walk or ride on the tread.

The dimensions for the trail tread, shoulders, upper clearing limits, lower clearing limits, and the clearing height are determined by the needs of the target users and the level of difficulty established for the trail.

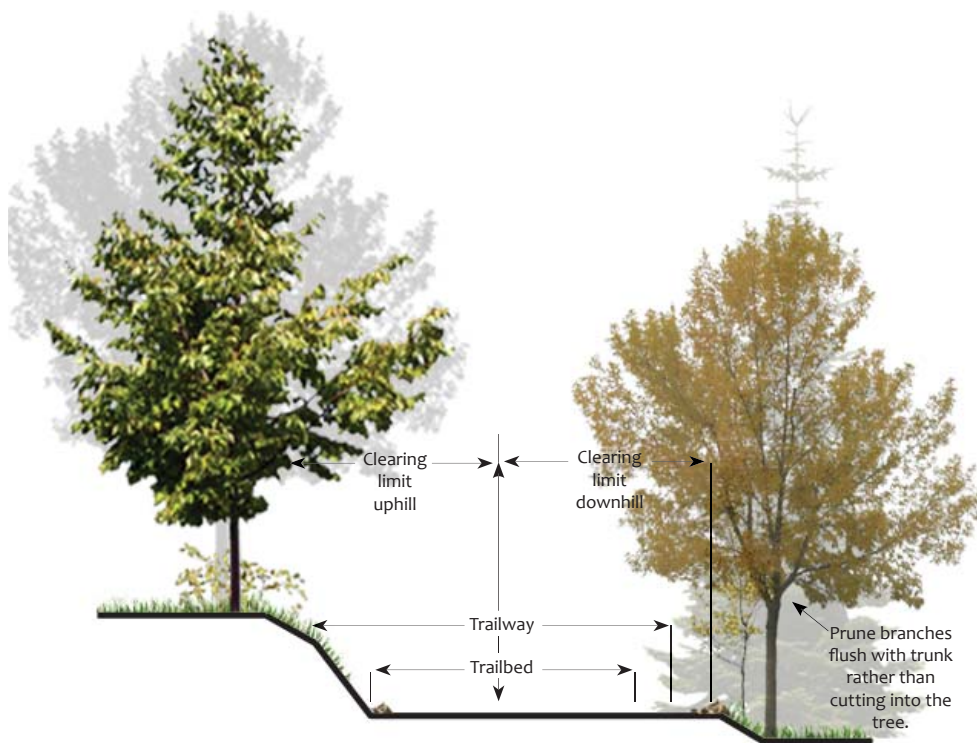


Figure 3-1: Typical Trail Corridor Elevation

Tread Design

Bench Cut Trails

After clearing the passage, clear the treadway of organic material and then shaped into a slightly outsloped walking surface. In most cases your work will occur on a slope, so excavation will occur across the side of a hill. A “sidehill” trail has a near level, slightly outsloping bench for the trail surface. In many cases this type of design allows for trail construction to occur without manmade structures. There are two ways to build a sidehill trail: full bench and partial bench.

Full Bench Trail Tread

A full bench trail tread has its entire width excavated into the hillside. Trail designers generally prefer a full bench tread for most trails, especially on steeper terrain with poor soils. The steeper the side slope, the closer the construction will resemble a full bench.



Full Bench Cut

Entire tread width is cut into firm mineral soil. Tread compacts uniformly and is sustainable. 5% outslope ensures water sheets across tread. Back cut is blended into back slope. Proper back slope, out slope, and full bench cut minimize maintenance.



Incomplete Full Bench

Left with unfinished vertical back cut, soil will slough off, making tread narrower. Obtrusive back cut also forces rider to outside of tread.



Half Bench

Relatively easy to construct, but fill soil may collapse and creep downslope, requiring excessive maintenance.



Partial Bench with Cribwall

In some cases, tread must be supported by a cribwall. Wall holds fill soil in place and tread is outsloped over top of wall.



Common failures resulting from improper construction of trail treads include:

- Fill material slipping downhill resulting in sliding of the trail.
- Fill material compacting and creating a berm on the downslope side of the trail.

Use full bench tread construction where possible. If not possible, construct a partial bench tread and reinforce it with a retaining wall on the downslope side of the fill, if necessary. Typically it is more costly and time consuming to construct a partial bench tread as compared to constructing a full bench tread.

Partial Bench Trail Tread

Partial bench construction is when part of the tread is excavated and the soil that is removed is placed on the lower edge of the trail corridor to fill and build up the tread. The fill should make up no more than half of the tread width. Remove all organic material from the tread to mineral soil and place organic soil on the outside of the bench.

Trail Tread and Materials

Selecting the tread surface is one of the most important decisions when designing your trail. When several different types of users will use the trail the surface material selected must meet the needs of all users. For example, when selecting the tread surface for a rail trail that will also accommodate cross-country skiing, consider constructing a compacted stone surface in lieu of asphalt. Dark asphalt surfaces absorb energy from the sun and therefore, tend to melt snow quicker than a surface, such as compacted stone, which has a higher coefficient of reflection (albedo).

Construct the trail tread on stable and compacted soils to achieve structural stability. Trail treads can be constructed from a variety of materials. Materials should be selected based on:

- User needs
- Maintenance needs
- Construction costs

Designing and selecting trail surfaces is similar to designing and selecting highway pavement sections. Consideration must be given to the loads being placed on the surface. Conduct a soils investigation to determine the load bearing capabilities of the native soil, or former railroad bed (if ballast has been removed), and the need for any special treatments. A soils investigation will also help determine whether subsurface drainage may be applicable and how freeze-thaw cycles may affect a trail. Consider using geotextile fabrics to reinforce weak sub-grades or subsoils.



While loads on trails will be much less than roadways, trail managers may require the tread design to sustain the wheel loads of occasional emergency, patrol, maintenance and other motor vehicles that may travel on or cross the trail.

When motor vehicles drive on trails, their wheels often will be at, or very near, the edges of the trail. This can cause edge damage that, in turn, will reduce the effective operating width of the trail. Therefore, when designing a trail to accommodate emergency response vehicles, construct it wide enough and with adequate edge support to accommodate vehicles. You can reinforce the edges of the tread by stabilizing the shoulders with geotextile and/or grid products.

On shared use paths and rail trails, it is important to construct and maintain a smooth riding surface. Use machines to lay aggregate and asphalt pavements; use soil sterilizers where necessary to prevent vegetation from erupting through the pavement. On concrete pavements, saw cut the transverse joints necessary to control cracking, rather than tooling the joints to provide a smoother ride.

Do not sacrifice skid resistant qualities for the sake of smoothness. Users prefer a broom or burlap drag concrete finishes. Where a shared use path crosses an unpaved road or driveway, pave the road or driveway a minimum of 20 feet on each side of the crossing to reduce the amount of gravel scattered onto or along the path by motor vehicles. The pavement cross section at the crossing should adequately sustain the expected loading at that location.

As discussed in Chapter 1 – Planning, the design of all trails should meet the Americans with Disability Act requirements. Therefore, upon construction the tread must be stable and firm. The following table lists commonly used tread surface materials and summarizes each surface’s ability to provide a firm, stable, and slip resistant surface.

Tread Surface Options

Surface Material	Firmness	Stability	Slip Resistance
Asphalt	Firm	Stable	Slip Resistant
Concrete, Broom finish	Firm	Stable	Slip Resistant
Soil with Stabilizer	Firm	Stable	Slip Resistant
Compacted Aggregate, 3/4” minus, with Stabilizer	Firm	Stable	Not Slip Resistant
Compacted Aggregate, 3/4” minus, without Stabilizer	Firm	Stable	Slip Resistant
Wood Planks	Firm	Stable	Not Slip Resistant
Grass or Vegetation/Groundcover	Soft	Moderately Stable	Not Slip Resistant

* from Designing Sidewalks and Trails for Access, Part II of II, Table 15-1.



Natural Surface Trails

As noted in Chapter 1 – Planning, sustainable trail design begins in the planning phase and continues through the design, construction, and management of your trail. To design a natural surface trail you must have an understanding of:

- The characteristics of the soil where you will construct the trails;
- The grades of trail;
- The watershed above your trail and its impact on the trail; and,
- The trail users and how they will impact the trail tread.

Soil Characteristics and their Impact on Sustainability

Soil scientists use a soil texture triangle to classify the texture a soil. Soil is comprised of clay, silt, and loam. To determine a soil's texture, soil scientists analyze the soil's make-up to determine the percentages of sand, silt and clay.

Soil Component	Particle Size
Clay	less than 0.002 mm
Silt	between 0.0002 mm and 0.05 mm
Sand	0.05 to 2 mm

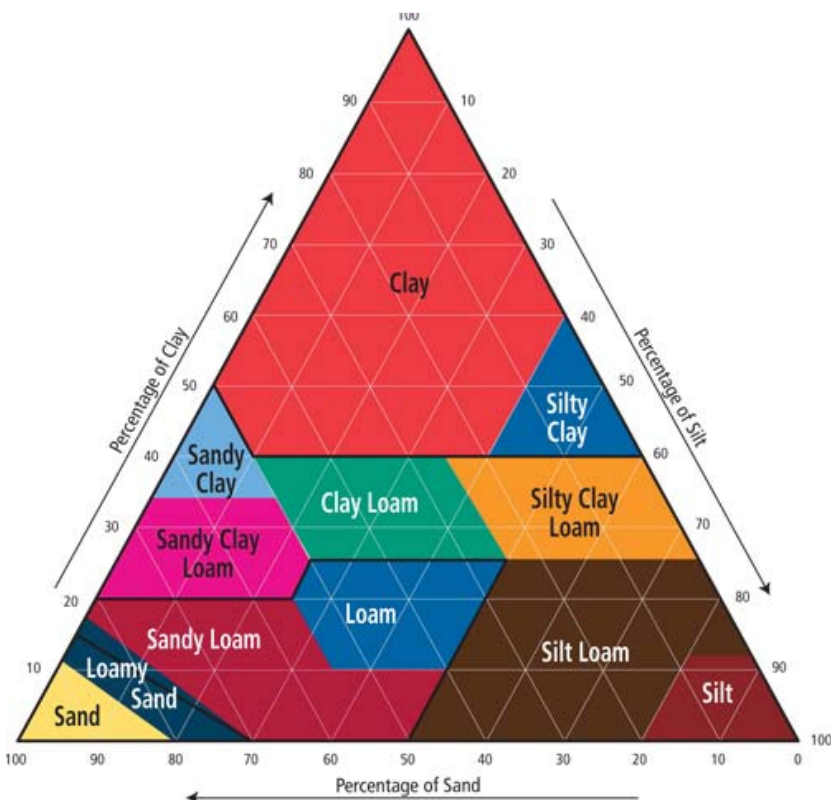
Components greater than 2 mm are further designated as gravel, stone, and boulders.

- Clay compacts easily, but holds water for long periods given the small void space between its fine particles.
- Silt drains well, can be easily compacted, and easily rutted.
- Sand has large pore spaces between particles and drains well, yet it does not compact easily.



USDA Soil Texture Triangle

Soils as classified according to the percentage of clay, silt, and sand that they contain by plotting the percentages of each on the USDA Soil Texture Triangle.



Soils that are a combination of silt, sand, and clay are classified as loam soils. A loam soil comprised of equal parts clay, silt, and sand provides a better natural trail surface than a trail comprised of just clay, silt, or sand.

Soil Texture can be estimated in several ways:

By Measurement

1. Spread soil on a newspaper to dry. Remove all rocks, trash, roots, etc. and crush lumps and clods.
2. Finely pulverize the soil.
3. Fill a tall, slender jar (like a quart canning jar) 1/4 full of soil.
4. Add water until the just is 3/4 full
5. Add a teaspoon of non-foaming dishwasher detergent.
6. Put on a tight fitting lid and shake hard for 10 to 15 minutes. This shaking breaks apart the soil aggregates and separates the soil into individual mineral particles.
7. Set the jar where it will not be disturbed for 2-3 days.
8. Soil particles will settle out according to size. After 1 minute, mark on the jar the depth of the sand.

Soils Desktop Analysis

To determine the types of soil present in your trail corridor, begin by reviewing your site on-line through the USGS' Web Soil Survey, <http://websoilsurvey.nrcs.usda.gov>.

Here you will find information related to soil limitations, soil stability, and many more factors important to trail designers. Once you complete the desktop analysis you must then evaluate the trail corridor's soils in the field.



9. After 2 hours, mark on the jar the depth of the silt.
10. When the water clears mark on the jar the clay level. This typically takes 1 to 3 days, but some soils may take weeks.
11. Measure the thickness of the sand, silt, and clay layers.
 - a. Thickness of sand deposit ____
 - b. Thickness of silt deposit ____
 - c. Thickness of clay deposit ____
 - d. Thickness of total deposit ____
12. Calculate the percentage of sand, silt, and clay.
 - a. Clay thickness / total thickness = ____ percent clay
 - b. Silt thickness / total thickness = ____ percent silt
 - c. Sand thickness / total thickness = ____ percent sand
13. Plot the values on the soil texture triangle to determine the soil texture class.

By Feel

Feel test - rub some moist soil between fingers.

- Sand feels gritty.
- Silt feels smooth.
- Clays feel sticky.

Ball squeeze test – Squeeze a moistened ball of soil in the hand.

- Coarse texture soils (sand or loamy sands) break with slight pressure.
- Medium texture soils (sandy loams and silt loams) stay together but change shape easily.
- Fine textured soils (clayey or clayey loam) resist breaking.

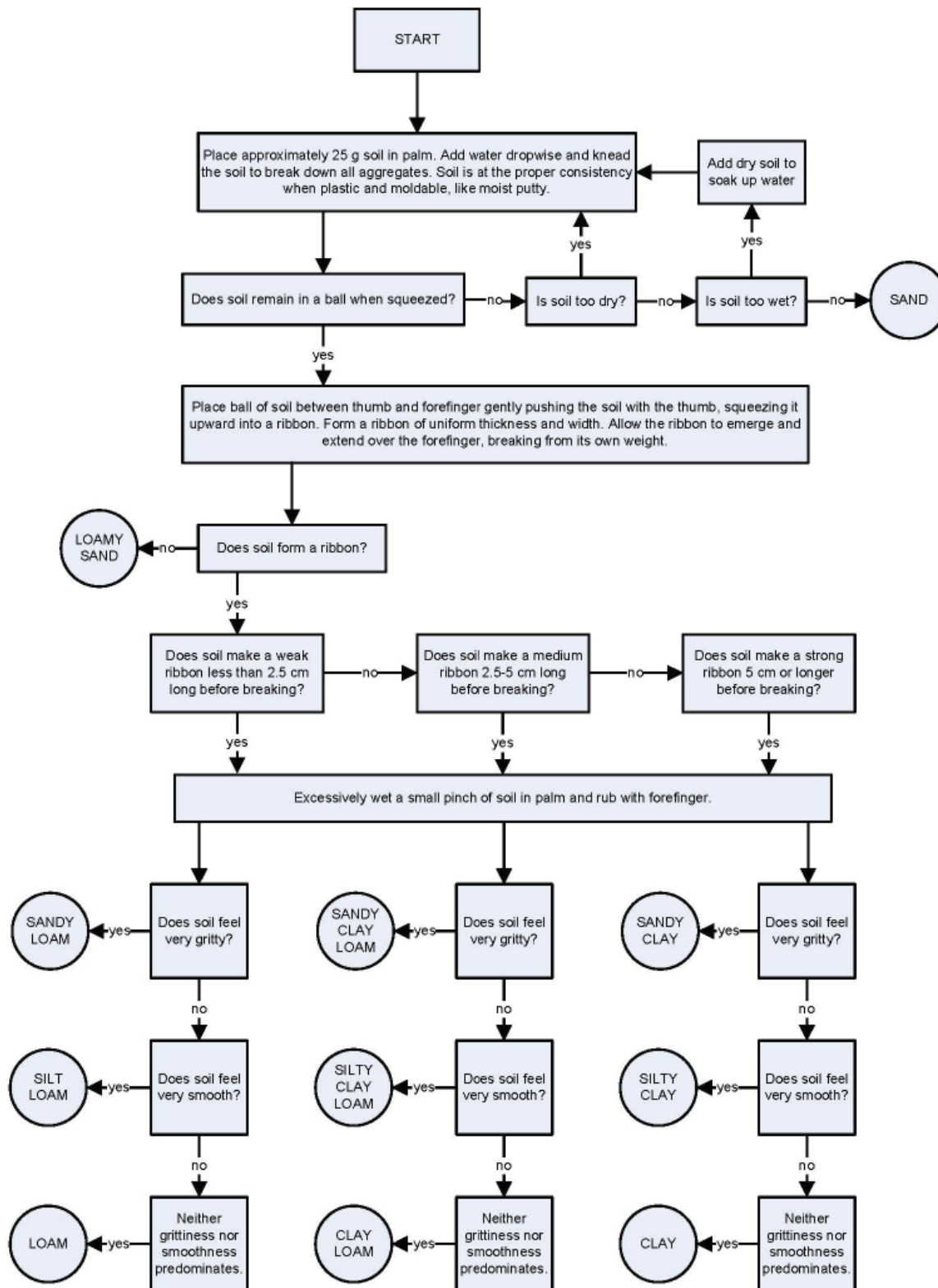
Ribbon test – Squeeze a moistened ball of soil out between thumb and fingers.

Ribbons less than 1"

- Feels gritty = coarse texture (sandy) soil
- Not gritty feeling = medium texture soil high in silt

Ribbons 1-2"

- Feels gritty = medium texture soil
- Not gritty feeling = fine texture soil
- Ribbons greater than 2" = fine texture (clayey) soil



Source: <http://soils.usda.gov/education/resources/lessons/texture/>



It is important for you to know the texture of your soils, so you can understand the limitations of the soils along your trail. Understanding these limitations will allow you to align the trail and design features to respond to these limitations.

The properties desired in soils as a base for trails include:

- Adequate strength
- Resistance to frost action
- Acceptable compression and expansion
- Adequate drainage
- Good compaction

Soils that do not exhibit all of these properties may still be suitable for trails if the missing properties can be supplied through proper construction methods. For instance, materials having good drainage characteristics are desirable, but if such materials are not available locally, adequate drainage may be obtained by diverting water away from the trail and by providing a base layer of aggregate. Another example is increasing low strengths in subgrade materials by increasing the thickness of overlying base materials or using a geotextile products.

For example, a loam soil with high clay content compacts well. However, it also tends to hold water. Therefore, it is likely that sections of a trail located on relatively flat terrain will not be sustainable. This is because relatively flat terrain does not provide enough slope for positive drainage, and the high clay content of the soil does not allow the water to infiltrate into the ground.

Soils containing gravel and stones, combined with loam soils are very sustainable. Rocky areas are the most sustainable, but are difficult to traverse because there usually are transitions between rocks.

Compacted Aggregate Trails

A popular tread material is a compacted aggregate surface. This material can be a practical solution popular when a tread cannot be constructed from natural soils, and on trails that need to support heavy loads. Many rail trails are constructed with a compacted aggregate cross section. Over the years, many formulas have tried to achieve the perfect mix of aggregate sizes to provide a stable, firm, and high-density trail surface. The Penn State Center for Dirt and Gravel Road Studies developed, and continues to refine a Trail Mix Aggregate (TMA) specification ideally suited for trail surfaces, www.dirtandgravel.psu.edu/Trails/trails.html.



**Shaver's Creek
Environmental Center**

Photo Credit: PSU Dirt & Gravel
Road Program



A compacted aggregate trail surface is considered to be an environmentally friendly alternative to an asphalt tread for the following reasons:

- The compacted aggregate trail has a higher rate of permeability than asphalt.
- The compacted aggregate trail has greater texture, and therefore, reduces the velocity of water run-off to a higher degree than asphalt.

In addition to the environmental incentives, a compacted aggregate trail is typically less expensive to install than an asphalt cross section. Another benefit of a compacted aggregate trail is that provides the users with a more forgiving tread due to its resiliency under foot.

The TMA specification allows trail surfaces to achieve very high densities that withstand traffic and erosion better than traditional aggregates. The Center designed the mix by “downsizing” its successful Driving Surface Aggregate (DSA) mix for use on roads. TMA functions as wearing surface for trails. It is different from traditional materials used to surface trails such as “number 10’s.” TMA has a uniform mixture of a range of rock sizes from 3/8-inch all the way down to fine material. This uniform mix allows excellent compaction to achieve a higher in-place aggregate density than commonly used aggregates to resist wear and erosion.

Material Specifications

Derive all trail mix aggregate from natural stone formations. Stone is defined as rock that has been crushed; rock is defined as consolidated mineral matter. THE TMA specification restricts material to mined or quarried rock from existing geologic bedrock formations. Derive all components of the aggregate mix, including fines passing the #200 sieve, by crushing parent rock material that meets TMA specifications for abrasion resistance, pH, and freedom from contaminants. Do not add or retain clay or silt soil after processing operations. Determine the amount of particles less than #200 sieve size using the washing procedures specified in PTM No. 100.

Trail Mix Aggregate Recipe

Combine existing aggregates and water in the ratio:

- 4 parts unwashed AASHTO # 10 (or B3 sand)
- 4 parts AASHTO # 8
- 1 part minus #200 fines (collector fines)

TMA has been designed so that it can be easily made by blending readily available aggregates to make it less expensive and easier to produce.



Centre Furnace Mansion

Photo Credit: PSU Dirt & Gravel Road Program



Aggregate Properties

Hardness: The acceptable limit as measured by weight loss is “less than 40% loss.” Use Los Angeles Abrasion test, AASHTO T- 96 [ASTMC131] to determine this property.

pH: The pH range of aggregate should be between 6.0 to 12.45 as measured by EPA 9045C.

Optimum moisture: Deliver and place material at optimum moisture content as determined by the supplier for that particular source. The optimum percentage moisture is determined by using Proctor Test ASTM D698, procedure C, Standard.

Installation Specifications

Mixing fines: Take care not to pre-wet the “minus #200” fines before mixing them with the other aggregates. Doing so may cause the fines to clump into balls that can be difficult to break apart by mixing. The resulting mixture will be light on fine material, and contain random balls of fines.

Transport: Use truck tarps to cover 100% of the load’s exposed surface from the time of loading until immediately before dumping. This requirement includes standing time waiting to dump. This avoids the excessive drying of aggregate during long hauls.

Site preparation & drainage: Address all trail drainage concerns, to allow for unrestricted water flow off of the finished trail surface, before placing TMA.

Crown: Place TMA with a crown or cross-slope of not less than ¼” per foot to encourage drainage. Establish the same crown or cross-slope in the base material before TMA placement.

Base material: The base material requirements for TMA depend on site conditions. Some existing compacted trails require no base modification. The ideal base course for TMA is a 2 to 6 inch thick layer of well-graded compacted aggregate such as PennDOT’s 2A. Do not place TMA over “open graded” stone (such as railroad ballast). Large open graded stone must be choked down in size until it can support the TMA without losing it in the voids. The use of separation fabric or even a “French mattress” under TMA is encouraged in wet areas. When placing TMA directly over fabric a minimum 6” uncompacted aggregate depth is recommended.

Placement with Paver: When possible, place TMA with a small paver. This will reduce the aggregate segregation by size that occurs when dumping and spreading the aggregate. The recommended minimum depth of placement is 4 inches loose, 3 inches compacted (in one lift).



Dump and Spread: On small trails, the use of a paver may be cost-prohibitive. TMA can be successfully placed using a “dump and spread” method on small trails. Ensure the aggregate is not “overworked” as it may cause some size segregation. Place TMA in one layer, progressively, starting from the furthest point and working back towards the source to avoid running equipment on the final trail.

Compaction: Use a 4-ton vibratory roller, minimum, to compact the final trail surface. The initial pass of the roller should be in static mode. Compaction of TMA must occur while it is at optimum moisture to achieve maximum density.

Properly placed and compacted TMA will meet the Americans with Disabilities Act requirements for gravel surface on properly designed trails by providing a stable and firm trail wearing surface. The following construction detail shows the compacted aggregate trail cross section.

Trail Aggregate Mix

4 parts unwashed AASHTO #10 (or B3 sand)
4 parts AASHTO #8
1 part minus #200 fines (collector fines)

Trail Mix shall meet the specifications for "Trail Aggregate Mix" as specified in the "Aggregate Handbook" published by the Penn State Center for Dirt & Gravel Road Studies.

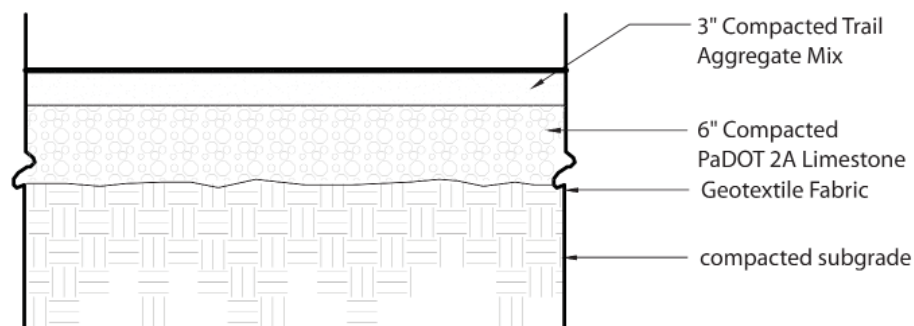


Figure 3-2: Compacted Aggregate Trail Tread

Asphalt Tread

Sometimes trail managers prefer hard, all-weather pavement surfaces over those of crushed aggregate, sand, clay, or stabilized earth, since these unpaved surfaces provide a lower level of service. On unpaved surfaces, bicyclists and other wheeled users must use a greater effort to travel at a given speed compared to a paved surface. Some users, such as in-line skaters cannot use compacted aggregate surfaces. In areas that experience frequent or even occasional flooding or drainage problems, or have moderate or steep terrain, compacted aggregate unpaved surfaces often erode.



Asphalt is a durable material for trail surfaces. Its placement on slopes and curves remains stable where native soils or compacted aggregate trails can wash out. Asphalt provides a more durable surface than compacted aggregate, requiring less maintenance and less frequent resurfacing. An asphalt surface eliminates the concerns over dust which can be associated with compacted aggregate trails. Further, asphalt's smooth surface is quieter, enhancing wildlife viewing experiences. Last, asphalt surfaces are softer than concrete, thus often it is often preferred by joggers.

Because of asphalt's dark color, asphalt trails absorb the sun rather than reflecting it, resulting in an increase in the "heat island" effect. The heat that reflects off the pavement increases temperatures on and near the trail and creates an undesirable environment for trail users and nearby vegetation and wildlife.

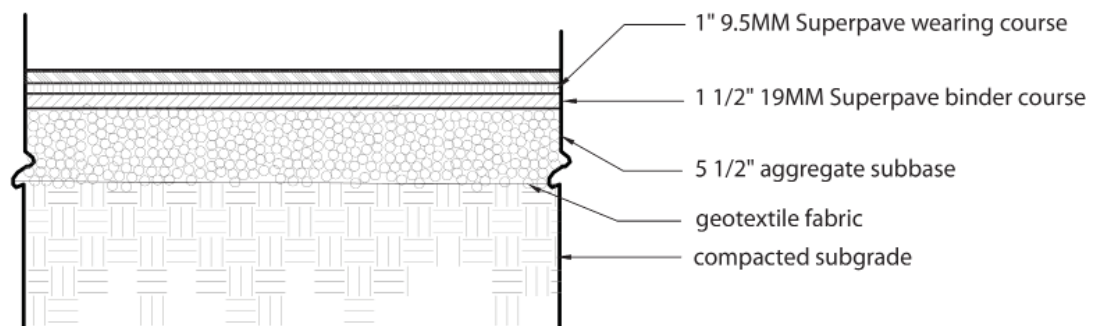


Figure 3-3: Asphalt Trail Tread

Porous Asphalt Tread

Porous asphalt is created by eliminating the smaller, graduated sizes of crushed rock and using a larger, uniform size, resulting in a rougher surface that has open voids. By maintaining pore space water will infiltrate through the tread surface instead of running off of it. Provided the trail's subbase has suitable infiltration rates, stormwater is conveyed through the tread and infiltrated into the ground below. If the subgrade cannot provide adequate infiltration, then porous asphalt should not be used for the trail tread. Some subgrades of old railroad berms or gravel roads may be impervious, so runoff will drain to the side of the trail rather than infiltrate under the trail surface. Avoid using porous asphalt in areas that flood or where debris will clog the voids in the pavement.

Concrete Tread

Concrete is the most durable material for trail surfaces, but it is the most costly. Concrete trail treads are used in urban environments. Advantages of concrete include longer service life, reduced susceptibility to cracking and deformation from roots and weeds, and a more consistent riding surface after years of use and exposure to the elements. The joints in concrete trail treads can degrade the experience of using the path for some wheeled users.



In addition, users can see pavement markings easier on asphalt than on concrete, particularly at night. Concrete's light color on a trail reflects the sun, rather than absorbing it, resulting in a reduction in the "heat island" effect.

The traditional concrete mix is impermeable. Like asphalt, porous concrete mixes are also available for consideration and have the same drainage qualities and concerns of porous asphalt.

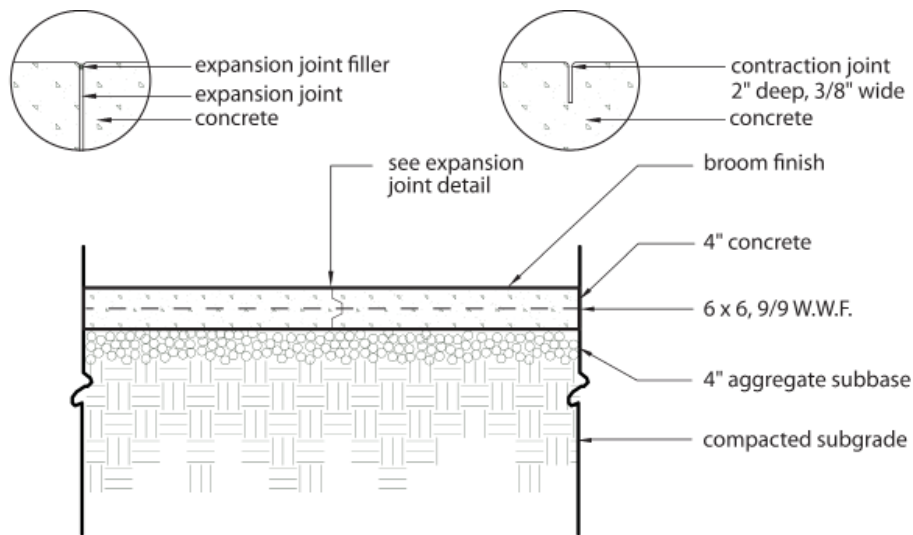


Figure 3-4: Concrete Trail Tread

Tread Drainage

Drainage is the most important aspect of trail design, regardless of the trail tread material selected. Without sufficient drainage, a trail will fail within the first 5 years. Although drainage improvements are designed on a case-by-case basis, some general rules of thumb apply.

Natural surface trails tend to catch and direct rain, as well as non-point-source runoff. When designing a trail system, it is important to consider all sources of water within the tread's immediate watershed. Some water sources may include rain or snowfall, seeps, springs, streams, ephemeral drainages, hanging or perched water tables, or floodplains.

Water typically drains on a treadway beginning at the top of the tread's immediate watershed and continues until water is directed away from the treadway. Therefore, the location of high points, such as crests, and dips in the tread further determine the length of the segment that drains between them. Since the locations of these features are not necessarily related to a site's topography, a tread's watershed size and location is flexible. Adding drainage dips, swales, waterbars, and other devices can reduce the length of drainage areas associated with the tread.



The larger the drainage area and the steeper the slope above the tread, the more likely runoff will occur. A steeper slope is more likely to produce runoff.

Watershed Above the Trail and Its Impact on Sustainability

Water runoff is the one physical force that contributes most to erosion. When you design a trail to direct water away from or around the trail, you are reducing the amount of erosion potential along the trail.

Therefore, analyze and evaluate the watershed located above each segment of the trail to determine how to direct drainage away from the trail. A watershed is the land area that drains to a given location. Typically, this location is a body of water such as a wetland, stream, or river. Traditionally designers associate the term watershed with streams. All of the land between the ridgelines on either side of a stream is located within the stream's watershed.

In the case of trails, however, look at watersheds at a much smaller and more detailed scale. For a trail, the watershed is the amount of land above the trail between a high point above the trail, and low point along the trail.

During the planning and design phases, locate the trail to take advantage of the topography. On a larger landscape scale, trails located at higher elevations typically have less watershed above them. At a smaller micro scale, align the trail so undulations in the landscape can serve as drainage locations for grade reversals and dips. This type of alignment utilizes the topography to limit the maximum tread length.

Flat areas with little slope in either direction are poor locations for trails. Remember the IMBA Half Rule. If an area has a cross slope of 5 percent and the trail has a running slope of 3 percent, the trail is a fall line trail that will not be sustainable. Further, there are fewer opportunities in flat areas to direct water away from the trail tread. Therefore, the tread becomes saturated; trail users go around the saturated areas, widening the tread, reducing vegetative cover, which in turn increases compaction and reduces infiltration. This typically results in a large mud puddle on the trail.

Trail Users and their Impact on Sustainability

The volume and type of use your trail receives will affect the sustainability of the trail. A hiking trail with steeper grades but few users can be just as sustainable as a trail with less grade but many more users. Equestrian trails with few users can be unsustainable if the trail is located on soils that cannot support the load of the horses on the trail.

Design the trail to disperse use along the length of the trail and throughout the trail network, rather than concentrate use at specific locations.



Runoff and Erosion

Tree canopy, vegetative cover, and forest litter help reduce runoff and absorb rainwater. On the other hand, bare soil and rock surfaces produce more runoff and increase runoff speed. Therefore, both the amount of vegetative cover, and the amount of trail tread are important considerations. The amount of runoff will depend on the proportions of each type of materials present in the trail corridor. Without vegetative cover, heavy rainfall will dislodge fine particles and erode part of a tread. The steeper the slope of the treadway the higher the runoff velocities and therefore the greater the erosion potential of stormwater runoff.

Weather, Climate, and Microclimate

When planning a trail consider the location's overall climate, seasons of use, and microclimates. Always assume intense rains or rapid snowmelt will occur and design trails to sustain these events. The only way to limit runoff caused by severe precipitation is to limit a tread's drainage area. On north facing slopes and in deep ravines, trails are generally cooler and wetter. These conditions affect an area's humidity, temperature, and speed of snowmelt.

Tread Grade and Length

A sustainable tread depends on its grade, length, tread texture, type, amount of use, and tread watershed factors. The steeper the tread grade, the more likely it is to erode. To prevent washouts on grades, install grade breaks into the trail alignment. Grade breaks should be integrated into the trail. The steeper the grade, the more often grade breaks are required. This further limits the continuous length of running grade on the trail, therefore, reducing the tread's susceptibility to erosion. Grade breaks are described in further detail later in this Chapter.

Cross Slopes, Side Slopes, Swales, and Culverts

Every part of the trail surface should pitch water at a minimum slope of 2%. Typically, a surface pitches downhill. When the trail is constructed on land with a cross slope of 30% or greater best practices recommend crowning the trail.

Side slopes, side swales, and culverts prevent water from reaching, and direct water away from, the trail surface and provide the water with a place to drain. Design side swales and culverts to correspond with a trail's grade and width, as well as the location's uphill watershed.

1. Where side slopes of the trail are less than 5%, the trail needs no swales unless construction occurs in a wet area.
2. Where side slopes are greater than 5% and/or when side slopes extend over 25 feet in length, these conditions typically require a swale on the uphill side of the trail to collect water, pipe it beneath the trail, and outlet the flow of water to daylight. Size all pipes to adequately



- contain a 10 year storm event, or as required by local ordinance.
3. Where trails are constructed on an embankment, designers need not to incorporate swales, assuming trail runoff drains away from the trail.
4. When trails are constructed in cut, swales are typically required on both sides and piping installed as required to drain water to daylight.
5. Where swales are required along trails, pitch the trail surface the swale on the high side of the trail.

Culvert crossings take the water from side swales and crowned surfaces and route the water beneath the trail surface. Each culvert crossing should have a headwall and endwall to prevent erosion at pipe openings, prevent flowing water from damaging the trail structure, and provide structural support for the trail. Furthermore, headwalls and endwalls increase flow capacity of the pipe by reducing turbulence and directing flow, as well as visually identify pipe openings to protect them from traffic and maintenance equipment.

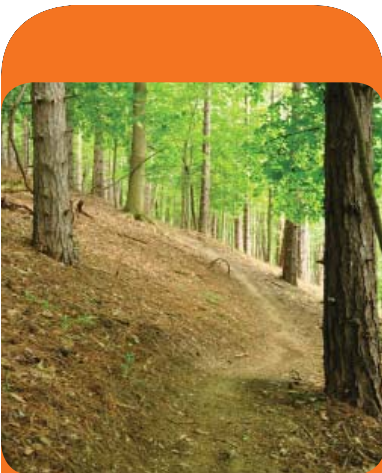
Waterbars and Drainage Dips

Constructed waterbars and tread dips prove problematic and often need maintenance. Some of these problems include ponding water, clogging, and slow or poor drainage. The most sustainable dips, those requiring minimal maintenance, incorporate several key characteristics in their design. These include a wide outflow, sufficiently sized dip, substantial crest, quick drainage, minimal erosion, as well as resistance against tread compaction and displacement. If a trail lacks any number of these characteristics, the more likely the feature will fail. In short, when designing a trail ensure the dips ability to handle both water and physical forces to minimize its need for maintenance. Further details concerning waterbars, and dips follow later in this Chapter.

Contour Trails

The contour trail is the most sustainable design, but how does one layout and create contour trails so they do not collect or channel water? As described in Chapter 1, sustainable contour trails should follow five best practices:

1. **The Half Rule:** A trail's grade should not exceed the half grade of the hillside or sideslope that the trail traverses. If grade does exceed half of the sideslope, consider it a fall line trail that will be susceptible to erosion.
2. **The Ten Percent Average Grade Guideline:** Generally, a trail with an average grade of 10 percent or less is most sustainable. This does not mean all grades must be kept less than 10 percent. Many sections of trail will have short steep sections greater than 10 percent, and some unique situations will allow average grades of more than 10 percent.



Contour Trail

Photo Credit: Pashek Associates



3. **Maximum Sustainable Grade Trails:** Maximum sustainable grade equals the steepest section of trail that is more than ten feet in length. When designing a trail, it is essential to determine early in the process the maximum grades the trail will be able to sustain given local conditions. Variables that impact the maximum sustainable grade include:
 - Soil Type
 - Rock
 - Annual Rainfall Amount
 - Grade Reversals
 - Type of Users
 - Number of Users
 - Planned Level of Difficulty
4. **Grade Reversals/Dips:** A grade reversal occurs at a spot where a climbing trail levels out and then changes direction, dropping subtly for about 10 to 50 linear feet before rising again. This change in grade forces water to exit the trail at the low point before it can gain volume, velocity, and erosive power. Other names for grade reversals include dips, grade breaks, drainage dips, or rolling dips.
5. **Outslope:** As the trail contours across a hillside, the downhill or outer edge of the trail tread should tilt slightly down and away from the high side. This encourages water to flow across and off the trail.

When designing sustainable trails, consider the level of trail development required based upon the location, use, and other factors. Some trails will consist of a natural surface, while others will consist of more developed surfaces.

Changes in Grade

Steep changes in grade prove difficult for both trail designers and builders to tackle, especially when they occur over short distances. Climbing turns, switchbacks, and grade reversals provide unique solutions for designers to accommodate trail users on challenging terrain. It is important to locate these design features in appropriate areas to maximize both their longevity and their ability to withstand the effects of erosion.

Trail Grades and their Impact on Sustainability

The grade of a trail also has a significant impact on the sustainability of the trail. As noted earlier steeper grades are less sustainable. Steeper grades also result in water runoff becoming concentrated in shorter distances, less water infiltration resulting in higher volumes of runoff, and water runoff flowing faster over the terrain. Each of these aspects contribute to soil erosion.



Climbing Turn

Photo Credit: PA DCNR

Climbing Turns

Climbing turns require trails to follow the fall line for a short segment. Therefore, in general, a climbing turn should only occur on cross slopes of no steeper than 7 to 10 percent, or less. When constructing a climbing turn designers must locate a grade reversal just above, and after a turn, to divert water away from the fall line portion of the trail.

Constructing climbing turns on cross slopes greater than 10% percent typically results in erosion of the trail. This is one of the most common trail-building mistakes.



Figure 3-5: Climbing Turn

Switchbacks

A switchback reverses the trail's direction by constructing a relatively level landing between trail segments traveling opposite directions. They are difficult to build but are more durable on steep slopes than climbing turns, because the trail does not follow the fall line. Moreover, construct switchbacks in lieu of climbing turns when sideslopes of a climbing turn exceed 10 percent.

Key features to designing switchbacks:

1. Drain water away from all sides of the turn.
2. Crown the platform slightly so that water drains in all directions.
3. Stay on contour for both approaches.
4. Combine bench cuts and retaining walls as needed to facilitate the change in grade.
5. Carefully construct retaining walls, when required, to ensure stability.
6. Use excavated material from the top leg as backfill behind the retaining wall along the bottom leg.



7. Inslope the upper leg to direct water away from the trail.
8. Outslope the lower leg to direct water off of the trail.
9. Design approaches to control user speed.
10. Construct grade reversals at the approaches to divert water.
11. Stagger switchbacks to prevent water accumulation.

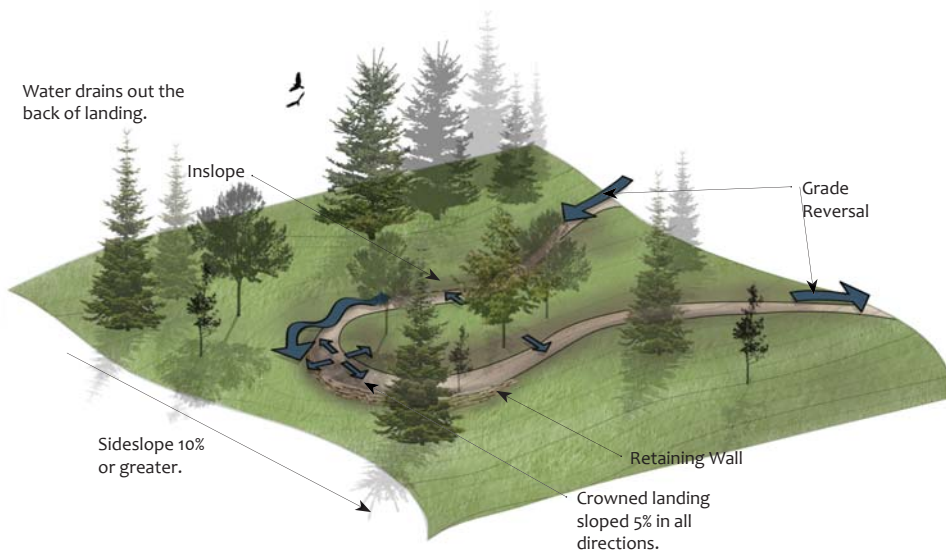


Figure 3-6: Switchback

Outslope

Outsloping is a method of grading the tread where the downhill edge of a trail is lower than the uphill edge of the trail to shed water. This deflection of water prevents concentration of flows that produce rilling, gullying, and rutting on trail treads. Users should barely notice the outslope which is generally between 2 and 5 percent.

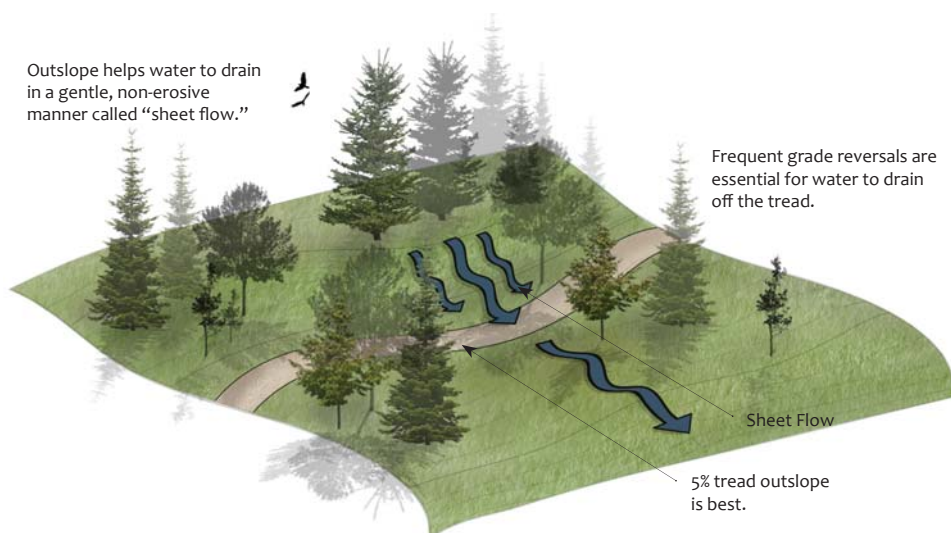


Figure 3-7: Outslope



Insloped Turn

Insloped turns can be constructed on either single use or shared use trails and provide a sustainable method of stabilizing turns on natural surface and compacted aggregate trails. Insloped turns reduce tread widening and displacement that can occur on flat or outsloped turns. A 5 percent slope to the inside of the turn helps stabilize the turn. When sideslopes are steeper than 25 percent a platform and/or retaining wall may be required.

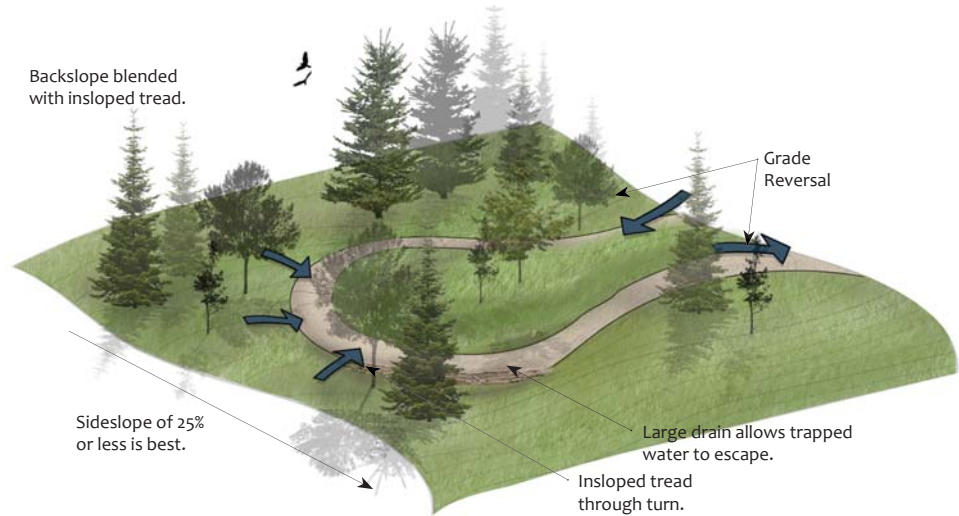


Figure 3-8: Insloped Turn

Drainage Solutions

Proper excavation, outsloping/insloping, and stabilization of a trail all contribute to create a stable treadway. However, even perfect construction cannot completely neutralize water's damaging effects on a trail over time. It is important to rely on contour trail construction to maximize a trail's sustainability. A well-constructed trail will incorporate a number of features in to construction of the trail to direct water away from and off of the trail. Consider installing more drainage than necessary to ensure proper water flow; even when one or more means of drainage fails.

Grade Reversals/Dips

A grade reversal is a reverse in the trail grade, usually a short dip followed by a rise, that forces water off the trail. Grade reversals are also referred to as grade dips, drainage dips, and/or rolling grade dips. Frequent grade reversals are a critical element of sustainable trail design. Most trails will benefit from grade reversals every 20 to 50 feet, depending on soil type and rainfall.

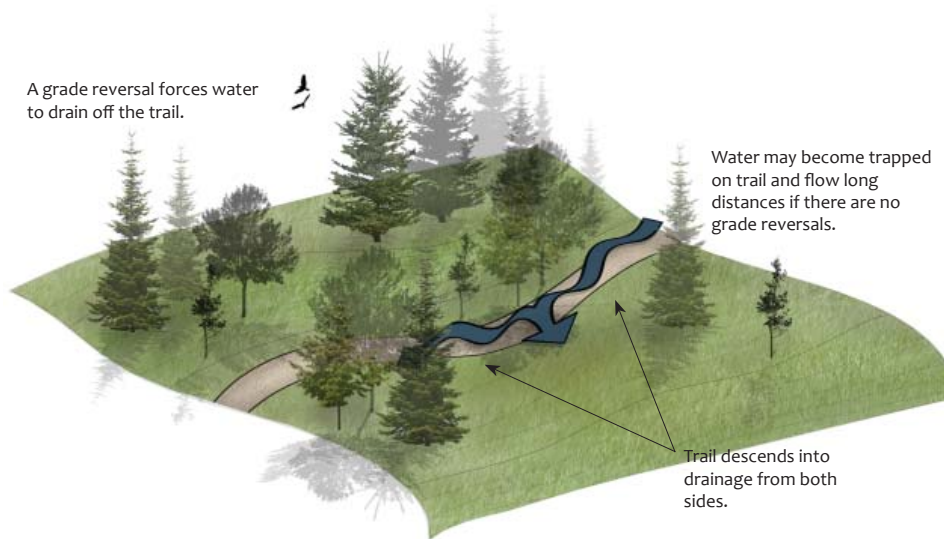


Figure 3-9: Grade Reversal

The table on the following page, adapted from *Natural Surface Trails By Design: Physical and Human Design Essentials of Sustainable Trails*, published by NatureShape, provides rules of thumb for maximum tread lengths for various soil textures.

These guidelines are based on the following assumptions:

1. The trail's grade does not exceed the half grade of the hillside or sideslope that the trail traverses.
2. Water reaching the trail drains along the trail to a dip.
3. The tread is well compacted.
4. Trail receives moderate use.
5. Watershed above the trail has moderate runoff potential.
6. Little to no tree canopy above the trail tread.
7. Intense rain events are limited to a few times a year.
8. There are no other contributing water sources present, i.e. high water table, spring seeps, etc.

Dips Are In, Bars Are Out

For existing trails with water problems, we encourage the use of rolling grade dips or knicks instead of waterbars. Here's why.

By design, water hits the waterbar and is turned. The water slows down and sediment drops in the drain.

Waterbars commonly fail when sediment fills the drain. Water tops the waterbar and continues down the tread. The waterbar becomes useless.

A good dip can be built quicker than installing a waterbar, and a rolling grade dip works better.

USDA Trail Construction and Maintenance Notebook



Maximum Tread Length between Grade Reversals/Dips by Soil Texture <i>Rule of Thumb</i>										
Soil Texture	0%	2%	4%	6%	8%	10%	12%	14%	16%	18%
Clay Loam with high quantity of gravel and stone	215'	160'	120'	90'	67'	50'	35'	24'	16'	10'
Gravelly clay	180'	132'	96'	69'	49'	34'	22'	14'	X	X
Loam with high quantity of gravel and stone	160'	117'	83'	57'	39'	26'	17'	10'	X	X
Clay	145'	104'	74'	51'	34'	22'	13'	X	X	X
Loam	135'	90'	57'	37'	23'	14'	X	X	X	X
Crushed stone, angular particles, 3/4" or less	125'	78'	49'	30'	17'	X	X	X	X	X
Organic soil	110'	68'	39'	22'	12'	X	X	X	X	X
Sand	100'	55'	30'	16'	X	X	X	X	X	X

This Rule of Thumb is not only modified by soil texture, but should also be modified as required to respond to:

- Amount of stone and rock in the tread.
- Annual rainfall amount.
- Location and frequency of grade reversals and dips.
- Type of users and their impact on the tread.
- Number of users, more users = more impact to the tread.
- Desired level of difficulty for the trail.



Knicks

Knicks are tapered, semi-circular sections of a trail that measure approximately 10 feet in diameter. Knicks are usually built on gentle, smooth sections of a trail tread where water tends to puddle. For the knick to be effective, its center must be outsloped at least 15 percent towards an area lower in elevation so that the water will have a place to drain.

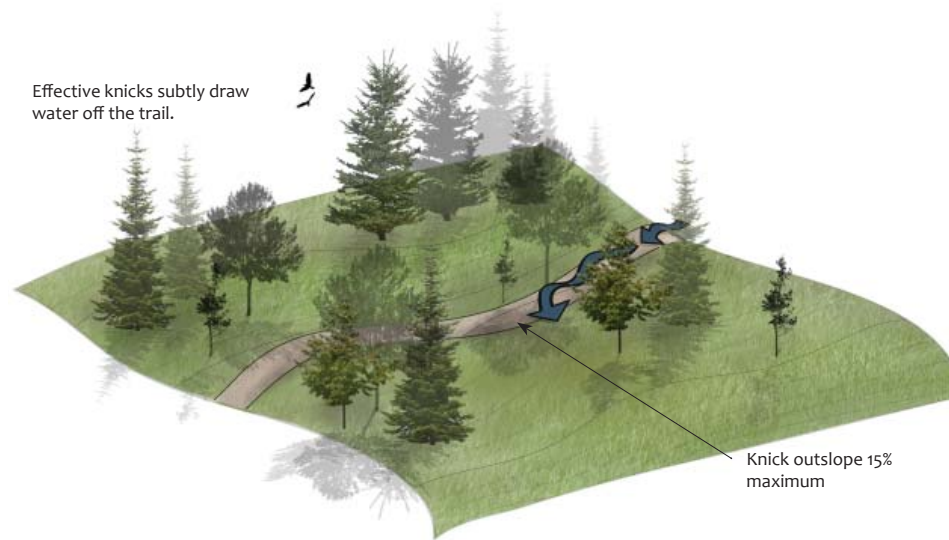


Figure 3-10: Knick

Drainage Lens

Low-volume water flow caused by spring seeps or cross swales can often be managed with a drainage lens. Beginning with a firm subbase, construct the drainage lens by placing large stone and progressively smaller stone capped with fine aggregate or suitable native fill. In wet areas, placing the drainage lens between two layers of geotextile material helps to stabilize the base. Perforated pipes can be placed at the base of the drainage lens to collect water that seeps through the stone, and outlet the water to daylight.

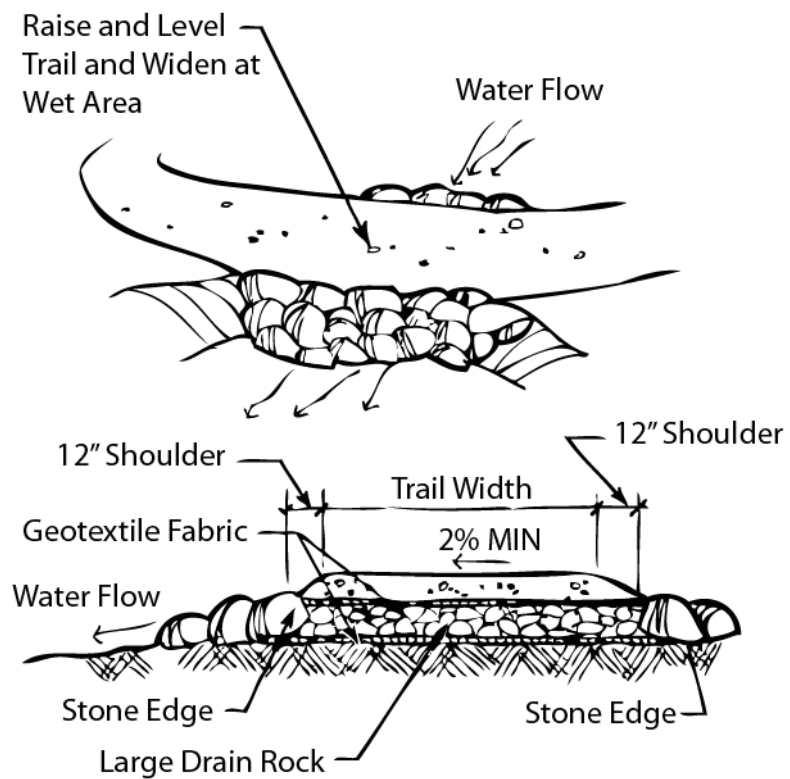


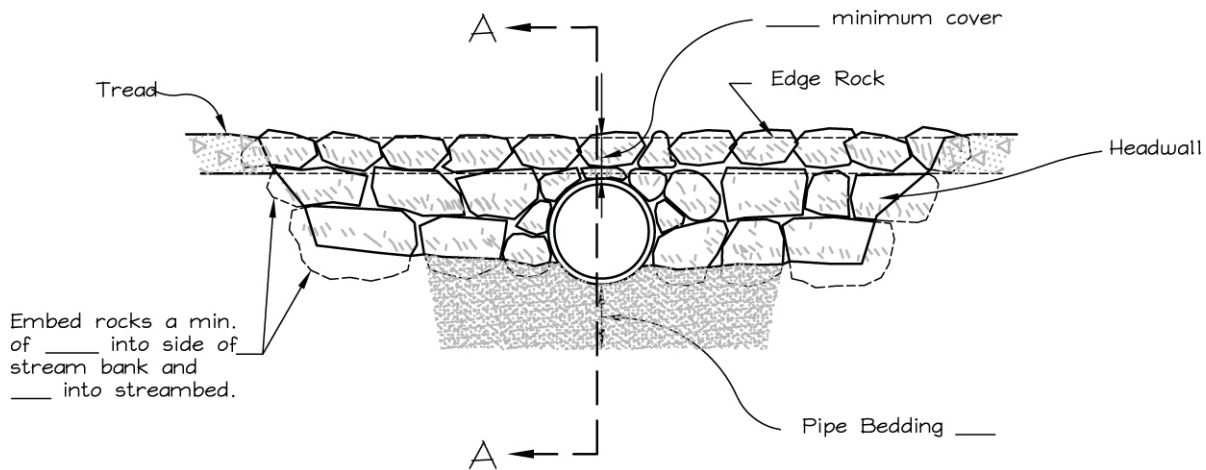
Figure 3-11: Drainage Lens

Bleeders

Bleeders are constructed swales angled to drain water off and away from the treadway. Build them by digging a shallow dip in the tread at a slant towards the outside edge of the treadway. Bleeders work best on sidehills, especially in spots where topography, roots, and rocks naturally facilitate drainage. Bleeders often work as effective backups where drainage dips will not fit. They also work well at the apex of turns and switchback corners. However, they work poorly along straight, unobstructed graded trails where they may clog with silt.

Culverts

Culverts are a form of drainage structure designed to convey water beneath a trail. Culverts can be constructed from natural rock, or plastic, metal, or concrete pipe. The fact that the trail tread extends across a culvert without interruption serves as an advantage over ditches and waterbars. They have relatively little visual impact on the trail, and are not easily displaced by trail users.



END VIEW

Headwall rocks:
 ____ minimum,
 50% larger than
 ____.

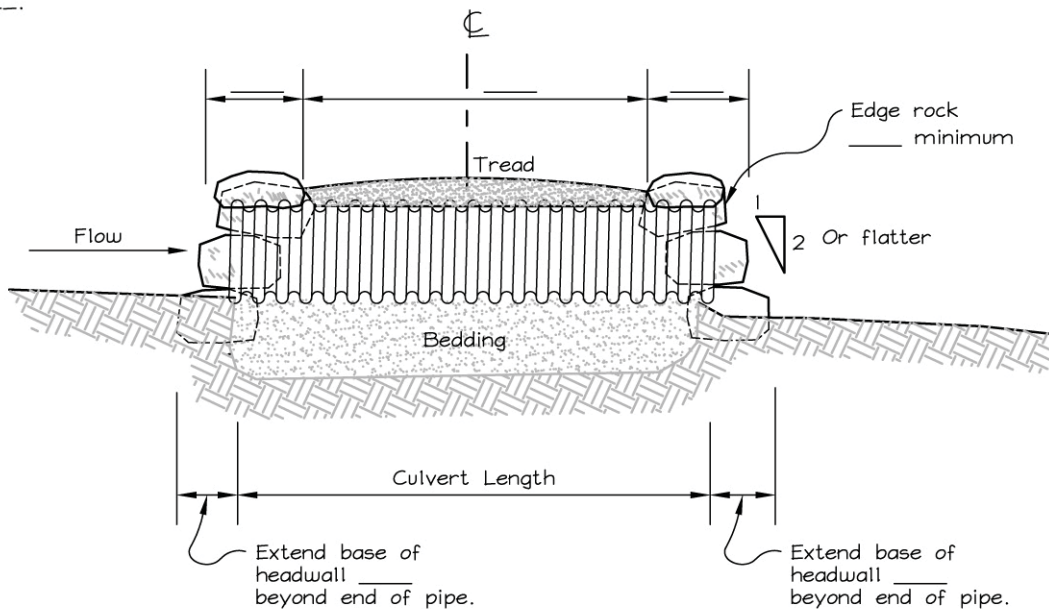


Figure 3-12: Culvert with Natural Stone Head/Endwalls

Culverts should be installed with a gentle down stream gradient of 2 percent and should be properly bedded to insure continued performance. Pipe diameters less than 12 inches may present frequent clogging and therefore are not recommended.



Tread Reinforcement and Trail Structures

Even well designed and constructed trails may need tread reinforcement or trail structures. Since these take a lot of time and effort to construct, they should be kept to a minimum in the initial trail design. However, in any trail section where the grade is steep enough to warrant concerns about erosion, tread reinforcement or trail structures can be constructed either preventatively or as erosion issues become apparent. In some cases, tread reinforcement or trail structures are creative solutions around obstacles on certain routes.

Geosynthetics

- **Geotextiles:** Geotextiles are fabric sheets of synthetic fibers that provide separation and reinforcement between soil and gravel surfaces. A single layer of geotextile fabric provides soil separation and encapsulates free-draining gravel that trail builders cover with a gravel drainage layer. A wrapped section, with two layers of geotextile, has more strength than the single-layer section alone. With the separation created by using geotextile fabric, the aggregate remains separated from the subgrade, provides a stronger surface that requires less gravel and resists rutting.
- **Geocells:** Geocells are plastic strips that are bonded together to create a honeycomb structure. Trail builders install individual cells in an excavated section, filled with gravel, and then covered with an added layer of gravel. This confinement system reinforces and restores areas easily eroded and reduces the load stresses over a unit area. Some features or benefits of the system include increased soil strength, increased soil bearing capacity, ability to vegetate, ability to utilize local soils, as well as vertical and lateral porosity.



Geogrid Being Backfilled with Stone

Photo Credit: PA DCNR



Armoring and Footpaths

- Fords and Step Stones:** Fords are used on swifter flowing streams in upper watersheds with rocks available for step stones. Naturally occurring or secured stepping stones retain gravel at the ford and can provide an interesting way for hikers, mountain bikers, and equestrians to navigate a small stream. When fords are not sufficient or need to be augmented, use step stones.

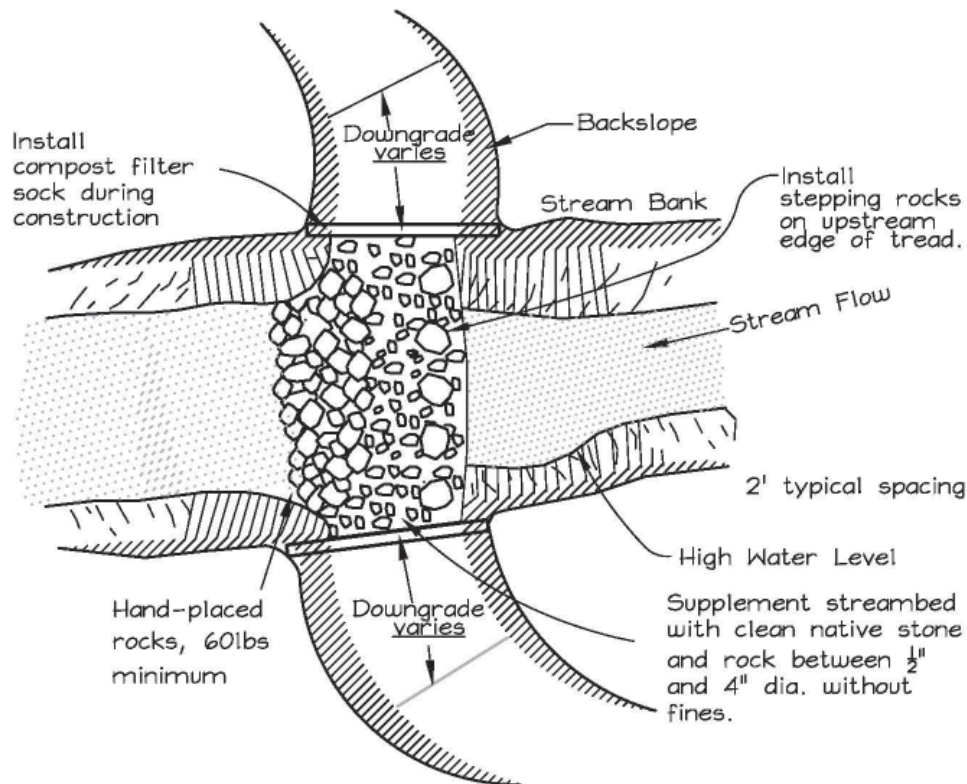
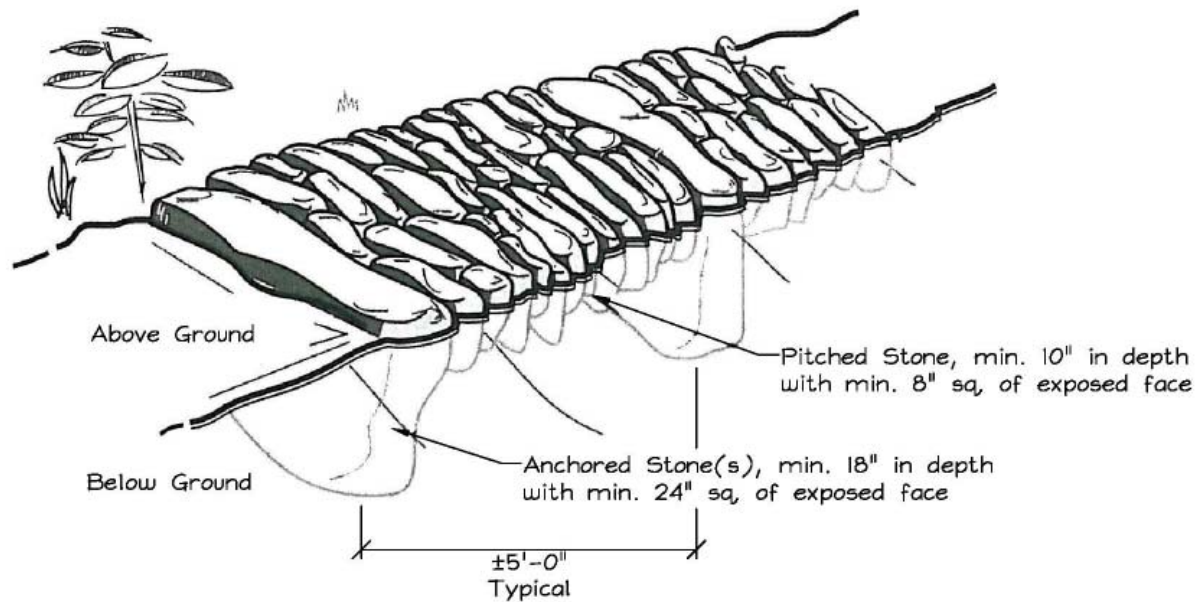


Figure 3-13: Shallow Ford Crossing

- Stone Pitching:** Stone pitching is used to create an elevated trail tread above soft terrain when no alternate route is available. Stone pitching provides a durable solution for hardening a section of a trail.



NOTES:

1. Stone for pitching shall be granite or other clean, hard dense and durable rocks free from cracks and organic or other impurities.
2. Stone shall be hand placed with closed joints on a layer of gravel or crushed stone to a minimum finished thickness of 10" or otherwise as shown on the Drawings.
3. The sides of stones shall be roughly trimmed with a spalling hammer to obtain a reasonably close fit and the interstices filled with clean coarse aggregate or gravel well rammed and wedged with spalls. The finished pitching shall show an even surface to the lines and levels shown on the Drawings.

Figure 3-14: Stone Pitching

Retaining Walls

- **Cribbing/Rock Walls:** Rock and log cribbing, sometimes called retaining walls, provide support for a treadway along gullied sections of trail or on steep side slopes. In areas difficult to drain, installing trail cribbing will allow regular drainage to occur. On steep sidehill cuts, cribbing stabilizes the upper or lower slopes along the trail. Rock cribbing functions as the most durable and aesthetically pleasing technique; however, designers may use logs to achieve the same effect.
- **Gabion Walls:** Gabion walls are retaining walls made of rectangular containers fabricated of thick, galvanized wire. They are filled with stone and stacked on one another, usually in tiers that step back with the slope, to stabilize slopes against erosion. Gabion baskets can be stacked in various shapes, resist being washed away by moving water, and drain freely. Their strength and effectiveness may increase with time as silt and vegetation fill the voids and reinforce the structure.



- **Soldier Pile Walls:** Soldier pile retaining walls provide an effective solution to addressing deep, vertical cuts in elevation. Because there is minimal soil displacement, cut and fill requirements are reduced along the trail. Wood, pre-cast concrete panels, or steel plates are used as lagging to temporarily hold back the soil. Depending on the geometry of the wall and soil conditions, builders may install tiebacks to provide lateral resistance of the soil and load.

Trailhead Design

Locate trailheads in key locations where trail users will likely enter the trail system, at intervals that provide convenience, yet protecting neighborhood privacy in urban and suburban areas. Typically, trail access points should be located at equal intervals along the trail corridor. The distance between access points is determined by the type of trail. Typically on rail trails through urban and suburban environments trail access points are located at intervals ranging between 5 and 8 miles. Access points for long distance trails, or rural trail systems are generally located further apart. Clustering of trail facilities provides for ease of maintenance, user convenience, and minimal vandalism. Be sure to locate trail access facilities in areas that are visible from adjacent roads, helping to provide basic security of the facilities.

Each access point should have enough parking to accommodate commuters or cyclists who may drive to trailheads to begin their journey. These areas should be well marked and provide a graphic map showing a trail user's location within the overall trail network.

Recommended trailhead facilities include:

- **Parking:** Consider the length of a trail, number of visitors, and proximity to population centers when determining the parking requirements for a trailhead. Also give consideration to the type of trail, and the typical vehicle used to transport persons and equipment to the trailhead. Visitors to equestrian trails need parking for their towing vehicles/trailers and prefer pull through parking spaces when available.

Standard parking stalls can be either angled or 90 degrees to the travel lane, measuring 10' wide by 20' long or as established by local zoning. AASHTO states that the minimum inside turning radius of a car is about 6 1/2' and the maximum outside turning radius is 25'. Travel lanes in a one-way parking lot should be a minimum of 12' wide and 24' wide when designed as a two-way lane.

Specialized parking areas for tow vehicle and trailer combinations between 18 to 28 feet wide and between fifty-five to seventy-eight feet in length.



Ernst Trail
Crawford County

Photo Credit: Pashek Associates



Characteristics of a Good Bike Rack

- Do not bend wheels or damage other bicycle parts.
- Accommodate high security U-shaped bicycle locks.
- Accommodate locks securing frame and both wheels.
- Do not impede or interfere with pedestrian traffic flow.
- Are easily accessed from the street and protected from motor vehicles.
- Are visible to passers-by to promote usage and enhance security.
- Are covered in locations where users will leave their bikes for a long time.
- Have as few moving parts as possible.

- **Loading/Unloading:** Areas for loading and unloading horses make nice additions to any trailhead. These areas should measure about 20' wide by 55 to 78 feet long and should be separate from other areas of the parking lot.
- **ADA Accessibility:** ADA Accessibility Guidelines for Buildings and Facilities (ADAAG) establishes the design requirements to provide accessibility for individuals with disabilities as classified under the ADA. The ADAAG outlines requirements for the number and size of parking stalls and access isles, as well as ground surface conditions and slopes for the parking areas. At least one accessible route must be provided from accessible parking stalls to all accessible facilities. Further, the parking area must should have a stable and firm surface. The minimum ratio of accessible parking spaces to standard parking spaces is one accessible parking space for every 25 spaces.
- **Bicycle Rack:** Choose a bicycle rack based on its ability to secure a bicycle while protecting it from vandalism. In addition, consider any potential damage that a bicycle may incur while it is in the rack. The preferred style of rack are those that secure the bike in two locations on the bicycle frame. Traditional bicycle racks, like the comb or toast racks, are also known as wheel benders because of the ease with which one could damage the bicycle by bending the rim. Where an aisle separates the bike racks a minimum width of 48 inches should be provided.
- **Air Station:** For the convenience of bicyclists consider providing an air station in urban and suburban locations, when and where restroom facilities are located, when feasible.
- **Security Lighting:** Limit trail to daylight hours unless the trail manager intends to light the trail corridor. Where feasible, provide at least one dawn-to-dusk security light at each trail access point. If electric service is not available, solar panels can be utilized to generate the electricity necessary for security lighting.
- **Drinking Water:** Select a cost effective frost-free design to provide a water source at all trailheads. Where municipal water is not available, consider providing a well and hand pump with a water purification system. In these instances trail managers must the staff and financial resources available to test the water supply for public use.
- **Toilets:** Toilets are a necessary amenity and should be provided at trailheads when feasible. Construct restrooms from materials that are in character with the surrounding setting. In urban and suburban areas construct restrooms from vandal resistant materials, and connect them to municipal sewer and water lines when available. If budget is a concern provide one unisex unit. Regardless of the arrangement, all restrooms



should comply with ADA accessibility standards. In suburban and rural areas consider sealed vault toilets and composting restrooms when a municipal sanitary sewer system is not available.

- **Benches:** The location of rest opportunities is crucial to ensuring a positive trail experience. Benches should have backrests and at least one armrest to provide support as a user returns to the standing position, as required by ADA.
- **Picnic Shelters:** Locate picnic shelters at trailheads. The minimum size of a shelter should be 20' x 28', housing 4 accessible picnic tables, to provide adequate cover from wind and rain. Consider using laminated wood beam shelters, or shelters with a similar roof truss design, as they eliminate roosting opportunities for birds and subsequently are much easier to maintain.
- **Trash and Recycling Containers:** Trash and recycling containers should be located at trail heads, where volunteers or municipal services agree to empty them at regular intervals. Use a 50 gallon drum with a removable plastic liner, as they prove the most cost efficient. The trail's logo can be painted on the container. In areas near populations of black bears, install bear-safe lids on the containers as well.
- **Trail Information Kiosk:** Information kiosks should remain functional and provide enough room for an overall trail map, as well as rules and regulations of the trail, and a map box.
- **Landscaping:** Screening along the trail provides a finished appearance to a trail, as well as privacy for wildlife habitat, streamside buffers, erosion control, windbreaks, and separate areas of different uses. Limit landscaping along the corridor and at trailheads to the use of native plant species. When selecting trail amenities consider an item's required maintenance, quality, affordability, and construction details. Use high quality, yet affordable items of simple design, reflective of the heritage of the area.
- **Trail Barriers:** There are many options for installing barriers along a trail corridor to limit access. They include: boulders, fences, and gates. They can be ornate and functional by corresponding to the design aesthetic/theme of an urban/suburban rail trail, or they may be purely functional, such as a gate constructed of steel tubing, to prevent motorized access to a trail corridor. Barrier types are discussed and evaluated in the U.S. Forest Service Publication *Vehicle Barriers: Their Use and Planning Considerations*, www.fs.fed.us/t-d/pubs/pdf/06231201.pdf.



Trailhead Kiosk

Photo Credit: PA DCNR

Kiosks

Kiosks introduce users to a trail. Signs and maps, as well as safety concerns are valuable to both inexperienced and experienced outdoor travelers and are typically posted at kiosks. The kiosk should provide a modest roof to eliminate glare and protect posted information from direct exposure to the elements. Maps and signs should be manufactured from weatherproof materials, designed for the intended use. Encouraging the responsible use of the outdoors conveys an important message for users, setting the stage for future generations utilizing public lands.

Well-designed information kiosks that communicate to persons of all skill levels are important for the continued enjoyment of outdoor recreation areas.

Kiosk Design

Kiosks can range from a simple, single panel containing a limited amount of information about an area, to three or more panels describing the location in more detail. Kiosks are constructed in either an angled or linear fashion. The size of the kiosk is determined based on the amount of information to be displayed, average number of visitors, and the type of use an area receives. Trail designers generally locate multi-sided kiosks with roofs at prominent trailheads. At less-visited trailheads, single-paneled kiosks relay key points and messages.

The design of a kiosk must match the character of the location, feel of the trail, and the needs of the user group to contribute to a user's overall experience. In remote areas, a rustic, simple design containing basic information remains the most appropriate choice. In residential or highly populated areas, an easily accessible space may need a larger kiosk with more detailed information.

Selecting appropriate materials for the kiosk is as important as the design itself. Use materials that fit the character of the landscape and meet the expectations of the user groups.

It is also important to consider the potential for vandalism or theft when designing a kiosk. Select durable materials that anchor kiosks firmly in the ground. Use vandal-proof screws and bolts secure the structure and information panels.

A map of a large trail could be displayed across one entire side of the kiosk. The trail information kiosk may also include brochures and maps, provide a location for a trail register, and provide information such as trail conditions and amenities, trail etiquette, area characteristics, local history, trail organizations, degree of accessibility, rules and regulations, interpretive programs, and upcoming events.



Place kiosks at all trailheads. Kiosks should include the following basic information:

- Trail name and logo
- Map showing the location of the trail and associated trail amenities
- Trail distance (in time and mileage)
- Trail level of difficulty and special instructions
- Trail accessibility information
- Rules of the trail
- Warnings of dangers, safety messages and trail closures
- Contact phone numbers for trail manager and emergency services
- Statement indicating the information posted reflects the conditions of the trail when it was last evaluated, events beyond the control of the trail manager can present hazards and make the trails temporarily inaccessible

The kiosk/bulletin board trailhead sign may also include:

- Carry in/out, Leave No Trace principles that suggest ways to decrease user impact
- Hours trailhead is open, if applicable
- Trail map handouts
- Trail register
- Information about trail organization partners and/or trail care crews
- Information about how users can get involved in volunteering
- Information about agency partners
- Interpretive information
- Types of trail uses allowed
- Recognition organizations providing volunteer services



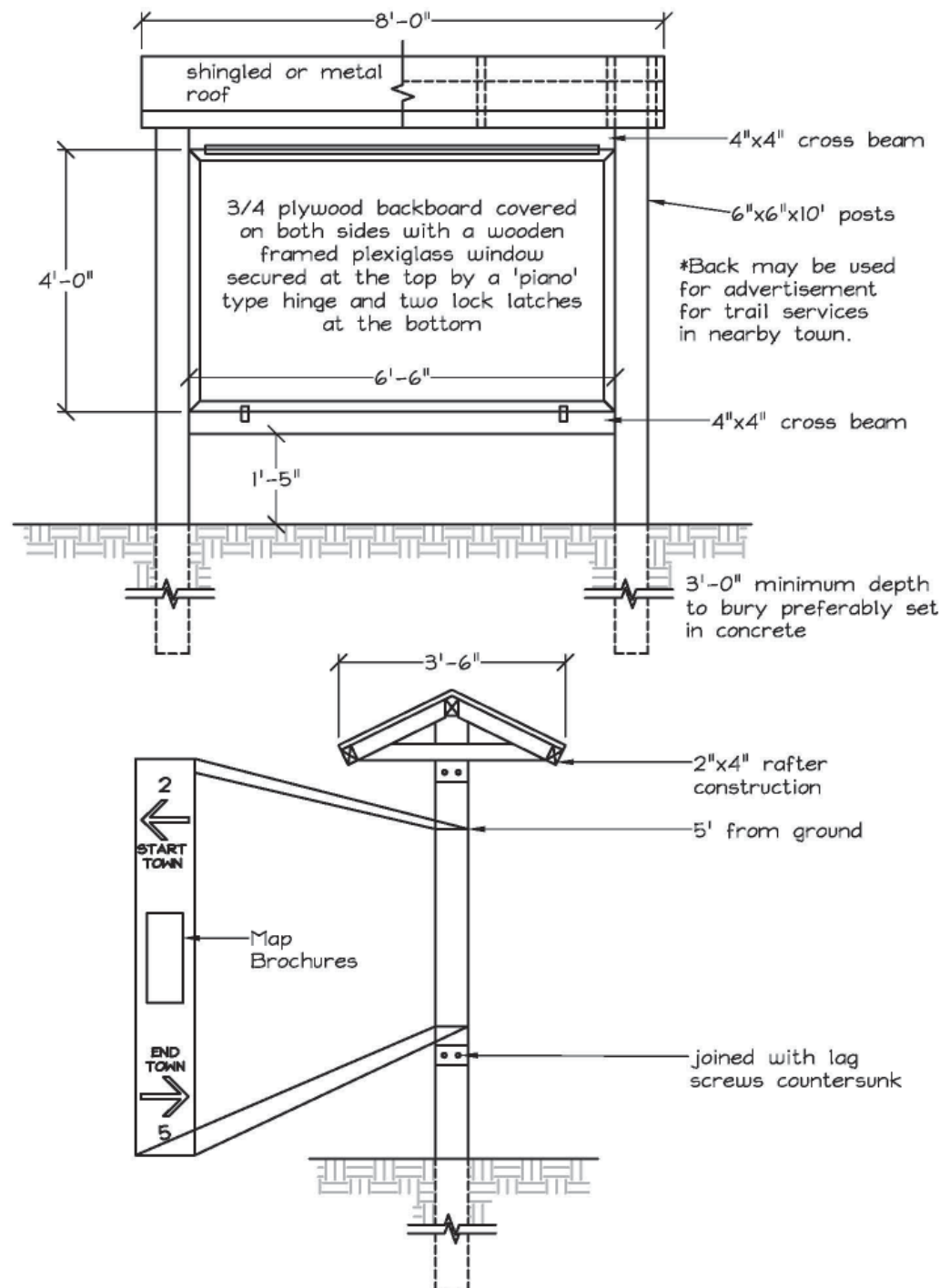


Figure 3-15: Kiosk

Letter Size

Determine the size of letters pedestrian signs by considering the location, volume, and type of visitors using the area. If the sign is intended for people



with disabilities, letter point size should comply with accessibility guidelines. To be easily read, the sign should have sufficient color contrast between letters and background.

The following rules of thumb pertain to the sign's primary message only. Letter size for a sign's secondary message will be two-thirds the height of the primary message.

1. Short messages to be read from some distance: lowercase height and thickness of arrow shaft, 2-1/2 inches.
2. Direction, distance, instructions, listings in moderate visitor use areas where pedestrian traffic is channeled by walks, etc.: lowercase height and thickness of arrow shaft, 2 inches.
3. Plaques, markers, and object identification: capital height, 1-1/2 inches; lowercase height and thickness of arrow shaft, 1 inch; 8-inch recreation symbol.
4. Description sign texts: capital height, 1 inch; lowercase height and thickness of arrow shaft, 5/8 inch; 8-inch recreation symbol.

As a rule, one inch of letter height can be read from a distance of 50 feet; 2 inches from 100 feet, and so on.

Construction and Installation

To begin the design and construction of a kiosk, first select its location. Common topics showcased in kiosks include local history, safety information, and maps. After determining its location consider the specific placement of the kiosk. Best practices include placing kiosks at trailheads, parking areas, or campgrounds to address a trail's user group before beginning their trip on the trail. Because the area around the kiosk will receive plenty of visitors, place stable and firm materials around the kiosk.

Next, construct detailed plans and an illustrated plan to determine the style, colors, and material choices that will work best for this location. Generally, trail designers will present construction drawings for kiosks to land managers for approval before construction. Construction can be time consuming and expensive. Pre-construct the kiosk to avoid kinks in its construction before transporting it to the site. After completing this step, piece together and install the final product.

User Education

Consider the information to be placed on the kiosk before finalizing its size and style. Since the trailhead indicates the beginning of the trail, it is important to introduce visitors to unique aspects of a location and quickly convey safety and stewardship information. Concise and interesting signs attract the attention of all users. Incorporating artwork, photos, and maps can keep the attention of trail users. Consider what messages the kiosk will convey to users before they head out on the trail. The real challenge involves

Basic Kiosk Information

Basic information that should be included in all informational kiosks:

- A map or layout of the land, including trail descriptions if appropriate.
- A short description of unique historical, wildlife, and other characteristics unique to the location.
- Safety information to warn visitors of dangers and suggestions for safe travel.
- Stewardship information of how visitors can get involved.
- Carry In/Carry Out and Leave No Trace principles that suggest ways to decrease user impact.
- Information on local services; eat, sleep, fuel, etc.
- Contact information for organizations and land managers.



keeping the information simple, relevant, and interesting.

When trail designers and trail managers must take the time and give the consideration necessary to plan and construct a kiosk, as it will serve as a source of information for years to come. Further, the information contained at the kiosk should be regularly reviewed and updated to avoid creating any confusion on the part of the trail users.

Trail Accessibility Information

When available, the following additional information should be included on trailhead signs:

- Trail-specific trail symbol
- Running slope (average and maximum grade)
- Cross slope (average and maximum)
- Cumulative elevation change (gain and loss)
- Profile of the trail grade showing changes in surface type and accessibility
- Clear tread width (minimum and average)
- Tread obstacles (magnitude and frequency)
- Any major height obstacle, such as boulders, in the trail tread

Accessible Trail Information

Signs identifying trails and trail segments that have been officially assessed and designated as accessible to persons with disabilities should be placed at the trailhead and at all designated access points. Display the official symbol designating that the trail or trail segment is accessible, and include the total distance of the accessible trail or trail segment and the distance to the location of the first point of exception to accessible standards.

Use marker posts to display accessibility information at access points without trailhead signs. Decals are readily available to attach to marker posts.

The size of the trailhead sign should be such that both text and graphics are easily readable. The minimum size should be 12"x18". Background colors, margins, and sizes of text and images are subject to change.

Materials for Signs, Trail/Sign Markers, and Posts

Signs can be constructed from a wide range of materials, including cast aluminum, fiberglass, phenolic resin, porcelain enamel, and vinyl on aluminum. The following is a description of common sign materials, as well as the pros and cons for each.

- **Fiberglass:** Fiberglass panels imbed a digitally produced paper graphic into the sign's surface. These signs are generally easy to maintain and clean, but have a relatively short lifespan of 5 to 8 years. Fiberglass panels will fade and become brittle over time, but are low enough in price to be replaced as needed.
- **Phenolic Resin:** Phenolic resin panels are a high-pressure laminate that encapsulates a digital paper graphic. These panels are appropriate for all locations and require waxing once every year. These colorfast panels are an excellent choice for reproducing high-resolution photographs.
- **Porcelain Enamel:** Porcelain enamel markers are made by fusing glass to steel. They are durable and perform well in most environments such as zoos and aquariums. However, if damaged this material will rust and cause surface stains.

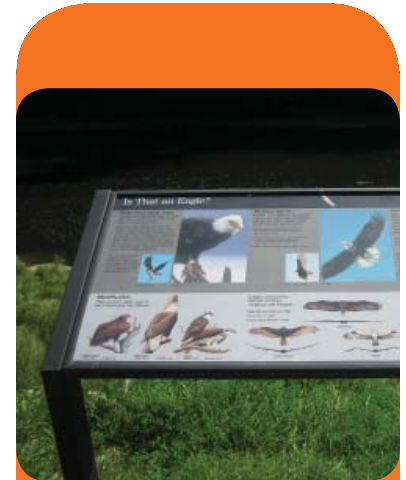


- **Vinyl on Aluminum:** Commonly used for highway markers, vinyl graphics on aluminum backers are ideal for semi-permanent wayfinding signs. This type of signage looks professional and offers flexibility, minimizing season-to-season costs.
- **Cast Aluminum:** Durable and heavy, cast aluminum signs require minimal maintenance and can withstand adverse weather conditions. Unfortunately, cast aluminum signs cannot be used for color photographs and require time and money for production.

Fabricating the Posts and Frame

While the graphics are being produced, use the time to fabricate the posts and frames that will hold them. Plastic, wood, steel, aluminum, and steel can be used for posts and frames. The choice will depend on location of the signs, budget, and required annual maintenance.

- **Wood:** Wood signs and posts are visually appealing and the most readily available material for trail posts and frames. When choosing to use wood, select a rot-resistant form and consider using paints and sealers to protect them from the elements. Wood frames are the most easily damaged and require the most maintenance.
- **Steel:** Steel is a durable, strong material for fabricating posts and frames. However, it is prone to rusting and must be galvanized to resist corrosion. Steel signs are also susceptible to vandals who use them for target practice.
- **Aluminum:** Aluminum frames and powder-coated finishes are more expensive at the outset but will require less maintenance over time. Furthermore, aluminum will not rust or stain when left outside and are usually half the weight of their steel counterparts.
- **Stone:** Stone is durable, but it limits the amount of information that can be carved into the sign. However, it can be a cost effective material to use if found on-site.
- **Plastics:** Plastics vary in quality and durability; therefore, it is important to research the various types before making an investment in them. Fiber-reinforced plastics are a popular choice for marking trails as they are durable and affordable.



Cook Forest State Park
Bald Eagle Interpretive Sign

Photo Credit: Pashek Associates



Allegheny River Trail Map

Photo Credit: Pashek Associates



Bridges

Building a bridge poses one of the biggest challenges when constructing a trail. It requires a strong understanding of engineering and hydrological principles, as well as sustainable design principles. The first step in determining the best way to handle a specific situation involves assessing likely users of the trail, its location, and alternative routes before concluding if a bridge is necessary. It is important to carefully plan and explore all options for a site during the design phase, and before construction begins. Further, the designer must be aware of permitting requirements to design the bridge in compliance with them.

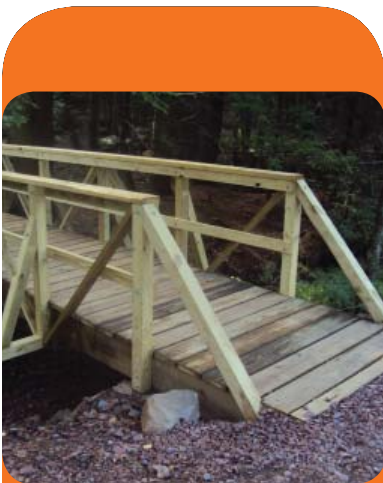
A wide range of bridge designs exist, from simple designs constructed on-site with native materials to expensive, factory-built models. Only the materials available and creativity of the builders involved in the project limit the variety of bridge designs possible. It remains important to base a bridge design on its aesthetic value, size, and the intended user experience proposed for the trail.

Almost all trails cross one or more waterways along their route. Narrow streams can be crossed in several ways depending on the size and the type of the stream. Appropriate crossings can include placement of flat stone along a narrow streambed or the use of culverts. For perennial streams, anchored bridges or culverts typically provide the user with the necessary means to cross the stream. By placing the trail above the water, the impact to the stream is decreased.

The thought that goes into the planning and design of a bridge remains every bit as important as the effort invested in its construction. Location is a key determinant of the durability and ease of construction for a bridge. Some important aspects to consider include: stable banks, banks close together, a sunny location, acceptable approach trails, and a bridge's potential environmental impact. Using the right size of lumber for the span and building supports, if necessary, remain important. Remember to build bridges to meet the needs of the intended users, and to prohibit other uses.

It also remains important to consider other aspects of a design before constructing a bridge or water crossing. By incorporating flat, skid-resistant surfaces, trail designers provide a safe and inviting approach is provided for the bridge. Designers should carefully select bridge types and construction materials for site conditions, building them to support emergency and maintenance vehicles where appropriate. It is also important to avoid sharp and blind curves, as well as drastic changes in elevation, on immediate approaches to minimize adverse effects on sight distance, drainage, and footing for users.

The USDA Forest Service identifies 6 general bridge types in their Trail Bridge



Nescopeck State Park
Stringer Bridge

Photo Credit: PA DCNR

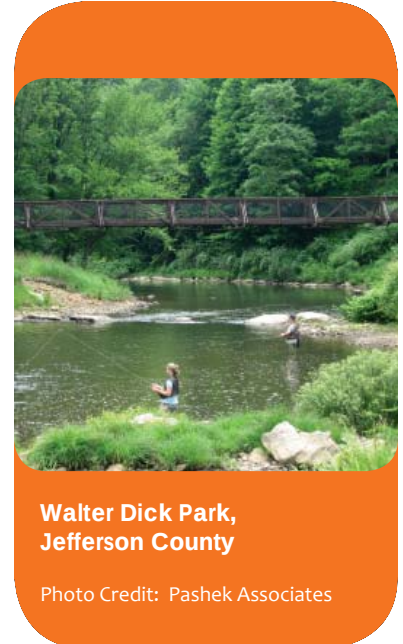


Catalogue, www.fs.fed.us/eng/bridges/, typically used along non-motorized trails, including:

- **Cable Suspension Bridge:** Cable suspension bridges have two main steel cables for support. Builders create the deck of the bridge using sawn timber planks typically hung from suspended cables or steel rods. They anchor the cables into the streambanks and use intermediate towers for support when necessary.
- **Deck Girder/Truss Bridge:** Deck girder bridges are supported by two or more longitudinal girders (beams). Decks support comes from the tops of the girders and usually consist of timber (log, sawn, or glued laminated timber), but may be concrete or steel.
- **Side Girder/Truss Bridge:** Side girder bridges are supported by two longitudinal girders or beams. The deck hangs on the interior side of the girders on either the floor or ledger beams, attached to the main girders.
- **Arch Bridge:** Longitudinal beams or walls typically support the deck of an arch bridge. Designers use a variety of materials including timber, masonry, concrete-filled, and open spandrel concrete in the design of these bridges.
- **Single Unit Bridge:** Consists of a single, self-supporting bridge. A foot log, nailed or glued-laminated timber, and prestressed concrete are typically used to construct this type of bridge.
- **Covered Bridge:** Traditional covered bridges essentially use a side truss bridge design to span up to 300 feet. Most modern bridges use side girders or deck girders with the covering simply added to the top of the bridge.

Several other types of bridges that can be considered when designing a non-motorized trail include:

- **Puncheon Bridge:** Puncheon bridges consist of wooden walkways used to provide passage over small streams. A puncheon refers to two flattened logs nailed side by side on sills to form an elevated walking surface. Puncheon for a horse trail consists of a deck 48 inches in width with a minimum thickness of 4 inches, laid on stringers with a diameter of at least 10 inches. The design of these bridges accommodate both pedestrian and equestrian uses.





Standard Dimensions of Puncheon and Bridge Decking

Trail Type	Deck Width	Deck Thickness	Bull Rail
Hiker	36"	3"	Optional
Cross-Country Skier	60" minimum	3"	No
Equestrian	48" minimum	4"	Yes

* Nails- For 3-inch decking, use #50 to #60 galvanized nails.

** For 4-inch decking, use 3/8 inch x 8 inch galvanized spikes.

- **Stringer Bridge:** Similar to the USDA Forest Service's Single Unit Bridge, a stringer bridge typically uses a single log or a timber beam to cross a span. These bridges span crossings of 50 feet or less. Two-stringer bridges (the width of two logs or beams) usually prove sufficient for use by hikers, while bridges intended for livestock or cross-country skiers will have 3 or more stringers. Depending on user needs, designers can equip stringer bridges with railings stabilized by longer planks of decking.
- **Ladder Bridge:** A ladder bridge is an elevated boardwalk put in place to get over an obstacle or wet area, or simply to add an interesting trail feature to a mountain biking trail. Elevate these bridges using large rocks, strong tree stumps, or well-designed wooden posts. The trail surface at the entrance or exit of a ladder bridge will require hardening, especially on steeper grades and landing areas.
- **Reinforced Fiberglass Bridge:** A fiberglass bridge system offers significant design and erection advantages. Manufacturer's ship components in sections that typically weigh less than 90 pounds, which trail crews can easily carry to any trail construction site. In a matter of a few days workers can easily assemble these bridges with hand tools. The design of these bridges withstand harsh weather that can quickly deteriorate traditional materials.

Signs

These trail signing guidelines provide standardization for signs and uniformity in the use of signs. Installing trail signs assists in managing trail use, warning trail users of trail conditions or characteristics, locating the trail, and providing general information to trail users.

Standardizing the size, shape, color, and content of signs improves recognition and safety through trail user familiarity, no matter what trail users visit.

Uniformity in the application of signs remains as important as standardization of sign design and placement. Always mark identical conditions with the same type of sign, regardless of where those conditions occur. Uniformity also enhances safety and the comfort level of trail users.



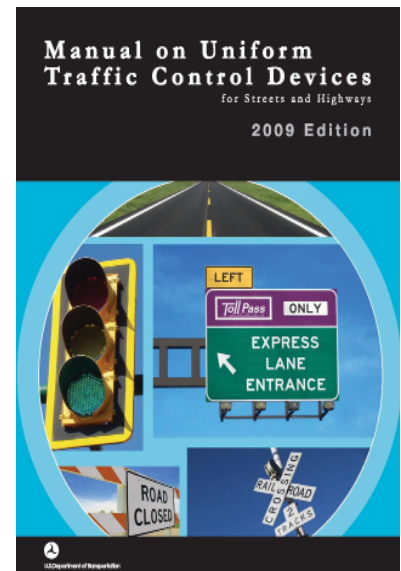
Designers typically classify signs into one of the following groups:

1. **Regulatory and Warning Signs:** Used when trail users must perform an action or to provide warning or caution and promote safety of users and property. All regulatory and warning signs installed along vehicular roadways shall comply with the requirements of Part II of the U.S. Department of Transportation's Manual on Uniform Traffic Control Devices for Streets and Highways. Only place signs where potential conflicts are unclear, or to emphasize the significance of a potential conflict.
2. **Identifier Signs, Reassurance Blazes, and Trail Markers:** to identify the type of trail use permitted or prohibited as well as indicate the trail's level of difficulty. These signs reassure trail users that they are following the correct path and provide them with a point of reference along the trail. These signs provide a trail user with a level of comfort that adds to the enjoyment of the overall trail experience.
3. **Information Signs:** Used to indicate location, direction, and/or distance to a point of interest; wayfinding signs to direct trail users to features off the trail corridor; and interpretive signs to uncover points of interest along the trail such as cultural, historic, or environmental features.

Regulatory and Warning Signs

The **Manual on Uniform Traffic Control Devices (MUTCD)**, <http://mutcd.fhwa.dot.gov/>, is recognized as the national standard for all traffic control devices installed on any street, highway, bikeway, or private road open to public travel.

The MUTCD defines traffic control devices as all signs, signals, markings, and other devices used to regulate, warn, or guide traffic, placed on, over, or adjacent to a street, highway, pedestrian facility, bikeway, or private road open to public travel by authority of a public agency or official having jurisdiction, or, in the case of a private road, by authority of the private owner or private official having jurisdiction.



Manual on Uniform Traffic Control Devices
U.S. Department of Transportation



Common Regulatory Signs

Regulatory signs regulate movement of trail users along the trail in a safe and orderly fashion. Failure of trail users to obey these signs can lead to legal consequences.

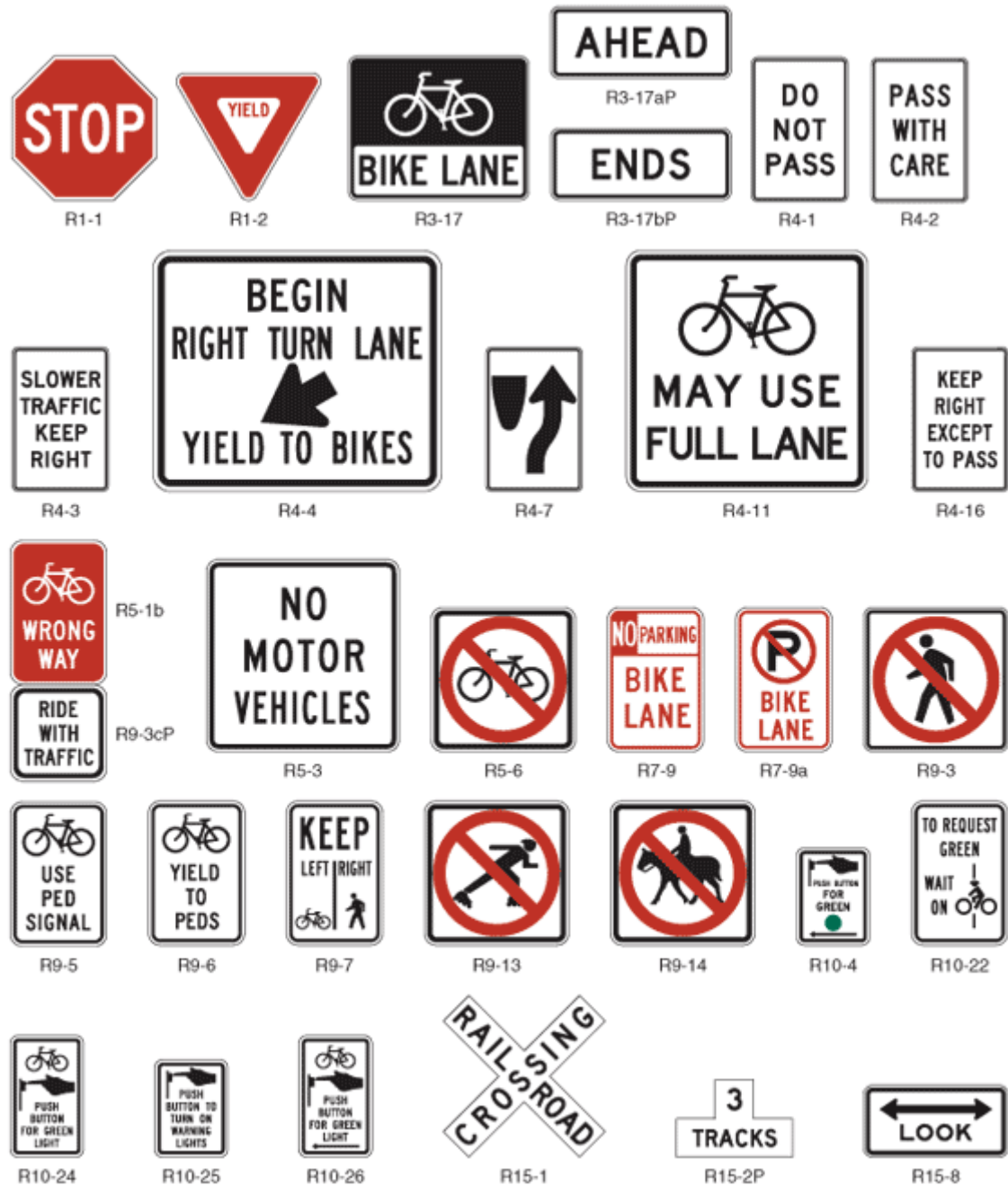


Figure 3-16: Common Regulatory Signs, US Department of Transportation MUTCD



Common Warning Signs

Warning signs alert trail users to hazards or conditions requiring special attention. Signs are located at a distance suitable to provide ample time for the trail user to react.



* A fluorescent yellow-green background color may be used for this sign or plaque. The background color of the plaque should match the color of the warning sign that it supplements.

Figure 3-17: Common Warning Signs, US Department of Transportation



MUTCD

Identifier Symbols, Reassurance Blazes, and Trail Markers

Identifier signs indicate the type of trail use permitted or prohibited, the trail's level of difficulty rating, the trail name or number, and the specific location along the trail. Use of these signs provides a trail user with a level of comfort and adds to the enjoyment of the overall trail experience.

Identifier Symbols

Use recreation symbols on trailhead signs, at trail junctions and road crossings, and on maps to indicate permitted and prohibited uses of the trail.

Permitted Use

Hiking

Equestrian

Biking

Cross-Country Skiing

Symbols



Level of Difficulty Symbols



Level of difficulty ratings are based on the degree of challenge a trail presents to an average user's physical ability and skill by using trail condition and route location factors such as alignment, steepness of grades, gain and loss of elevation, and natural barriers that must be crossed. The sign should note that conditions are subject to change due to weather and other factors.

Trail Markers

The most basic trail signs are those that identify the trail. These signs may provide the name of the trail, mark the route of the trail, and include simple information, such as difficulty rating, mileage point, and symbols that show allowable uses of the trail. Where directional signs are used, identification of the trail is often incorporated into the directional sign.

Trails can be marked in many ways. Blazes can be painted on trees, stakes, or other objects, posts can be set into the ground, markers can be nailed to



trees or posts, or cairns (piles of rocks) can be carefully erected. Regardless of the method used or type of trail being marked, each specific trail should be marked clearly and consistently and the marking should conform to a standard color, shape, and size. Where trail conditions prevent the use of the standard marker, an alternative can be used.

One method to mark trails is to use colored plastic or metal markers attached to posts. Plastic markers are less expensive but may not last as long as metal markers. Markers of various colors, shapes, and sizes are useful for distinguishing between multiple trails and between different types of trails. However, it should be noted that a large number of people are unable to distinguish between different colors; therefore, trail intersections should be well-labeled using different symbols or text (on markers or signs) to signify different trails. Use markers with arrows to indicate major changes in direction along a trail.

Use distance markers to show the mileage from either end of the trail or from a designated trailhead. These types of markers can be very useful in emergency situations and for maintenance purposes. Distance markers are often placed every quarter mile, although placement at tenth-mile intervals may be preferable.

Use of distance markers is encouraged along all trails and should include a unique identifier, such as the trail name. Where distance markers are used, other methods of marking the trail are usually unnecessary. Distance markers are often embedded in a post that is placed into the ground alongside the trail using vandal resistant hardware. Alternatively, distance markers can be metal or plastic markers nailed to trees or attached to posts; however, these types of specialized markers are prone to theft. Whichever method is used, the design of distance markers should be consistent along the entire length of the trail on which they are placed.

On long distance trails, it is not cost effective or desirable to use official trail markers as the sole method of marking a trail. Other less costly, less visually obtrusive, and more vandal-resistant methods should be used between widely-spaced markers. The preferred method is to use paint blazes, typically vertical rectangles painted on trees, posts, and other objects along the trail.

Blazes

Painted blazes along any given trail should eventually conform to a standard color, shape, and size, namely painted rectangles six inches in height and two inches in width. When painted neatly with sharp corners and clean edges, blazes remain visible to users at a distance and distinguish themselves from natural occurring marks.

Place the blazes on trees at approximate eye height. Remember, the trail should be marked for the benefit of users traveling either way, so place

PA DCNR's publication Guidelines for Marking Recreational Trails provides recommendations for marking recreational trails on State Forest and State Park lands. To download it visit: www.dcnr.state.pa.us/ucmprd2/groups/public/documents/document/dcnr_007076.pdf



blazes facing in both directions. If you can't find a suitable tree next to the trail, paint blazes on ledges or trail side rocks. Check with and receive permission from the land manager before painting blazes.

Frequency

Blazing needs to be continuous, even along roads or unmistakable parts of the trail corridor. Immediately beyond any junction, paint a blaze even if there is a direction sign. Place a second "safety blaze" 50 to 100 feet beyond.

Normally, you should change blazing frequency naturally with changes in trail terrain, forest cover, or the clarity of the footpath. When the trail is conspicuous, place one blaze for every five minutes of hiking time, or about six per mile in each direction (800 to 1000 feet apart). Where you run into hard-to-follow sections, often in transitions between field, forest, balds, and other environments, blaze more frequently.

Be careful not to over blaze. Too many single and double blazes can mar the primitive character of a trail. This is a special concern in wild and natural areas, where blazing should remain minimal, or six per mile. Elsewhere, you should place blazes so that no more than one is visible in either direction. In other words, except near trail junctions, keep blazes at least 150 feet apart.

Double Blazes

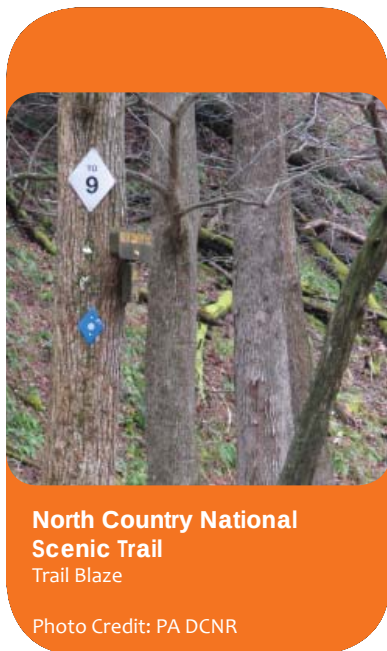
A double blaze means "caution". Place a double blaze 25 to 50 feet before abrupt turns and highway or trail junctions. Remove painted arrows, or slanted blazes, and replace them with standard double blazes. Double blazes should be placed one over the other, and about one to two inches apart. Where the double blazes are alerting a trail user to a turn, the top blaze, tree size permitting, can be offset in the direction that the trail will turn.

As with the single blaze, place the blazes sparingly. They are unnecessary at most turns in the trail, and they become unsightly and meaningless with frequent use. Only use double blazes where necessary for the safety of trail users. On switchbacks, for example, use only single blazes, but paint them near the switchback corner, one above the corner and one below. If needed, pile brush, logs, or rocks at the corner to define the trail and guide users around the turn.

Where the route remains ambiguous even with blazes, avoid the urge to paint an arrow to direct the user. The extra, nonstandard paint may hurt the trail's primitive character, detract from the trail users' sense of exploration, or set a precedent for painted arrows. Try to use small directional signs, posts, or cairns, instead.

When Two Differently Marked Trails Share the Same Path

Sometimes two or more trails briefly share the same path or corridor. When blazing in these areas, avoid confusing over-blazing and consider combining different blaze colors into one blaze.



**North Country National
Scenic Trail**
Trail Blaze

Photo Credit: PA DCNR



Placement

Paint blazes on trees that will easily be seen by trail users. Look down the trail to find a tree that will catch hikers' attention in all seasons. If the tree is far enough away, and within one to three feet of the right side of the trail, you've found your next blaze tree. Try to make sure that leafy summer growth or branches weighted with snow or rain will not later hide the blaze. Clear any interfering growth with lopping shears or hand pruners, if permitted by the land manager.

Suggested Paint Colors

The following paint colors are recommended in PA DCNR's *publication Guidelines for Marking Recreational Trails*. Following these recommendations provides uniformity and consistency when blazing trails. Each color includes the Pantone Color ID Number to maintain consistency when purchasing paint from various manufacturers.

- Red PMS 485 2X
- Orange PMS 165 2X
- Yellow PMS 102
- Blue PMS 300
- White
- Brown PMS 161

High quality latex paint is recommended. Enamel is not recommended as it may damage thin barked trees.

Trail/Sign Markers

There are many materials that can be used to construct trail signs and markers. They include:

1. **High-Density Overlay (HDO) Plywood:** Marine-quality, 3/4-inch plywood with one side covered with a high density, slick material (the overlay), to which adhesives cling quite strongly. Commonly used as the substrate for pressed-on materials such as reflective vinyl. This substrate should be used extensively for the larger signs. It weathers well, and holes in the vinyl can be easily repaired.
2. **Medium-Density Overlay Plywood:** Marine-quality, 3/4-inch plywood, with one side covered with a smooth but more porous overlay than HDO. This substrate accepts paint much better than HDO. The porosity of the overlay allows the paint to bond with the substrate better.
3. **Medium-Density Fiberboard (MDF):** A pressed-particle board product that accepts paint (silk screening) very well and weathers well. MDF is gaining adherents among transportation departments for large highway signs (green, blue, and brown).
4. **Plastics:** Sign making can involve a variety of plastics:

Keeping Blazing Primitive

Blaze Less

- Along well-defined Trail
- Along highly constructed trail in dense vegetation (example: mountain laurel or rhododendron)
- Along sharp ridgelines

Blaze More

- At turns, both on and off roads
- Along obscure footpath
- In open forests &
- In open areas

Other points to remember:

- When choosing a blaze tree, remember that one well placed blaze is better than several that are poorly placed or partly hidden.
- Most importantly, avoid defacing trees and rocks that form distinctive and pleasing elements of the scenery.



- Acrylic, or Plexiglas: is a hard, rigid material that withstands abrasion well but breaks easily. It is often used as a clear protective covering over another sign.
- Polycarbonate, or Lexan, is similar to the acrylic panel but is softer, with a greater flex. Its softness makes it more likely to be marred by dust and blowing sand.
- Polyethylene and polypropylene are fairly common materials that are suitable for most routine sign applications. They are soft materials that have sufficient rigidity to stand up as small signs, but not so rigid that they are easily broken.

They come in basic colors, and accept paint (silk screening) well. Generally, they weather well; however, their softness makes them easy prey to vandals wielding sharp or pointed instruments. Initial and replacement costs are low.

5. **Aluminum:** A long used, common substrate for routine, smaller signs. Message usually silk screened onto substrate. Easily and significantly damaged by bullets and other forms of vandalism. It has good weather resistance. Medium initial and replacement costs.
6. **Aluminum-clad plastic:** Similar in character to aluminum signs. The plastic core adds strength; this substrate is highly durable and light weight, making it ideal for kiosk panels or other signs mounted with a backing. The cost of this product is moderate.
7. **Aluminum-clad plywood:** Similar in character to aluminum signs. Plywood backing adds support to the aluminum to provide stability/rigidity for larger size signs. Initial and replacement costs are moderate to high.
8. **High-Density Foam Boards:** Three-dimensional signs made by cutting a matte and sandblasting to the desired depth. Sand blasting and mattes can also be used to make three-dimensional wood signs from 2-inch-thick material.
9. **Reinforced Fiberglass Trail Marker (RFT):** Recreational trails can be marked using dual sided reinforced fiberglass trail (RFT) marker posts with decals to identify type of trail (horse, hiking, etc.) and the individual trail identification (name, number, or color code).



**Boilinger Trail, Prince
Gallitzin State Park**
Reinforced Fiberglass Trail Marker

Photo Credit: PA DCNR

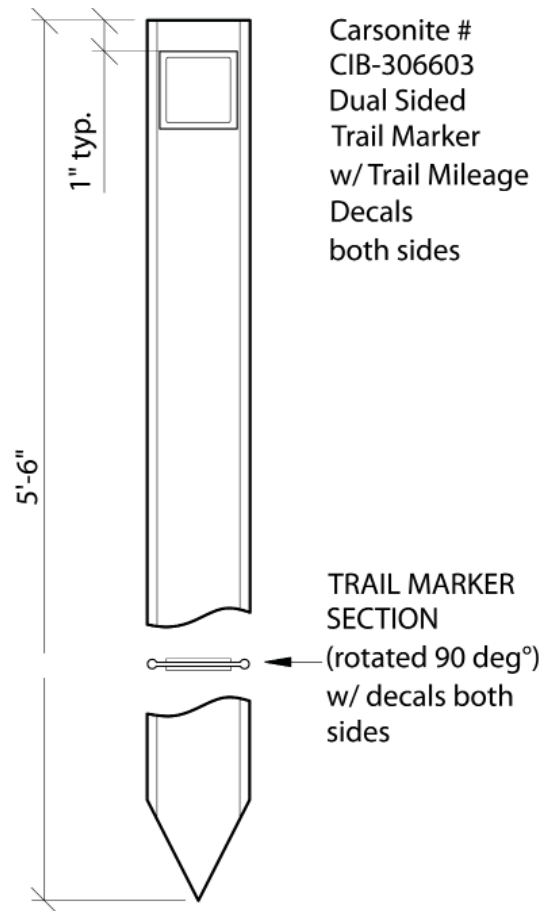


Figure 3-18: Reinforced Fiberglass Trail Marker

Marking other Property and Recreational Facilities

You may use any color of RFT posts, with the following exceptions to mark other property facilities.

1. Red: traditionally used to identify safety zones or other danger areas.
2. Orange: traditionally used for boundary marking.
3. Yellow: traditionally used to identify survey marker locations (corner stones, etc.) with appropriate decal (“Survey Marker”). Survey markers may be on property lines or off line.
4. Blue: traditionally used for marking firewood cutting areas.



Bellefonte Central Rail Trail
Split Rail Fence

Photo Credit: Pashek Associates

Barriers

Barrier design depends on an area's intended use, native materials, and character. Consider vandalism as a factor in barrier planning. Generally, wooden barriers are more vandal prone because they can be carved or burned. Concrete and metal barriers are more durable, but still succumb to vehicular ramming.

Some barriers serve as a deterrent for access to a road or a trail, while others keep vehicles on a road or in a parking lot. Most visitors recognize and respond to design features such as a simple curb used to deter vehicles from driving off-road.

Use barriers or fabricate specialized barriers for areas where user or vehicle control problems exist.

To make barriers more difficult to remove, use:

- Steel instead of wood
- Large boulders
- Larger diameter posts
- Posts buried deeper
- Posts set in concrete
- Posts anchored with rebar spikes

Movement Barrier Types

Generally, there are 5 types of barriers: bollards, fences, gates, large rocks, and wooden guardrails. The barrier size and its materials vary depending on the problem's severity and the proper scale to fit with the site's resources.

Bollards

A bollard consists of a large post with no stringer or rail, commonly 1 to 4 feet tall, and used singly or grouped to block vehicle entry from pedestrian right-of-ways. Materials for bollards consist of wood, concrete, steel, or plastic lumber posts.

Fences

A fence functions as a boundary or barrier, usually made of posts, boards, wire, or rails. Fences are designed to keep people and stock in or out of an area, but generally cannot withstand vehicular impact.

- **Split Rail Fence:** Split rail fences are desirable from a cost and aesthetic standpoint. However, in isolated locations they may be prone to vandalism.
- **Rolled Wire Boundary Fence:** A rolled wire boundary fence also is a cost effective solution. Generally, this fence disappears from sight from a distance. Therefore, when constructing this fence install appropriate



Ernst Trail, Crawford County

Photo Credit: Pashek Associates



signage or flagging to ensure trail users recognize the location of the fence.

Gates

Gates made of wood or steel allow passage and may or may not swing open. Secure gates to a fence or a large natural feature to prevent people from navigating around them. Use gates to allow pedestrian and wheelchair access, while blocking stock, motorcycle, and ATV access.

- **Right-Angle Gates:** A right-angle gate creates a passageway for pedestrians and stock. This type of gate is generally constructed from wooden or steel posts with wooden or pipe rails.
- **Steel Road Gates:** Steel road gates prevent vehicle or ATV entry to roads, trails, administrative roads, and campgrounds. Consider using large rocks and berms to block entry around the gate.
- **Steel Trail Gates:** Steel trail gates limit access of 4 wheel vehicles and ATVs while allowing motorcycle access. For safety reasons, this gate must be visible at the trailhead and as one approaches it while riding.
- **Non-motorized Trail Barrier (Stock) Gate:** Non-motorized trail barrier gates are located at trail access points to allow pedestrian and stock access to pass through. Flank the gate with a steel rail fence to prevent people from skirting the gate.

Large Rocks

Rocks prove difficult to drive over in a standard automobile and generally deter trail users from navigating over them. Mimic nature by placing rocks, that weight 200 to 400 pounds each, in groups and vary the space between them to create a natural looking barrier.

Wooden Guardrails

Wooden guardrails consist of a series of low posts tied together by wooden rails to block and control vehicular access. Combine guardrails with a curb, bumper stop, wheel stop, or shrubbery to increase effectiveness against vehicle intrusion.



Landscaping with Native Plants in Pennsylvania

is a good resource for identifying which native plants may be appropriate for your situation.

Addition information on:

- Rare, Threatened and Endangered Plants
- Common Trees of Pennsylvania
- Landscaping with Native Plants
- Invasive Plants
- Plant Communities
- Wild Plant Sanctuaries

You can download it from:
www.dcnr.state.pa.us/forestry/plants/index.htm

Landscaping with Native Plants

When landscaping along trails, whether to stabilize recently graded areas, to create natural barriers, or for aesthetics, native plant materials should be used.

A native plant is one which occurred within this region before settlement by the Europeans. Native plants include ferns and clubmosses; grasses, sedges, and rushes; perennial and annual wildflowers; and woody trees, shrubs, and vines which covered “Penn’s Woods” when the first settlers arrived. Over 2,100 native plant species occur in Pennsylvania.

An introduced or non-native plant species is one that travelers brought into the state and escaped cultivation to establish itself in the wild. At the turn of the 21st century, about 1,300 species of non-native plants exist today in Pennsylvania outside of gardens, parks, and agricultural lands. That is 37 percent of Pennsylvania’s total wild plant flora, with more identified every year.

Basics of Plant Conservation

Protect native plant communities and minimize habitat destruction

The most important guideline involves conserving existing areas of native vegetation as a whole, functioning unit. The easiest, least expensive, and best way to conserve Pennsylvania’s plant heritage is to protect existing native plant communities from further disturbance. If disturbance must occur, strive for minimum habitat destruction. Some cases may require ecological restoration, which can include planting native species, removing invasive introduced species, controlling erosion, and loosening soil compaction.

Landscape with native plants

The destruction of native plant communities has occurred in many areas. Intelligent landscaping in parks, yards, and campuses can help redress this loss. Well-chosen native plants can flourish in these landscapes. The Department of Conservation and Natural Resources recommends avoiding rare, endangered, and threatened plants and instead choosing native plant species, which grow commonly throughout the state. If you do not want all natives, select adapted introduced plants suited for the site, colorful annuals, or flowering plants that will not escape and become environmental weeds.

Learn more about native plants

Learn native plants in your area. Many field guides focus on the topic of wildflowers that can get you started.

Buy nursery-propagated native plants

Most retail nurseries and mail-order catalogs now offer native plants. If you want guaranteed ornamental characteristics, in some cases cultivars (named varieties) are available; for instance, a cultivar of New England Aster named



‘Purple Dome’ was selected for shorter height and showier flowers. Cultivars have predictable attributes like height, color, blooming period, or absence of seed pods/thorns—qualities many gardeners want. If your goal is genetic diversity, however, ask for straight species, not cultivars, grown from local seed sources. Plants grown from seeds have much more variety than cloned cultivars.

Do not remove native plants from the wild

Taking native plants from the wild depletes native populations. In addition, many wild-collected plants do not survive transplanting. Prevent wild collecting of plants by making sure a nursery propagates the plants that you buy, or by starting plants yourself from a local seed supply. Before you collect seeds always obtain the property owner’s permission.

Practice responsible landscaping techniques

The first rule of responsible landscaping is planting the right plants in the correct environment: never introduce invasive plants to your landscape that will aggressively spread off your trail and invade native plant communities. They can drastically alter ecosystems and give you and your neighbors maintenance headaches for years to come.

When landscaping with native plants it is important to choose plants that will grow well at the site: wet or dry, shade or sun, acid or neutral soil. By noticing which native plants thrive nearby, you can use those clues to guide plant selection. Other information can be found from plant nurseries, catalogs, books, or the Internet.

For soil fertility, composting and mulching leaves or grass clippings provides a slow release of nutrients. Chemical fertilizers often provide too many nutrients too quickly for native plants, and this flush of nutrients gives weeds a competitive edge. Proper site preparation begins with a soil test before applying fertilizer. Try organic pest control. Keep the soil covered to prevent weeds. Remove invasive plants nearby. Take out severely diseased plants, or ones with insect infestations. Many native plants attract beneficial insects, which help control pests, so try creating habitat for “good bugs.”

Design Process

Complete the design process taking the following steps to advance the design of your trail from the planning stage through final design.

Develop Planning Inventory Map

At this point you should have completed the planning inventory and analysis as described in Chapter 1 - Planning. Your trail designer should review the map and identify anything that may have changed since the completion of the inventory that occurred in the planning phase.



Develop Trail Management Objective Sheet(s)

You have also established the trail management objectives by completing the Trail Management Objective sheet(s) for your proposed trail(s) during the planning phase. This sheet provides your trail designer with the information they need to develop the preliminary alignments and prepare the specifications for each segment of trail. This information documents decisions made about:

- Designed use(s)
- Season(s) of use
- Trail system layout
- Trail specifications
- Managed use(s)
- Prohibited use(s)
- Sensitive areas
- Destinations
- Structures
- Trailhead(s) & access point(s)
- Amenities
- Maintenance requirements & frequency

Identify Trail Corridors

During the planning phase, as described in Chapter 1, your designer has also identified potential corridors and control points the trail will pass including destinations, general play area locations, and general trailhead and trail access locations. The widths of these corridors can vary greatly, indicating where your designer can best place the trail.

Develop Preliminary Trail Alignments

With the above information, your trail designer will confirm the level of difficulty for each trail segment, which will provide the maximum grade for that segment and establish the general guidelines for the horizontal alignments. Your trail designer will then locate the preliminary horizontal alignments in each trail corridor on a base map.

Flagging and Refinement of Preliminary Alignments

After establishing the preliminary alignments on paper, your trail designer will flag the alignments in the field. Then they will refine the alignments to avoid obstacles and previously unidentified constraints, and flag the locations of specific construction details. Your designer will also adjust the alignments to provide a flow for the trail and to take advantage of interesting visual features. These features may include rock outcrops, waterfalls, habitats, and scenic views.

Your designer should flag the trail alignments by placing flags no greater than 50 feet apart, at any change, such as horizontal curves, vertical grade, and



cross section, and where the construction of specific trail features will occur. The designer should also flag the clearing limits, and any trees that need to be removed. If the designer cannot see adjacent flags along the alignment, your designer should place more flags along the alignment. Placing too few flags along the alignment is a common mistake in flagging trails. Your designer will document the final alignments on the base map.

Revise and Finalize Trail Management Objectives

Your trail designer will revise the Trail Management Objective Sheets as necessary, based on the refinement of the preliminary alignments.

Document Final Alignments on the Plans

Once your trail designer is satisfied with the final alignments, they will record the alignment utilizing GPS equipment to ensure properly located trail alignments on the plans.

Normally, trail design and construction does not require a detailed survey. Your trail designer will identify when you may need a property boundary survey. A property survey establishes the locations of the property boundaries and ensures that you build the trail on your property. You may also need to have a topographic survey prepared in certain situations. Generally, the use of topographic surveys occurs to show elevations and geometry when your trail will intersect with a state or local road. Further, topographic surveys prove useful when designing trailheads, and for designing structures, such as bridges and retaining walls.



Construction Documents

After completing the final alignment(s) and recording their locations, your designer will prepare the construction drawings. On these drawings, the designer identifies the locations and establishes the details for each section of the trail. This generally includes typical cross sections, horizontal and vertical trail alignments, tread materials, location of water control structures, location of other structures and specific details for trail facilities such as trailheads and amenities. The trail designer should prepare an opinion of probable construction cost to compare the projected cost of constructing the trail to the available funding.

If competitive bidding will occur on your project, your designer will prepare bid documents that also contain regulatory requirements, conditions of the contract, and technical specifications.

Construction drawings typically include:

- **Erosion and Sedimentation Control Plans:** To document the locations of temporary and permanent erosion and sedimentation control features.
- **Layout Plans:** Existing conditions with proposed improvements overlaid, GPS coordinates, stationing, and/or dimensions to locate the proposed improvements, detailed call-outs identifying locations of various construction requirements that a construction detail further defines.
- **Grading Plans:** Existing and proposed topography which indicates the amount of cut and fill needed to establish the trail bed. Grading plans also typically identify locations and the extent of permanent erosion and stormwater control features.
- **Planting or Landscaping Plan:** Identifying existing vegetation to remain; location, size, and quantity of proposed plant materials; and extent of seeding.
- **Construction Detail Plans:** Both graphic and written information providing detailed information about specific parts of the construction. A drawing showing the area of construction at a larger scale in order to clearly show the materials, dimensions, method of building, desired joint or attachment, and so on. Commonly, shows details in various sections.

Depending on the size and scope of your project designers can combine information on fewer plans, or separated into more plans.



Standard Drawings for Construction

The U.S. Forest Service (USFS) has National Trail Drawings and Specifications. These construction details and specifications are available for professional use by those qualified to adapt these drawings to local conditions. The user accepts full liability for their use. The USFS does not provide supporting data for these drawings and specifications. The USFS makes them available on an as-is basis. The details and specifications can be previewed and downloaded from www.fs.fed.us/database/acad/dev/trails/trails.htm.

National Trail Drawings

This includes fifty-three standardized set of model construction details that designers can adapt and modify to fit many trail situations.

- | | |
|---|--|
| 1. Clearing Limits | 28. Underdrain |
| 2. Typical Trail Cross Section | 29. Foot Log Trail Bridge with 2 Handrails – Side View |
| 3. Outsloped Climbing Turn | 30. Foot Log Trail Bridge with 2 Handrails – End View |
| 4. Trailbed and Slope Finish | 31. Optional Deck and Handrails |
| 5. Talus and Rubble Rock Section | 32. Puncheon without Decking |
| 6. Grade Dip | 33. Puncheon with Decking |
| 7. Rolling Dip | 34. Plank Stairway |
| 8. Turnout and Passing Sections | 35. Crib Ladder Stairway |
| 9. Shallow Stream Ford and Gully Crossing Rock Structure | 36. Rock Stairways |
| 10. Shallow Stream and Ford or Gully Crossing Log Structure | 37. Pinned Stairway |
| 11. Insloped Climbing Turn | 38. Log and Treated Timber Riser Stairway |
| 12. Turnpike - Type I | 39. Log Retaining Wall |
| 13. Turnpike - Type II | 40. Rock Retaining Wall |
| 14. Switchback - Type I | 41. Aggregate Surfacing |
| 15. Switchback - Type II | 42. Bituminous Surfacing |
| 16. Switchback - Type III | 43. Grid Pavement Units |
| 17. Existing Trail Restoration | 44. Sign and Post Installation |
| 18. Check Dams | 45. Rock Cairn Construction |
| 19. Culverts with Headwalls | 46. Log Barrier |
| 20. Culverts without Headwalls | 47. Log Barrier on Posts |
| 21. Rock Culverts | 48. Treated Timber Barrier |
| 22. Treated Timber Box Culvert | 49. Treated Timber Barrier on Posts |
| 23. Treated Timber Box Culvert Details | 50. Rock Barrier |
| 24. Rock Water Bar | 51. Trail Obliteration |
| 25. Log or Treated Timber Waterbar | 52. Seeding and Fertilizing |
| 26. Rubber Belting Waterbar | 53. Illustration of Trail Structure Terms |
| 27. Rock Spillway | |



Standard Specifications for the Construction and Maintenance of Trails

Further, the companion document *Standard Specifications for the Construction and Maintenance of Trails* provides the technical specifications for each item, along with maintenance specifications for existing trails. These specifications include the following sections:

General Specifications

- 901 Abbreviations, Acronyms, and Terms
- 902 Definitions
- 903 Intent of Contract
- 904 Maintenance for Traffic
- 905 Control of Materials
- 906 Measurement and Payment
- 907 Quality Assurance and Quantity Measurement
- 908 Staking, Flagging, and Cleanup

Construction Specifications

- 910 Earthwork
- 911 Clearing and Grubbing
- 912 Excavation and Embankment
- 913 Turnpike
- 914 Switchbacks
- 915 Existing Trail Restoration
- 916 Removal of Structures and Obstructions
- 920 Drainage
- 921 Culverts
- 922 Waterbars
- 923 Rock Spillways
- 924 Underdrain
- 930 Structures
- 931 Log Stringer Bridge
- 932 Puncheon
- 933 Trail Stairways
- 934 Log Retaining Walls
- 935 Rock Retaining Walls
- 936 Wire Baskets
- 940 Surfacing
- 941 Aggregate Surfacing and Base Course
- 942 Hot Bituminous Plant Mix Trail Surfacing

- 943 Cold Bituminous Mix Trail Surfacing
- 944 Grid Pavement Units
- 950 Incidental Construction
- 951 Mobilization
- 952 Sign, Post, and Cairn Installation
- 953 Barriers
- 954 Obliteration of Abandoned Trailways
- 955 Seeding and Mulching
- 960 Materials
- 961 Rock, Grid Pavement Units, and Aggregate
- 962 Material for Timber Structures
- 963 Drainage Pipe
- 964 Geosynthetics
- 965 Wire Baskets

Maintenance Specifications

- 981 Brush Cutting
- 982 Logging Out
- 983 Danger Tree Removal
- 984 Loose Rock Removal
- 985 Rock and Root Removal
- 986 Borrow
- 987 Slide Maintenance
- 988 Slough and Berm Removal
- 989 Turnpike Maintenance
- 990 Switchback Maintenance
- 991 Drainage Maintenance
- 992 Rock Retaining Wall
- 993 Sign Repair and Placement
- 994 Barrier Maintenance



Construction

We build trails to create and to connect points of interest, and to create experiences along the way. As we build trails we strive to reduce the impact on the environment by building them to ensure trails are stable, safe, and constructed in a manner which prevents erosion.

Trail building is tough. Trail building needs long, strenuous days requiring a great deal of physical effort.

Whether you are building trails with volunteers, organizational staff, or paid contractors, this Chapter provides you with the information necessary to guide your project from the planning and design phases, into construction.

In this Chapter we discuss:

- Trail Construction Options
- Trail Crews
- Safety and Preparation
- Trail Construction Tools
- Field Layout
- Markings and Signage
- Trail Clearing and Excavation
- Tread Construction
- Construction Management

Further, realize that trail construction and reconstruction is an on-going process because a trail is dynamic. Over time, use and environmental factors will *wear in* the trail. As time goes on you will find areas that may need to be re-routed, reconstructed, or otherwise improved to address a particular concern.



Bench Construction with a SWECO

Photo Credit: PA DCNR



Hoisting Rocks on the North Country Scenic Trail

Photo Credit: Andrew Bashaw



Trail Construction Options

Trails can be constructed in several ways. Before you can begin construction, you'll need to determine how and who will be constructing your trail:

- If a project involves public funds you will most likely need to follow public bidding requirements to procure materials, or labor and materials, to construct your trail.
- If not using public funds, you may be able to procure materials, or labor and materials, without following public bidding requirements.
- The project may be completed with labor supplied by your agency or organization.
- The project may be completed by volunteers.
- Materials may be donated for use in constructing your trail.
- Or, your trail may be constructed using a combination of the above options.

Your organization will need to evaluate which option, or combination of options will both meet your needs and meet any state or federal requirements. If you are unsure whether you need to competitively bid your project, contact the local municipality or governmental organization having jurisdiction to advise you.

Public Bid

If a project involves public funds and you desire to hire a contractor to construct your trail, you will likely need to follow public bidding requirements. In Pennsylvania, purchases by local governments fall into two general categories:

1. those where competitive bidding is required by law
2. those where competitive bidding is not required

In general, purchases over \$18,500 require competitive bidding. Purchases where competitive bidding is not required fall into three general areas:

1. those where the amount is less than the \$18,500 limit established in the municipal codes
2. categories of purchases specifically exempted from bidding requirements although the amount of the purchase may exceed the limit
3. and, purchases made under emergency conditions

Detailed information on public bidding requirements can be reviewed in the Purchasing Handbook, published by the Center for Local Government Services. You can download this handbook by visiting www.newpa.com/webfm_send/1530.



Further, depending on the value of the work, and the source of funding, additional requirements may come into play. The following are examples of common requirements that often need to be met:

Bid Bond

A bid bond (or certified check, bank cashier's check or treasurer's check) is to be provided with the bid proposal to ensure the bidders financial responsibility for all bids over \$18,500 as applicable.

Performance Bond and Labor and Material Payment Bonds

The Public Works Contractors' Bond Law of 1967, as amended, 8 P.S. §§ 191-202, requires that all construction contracts in excess of \$10,000 be accompanied by a Performance Bond and a Labor and Material Payment Bond, unless current municipal codes are more restrictive.

Each bond must be in the amount of 100% of the contract award price. These bonds must be secured from an agency or firm approved by the Commonwealth's Department of Insurance to do business in Pennsylvania. These bonds may be substituted with other acceptable forms of security by Municipal Grantee (Act 171 of 1991, 8 P.S. 193, 1967, P.L. 869 No. 385).

Maintenance Bond

A maintenance bond is also required to protect Municipal Grantee/Non-Profit from faulty or defective work by the contractor. This bond should cover a period of one year following the completion of the project and should be in an amount of at least 10% of contract price.

Workers' Compensation

The municipal codes require all contracts involving the employment of labor include provisions for workers' compensation. All contractors must produce proof they accept the provisions of the Workmen's Compensation Act and have insured their liability under the Act.

Prevailing Wage Rates

Pennsylvania prevailing wage rates are required if the total project costs is estimated to be in excess of twenty-five thousand dollars (\$25,000). If the Municipal Grantee/Non-Profit will be using Federal Funds as a part of the local match, and the contract(s) or subcontract(s) exceed two thousand dollars (\$2,000), compliance with Davis-Bacon Federal Wage Rates may be required. In all cases, the appropriate wage rates must be followed.

Davis-Bacon Act

When using federal grant or loan monies for construction projects, wage rates and record-keeping procedures established by the Davis-Bacon Act must be followed. Davis-Bacon requirements apply in all cases of





construction projects exceeding \$2,000 performed with contracted labor and where federal funds supply more than 25 percent of the total project costs. For more information about the Davis-Bacon Act, visit the U.S. Department of Labor website at www.dol.gov. Use the Search tool to find Davis-Bacon and Related Acts.

Steel Products

Under the terms of the Steel Products Procurement Act, all public works contracts, including construction, maintenance, and repair, must specify use of steel products made in the United States. Each municipal code also contains a provision referring to the Act. The Act requires municipal officials to obtain certification of the source of steel products used before payments are made under the contract.

Award of Competitive Bid Contract

When a materials and/or construction is procured through a competitive bidding process, the owner, if awarding a contract, must award it to the lowest responsible bidder.

Purchases Below Bidding Limit

You are permitted to award negotiated contracts for small purchases. As exception in the law requiring competitive bidding permits negotiated contracts when the amount of the project does not exceed \$18,500. There are criminal penalties in the law as well as the possibility of surcharges for making purchases piecemeal to avoid competitive bidding. The entire project or purchase must be \$18,500 or less to eliminate the need for competitive bidding.

Except for cities, second class counties and some home rule municipalities all other counties and municipalities must obtain at least three written or telephone price quotations for all contracts between \$10,000 and \$18,500. A written record of telephone price quotes must be made, including the vendor's name, the purpose of the contract, and the price. Records of informal written or telephone price quotes must be retained for three years.

If fewer than three qualified vendors exist in the market area, a memo must be placed in the file explaining why it is impractical to obtain at least three price quotes. Informal price quotes are not required for purchases below \$4,000, but are a sound business practice.

Constructing Trails with Agency or Organizational Staff

If your trail is going to be constructed with agency or organizational staff, you must comply with applicable requirements of your agency or organization.



Constructing Trails with Volunteers

If your agency or organization will be utilizing volunteers to construct trails ensure your organization has done everything within its powers to minimize the potential for liability before agreeing to proceed. This includes evaluating the need for and implementing:

- A release, hold-harmless, and indemnification agreements as discussed in Chapter 5 - Management.
- Insuring your organization with:
 - Nonprofit Directors and Officers Liability Insurance
 - Property Insurance
 - General Liability Insurance with volunteers as insured, with abuse/molestation insurance rider
 - Automobile Insurance with employees and volunteers as insured and non-owned auto liability coverage
 - Accident Insurance to cover injuries incurred by volunteers

When considering insurance policies consult with an attorney who can advise you on the types and amounts of insurance coverage you should retain.

Trail Crews

Crew members are expected to work with great diligence and attention to detail, as the trails that are being built are designed to be long lasting. Crew members are expected to observe and participate in group wilderness safety, good judgment, and motivation activities.

Trail crews can consist of paid or volunteer staff. To work on a trail crew workers must be at least 16 years of age and in good physical condition; those younger than 16 can work if accompanied by an adult. Discourage volunteers from bringing children to participate in all-day trail building, as most find it to be boring. Crew leaders should be experienced in trail building, but no previous trail building experience is required for crew members.

Education

A crew leader should participate in crew leadership training and have previous experience as a crew member. Crew leadership training prepares individuals to begin achieving proficiency in the design, layout, construction, and leadership of recreational trails projects. While most courses are detailed and technical, they typically provide only an initial foundation on which to build proficiency through additional experience within volunteer, conservation corps, commercial, and agency trail construction and maintenance organizations. Crew leadership courses typically cover the following topics and have the following objectives:



French Creek Tuesday Night Trail Crew

Photo Credit: PA DCNR



Crew Leadership Training

Course Topics

- Design Concepts
- Trail Project Leadership
- Layout Principles
- Clinometer Orientation
- Topography Map Exercise
- Field Layout Exercise

Course Objectives

- Understand the broad requirements for creating a recreational trail
- Understand trail corridor design concepts
- Understand trail tread layout principles
- Use a clinometer to determine gradients
- Use a topo map and ruler to determine corridor parameters
- Layout and flag a trail project in the field.

Pennsylvania Child Labor Law

When working with trail crews and volunteers it is important to have an understanding of the Pennsylvania Child Labor Law and how it influences what a child can and cannot do. The following is a brief synopsis of the Child Labor Law in Pennsylvania.

The Pennsylvania Child Labor Law applies to minors (persons under 18 years of age) who have not yet graduated from high school. Under the Child Labor Law:

- Minors must have a valid employment certificate or transferable work permit before beginning the volunteer work.
- Certain restrictions apply to the number of hours minors may work and the starting and ending times of a workday.
- Minors may not perform hazardous or strenuous work. The following are occupations, that may be required during trail construction and maintenances, which are prohibited occupations for minors:

Under 18

- Crane Operators: Cranes and hoists.
- Electrical Work: All electrical wiring, installing, removing and testing electric meters.
- Excavating: Within tunnels or shafts and trenches more than four feet in depth.
- Explosives: Where explosives are manufactured, handled or stored.
- Machinery: Includes all types of powered driven machinery, i.e. chain saws, brush cutters, string trimmers, mowers, etc.
- Paints, Acids, or Poisons: In any capacity in the manufacture of paint, color or white-lead, poisonous dyes, or compositions using dangerous



lead or acids.

- Pits and Quarries: Most occupations.
- Printing Industry: Operate power driven paper cutters or circular saws.
- Roofing: All occupations.
- Welding: Acetylene or electric welding.
- Wrecking and Demolition: All occupations.

Under 16

- Building Trades: Heavy work.
- Highways: Sections of the highway open to the public for vehicular travel.
- Tunnels: In any capacity.

The Child Labor Law does not apply to conservation and cleanup activities. This means that an employment certificate is not required for a minor who performs conservation and clean-up services, such as litter pickup, trail brushing, trail construction, tree planting, signing, or boundary maintenance as this type of work is typically not hazardous or strenuous.

Field volunteer coordinators must review proposed projects for minors to determine whether they require an employment certificate and to ensure the activity is not hazardous or strenuous.

Field coordinators can obtain additional information from the Bureau of Labor Law Compliance in Pennsylvania's Department of Labor and Industry. They may also consult DCNR's Office of Chief Counsel.

The *PA DCNR Conservation Volunteer Manual* typically governs volunteer work conducted on Pennsylvania State Forest land or within State Parks. It requires parental or guardian consent before a minor is permitted to engage in volunteer work. However, when minors are part of a group or organization (e.g., scouting or church group) that is providing volunteer services, the organization may sign on behalf of its minors if the volunteer coordinator has ensured that the organization has obtained parental or guardian consent for the minors to engage in the activity.

Pennsylvania Child Protective Services Law

Pennsylvania Child Protective Services Law requires private and public agencies to secure both child abuse and criminal history background clearances for a broad range of individuals who are in paid employment capacities with children and youth. This clearance process is one mechanism used to address issues of child safety. The background clearances identify substantiated incidents of child abuse and any relevant criminal offenses.

Many private and public agencies also extend these requirements to volunteers and others for whom they are not statutorily required but who are



involved in activities with children and youth. FBI clearances are also required for potential employees residing out of state.

These clearances typically include:

1. Pennsylvania State Police Request for Criminal Records Check (Act 34)
2. Department of Public Welfare Child Abuse History Clearance (Act 151)
3. Federal Criminal History Record Information (CHRI)

Pennsylvania Conservation Corps Crews

The Pennsylvania Conservation Corps (PCC) is a statewide program that offers work experience, job training, and educational opportunities to young people who complete conservation, recreation, and historical preservation projects on the state's public lands. PCC corps members are between the ages of 18 and 25; they work in crews under the guidance of skilled, adult crew leaders.

Political subdivisions and certain state agencies may apply for PCC grants to carry out projects related to recreation, conservation and historical preservation. Non-profit agencies in cities of the first class may also apply, but only for projects involving the removal of graffiti and the repair of institutional vandalism.

Agencies selected to receive grants get the services of a PCC crew--all wages paid--for one year. They may also get funds to pay for the materials and contracted services needed to complete their proposed projects (political subdivisions must provide a cash match).

A PCC crew members must be a Pennsylvania resident between the ages of 18 and 25 (16- and 17-year-olds may participate in special circumstances), not be enrolled in school full time, and be physically and mentally capable of performing labor-intensive work.

PCC crews have been used to construct trails on state lands, as well as, at the municipal level. For more information on this program contact:

Pennsylvania Conservation Corps
1405 Labor & Industry Building
651 Boas Street
Harrisburg, PA 17121
www.dli.state.pa.us/pcc
pcc@state.pa.us



Crew Leaders

Above all else is attitude! Crew leaders should not make the assumption that everyone thinks the way they do. Crew leaders should spend a considerable amount of time going over safety rules before each work-outing.

Safety Rules

1. Carry tools in the safest way. Grip tools by the handle about 6 inches behind the head, or at the balance point, and carry it to the side, on the down-slope side of the body rather than over the shoulder or as a walking stick. This prevents injuries due to falling on the tool, since it can be easily tossed away when carried correctly. Carry tools with sharp blades by facing the blade to the ground, and equipped with a sheath to prevent accidental cuts and to retain their sharp edge. Keep the sheath on the tool while carrying it to the worksite and only remove it when it is being used. Hold bulky or clumsy items with two hands or have two people carry them.
2. Leave plenty of room between volunteers for walking and working, generally 10 feet between each crew member.
3. Crew members should always be aware of what others are doing and take full responsibility for their own safety and the safety of others.
4. The right tool should be used for the job.
5. Implement the “Scan-Shout-Swing” order of doing things. Crew members should look around to make sure no one is in harm’s way and to ensure there is plenty of room to swing safely. If necessary, clear brush or limbs first to avoid injury from a deflected tool. Second, communicate intentions and third, crew members may proceed when all is clear.
6. Remove trail hazards as they are encountered, or communicate their presence to other workers down the line, either verbally or with a temporary sign (for instance, a temporary sign could warn others of a nearby yellow-jacket nest or a poorly supported leaning tree). Remove hazards as soon as practical to prevent others from being harmed.
7. Dehydration, heat stroke, lack of energy, and hypothermia are life-threatening concerns. Keep first aid supplies on hand and ensure every crew member knows what is available and where it is kept. When working in remote locations, let someone, park ranger/local law enforcement, etc. - know the crew’s location and expected time of return.



8. Machismo should be saved for the football field, it's easier to be carried off a football field than it is to be carried out of the woods.
9. Crew members should be aware of their physical condition and limitations, weariness can lead to accidents.

Leading a Work Outing

Safety is the number one priority in all trail operations. Crew leaders are responsible for briefing crew members to maintain a safe working environment and instilling in them a sense of responsibility. Every work leader should learn and teach his/her crew safe work habits and see that these practices are adhered to. Every tool is a potential source of injury and everyone can not be watched, at all times. Therefore, the crew leader must establish ground rules at the beginning and teach by example.

Select and inspect the correct tools for the job before using them. Ensure blades are sharp, handles are smooth, and heads are securely fastened. Properly care for tools and use them correctly. Crew leaders should demonstrate proper carrying and handling techniques before leaving the parking lot.

Careful planning will prevent problems during the outing. A checklist of supplies and safety-briefing points is a must.

Sample Checklist

- | | |
|-------------------------------|----------------------------|
| • Tools (list) | • Personal Protection |
| • First Aid kit | Equipment |
| • Insect repellent | • Compass |
| • High energy food (list) | • Topographic maps |
| • Water (adequate amount | • Project maps and plans |
| for conditions and crew | • Trail brochures |
| members) | • Membership information |
| • Individual water containers | • Copies of "Safety Rules" |
| • Extra rain gear or plastic | • GPS with extra batteries |
| garbage bags | |
| • Extra gloves | |
| • Hard hats | |
| • Eye & ear protection | |
| • Other personal protective | |
| equipment | |

Tell all potential participants what they are expected to bring prior to the work-day. Normally, participants are responsible for their own footwear, rain gear, and gloves. There will always be a few who need an item, so crew leaders should bring extra, if they are available.



Sometimes people will bring their own tools. This is fine as long as their tools are in good condition and appropriate for the project. Inexperienced workers, for instance, seem very fond of hatchets. They are ineffective for trail work when compared to loppers, bow saws and other trail tools. Crew leaders should be firm about leaving such tools behind, and should check all personal tools for soundness.

Just because a person has brought a tool from home does not mean that he/she knows how to use it correctly. Take time to discuss proper use of all tools that are going to be used that day. If there are more tools than people, the determine which ones are really necessary and which ones should be left behind. In most cases crew members should not carry more than one tool, except when walking in to major project sites. An exception to this is when small tools such as wire cutters for old barbed wire can be carried in a pocket or day-pack. In addition to teaching basic rules, a crew leader must also discuss other precautions, such as being aware of poisonous pests or plants suspected in the work area.

Crew leaders must be aware of any signs of fatigue, dehydration, or heat stroke among crew members. If someone seems to be having trouble, crew leaders do not need to draw undue attention to it, but should not ignore it either. Crew leaders should ensure that everyone is getting enough rest periods and water.

Trail building should be a good experience for everyone. If the experience is positive, people will come back for the next work-outing.

Special Considerations When Working with Youth

If supervising young people, especially teens who have a history of problems, crew leaders should take a deep breath, sit back, and relax. Crew leaders of youth are to be congratulated for having accepted such an important and challenging endeavor. The first prerequisite is to learn stress reduction and relaxation techniques. The following tips may be helpful:

1. Safety is priority from the very beginning. Crew leaders should let members know that mishandling of tools or abusive behavior toward others is not tolerated. The reason being that they genuinely care about the crew members, the continuation of the program, and performance quality.
2. Crew leaders should establish a rapport by doing things with the teens that are fun, like jumping in a lake, or picking berries, and by being tolerant of other bothersome things the youth might do like making fun of the functional clothes crew leaders wear.
3. It is essential for crew leaders to have a sense of humor.
4. Crew leaders should be impeccable role models, but not afraid to admit their own mistakes.
5. Crew leaders should empower the youth with a can-do spirit, not



a spirit of invincibility but one of humble self-reliance built upon cooperation.

6. Working with youth, especially those often referred to as “at-risk,” is not something that is for everyone. It is a job that can be both extremely rewarding and extremely frustrating. However, it is clear that teaching youth the importance of respecting themselves and each other, by practicing safety and strong work ethics, can make a real difference in their lives and in the life of the trail.

Estimating Work

Before heading out into the field to begin construction, crew leaders need to understand the goals and expectations for the crew’s work, estimate how much work is required and how much work their crew can accomplish in a given time. Crew leaders assist organizational leadership with determining larger management plans and crew leaders must motivate their crews.

Many conditions influence the amount of work that crews can complete during an outing, including: ability of workers, availability of tools and materials, weather conditions, terrain, and time required getting to the work site. Estimating work is an acquired skill mastered with experience.

Visiting a site in advance of a crew’s arrival is another essential part of estimating work. Crew leaders must visit a site to find answers to the following questions:

- How long will it take the crew to travel to the work site each day?
- What tools are required to complete the work and how will they be transported to the work site?
- What building materials are required and are they available on-site? If materials need to be moved, will the crew have wheelbarrows, motorized wheelbarrows, or rigging gear, and do they know how to use them?
- Will the trail sponsor have personnel on-site?
- Will the work site be closed to public use during construction? If not, the crew is likely to lose production time allowing trail users to pass through.
- When the work is complete how much time will be required to close down the project, clean up the work area, carry out tools and gear, and store or return equipment?
- Are there seasonal patterns of heat, cold, rain, or snow that will impact production?
- Can the crew complete the work safely?
- Will the project interest crew members and give them a sense of accomplishment?
- Where will the crew stay? Backcountry camp, modern camp, etc.? Will they transport tools and gear to their camp?
- Can the crew stay at the camp for the duration of the project?



- Is there a backup project for the crew if they finish their primary work ahead of schedule?

Crew leaders should take into consideration the age, fitness, ability, experience, and motivation of their crew members. Complex, intensive, or extensive projects may be right for an experienced crew, but too demanding for inexperienced crews.

Attention to the final details of construction and clean up of the work site makes a great difference in the appearance and quality of the project. Trail crews often overlook this final 10 percent of a project in the haste to complete it. Crew leaders need to budget time to allow this work to be completed, to motivate crew members to take pride in their work, and to finalize the work in a manner that makes the crew proud.

For additional information and resources on crew leadership and working with crews refer to [Lightly on the Land: The SCA Trail-Building and Maintenance Manual](#), published by the Student Conservation Association.

Safety and Preparation

Trail work can be challenging and exhausting, and danger is always present. Therefore, trail crews must be properly equipped and prepared to respond to the hazards of the work environment.

Personal Protective Equipment (PPE)

By using personal protective equipment (PPE) trail builders protect themselves from hazards and injuries. Having appropriate, functioning PPE for an activity should be at the forefront of any trail maintainer's mind before starting a project.

Boots and Socks

Well-made, sturdy leather boots are one of the most important pieces of gear a trail worker must have. Boots suitable for hiking will last much longer and provide better protection than traditional hiking boots made of lightweight cloth. Steel toe boots offer an additional layer of protection, but can be uncomfortable for long hikes.

Use heavyweight wool or wool blend socks to help protect feet from blistering. Do not wear cotton socks that will absorb moisture and increase chances of blistering. Remember as soon as you feel the onset of a blister immediately remove your boots. Allow feet to dry and apply duct tape, molefoam, or moleskin to the injury to prevent it from worsening.



Gloves

Sturdy gloves can help protect hands from potential injury. Although certain tasks require finger dexterity that is limited by using gloves, it is important to wear gloves for most trail activities. Trail workers often choose leather gloves over cotton gloves, because they withstand heavy use and provide better protection from injury.

Eye Protection

Safety goggles and glasses provide an inexpensive, lightweight protective barrier during trail construction. It is important to choose ones that are shatterproof and impact-resistant, as well as meet or exceed ANSI Z87.1-2003 standards. Helmet mounted, mesh facemasks may also serve eye protection while operating a chain saw or brush cutter. When using a chainsaw, users should wear an additional pair of safety glasses underneath a mesh face guard.

Ear Protection

Ear protection is required during some trail activities that include the use of chain saws, rock drills, and other forms of gas-powered equipment. Plastic earmuffs and foam earplugs are two types of ear protection designed to reduce external noise to a level that prevents or limits hearing loss. It is important to know the decibel level of a piece of equipment to choose the appropriate hearing protection. Use hearing protection that meets ANSI S3.19-1974 standards or higher.

Hard Hats

Wear hard hats when doing trail work activities, especially in situations where there may be falling or flying objects. Hard hats should meet all OSHA-ANSI Z89.1-1997 safety standards in its design for an intended activity. Only wear helmets designed or approved for construction activities. Remember to replace damaged hard hats or those that have met their expiration date.

Long Pants and Chain Saw Chaps

When maintaining or building a trail, it is important to wear long pants to protect from cuts, scrapes, and bruises. During warmer months, consider hiking to the work site in shorts and changing into longer pants before beginning work.

Wear chain saw chaps over pants, covering a person's legs and torso to prevent cuts or more serious injuries from occurring. The fibers of chaps will clog drive sprockets and stop chain rotation quickly, before a serious injury can occur.

Dust Masks and Respirators

It is important for trail workers to wear dust masks and respirators to protect against breathing undesirable contaminants from the air. Moreover, trail works should receive training on hazardous material identification and safety



practices to choose the proper mask. For example, dust or particle masks filter airborne particulates, while respirators have a filtration component that filters hazardous fumes.

Rain Gear and Insulation

It is important to prepare for all types of weather during the construction or maintenance of a trail. Pack rain gear and insulating layers appropriate for your location, activity, and worst-case weather scenario. This might require packing several layers of clothing, hats, and gloves even during summer months. Choose synthetic or wool blends designed to keep warm even when wet.

Other Considerations

Clean, inspect, and store all personal protective equipment on a yearly basis. It is a good idea to label and date all PPE at the time of purchase and keep records of the number of uses and stress placed on equipment. Replace personal protective equipment when an item has reached its expiration date or sooner, depending on wear and tear.

Every trail crew or worker should have a first-aid kit handy at all times. Its contents should be based on the hazards posed by the work, remoteness of location, and number of crew members. It is important to become educated on local poisonous plants and pests. Bring soaps, repellents, snake bite kit, and other kits for individuals susceptible to allergic reaction from bee stings and poisonous plants.

Workers should be trained in backcountry sanitation practices.

Workers who stay properly hydrated are safer, happier, and more efficient. Bring plenty of food and water on the trail and eat it. By eating and drinking appropriate amounts, it is easier to avoid hazardous health risks including exhaustion, heat stroke, hypothermia, impaired judgment, and injuries.

First Aid

Ideally, all crew leaders should be certified in first aid by the American Red Cross. A first aid kit should be checked, complete, and large enough for the crew and the job at hand. Above all, it should be taken along on the job, and crew members advised of its location. Professional assistance may be hours away.

Poisonous Pests

Ticks

Wood ticks are a nuisance, but seldom pose medical problems. Check skin and clothing and if any ticks remove them.

Deer, or bear ticks, on the other hand, are potentially very dangerous.

Personal First Aid Kit

- a package moleskin/molefoam
- aspirin or ibuprofen
- a triangular bandage and safety pins
- an assortment of band-aids
- an ace bandage
- 3, 4" x 4" gauze dressings
- a roll of tape or kling bandage
- gloves
- pocket knife
- antihistamine tablets
- iodine antiseptic or iodine antiseptic wipes
- shears or scissors



8 Person Crew First Aid Kit

- a roll of tape
- 4 triangular bandage and safety pins
- 3 - 3" kling bandages
- 6 - 4" x 4" guaze pads
- 15 band-aids
- 2 packages of moleskin or molefoam
- 1 pair of bandage scissors
- 1 pair of tweezers
- 2 - 5" by 9" combine dressings
- 1 sam splint
- 2 ace bandages
- 1 pen
- 5 "SOAP" notes or incident report forms
- 1 pocket mask
- 1 eye wash
- 1 bottle iodine antiseptic
- 12 ibuprofen
- 6 pairs of gloves
- 1 tube of glucose
- 1 cold pack

These ticks, which are much smaller than wood ticks, can carry the bacteria that causes Lyme Disease. Generally, only about one percent of all deer ticks are infected with the Lyme disease bacterium. However, in some areas more than half of them harbor the microbe. Most people with Lyme disease become infected during the summer, when immature ticks are most prevalent.

Except in warm climates, few people are bitten by deer ticks during the winter months. Remove ticks from clothing and the body as soon as they are observed. Research suggests that a tick must be attached for many hours to transmit the Lyme Disease bacterium, so prompt removal can prevent the disease. A thorough tick-check at the end of the day is advised. Immature deer ticks are only about the size of a poppy seed and may easily be mistaken for a freckle or a speck of dirt.

The risk of developing Lyme Disease from a tick bite is small even in heavily infested areas. Most physicians prefer not to treat patients bitten by ticks with antibiotics unless they develop symptoms of Lyme Disease.

Tips

- Wear light-colored clothing so ticks can be easily spotted.
- Wear long-sleeved shirts and closed shoes and socks.
- Tuck pant legs into socks and tuck shirt into pants.
- Apply insect repellent containing permethrin to pants, socks, and shoes, and compounds containing DEET on exposed skin. Do not overuse these products.
- Walk in the center of trails to avoid overgrown grass and brush.
- After being outdoors in tick infested areas, remove, wash, and dry clothing.
- Inspect your body thoroughly and carefully remove any ticks.
- Inspect pets for ticks.
- Your local health department and park or agricultural extension services may have information on the seasonal and geographic distribution of ticks in your area.

How to Remove a Tick

- Tug gently but firmly with blunt tweezers near the "head" of the tick until it releases its hold on the skin.
- To lessen the chance of contact with the bacterium, try not to crush the tick's body or handle the tick with bare fingers.
- Swab the bite area thoroughly with an antiseptic to prevent infection.

Mosquitos

Like wood ticks, mosquitoes are a nuisance but they are not particularly a safety hazard. Use repellents and wear long sleeved shirts and long pants. If workers prefer herbal repellents can be used in lieu of products containing DEET.



Poisonous Snakes

Poisonous snakes are infrequently encountered. Their presence along the trail varies tremendously, depending on the location within the state. Generally, snakes will move away from people if they hear them coming. Poisonous snakes are most dangerous when they are surprised. When you are in poisonous snake country, you can take several precautions to decrease the chance of being bitten. Wear gloves when moving fallen logs or other debris, wear high-top leather boots and avoid stepping over large logs, and take care where you put your hands and feet.

Before working in poisonous snake country, crew leaders and crew members should take a first aid class or read first aid books so appropriate action can be taken in case someone is bitten. Further, before heading into the field, crew leaders and members should determine the location and telephone number of the nearest location to obtain medical treatment. Generally, if a doctor can be reached within one to two hours, you should be able to avoid drastic measures of treatment such as incisions and tourniquets. If a person is bitten, keep them calm and escort them to the nearest doctor immediately. The injured should not run as this will cause the heart to more quickly pump the poison throughout the body. If the crew is working in a remote area, members should be prepared to administer first aid.

Spiders

Like snakes, poisonous spiders are infrequently encountered and are no more common in the woods than in a typical backyard. However, the brown recluse spider may be present in some areas. Therefore, you should inspect clothing before dressing. Also, shake out sleeping bags before use.

Deer Flies and Black Flies

Wearing hats and long sleeve clothing is the best preventive measure for these nuisances.

Yellow-Jackets and Hornets

These pests can cause painful stings and more serious allergic reactions. Persons who know they are allergic should carry a complete sting kit including antihistamine inhalants and other supplies. Crew leaders should check with crew members to see if anyone is allergic and to be sure they have their kit with them. The crew first aid kit should also be equipped with both “sting-kill” ampules and treatment for allergic reactions.

All members should be aware of insect nests and warn others of their presence. It is often the second and later crew members who get stung, as the first person who steps into or bumps a suspended nest has moved out of harms way by the time the insects attack. Yellow jackets typically nest in the ground while hornets build their nest in shrubs and trees. If



you discover a nest mark the area with flagging or a temporary sign to alert others.

At certain times of the year and in certain kinds of weather, yellow jackets seem to be more aggressive. Generally this is in late summer and fall and during periods of dry weather. During these times, yellow jackets are attracted to food and drink. All members should be alert when eating or drinking soda to avoid getting a yellow jacket in the mouth. A sting in the tongue is particularly serious as the tongue can swell and quickly block the airway.

Poisonous Plants

Poison Ivy, Poison Oak, and Poison Sumac

Some people are extremely allergic to these easily identified plants. If seen, others should be alerted to its location so they can avoid it. A line of Technu® products, available in most pharmacies, can help prevent contracting the poison.

Simple Precautions

Crew members should work in clothing that covers most of the skin. Good quality work boots, best if they have a steel toe, a hard hat and gloves should be worn. With power equipment, like a chain saws, safety goggles, ear protection, and high quality chain saw chaps should be used.

Crew members should take frequent water and rest stops. Crews should use common sense and not try to do more than is prudent.

Trail Construction Tools

By planning and preparing for work outings ahead of time, you will be able to estimate the type and quantity of tools necessary to complete the work. Trail construction tools fall into one of three categories:

1. Hand Tools
2. Gas/Electric Powered Hand Tools
3. Mechanized Equipment

The type of tool you use will depend on the type of work you are performing. Typically, agencies and organizations responsible for trail construction will use hand tools, and/or gas/electric powered hand tools. Professional trail builders, contractors, and sometimes municipal parks departments, expand their palette by including mechanized tools and equipment where appropriate.



Determine what tools to bring with you ahead of time by planning what type of work your trail crew will be performing and where the work is located. Focus on one task during each outing and only bring tools required to complete that task. Bringing too many different types of tools or trying to accomplish too many tasks will reduce the crew's efficiency and slow down construction.

Before heading to the work site:

- Select high quality tools
- Ensure crew members are properly trained to use the tools
- Review with crew members the safety requirements and procedures related to the tools
- Review the recommended personal protection equipment that should be worn when operating the tools

Hand Tools

The following is a summary of the various hand tools available for the respective tasks. Additional information on hand tools can be obtained from the U.S. Forest Service publication *Handtools for Trail Work: 2005 Edition*, http://www.fhwa.dot.gov/environment/recreational_trails/publications/fs_publications/05232810/lc05232810.cfm.

Trail Clearing Tools

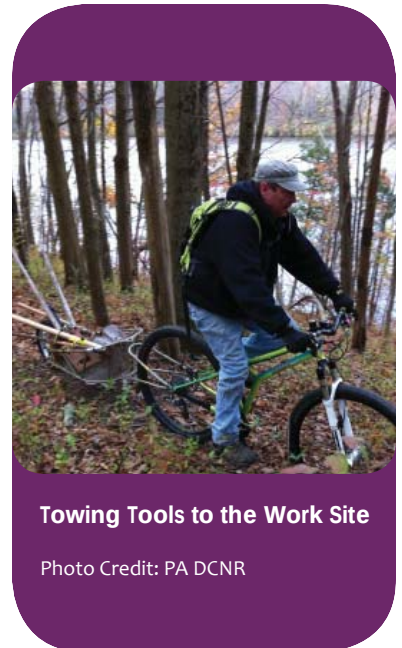
- **Swizzle Stick:** Also known as a weed whip. It is used in a swinging motion, like a golf club, to clear brush and low growth.
- **Lopping Sheer:** Essential for trail corridor brushing involving nipping small branches and clipping vegetation flush to the ground.
- **Machete:** Most effective when wielded with a vertical stroke to clear brush and vegetation.

Timber Tools

- **Axe:** Double bit, single bit, and three-quarter axes are effective for making deep cuts when felling trees and chopping through logs.
- **Brush and Bow Saw:** Suited for making vertical cuts involved in clearing modest-sized logs from trails and bucking firewood in camp.

Trail Tread Tools

- **Pulaski:** Preferred by many trail groups for loosening compacted soil, grubbing tread, chopping roots, and removing slough.
- **Fire Rake:** Suited for cutting, pulling, and scraping debris, mulch, leaves, small roots, and small bushes when clearing a trail.
- **Hoe:** Useful for breaking up sod when constructing a new trail or when leveling an existing trail tread.
- **Shovel:** A spade shovel is most effective for digging holes, moving loose dirt, and digging in tight spaces. As a rule, the longer the sleeve



Towing Tools to the Work Site

Photo Credit: PA DCNR



the stronger the shovel will be.

- **Mattock:** Versatile tool for digging, prying, and moving dirt, especially when unearthing rock, digging through roots, and breaking through rock.

Rock Tools

- **Rock Bar:** The rock bar, also known as a crowbar, is necessary in heavy trail work to obtain enough leverage to move large boulders and unearth rocks.
- **Hammer:** Hammers for shaping rock may have a wedge-shaped peen needed to easily chip away parts of a rock.

Wood Structure Tools

- **Broad Axe:** The beveled head of the broad axe cuts with the grain and hews a flat face in a log.
- **Splitting Wedge:** Wedge for splitting firewood or for laying open logs to support walkways, usually made of soft steel or aluminum.

Bark Removal Tools

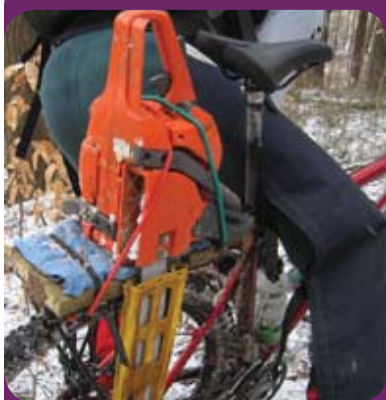
- **Drawknife:** When pulled toward the body, this tool is useful for easily removing dry, hard bark and shaving wood.
- **Barking Spud:** One of the safest tools for removing bark, the spud is valuable for removing green bark.

Hoisting Tools

- **Grip Hoist:** A lightweight tool capable of dragging tremendous weights on wire rope in rigging systems designed for moving timber, rock, and other building material.
- **Ratchet Winch:** Useful for moving rocks and logs, as well as pulling stumps.

Power Tools

- **Chain Saw:** A motorized tool effective for felling and bucking trees without the use of an axe.
- **Gas Powered Brush Cutters:** Similar to a heavy duty string trimmer, with a metal blade for cutting wood. A brush cutter can be useful for extensive trail clearing through young/heavy growth.
- **Rock Drill:** Useful in specific construction applications for drilling holes in rock for pinning rock, installing metal rungs or ladders, or splitting rock with feathers and wedges.



**Transporting a Chain Saw
by Bike**

Photo Credit: PA DCNR



Mechanized Equipment

Additional information on mechanized equipment can be obtained from the U.S. Forest Service publication *Mechanized Trail Equipment*, www.fhwa.dot.gov/environment/recreational_trails/publications/fs_publications/

Excavators

The most common excavator configuration is a boom and shovel with a dozer blade mounted below the boom. A tracked undercarriage is usually used. The primary earthworking implement is the bucket used to dig and move earth. If the excavator is equipped with a small blade, the blade is useful in grading off excess material, leveling and back filling. Most excavator manufacturers have grubhoes or different size buckets available as attachments. The smallest excavator evaluated has a track width of 19 inches and the largest has an overall width of 63 inches. On some models the track width is variable and can be adjusted depending on trail conditions and operator experience. Horse power rating varies between 3.5 hp and 43.4 hp.

Dozers

Dozers are self-propelled machines mounted on either a wheeled or crawler tractor to exert a push/pull force through a blade, drawbar, or ripper and are used to move objects or materials. The primary trail building implement is the dozer blade. Dozers considered for this section are machines with a width under 72-inches, not including the width of the implement.

The advantages and disadvantages of dozers compared to excavators are listed below. Other advantages/disadvantages may be observed depending on use, terrain, and locale.

Advantages

- Large dozer blade is faster cutting a rough trail tread than using a backhoe bucket
- More stable than the backhoe because the center of gravity remains constant. These machines are usually wider than the compact backhoe thus making them more stable
- Moving large boulders or tree stumps is easier with a dozer. The larger engine available in these machines makes pushing obstacles out of the way faster than digging them out
- The auxiliary hydraulics available makes the dozer an excellent platform to power other hydraulic implements

Disadvantages

- Larger than most compact backhoes; and cuts a wider trail
- Larger trailers and towing vehicles are required for transport to the job site



Mini-Excavator

Photo Credit: PA DCNR



Loader

Photo Credit: PA DCNR

Loaders

Loaders are mounted on either a wheeled or tracked undercarriage. These machines have loader buckets up front with engines mounted in the rear. Loaders are ideal for moving rocks, soil, or other trail building material to the job site. They are not designed to cut trail tread, however an angled blade and a york rake attachment are available for that purpose. The smallest loader has a width of 35.5 inches, the largest width of 74.1 inches. Some loaders come standard with wheels, but a track to fit over the wheels is available.

The size of the machine dictates the width of the trail on which the machine can operate. Tracks on the vehicle provide more traction because of a large contact area with the ground. The larger contact area also provides better resistance to inadvertent slipping, making it more stable. The larger contact area also provides more “float” over plastic soil conditions. The damping action provided by the pneumatic tires gives the wheeled vehicle a smoother ride. The vast assortment of available attachments for these loaders makes them versatile trail machines. Loaders can be used to haul or move trail materials closer to the job site.

The advantages and disadvantages of the loader compared to the excavator, dozer, and hauler are:

Advantages

- Versatile with the available attachments the loader performs many other functions. Able to move large volume of material
- Higher travel speeds reduce cycle time
- Parts and service are widely available

Disadvantages

- Performance is compromised when used as other than a loader or hauler
- Additional cost to procure attachments



Hauler

Photo Credit: PA DCNR

Haulers

The basic hauler has two components: cargo area and power unit. The cargo area of the haulers listed in this document range from a polyethylene tub to a hydraulically tilting dump bed. The power units range from an electric motor to a 4-stroke, liquid-cooled 90°V-Twin.

Haulers are used to haul material to and from a job site. The hauling capacity and travel speed becomes important when hauling to and from a remote location. Although these machines were designed to traverse unimproved areas, in most cases the machines travel on trails being maintained or trails with a rough cut tread.

The advantages and disadvantages of mechanized haulers versus hand



haulers are:

Advantages

- Greater hauling capacity with shorter cycle
- Reduced physical demand to the user
- Reduced injury due to manual labor
- Traverses rougher areas than a garden cart or wheelbarrow

Disadvantages

- Requires more maintenance
- Requires transport
- Need to carry gasoline or batteries
- High initial capital investment when compared to manual methods

All-Terrain Vehicles (ATV) and Utility Task Vehicles (UTV)

There are many ATV and UTV models on the market. The models come in either a two or four wheel drive. Both typically have a high ground clearance, and ATV's typically have a short wheel base.

ATV's are versatile trail machines used for hauling material and shuttling personnel to and from the job site. An ATV can tow a trailer to haul material and equipment to the job site. There are numerous attachments which can be purchased or fabricated to adapt the machine to all sorts of tasks. The ATV's are used with a drag/plow or a harrow to groom trails. After market kits are also available to improve the performance of ATVs and UTVs.

Advantages

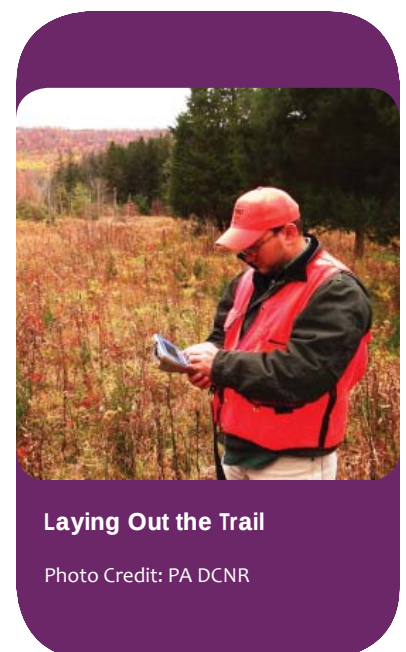
- Versatile the ATV can be fitted with implements used for trail building
- Readily available
- Accessibility most ATV's are small and rugged and can travel over rough unimproved terrain or narrow trails
- Able to haul larger loads
- Higher travel speed thus covering a larger area. When used to haul tools and material, the higher travel speed reduces the cycle time

Disadvantages

- Need to transport
- Need to carry fuel
- Operator has to be trained to operate and handle the machine

Field Layout

The objective of a field layout aims to provide safety, protect the environment, improve recreation, and minimize conflicts with adjacent properties. After mapping and scouting a future trail route, establish the flag line. A flag line marks where trail builders will cut down trees and brush, excavate a trail's treadway, and construct the trail.



Laying Out the Trail

Photo Credit: PA DCNR



During this part of the process, many small decisions may shift the location of the route several times. These details of layout ultimately determine the quality of a trail. A trail's layout should be safe, appealing, and free of conflicts. When a new section of trail opens, it introduces new pressures to primitive lands, so it is important to keep pressures as light as possible.

Trail layout is generally not complicated. Through a trial and error process, trail designers eventually settle on the most stable, sustainable, and attractive route. Using planning maps, first scout for and flag feasible lines through the woods. Then on return trips, add more flags to refine the line of the trail. Finally, coordinate with agency partners to adjust and approve a final trail route.

Scouting

The best time of year for field scouting is when trees are bare throughout the fall and spring. During these times, an entire sweep of the proposed route can be seen that otherwise may be hidden by foliage during warmer months. Avoid scouting when the ground is snow covered because the quality of soil and drainage underneath are not visible. Its best to scout during the wet season to determine where the ground may be too wet to support a stable treadway.

As discussed in Chapter 1 - Planning, it is important for designers to study maps of the area before scouting in order to confirm all property boundaries discovered during planning. Recent advances in satellite technology have provided trail designers with a new design tool known as handheld global-positioning system (GPS) units. This tool calculates the user's position in latitude, longitude, and elevation by timing signals from satellites maintained by the United States. To prepare to scout a new location, upload flag locations to your GPS unit. The GPS unit will guide you to these locations in the field.

Fieldwork can be dangerous, especially when alone. Scouting is easiest with two or three people who can walk on and assess terrain in multiple areas. By analyzing various routes, as well as taking field notes, designers are able to discover a route that provides the best continuous course.

Flagging

Trail flagging is essential for accurate trail layout. Flagging is the process of tying plastic ribbons around trees to indicate the center line of a route. Surveyor's flagging tape may be used to wrap around living trees and branches, while wire flags are commonly used in open, treeless terrain. Both surveyor's tape and flags come in various colors and lengths, allowing designers to choose colors that contrast with the prevailing foliage.

Do not skimp on delineating the final flag line. Closely tie flags approximately every 10 feet to indicate fine points of design and provide insurance if one



or two flags are blown away. Use different colored flags when flagging alternative routes. During layout, flag carefully at junctions and switchbacks to make the footpath the shortest, easiest, and most logical route for trail users. Use distance, rugged terrain, or vegetation to prevent shortcutting through these areas.

Remember that patience is often the determinant of the quality of a trail's relocation. Small and large obstacles sometimes lead to significant changes in the trail route and can result in a reroute well ahead or behind an obstacle. Willingness to keep making changes to this route often distinguishes the best trails and their designers.

Always attempt to locate a line that keeps the final treadway free of structures such as waterbars, cribs, steps, and other rock and log improvements. Allow the treadway to maintain a primitive quality by laying out the trail center line in a location that will require minimal maintenance over time. For instance, locate treadways upslope of large trees so excavation does not damage the root system. Large roots extend downslope of trees near the surface, making it difficult to excavate the edge of the treadway.

Sometimes there is no other choice than to navigate treadways through steep terrain that may require heavy trail work. This may occur in areas with a narrow corridor or those surrounded by steep slopes. When this occurs, consider including steps, waterbars, punchon, and cribs in the layout.

When laying out accessible sections of a trail, exclude the above mentioned elements within the design. Locate a center line that allows trail builders to excavate a sidehill treadway with grade dips to provide drainage. When constructing installing accessible trails, anticipate use by designated users and even trespassers that may negatively affect a trail's appearance and accessible character.

Final Design Work

Once flagging is finished, a local management partner must review and, if necessary, adjust a flag line. The local management partner must give a trail its final approval, which sometimes includes an environmental assessment. Agency partners are responsible for protecting the environment and must survey the site to ensure that adjacent plants, animals, and other resources are not affected.

If a new trail line poses unusual challenges for trail builders, consider using wire flags to mark the trail line every 10 feet. Flags are useful on fragile terrain and along routes requiring extensive construction. To remind trail builders of elements critical to the design, write notes with a permanent marker on the flags. Walk along a trail route using a measuring wheel and note where



important structures, drainage, as well as notable features such as boulders and streams are located along the trail route.

Before construction begins, walk the flag line with the trail builders to make sure they understand a trail's overall layout. Ensure the crew leader and all trail builders understand the important aspects of the layout. Without the establishment of good communication, a trail will most likely differ from what was originally proposed.

Protecting the Environment

Understanding how and why trails break down over time makes it easier to establish a trail layout that can withstand heavy traffic. Erosion and saturation processes create the most destruction on a trail. Therefore, it is important to minimize the potential for both during every step of the layout process.

Foot traffic erodes the soil on all trails to some extent. This breakdown follows a series of stages, the first being compaction. Compaction of a trail results in loss of pore space and the ability to absorb water. While compaction is normal, it sets the stage for increased deterioration of a trail. Runoff from melting snow and rain empties onto the trail, sweeping loose soil and gravel downhill. Gravel and pebbles then collect in the center of the treadway until the tread becomes so eroded it turns into a gully. If a gully forms, hikers may choose an alternate route or trample vegetation and soil alongside of a path.

Although less common, saturation caused by standing water in poor drainage areas can cause as much damage as erosive processes. Saturation challenges designers in areas near swamps, low-lying terrain, streams, and areas with impermeable soils. Trails routed through saturated terrain create an unpleasant experience for hikers, ruin the natural appeal of the wet area, and damages the health of surrounding plant communities. Fortunately, deterioration from saturation does not permanently damage soils.

Soil Characteristics

Soils play an important role in a trail's layout. Subsoils underlying the top layer of organic soil should be approximately 1 to 3 feet in depth to accommodate heavy foot traffic. The best soils for trails are those that include both inorganic and organic matter with different sized particles. Inspect and examine soils by feel and a hand lens.

Avoid soils composed of silt and clay that erode easily, as well as soils that are sponge-like with a lot of organic content commonly found in wetlands or bogs. Hardpan soils, underlain by an impervious layer, may retain moisture no matter what their content is. If it is necessary to construct a trail in these areas, consider building a puncheon or applying trail hardening techniques.

Avoid locating a route through rocky terrain to make the trail more rugged.



Where soils are unstable, consider using exposed bedrock or rocky soil to minimize or eliminate erosion. Where soils are thin, remove the exposed rock and organic layer. Consider the use of a ledge as another alternative since it provides the opportunity for a route to navigate steep grades. If the ledge is steep and wet, it may become dangerous during winter months.

Drainage

It is important to study both topography and drainage patterns to understand how a soil drains. The drier the terrain, the less likely it is to be damaged from saturation and erosion. Often ridgecrests provide the driest locations for the treadway, since they are located at the uppermost limit of drainage basins. Avoid surface channels that would direct runoff onto the proposed route. Visit the route in wet seasons at various times throughout the year to indicate where water flows and collects.

After examining surface flow, dig a soil pit or take a sample with a soil auger to check a soil's depth, composition, and wetness. If the pit fills with water, or an auger is covered with wet or dark colored soils, it may indicate harmful water problems. The ideal layout crosses soils whose seasonal high water table is at least 2 to 4 feet below the surface.

Slopes and Topography

Always locate trails on a moderate grade that naturally follows gently sloping features of the topography. It is important to study the lay of the land. It may be helpful to pace an entire route to avoid steep slopes and to find the most feasible route.

In years past designers estimated trail grades visually; now, clinometers eliminate this guesswork. A clinometer is a handheld instrument that allows designers to read the grade in both slope and percent of various layouts. To calculate slope, divide the rise (vertical feet) by run (horizontal feet). Then multiply by 100.

Keep trail grades below 10 to 15 percent and avoid all slopes exceeding 30 percent. Grades over 30 percent can erode quickly unless fortification of a tread occurs during construction. Do not rule out steep grades up to 50 percent for short distances. These steep pitches can add variety and provide flexibility in designing the best overall route.

Using the clinometer ensures that you establish flaglines with consistent, predictable grades and build trails with minimal risks of erosion. Consult the instrument's instructions to provide guidance when determining percent slope for the trail. If the reading is wrong or the sight distance is not the same distance about the ground as your eye, your flagline is incorrect.

Flagging with a clinometer is often easiest with two people of roughly the same height. One person walks ahead and the second person stands along



the route's centerline, sights the person, and asks him or her to move along the slope to find the appropriate grade. This person then ties the flag on a branch where the centerline is located and moves ahead to repeat the exercise.

Trail Construction

When designing trails, often it is difficult to layout the trail without having problems in achieving a sustainable route. Especially in mountain environments, it is very difficult to determine a route that follows the lay of the land, encountering few risks of erosion and saturation. Long, steep slopes are a common issue. Designers must compromise in these circumstances and build contour trails while incorporating climbing turns or switchbacks to overcome changes in grade.

When laying out your trail for construction follow the techniques and best practices for laying out a sustainable trail as discussed in Chapter 2. At this point your trail is ready to be constructed.

Trail Clearing

The first step of construction is the clearing of a trail corridor to adhere to a specific project's design. Cutting new trails can be dangerous, therefore ensure to follow all safety procedures.

Trail crew leaders and their crews must understand that clearing too much can create as many problems, if not more, than clearing too little. Clear a trail corridor to the appropriate design specifications to provide an adequate clearance buffer along the edges of the trail tread. Further, the crew leader needs to understand and communicate to the crew that the designer used trees and other features to accentuate turns, limit speeds, prevent trail widening, and screen adjacent trails from view. Therefore it is critically important for them to follow the established flag line.

Before clearing the trail corridor, install all temporary and/or permanent erosion control features that the designer specified for the trail. Clearing involves removing blow downs and brushing the corridor. During the clearing process, avoid cutting large trees to minimize the need for stump removal. Trees left on the side of the trail define the trail and often are stabilizing the soils. Whenever you need to remove a sizable tree consider using the trunk as material for bridges, retaining walls, or other rustic features.

To clear the corridor the trail builders should use the door technique. The door technique involves walking the trail and imaging you are carrying a door vertically, like a shield, along the corridor. The size of the door corresponds to the vertical and horizontal clearing requirements identified in Chapter 2.



Everything that falls within the outline of the door should be removed. Look for branches or limbs that touch the door. They should be removed as well to reduce future maintenance requirements. Once the corridor is cleared and brushed, the tread can be constructed.

Tread Construction

At this point your trail corridor should be clear so you can safely and easily work on the tread. To construct the tread the ground must be cleared of organic materials and topsoil so the mineral soil can be shaped into the trail tread. It is necessary to remove the organic materials and soil to provide a firm base for the trail tread. On very flat trails it may be necessary to fill the void created by removing this material, or if borrow material is not available it may be necessary to leave the organic material and soil so the tread is not entrenched into the surrounding grade.

To remove organic material and soil use a mattock, fire rake or other scraping tool. Be sure to mark the sides of the treadway and follow the marks to ensure a consistent tread width. Remove all roots exposed during this process with a fire rake, or a root ax.

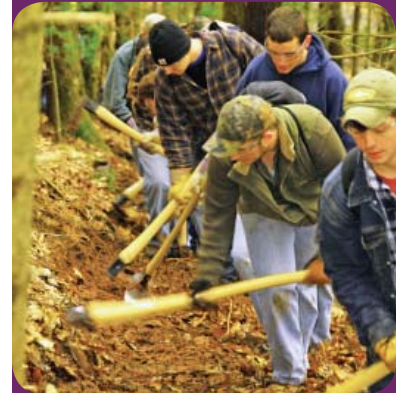
All organic matter must be placed on the downslope side of the trail, far enough way from it that it does not impede drainage. If the trail is near a stream, internment water course, or other body of water, transport the organic material away from the tread and the water course to ensure it does not enter the water.

Where the designer has specified the use of geotextile, and/or specifies a form of tread hardening, remove saturated mineral soil to firm soil.

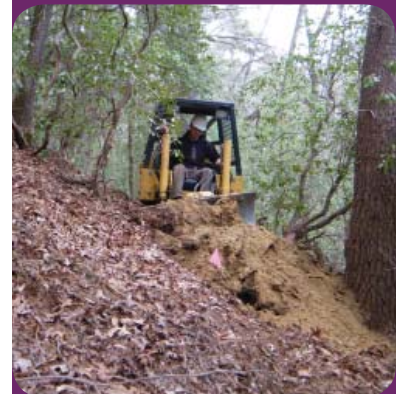
After exposing the mineral soil follow the prescribed tread cross section as selected by the designer. You may be constructing the trail on the existing grade, or you may need to construct a bench to accommodate the trail. If a bench construction is prescribed by the trail designer, ensure that all organic material and soil has been removed so you have a sound foundation to work from.

With the mineral soil exposed, begin cutting the bench in starting on the uphill side, and pulling excess mineral soil to the downhill side of the tread. Continue working your way across the tread until you have achieved the desired elevation and width for the tread. Ensure the stability of the both the upslope and downslope banks you are constructing on the trail.

Typically the sideslopes supporting the tread should be no steeper than 45 degrees - less is better. As you construct the sideslopes carefully smooth and finish them. Time spent fine tuning their shape will reduce future maintenance requirements. Once you complete construction of the tread, and the sidehills are finished, cover them with humus and leaf litter, or seed



Tread Construction by Hand



Tread Construction with Sweco



Rail Trail Tread Construction with Road Grader

Photo Credits: Pashek Associates



them with a native seed mix. This step begins the revegetation process, and in the meantime stabilizes the soils to reduce rilling and splash erosion.

As you progress along the length of the trail follow the designer's specifications for install drainage features such as dips, drainage lens, culverts, and any other trail structures, following the designers specifications for all features.

Once the tread has been established it must be finished. This may mean you need to rake the tread smooth while the specified cross slope to direct water off of the tread. Or it may mean the tread is ready to receive a specified cross section to harden its surface.

Ensure you follow the specified requirements for compacting the tread base when installing geotextiles, and/or when hardening a tread.

Once construction is completed review the construction with the trail manager to review the work and address any areas that are unstable, unsafe, have eroded, or that otherwise do not meet the designer's specifications. Complete any required remedial work as quickly as possible.

Ideally, natural surface trails should not be opened immediately following construction. To improve stability, the tread should be allowed settle and bond before use. Ideally the new trail should remained closed to use for three months.

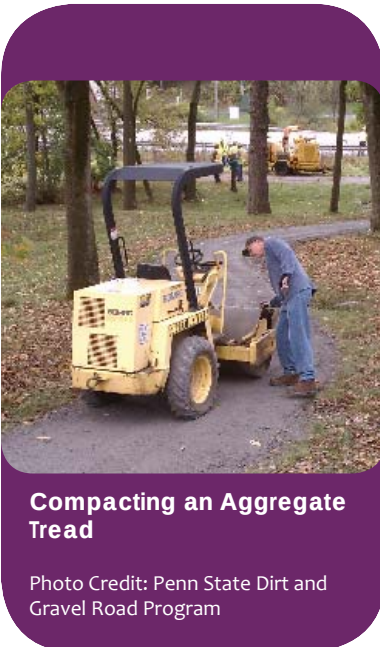
The process is the same whether the trail is constructed by hand or with mechanized equipment.

Trail Closures and Restoration

The closure and restoration of degraded trails allow for the rehabilitation of the landscape, provided humans cannot access these previously accessible areas. Detailed restoration plans require the planning and implementation of long-term environmental repair efforts. These efforts include site monitoring, site and soil preparation, plant revegetation, as well as maintenance efforts.

Take these steps before closing or rerouting degraded natural trails:

1. Talk before rerouting
2. Educate trail users
3. Design a sustainable, more enjoyable trail
4. Design a smooth intersection
5. Break up the old tread
6. Control erosion on the closed trail
7. Transplant vegetation



Compacting an Aggregate Tread

Photo Credit: Penn State Dirt and Gravel Road Program



8. Disguise the corridor
9. Redirect trail users
10. Block the corridor – Last Resort

There are several resources available that provide detailed and in-depth information on trail construction, they include:

- *AMC's Complete Guide to Trail Building*, published by the Appalachian Mountain Club, 2008
- *Appalachian Trail Design, Construction and Maintenance, 2nd Edition*, published by the Appalachian Trail Conference, 2000
- *Lightly on the Land: The SCA Trail Building and Maintenance Manual*, published by the Student Conservation Association, 1996

Signage

In Chapter 3 we reviewed the three classes of signs that are typically associated with trails:

1. Regulatory and Warning Signs
2. Identifier and Reassurance Blazes and Trail Markers
3. Information Signs

Locate signs and markings in accordance with the trail management plan and as prescribed by applicable guidelines and standards for the respective trail use.

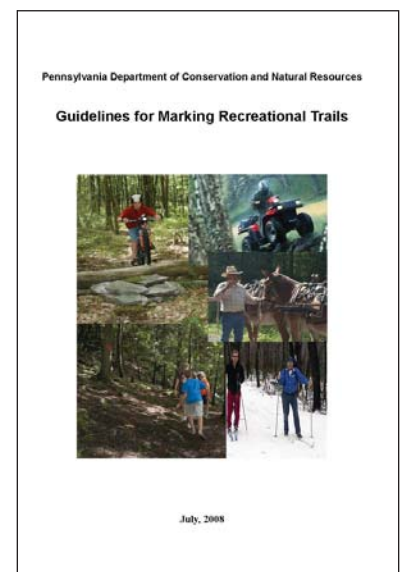
Non-Motorized Trails Marking Guidelines

PA DCNR has established the following guidelines for non-motorized trail markings, as published in *Guidelines for Marking Recreational Trails*, available from http://www.dcnr.state.pa.us/ucmprd1/groups/public/documents/document/dcnr_007076.pdf.

International symbol signs should be used to indicate permitted or prohibited trail use. Always use minimal signage to avoid sign pollution. Dimensions of signs should be 4"x 4", 6"x 6", or 6"x 8" unless stated otherwise.

State Forest and Parks shared-use trails will be marked at trail and road intersections, with a brown 6" even triangle containing three permitted use symbols. Shared-use trailheads will display a brown 12" even triangle with the three permitted use symbols:

- Shared-use trails open to horseback, mountain bike and foot travel, are to be marked with vertical red 2"x 6" reassurance blazes. Red



Guidelines for Marking Recreational Trails
PA DCNR



should not be used on any other trail.

- The State Forest Hiking Trails (presently 18) are to be marked with vertical orange 2”x 6” reassurance blazes. Orange rectangles should only be used on State Forest Hiking Trails.

Accepted exceptions to the previous rule are as follows:

1. The Tuscarora Trail, part of the Potomac Heritage National Scenic Trail, is marked with blue blazes.
2. The Appalachian Trail, a National Scenic Trail, is marked with white blazes, and blue blazes are used on Appalachian Trail side and connecting trails.
3. The Laurel Highlands Trail, a National Scenic Trail, and the Horseshoe Trail, connecting Valley Forge to the Appalachian Trail, are marked with yellow blazes.
4. The Loyalsock Trail, a State Forest Hiking Trail, is marked with red and yellow blazes.
5. The Baker Trail, a State Forest Hiking Trail, the North Country Trail, a National Scenic Trail, and the Mason Dixon Trail, connecting the Appalachian Trail to the
6. The Standing Stone Trail (south central PA) is marked with orange blazes.
7. The Lost Turkey Trail is marked with red blazes.

Cross-country Ski Trails should be posted at the trailhead with the international symbol of a white skier on a brown background along with the Trail Level of Difficulty Rating. Trail reassurance markers will be blue 2” x 6” rectangles. It is not necessary to use Trail Difficulty Rating symbols at all hiking trailheads, however, they may be appropriate for day hiking loops, etc.

Local Hiking Trails (Bureau of Forestry-posted CLOSED to all other uses) can be marked with yellow 2”x 6” reassurance blazes along the trail. Trailheads and other major intersections can be marked with the international symbol of a hiker on a brown square background.

Non-Motorized Trail Marking Techniques and Theory

Some of the following material is adapted from Chapter 10 in “*Trail Design, Construction, and Maintenance*,” published by the Appalachian Trail Conservancy.

Paint Blazes

The intent of this document is to develop, over time, a consistency in trail marking techniques throughout the Commonwealth’s publicly owned lands. Therefore, painted blazes along any given trail should eventually conform to a standard color, shape, and size, namely painted rectangles six inches in height and two inches in width. When painted neatly with sharp corners and clean



Painting Blazes

Photo Credit: PA DCNR



edges, blazes remain visible to hikers at a distance and distinguish themselves from natural occurring marks.

On hiking trails, place the blazes on trees at approximate eye height. Remember, the trail should be marked for the benefit of hikers traveling either way, so place blazes facing in both directions. If you cannot find a suitable tree next to the trail, paint blazes on ledges or trail side rocks. Check with appropriate staff before marking objects other than trees. Other options for blazing are available, such as wooden or Carsonite posts.

Note: Additional colors can be used for blazing State Park Hiking Trails only, based on the exception procedure described in the “State Park Trail System” addendum found at the end of these guidelines.

Frequency

Blazing needs to be continuous, even along roads or unmistakable parts of the footway. Immediately beyond any junction, paint a blaze even if there is a direction sign. Place a second “safety blaze” 50 to 100 feet beyond. Where State Parks and State Forests meet, check that blazes extend into the next section. Eliminate all gaps in marking, and avoid suddenly varying the spacing of blazes (in similar terrain) in a way that confuses hikers.

Normally, you should change blazing frequency naturally with changes in trail terrain, forest cover, or the clarity of the footpath. When the trail is conspicuous, place one blaze for every 5 minutes of hiking time, or about 6 per mile in each direction (800 to 1000 feet apart). Where you run into hard-to-follow sections, often in transitions between field, forest, balds, and other environments, blaze more frequently.

Be careful not to over blaze. Too many single and double blazes can mar the primitive character of the trail. This is a special concern in wild and natural areas, where blazing should remain minimal, or six per mile. Elsewhere, you should place blazes so that no more than one is visible in either direction. In other words, except near trail junctions, keep blazes at least 150 feet apart. You may degrade the primitive trail experience by blazing too often.

Double Blazes

Remember, a double blaze means “caution” or “heads up.” Place a double blaze 25 to 50 feet before abrupt turns and highway or trail junctions. Remove painted arrows, or slanted blazes, and replace them with standard double blazes. Double blazes should be placed one over the other, and about one to two inches apart. Where the double blazes are alerting a trail user to a turn, the top blaze, tree size permitting, can be offset in the direction that the trail will turn.

As with the single blaze, place the blazes sparingly. They are unnecessary at most turns in the trail, and they become unsightly and meaningless with



frequent use. Only use double blazes where necessary for the safety of trail users. On switchbacks, for example, use only single blazes, but paint them near the switchback corner, one above the corner and one below. If needed, pile brush, logs, or rocks at the corner to define the footpath and guide hikers around the turn.

Even in the rare instances where the route remains ambiguous even with blazes, avoid the urge to paint an arrow to direct the trail hiker. The extra, nonstandard paint may hurt the trail's primitive character, detract from the trail users' sense of exploration, or set a precedent for painted arrows. Try to use small directional signs, posts, or cairns, instead.

When Two Differently Marked Trails Share the Same Path

Sometimes two or more trails briefly share the same path or corridor. When blazing in these areas, avoid confusing over-blazing and consider combining different blaze colors into one blaze:

Keeping Blazing Primitive

Blaze Less

- Along well-defined footpath
- Along highly constructed trail in dense vegetation (mountain laurel or rhododendron)
- Along sharp ridgelines

Blaze More

- At turns, both on and off roads
- Along obscure footpath
- In open forests
- In open areas (balds, boulder fields, alpine zones)

Placement

Paint blazes on trees that will easily be seen by trail users. Look down the trail to find a tree that will catch hikers' attention in all seasons. If the tree is far enough away, and within one to three feet of the right side of the footpath, you've found your next blaze tree. Try to make sure that leafy summer growth or branches weighted with snow or rain will not later hide the blaze. Clear any interfering growth with lopping shears or hand pruners, if permitted by the state park or forest district manager.

Bear in mind a couple of other pointers: When you are choosing a blaze tree, remember that one wellplaced blaze is better than several that are poorly placed or partly hidden. Most importantly, avoid defacing trees and rocks that form distinctive and pleasing elements of the scenery.



Surface Preparation

Apply paint to as smooth and dry a surface as possible, preferably during fair weather above 50° Fahrenheit.

On trees with thick, rough bark, such as oak and ash, smooth the surface by scraping with firm strokes of a hardwood floor scraper, also known as a paint and varnish scraper. Never cut through the bark and into the cambium; such a wound will cause the tree to bleed, the blaze to run, and injury to the tree.

On other trees, smooth the bark by simply rubbing with a wad of steel wool, a nylon dish pad, or a canvas-gloved hand to remove dirt, lichen, and loose bark. Only scrape if absolutely necessary. If you scrape conifers, such as white pine or balsam fir, they will bleed. White birch and black cherry will fray. Beech and red maple already have smooth bark that you can paint without scraping. On rocks, minimally clear lichens, moss and other debris with a stiff wire brush. The surface must be clean and dry or the paint will not adhere.

Painting Technique

Before you head out on the trail, take a moment to consider which of several blazing techniques to use. Some people prefer using a stencil. They apply the paint with a brush or spray can. Sometimes, they trace the blazes outline through the stencil with a felt-tip marker and then fill in the rectangle with a paint brush. Others use a two-by six stamp, made from a sponge, and simply press the blaze to a tree (spare sponges may be needed). Still others simply use a straightedge. But most people blaze freehand, gauging the size of each blaze with a cardboard template.

The object, in any case, is to master a technique that you can perform neatly and consistently, leaving standard size blazes. Always avoid blemishing trees or rocks with pudgy blotches or swollen, oversized rectangles. Be careful to avoid dripping paint on trees, rocks, and leaves.

You'll save yourself a lot of trouble if you avoid blazing light-colored trees, such as white birch, light gray birch, or young poplar. Where you must blaze such a tree, paint the blaze as usual, and then frame it with a narrow line of natural-colored dark paint, to make the blaze stand out.

To ensure your blazes are durable, always stir the paint vigorously for ten minutes at the start of the day. Whip all pigment on the bottom of the can into suspension. Along the trail, remix the paint regularly.

Even when you prepare the paint and the surface properly, you'll have to repaint blazes on a regular schedule. Tree growth splits blazes, dirt dims them, animals scratch them, and trees themselves fall. In most circumstances, a blaze will last only three to five years, depending on the surface, type of paint, and weather. For example, on black cherry trees, paint usually drops off in a year or two.



Markings and Blazes

Photo Credits: PA DCNR



Plan to renew blazes every two years. For blazes that are still in good condition, repaint after scraping the surface lightly to remove paint flakes and dust. For faded, widened, or split blazes, repaint after scraping the surface as if for a new blaze. Paint over any part of the old blaze still showing with neutralizing paint as close to the color of the surrounding surface as possible. Use brown paint in conifer forests, light grayish-green in hardwoods.

Blaze Obliteration

Sometimes you'll have to eliminate blazes because they are sloppy, too frequent, and/or in the wrong places, or more often, because they may threaten to lead hikers down an old trail following a trail relocation. In the later situation, you should obliterate all former blazes, end to end.

To obliterate blazes, scrape off as much of the old paint as possible, using care not to damage the tree. Lightly cover any remaining paint with neutralizing paint. Use mixes of brown, green and gray to match the background. On rocks, apply the paint carefully, and sparingly; otherwise, it may merely make a conspicuous mark of another color. Spray paint makes good neutralizer because it can be layered and feathered to obscure the old blaze.

Equipment and Supplies

Use a scraper with a 1 1/2 inch blade and a six-inch handle. Buy high-quality paint, gloss exterior house paint, or boundary-marking ink. Latex paints are easy to apply, thin with water, dry quickly, and are less harmful to the environment. Oils are thicker, dry slower, and require thinner for clean-up. If you choose oils, buy the brand with the most pigment (white) compared to vehicle (oil): 65 percent pigment is most durable. Boundary ink, which comes thick and dries quickly, can be specially ordered.

Use a plastic squeeze bottle to apply just enough paint to the brush for each blaze. The bottles, old catsup and mayonnaise containers work well, keep the paint fresh and cut down on spills and drips. Carry only a small can of neutralizing paint, sufficient for a day of blazing. To match different tree species, bring several colors to mix in the field.

Suggested Paint Colors

The following list refers to the paint colors described in this document. This list provides uniformity when selecting paint colors to use while blazing trails. Each color includes the Pantone Color ID Number so that consistency is maintained when purchasing paint from various manufacturers (no need to purchase "Pantone" brand paint). High quality enamel or latex paint is recommended (enamel may damage thin barked trees).



The paint colors are:

- Red PMS 485 2X
- Orange PMS 165 2X
- Yellow PMS 102
- Blue PMS 300
- White
- Brown PMS 161

Bring several one-inch brushes for the white paint and one two-inch brush for the neutralizing paint. Note that nylon bristles work poorly in oil-based paint. Two small cans with plastic tops work well as receptacles for the brushes. Cut a hole in the plastic top for the brush handle for a secure method of carrying paint.

State Park Trail Systems

In an effort to retain quality trail network systems in State Parks that have served visitors' trail experience expectations for years, the following exceptions to the DCNR standard trail marking guidelines will be considered at State Parks. The Bureau Director or their designee must approve all exceptions.

Every effort should be made to follow the guidelines without exceptions.

One of the main reasons for the trail marking guidelines is to make our Department's trail systems more synonymous across parks and forestry so that our visitors have a more consistent trail experience and expectation from one facility to another. Therefore, no exceptions will be allowed for trails that cross park and forest boundaries. Rather, these trails should have the highest priority to be brought in-line with the marking guidelines.

Exceptions may not impact forestry trails or their compliance with these guidelines. In addition, the end result of an approved exception should be to enhance the visitor's trail use experience.

State Park High Density Network of Trails

In the instance where a park has established a trail marking system using multiple colors to aid visitors' trail experience by making it easier to navigate a high density network of trails within a park, an exception to the colors listed in relation to particular trail uses in the guidelines will be allowed. In order to qualify for this exception, the park must submit its mini-map and an explanation of the trail system and the need to use multiple colors to their supervisor for review. Upon region approval, Region will submit to central office/RMPD for approval, and finally it will be forwarded to the director for approval. Specifically, the only exception is for the color of the blazing as to use and type of trail. The trail will still need to be marked appropriately according to the guidelines for all other aspects. For example, it still needs



Reinforced Fiberglass Trail Marker

Photo Credit: PA DCNR



proper trailhead use signage, and the blazing still needs to be the correct size and shape.

Other blaze color exception will be considered for trails with obvious paths such as the towpath at Delaware Canal State Park or a paved lake loop bicycle trail. Trails that are currently not blazed and where blazing would have no benefit to the trail user may be considered for an exception for the trail to remain without blazes. Trails that follow boundaries between Forestry and State Park lands may also be considered for an exception.

Construction Management

A construction management team is essential for a successful trail building project. This team may consist of an individual or a team whose skills are specifically tailored to meet the specific inspection needs of a project. Additionally, the team should have an individual with strong skills in field soil identification or soil preparation and compaction. This is especially important if the construction plans for a trail are not overly detailed.

The construction management teams represents the agency responsible for the trail and serves as a liaison between the design team, the contractor, and/or volunteers. The construction manager is responsible to ensure that the materials, procedures, and workmanship conform to appropriate standards and specifications. The construction manager should also ensure that the contractor or volunteers maintain a clean, safe work environment.

Erosion Control

User-Caused Erosion Problems and How to Solve Them

Boots, Hoofs, and Tires

- Problem: The trail is too steep and exceeds a maximum sustainable grade.
- Solution: Maximum sustainable grades fluctuate based on soil properties, length of the section, and types of users for each specific site. A designer may choose to reroute a trail to include sustainable grades or armor the trail to protect against user-caused erosion.
- Problem: The trail is located in a wet area.
- Solution: If a trail is located in low-lying, muddy terrain it may benefit from a reroute onto a drier hillside. If a trail is built on the sideslope but still holds water, a knick or grade reversal may need to be incorporated to enhance drainage. In areas with high water tables, build raised trail treads to avoid persistently wet trail treads.



Trail Widening, Trail Braiding, Tread Creep

- Problem: Obstacles are forcing users off the trail.
- Solution: If a trail is located on flat land, nothing can stop users from choosing alternative routes. The best way to keep users from straying off the path is using sideslopes and varying terrain. Consider removing vegetation or obstacles that may be causing users to leave the center of the tread.
- Problem: The trail intersection is not well designed.
- Solution: Install choke points on either side of a trail to monitor the flow of traffic at intersections. Attempt to minimize speeds at intersections and when possible have users climb to a crossing to avoid leaving the intended travel route.

Washboard and Braking Bumps

- Problem: Abrupt flow transitions or turns force users to brake excessively in one spot.
- Solution: Slow users down before turns by creating transitions with choke points and challenging terrain.

The Development of Social Trails

- Problem: The trail bypasses a potentially positive control point.
- Solution: Reroute a trail to include lake access, rock outcropping, or a scenic overlook to keep users on the trail.
- Problem: The trail comes within view of a potentially negative control point.
- Solution: Reroute the trail to avoid sightlines to sensitive ecological areas, as well as trash dumps or campgrounds, to avoid users from straying off the path.
- Problem: Users feel that they are not getting the experience they seek.
- Solution: Know the users.

Water-Caused Erosion Problems and How to Solve Them

Eroded Trails

- Problem: The trail is on the fall line.
- Solution: The best option for a fall-line trail is to incorporate a rolling contour design and full bench construction. If this is not an option, consider the use of armoring techniques.
- Problem: The trail lacks outslope and water is not sheeting off the trail gently.
- Solution: Restore the outslope of the trail to encourage water to sheet across the trail, instead of funneling down the center. It is also



important to incorporate grade reversal as a defense against water.

- **Problem:** The trail lacks grade reversals.
Solution: If a trail has conservative grades, consider retrofitting grade reversals into its layout. If grade reversals cannot be incorporated into a trail's design, several water-diversion techniques can be used. Further information may be found in Chapter 3: Design.

Inspection/Quality Assurance

The objective of quality control is to provide a safe, functioning and lasting project, with an acceptable appearance. The trail manager and the trail designer should provide continuous inspection as necessary to ensure the field crew's performance of work meets the standards required. The degree of inspection will depend on the experience of the crew, the complexity of the job, and the specific task being accomplished.

Permit Compliance

The trail manager should have a copy of all permits that have been issued for the project. In addition to ensuring compliance, the construction manager should monitor completion of reporting requirements. The trail manager should also coordinate and facilitate inspection by permit-issuing agency personnel, such as Conservation District personnel, DEP staff, and local municipal code enforcement officers.

Erosion Control

Proper installation of temporary erosion control measures during construction not only protects the environment, but also protects new construction until permanent measures have been established. The construction manager must frequently inspect temporary control measures to ensure they are functioning properly, particularly after storm events. Conservation District personnel will periodically inspect the project site to determine if selected best management practices are effectively controlling soil erosion. Best management practices that have failed or are not effectively controlling erosion as intended, must be replaced or repaired. Alternative measures may be required.

Final Inspection

When construction is nearly complete, the trail designer, the trail manager and the trail builder should make a final inspection as applicable. The inspection party should walk the trail on foot. A "final punch list" of items to be completed or corrected shall be prepared and agreed upon by the inspection party. Upon correction of the items on the punch list, the contractor shall complete and submit "As Constructed" plans.



Management

Trail management is the key to having a successful trail system. To have a successful trail system you must implement specific and consistent management strategies.

Successful trail management plans have two components. The first component addresses the management aspects, including policies and procedures to: ensure trail access, provide a quality experience for the user, and minimize maintenance impacts. The second component addresses the maintenance aspects, including policies and procedures for maintaining the trails to: provide for user safety, access, and convenience, protect adjacent resources, and preserve trail investment.

A maintenance plan should establish a risk management inspection schedule, a routine maintenance schedule, and a preventative maintenance schedule.

Trails must be maintained in a safe and usable condition at all times. All defects in trail surface, rough edges, depressions, uneven pavement, etc., must be corrected immediately by repairing the trail to a safe and usable condition. All potential hazards must be clearly identified and marked to alert users until they can be repaired.

Failure to properly manage and maintain a trail may result in an unsafe condition that has the potential to become a hazard to the trail users, and a liability to the trail manager.

Sometimes questions arise whether seldom used stretches of trail are worth the time and effort required to maintain them. However, if the trail is not maintained, why should anyone use it? All trails should be maintained in the best condition possible, well maintained trail is self-perpetuating.

In this Chapter we discuss:

- Management Structures
- Trail Management Objectives
- Trail Assessments
- Management Plans
- Trail Maintenance Plans
- Education
- Risk Management

Planning for trail management and maintenance of a trail begins during the planning stages, continues through trail design, and is implemented immediately as a trail is constructed and placed into service. Risk management is a continuous process that must be evaluated and refined on an on-going basis.



PEC Trail Care Crew
Black Moshannon Division Moshannon State Forest

Photo Credit: PA DCNR



Management Structures

In many instances an existing agency or organization will be responsible for managing a trail or trail system. However, in instances where there is not, there are many options to consider when exploring the appropriate management structure for a particular trail. If your trail extends through multiple jurisdictions and/or multiple entities are responsible for its management, a well defined management structure is necessary and important to the success of the trail.

Before assuming that you need to create a new organization or agency to manage your trail, the first step is to determine whether an existing organization or agency is capable and willing to manage your trail. To accomplish this, identify potential partners and discuss with them your trail management needs and expectations.

The framework identified capabilities within agencies and organizations can vary greatly depending on the regulatory, agency, and organizational requirements governing a particular structure. Therefore, it is important to define the capabilities needed to manage your trail. In other words, it is imperative to compare the management needs with the capabilities of existing agencies and organizations to determine if they have the ability and capacity to meet your needs.

Potential Management Needs

1. Organizational and financial administration
2. Organizational sustainability
3. Financial sustainability
4. Administration of volunteers and ability to build and retain pools of volunteers
5. Ability to secure federal, state, and or municipal funding
6. Ability to secure foundation funding
7. Geographic jurisdiction or potential area of jurisdiction
8. Project planning and design
9. Partnership and relationship building
10. Land acquisition, access easement, and rights-of-way
11. Negotiation/equitable ownership
12. Construction funding
13. Project mobilization and implementation
14. Other resources receipt and management (e.g. donated materials or equipment with operator)
15. Maintenance capabilities/needs or arrangements for maintenance
16. Insurance or insurability/risk management
17. Operations/maintenance funding
18. Accountability to public interest/fiduciary responsibility to public trust, and
19. Public relations and marketing ability



Your particular trail may have additional management needs. It is important that you identify them early in the process, and that you discuss all needs with potential partners who may assist in managing the trail.

If an agency or organization cannot be identified that can manage your trail, then you must determine if there is potential to establish an organization to provide your management needs. There are many types of management structures, each type providing opportunities and limitations regarding their ability to manage trails. Potential management structures to consider and evaluate include:

- Multi-Municipal Authority - as owner/operator
- Multi-Municipal Authority - as owner/ with contract operator(s)
- Regional Trail Corporation – 501c3 formed as per Intergovernmental Cooperation Agreement between corporation and original trail owners or sponsors or municipalities or counties
- Multi-municipality Authority – as owner/operator
- Private Community Board of Grassroots Volunteers – with no government involvement
- Private Non-Profit Supporting Charity
- Utility-Type Entity – Fee Based
- Non-Profit Greenways & Trails Partnership - as owner/with contract operators
- Non-Profit Greenways & Trails Partnership - as owner/with contract local trail chapters as operators
- Non-Profit Greenways & Trails Partnership - as supporting organization with local trail chapters as operators
- Greenways & Trails Partnership - as contract operator w/ counties as owners of segments within their jurisdiction
- County Recreation Authorities – each owns and operates/maintains their segment
- Private Company – with contracts with each county to design, develop, maintain and operate the trail
- Long-Term Lease for Trail(s)
- Municipal Authority
- Municipality Owned/Operated
- Regional Trail Commission – formed as per Intergovernmental Cooperation Agreement between commission and original trail owners, sponsors, municipalities or counties
- Regional Trail Land Trust
- Council of Governments - as owner with local trail sponsors as contract operators, and
- County – as owner of trail easements or right-of-way

Adapted from Alternative Organizational Models for Multi-County Trails as prepared for the SEDA Council of Governments by Jerry Wells, AICP



Should it be necessary for you to develop and evaluate alternatives for a new management structure for your trail, PA DCNR may be able to provide assistance through their Peer-to Peer Grant Program. Peer to Peer grants are awarded to help municipalities improve their park, recreation and conservation services through a collaborative process. Projects are accomplished through contracts with experienced park, recreation, and conservation professionals from nearby communities who will work closely with local leaders. For additional information on the PA DCNR grant program contact your regional DCNR adviser or refer to www.grants.dcnr.state.pa.us/GrantPrograms.aspx.

Trail Management Objectives

The Trail Management Objective (TMO) document synthesizes the management intention of a proposed or existing trail. This document provides a means of recording basic information for future trail planning, management, and reporting. TMOs are a prerequisite for completing an effective trail condition assessment, as well as planning future work required to comply with current standards.

After completing a TMO, trail specifications, level of difficulty, and the number and types of users can be determined. In the past, trails were maintained based on their type and level of use and not their intended designed use. By establishing TMOs for each trail system, a trail's design and management can better correspond with its intended type and level of use.

Developing Effective TMOs

Trail sponsors should review and approve TMOs before and after the trail design is completed. This ensures that objectives for the trail are consistent with the location's current and future land management practices. Furthermore it ensures consistent use between trails. TMOs should be updated if the management intent for the trail, special considerations, or other factors change.

It is essential to establish and document TMOs before you construct a trail, to ensure proper management of the trail in the future. The following instructions explain the importance of the information to be documented on the TMO worksheet:

Overall Trail Information

- **Trail Name:** Enter the official trail name, double-checking for correct spelling.
- **Trail Location:** Enter the trail's location based on the naming protocol of the trail sponsor.
- **Trail Length:** Enter the length of the trail in miles. Mileage accuracy recorded on the TMO should correspond to the method of collection: wheel, GPS, map, or unknown.



- Trail Beginning and Ending Segment: Enter the location of the trail's beginning and ending segment using coordinates or mileposts for each location.

Designed Use Objectives

- Designed Use: Enter the prescribed use for the proposed trail. The designed use is necessary to establish the standards for which the trail is designed, constructed, operated, and maintained.
- Season of Use: Enter the season(s) the trail will be open for use. Indicate the specific date(s) or frequency of seasonal closure.
- Level of Difficulty: Enter the prescribed level of difficulty for a specific trail based on its corresponding level of difficulty standards.
- System Layout: Enter the type of layout the trail will conform to.

Trail Specifications

- Maximum Grade: Enter the maximum grade of the trail. Indicate the maximum percent grade for a sustained length of tread, as well as a short length of tread along the proposed trail route.
- Minimum Clearances: Enter the minimum clearance of a trail. Indicate the width and height in feet based on a trail's design guidelines.
- Design Requirements: Enter the trail's proposed tread width, curve radius, and surface material. This information shall be based on the trail's suggested design guidelines.
- Managed and Prohibited Use
- Managed Use: Enter the mode of travel that is appropriate for a trail, based on its design and management.
- Prohibited Use: Enter any use or mode of travel prohibited by an official legal order or the trail sponsor.

Trail Characteristics

- Sensitive Areas: Enter all ecologically or hydrologically sensitive areas within the proposed trail corridor.
- Destinations: Enter all destinations accessible to or visible from the proposed trail route.
- Structures: Enter all structures currently located along the trail route or that will need to be constructed.

Trailheads and Amenities

- Trailheads and Access Points: Enter the name and location of all trailheads providing access to the proposed trail route.
- Amenities: Enter all current or proposed amenities located within a trailhead or along the proposed trail route.



Maintenance Requirements

- Routine Maintenance Schedule: Identify the anticipated frequency for completing routine maintenance tasks within this section. Use bullets to define which maintenance tasks should occur on a weekly and monthly basis.
- Special Considerations and Notes: Identify any additional considerations that trail managers, design, construction, or maintenance personnel should be aware of. Provide details or references for corresponding direction.



Trail Management Objectives

Trail Name _____ Begin Segment _____ identified by mileposts ☐
 Location _____ End Segment _____ or coordinates ☐
 Total Trail Length _____ Segment Length _____

page 1 of 2

Designed Use ☐ Hike / Pedestrian ☐ Bike ☐ Mountain Bike ☐ Equestrian
☐ Cross Country Ski ☐ Snowshoe

Season of Use ☐ Year Round ☐ Winter ☐ Spring ☐ Summer ☐ Fall

Level of Difficulty ☐ Easiest _____% ☐ More Difficult _____% ☐ Most Difficult _____%

System Layout ☐ Linear ☐ Single Loop ☐ Stacked Loop ☐ Multiple Loop
☐ Spoked Wheel ☐ Primary & Secondary Loop ☐ Maze

Trail Specifications

Maximum Grade Sustained %
 Short %
Minimum Clearances Width feet
 Height feet

Tread Specifications

Tread Surface ☐ Stable ☐ Firm
 Tread Material ☐ Native In-Place ☐ Native Borrow
☐ Manufactured, _____ ☐ Stabilized Soil
☐ Stone ☐ Pavement

Design Requirements

Tread Width feet
 Target Cross Slope %
 Minimum Curve Radius feet

Accessibility

☐ Complies with ADA – Draft final Accessibility
 Guidelines for Outdoor Developed Areas

Managed Use ☐ Shared Use Path / Rail Trail ☐ Multi-Use / Shared Use Trail
☐ Hike / Pedestrian ☐ Bicycle ☐ Equestrian
☐ Cross Country Ski ☐ Snowshoe

Prohibited Use ☐ All Motorized Use ☐ ATV ☐ Snowmobile
☐ Hike / Pedestrian ☐ Bike ☐ Mountain Bike ☐ Equestrian
☐ Cross Country Ski ☐ Snowshoe

Sensitive Areas ☐ Wetlands ☐ Seeps ☐ Habitat ☐ Species
☐ Cultural / Historical ☐ EV / HQ Streams ☐ Adjacent Land Use
☐ Other, describe: _____

Destinations ☐ Parks ☐ Connecting Trails ☐ Wildlife Areas ☐ Control Points
☐ Stream Crossings ☐ Trailheads ☐ Vistas ☐ Scenic Areas
☐ Others, describe: _____

Structures ☐ Culverts ☐ Bridges ☐ Rock Walls
☐ Crib / Gabion Walls ☐ Other Walls
☐ Other Structures, describe: _____

Notes

Prepared by _____ Date _____



Trail Management Objectives

Trail Name _____ Begin Segment _____ identified by mileposts ☐
 Location _____ End Segment _____ or coordinates ☐
 Total Trail Length _____ Segment Length _____

page 2 of 2

Trailheads & Access Points Name _____ Location _____ identified by milepost ☐
 or coordinates ☐

Amenities

☐ Parking, no. of spaces _____	☐ Restroom	☐ Potable Water
☐ Trash / Recycling Containers	☐ Picnic Shelter	☐ Kiosk
☐ Message Board	☐ Primitive Camping	☐ Warming Hut
☐ Bike Rack	☐ Air Station	☐ High Line
☐ Hitch Rail	☐ Corral	☐ Horse Stall
☐ Watering Trough	☐ Fire Ring	☐ Picnic Table
☐ Others, describe: _____		

Maintenance Requirements: Identify the anticipated frequency for completing routine maintenance tasks

Routine Maintenance Schedule												
Maintenance Activity	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
EXAMPLE- Mowing			•	••••	••••	••••	••••	••••	••••	•••		
Mowing												
Trimming												
Trash												
Pruning												
Weeding												
Invasive Removal												
Brush Hog												
Signs - inspect/repair/replace												
Blazes - inspect/re-paint												
Fence - inspect/repair/replace												
Culverts - inspect/repair/replace												
Storm Drains - inspect/repair/replace												
Gates - inspect/repair/replace												
Bridge - inspect/repair/replace												
Maintain Dips												
Grade Ditches												
Trail Grooming												
Trail Surfacing												
Landscaping												
Storm Damage												
Vandalism												
Repair Washouts												

Special Considerations	Notes

Prepared by _____

Date _____



Trail Assessments

Trail Assessment Forms (TAF) provide a means of recording basic information for current and future trail planning, design, construction, and maintenance. TAFs document current trail conditions and help identify subsequent work required to satisfy current trail standards.

After completing a TAF, a trail's intended use, overall condition, and necessary maintenance can be determined. Review of past TAFs will help identify areas or sections of trails that require or have experienced maintenance or improvements. In areas that require more maintenance than expected or are too costly to maintain, trail redesign or improvements may be considered.

Trail sponsors are responsible for reviewing TAFs and ensuring that any required maintenance on a trail or trailhead be addressed. A complete inspection of a trail can be performed in conjunction with scheduled maintenance by the maintenance crew or as a separate action to determine the need for additional maintenance. Trail assessment forms should be updated if the management intent for the trail, special considerations, or other factors change.

The instructions below explain the importance of each field on the TAF worksheet:

Overall Trail Information

- **Trail Name:** Enter the official trail name, double-checking for correct spelling.
- **Trail Location and County:** Enter the trail's location and county based on the naming protocol of the trail sponsor.
- **Trail Length:** Enter the length of the trail in miles. Mileage accuracy recorded on the TMO should correspond to the method of collection: wheel, GPS, map, or unknown.
- **Trail Beginning and Ending Segment:** Enter the location of the trail's beginning and ending trail segment. Identify whether these locations were determined by coordinates or mileposts.
- **Name of Conductor and Date:** Enter the name of the person who conducted the assessment. Identify the month, date, and year the assessment was taken.

Designed Use Objectives

- **Designed Use:** Enter the prescribed use for the proposed trail. The designed use is necessary to establish the standards for which the trail is designed, constructed, operated, and maintained.
- **Season of Use:** Enter the season(s) the trail will be open for use. Indicate the specific date(s) or frequency of seasonal closure.



- **Level of Difficulty:** Enter the prescribed level of difficulty for a specific trail based on its corresponding level of difficulty standards.

Trailhead and Access Points

- **Name:** Enter the name of all trailheads and/or access points.
- **Location:** Enter the location of each trailhead and/or access point. Indicate both the latitude and longitude of the site and whether these measurements are based on coordinates or mileposts.

Trail Tread/Surface

- **Condition:** Enter the current condition of the trail tread or surface.
- **Materials:** Enter the type of materials used to create the trail tread.
- **Average Width:** Enter the average width of the trail tread. Identify the minimum and maximum width observed along the trail.

Drainage

- **Drainage:** Enter the drainage conditions of a trail. Identify any and all drainage issues that need addressed.
- **Bridges and Culverts:** Enter the level of condition of bridges and/or culverts along the trail corridor.
- **Dips:** Enter the level of condition of dips along the trail corridor.

Road/Railroad Crossings

- **Condition:** Enter the level of condition of road and/or railroad crossings intersecting with the trail.
- **Sight Line:** Enter the level of condition of sight lines to and from road and/or railroad crossings intersecting with the trail corridor.
- **Accessible:** Enter if there are accessible road and/or railroad crossings intersecting with the trail corridor. Identify whether or not there are any exceptions.

Adjacent Land Uses

- **Adjacent Land Uses:** Enter all current land uses that are adjacent to the trail corridor.
- **Historical and Other Structures:** Enter the level of condition of all historical and/or other structure along the trail corridor.

Signage

- **Blaze/Marking Color or Style:** Enter the blaze and/or marking color and style at trailheads and along the trail corridor.
- **Overall Condition:** Enter the overall condition of blazes and/or markings at trailheads and along the trail corridor. Identify the condition of specific blazes and/or markings where applicable.
- **Replace Signs:** Enter all areas at trailheads and along the trail corridor where blazes and/or markings need to be replaced.



Trail Assessment Form

Trail Name _____ **Begin Segment** _____ identified by mileposts ☐
Location _____ **End Segment** _____ or coordinates ☐
County _____ **Total Trail Length** _____ Segment Length _____
Conducted by _____ **Date** _____

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Designed Use
☐ Hike / Pedestrian ☐ Bike ☐ Mountain Bike ☐ Equestrian
☐ Cross Country Ski ☐ Snowshoe ☐ ATV ☐ Snowmobile
☐ Others, _____

Level of Difficulty ☐ Easiest ☐ More Difficult ☐ Most Difficult

Trailheads & Access Points **Name** _____ **Location** _____ identified by milepost ☐
 Latitude N _____ or coordinates ☐
 Longitude W _____

Notes

Trail Tread / Surface **Condition** ☐ Good ☐ Fair ☐ Poor ☐ Needs Improvement

Materials _____ **Average Width** _____, min. _____, max. _____

Grade _____% **Cross Slope** _____%

Notes

Drainage ☐ Drains properly ☐ Draining onto or across trail surface
☐ Water staying on trail ☐ Needs drainage Structure

Bridges and Culverts ☐ Good ☐ Fair ☐ Poor ☐ Cleanout

Dips ☐ Good ☐ Fair ☐ Poor ☐ Cleanout

Notes

Road / Railroad Crossings

Condition ☐ Good ☐ Fair ☐ Poor ☐ Needs Improvement

Sight Lines ☐ Good ☐ Fair ☐ Need to prune ☐ Unsafe

Accessible (Note Exceptions) _____

Notes



Trail Assessment Form

Trail Name _____ *Begin Segment* _____ identified by mileposts ☐
Location _____ *End Segment* _____ or coordinates ☐
County _____ **Total Trail Length** _____ **Segment Length** _____
Conducted by _____ **Date** _____

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Adjacent Land Uses (Check all that apply)

☐ Forest ☐ Farm ☐ Residential ☐ Commercial ☐ Industrial ☐ Encroachment

Notes

Historical and Other Structures

Condition ☐ Good ☐ Fair ☐ Poor ☐ Needs Improvement
☐ Needs Replaced ☐ Needs Cleared Out

Notes

Signage (includes trailhead and reassurance markings, blazes, etc.)

Blaze / Marking Color or Style _____

Overall Condition ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Regulatory ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Wayfinding ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Interpretive ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Wooden / Routed ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Reassurance Markings ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Intersections ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Pavement Markings ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Others (fiberglass, etc.) ☐ Good ☐ Fair ☐ Lacking ☐ Needs Maintenance

Replace Signs:

Notes



Trail Assessment Form

Trail Name _____ *Begin Segment* _____ identified by mileposts ☐
Location _____ *End Segment* _____ or coordinates ☐
County _____ *Total Trail Length* _____ Segment Length _____
Conducted by _____ *Date* _____

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Other Notes and Summary

Please return this form to



Universal Trail Assessment Process

Until the recent release of the *Draft Final Accessibility Guidelines for Outdoor Developed Areas* information provided to users about trails has traditionally been very limited. Typically it consisted of a subjective rating such as easy, more difficult, or most difficult. Subjective ratings reveal very little about the actual conditions of the trail. For example, a hiker who uses a wheelchair cannot determine from an “easy” rating whether a trail is wide enough throughout to permit the wheelchair to pass other trail users. If the trail is too narrow, the user may have to turn back. This is frustrating for both the hiker using a wheelchair and any hiking companions.

Determining whether a route is accessible in a given situation is dependent on the ability and mobility of the visitor at a particular time. A person, temporally impaired by a broken leg and confined to a hospital style wheelchair, visiting a trail alone most likely would not be able to access a stable and firm trail with protruding rocks and roots. However, if the scenario were changed and this person were accompanied by an able adult, or by a power-driven off road wheelchair, the trail may be accessible.

The Universal Trail Assessment Process (UTAP) is an inventory tool that records accessibility and maintenance information on a trail. Further, it is designed to provide visitors to the trail with specific information that allows the visitor to determine at that point in time, depending on their capability at the moment, whether a particular trail will be accessible to them.

The UTAP was designed to meet the information needs of both trail users and land management agencies. The assessment process was developed through the collaboration of Federal, State, and local land management agencies, as well as trail organizations.

While the *Draft Final Accessibility Guidelines for Outdoor Developed Areas* requires the primary accessibility information to be presented, UTAP goes further by requiring not only the some objective trail data:

- Trail length
- Maximum and average cross slopes
- Maximum and average grades
- Surface type and firmness
- Minimum clear width
- Average tread width



UTAP also includes collecting and recording primary maintenance information about a trail, including:

- Tread condition
- Obstacle locations and magnitude
- Vegetation within the trail corridor
- Condition of drainage structures
- Presence of downed trees
- Washed out sections of the trail
- Condition of signage
- Condition of benches, tables, and other amenities, as well as whether or not these are accessible designs

Benefits of Assessment

Assessing the content and condition of existing trails is the first step towards providing access for all users. One of the greatest obstacles people face when trying to select a trail to hike is the lack of objective information regarding trail access. With enough specific information, people will be better able to travel independently with confidence in their ability to reach their destination safely.

Objective data obtained from trail assessments enables trail managers to create maps, signage, and other informational guides that will assist users in selecting the best routes for travel. Conducting trail assessments also allows trail managers to inventory trail conditions and plan for trail projects including:

- Determining if the trail meets original design specifications and guidelines
- Prioritizing trail maintenance projects
- Budgeting for trail projects
- Identifying portions of trails that can be made more accessible
- Developing maintenance schedules
- Quantifying the extent of repair work required
- Cataloging feature and maintenance information
- Sharing trail data and project plans with disability focus groups
- Creating objective trail information for users, such as signing, maps, and websites

The UTAP should be conducted by a certified Trail Assessment Coordinator. UTAP Coordinator Workshops are offered to teach the assessment process and to train people to become Trail Assessment Coordinators. The workshop involves two days of training that is conducted through classroom and on-trail learning experiences. Upon completion of the workshop, participants have the knowledge and experience that is needed to conduct assessments.



Data Collection and Processing

The UTAP relies on inexpensive and easy-to use tools to reduce costs and to simplify the process. An overview of the tools used to conduct the UTAP is provided below. For more specific information about how measurements are collected, it is recommended that an individual attend a UTAP Coordinator Workshop.

- A rolatape, which is a wheeled measurement device, is rolled down the center of the trail to measure trail length.
- Stations are marked to provide a location for trail measurements to be taken.
- A hand-held clinometer is used to measure running grade between stations.
- A digital inclinometer (level) is used to measure maximum grades, cross slopes, and maximum cross slopes. The inclinometer is 44 inches long, and it provides measurements over the same distance spanned by the length and width of the wheelbase of an average wheelchair, walker, or crutch span.
- A tape measure is used to record:
 1. Trail width at each station
 2. Height of the clear vertical passage when it is less than the specified design for the trail (vertical clearance)
 3. Distance and magnitude of trail segments narrower than specified for the trail (minimum clear width)
 4. Dimensions of features, obstacles, and protruding objects, that might obstruct passage or require maintenance or repair.

Trail surfaces are subjectively classified as paved, hard, firm, soft, or very soft. The trail data collected during the UTAP can be processed and summarized as Trail Access Information (TAI) into a software program called TrailWare™. The program creates tables listing the most severe measurements of each trail characteristic, calculates average trail grade, cross slope and width, tabulates surface type percentages, and produces a trail grade profile. In addition, the program provides a list of features and potential maintenance items that can be sorted and used to prioritize work projects. The program also generates reports, complete with graphics that can be used as trail signage or information sheets.



Presenting Trail Access Information

Trail Access Information (TAI) on signs, maps, and other trail guide products provides potential users with the information needed to determine which trails can best meet their desired experiences, interests, and abilities. A system of symbols and trail signage layouts has been developed to convey TAI in attractive and easy-to-use formats. Providing the information in multiple formats, such as large print or audible formats, benefits people of all abilities. The following formats are examples of how TAI can be disseminated:

- Trailhead signage - A trailhead map containing text, grade profiles with surface information, a top view map with symbols showing the location of major obstacles, and TAI.
- Trail Access Information strip - A trail map summarizing critical TAI with symbols and measurement numbers formatted as a slim strip that can be attached to trail posts and located at trailheads or trail intersections.
- Trail Information Sheets - An informational sheet that can be provided at the trailhead or visitor center summarizing TAI with symbols and measurement numbers, a trail grade profile, description of the trail, and location of trailhead.
- Audio descriptions - A short audible narrative with descriptions of trail conditions and details about the trail environment. This format may benefit individuals who have vision impairments or who have limitations reading in English.
- Pocket map - A trail map featuring trail descriptions, TAI, and a grade profile that folds up to fit into a pocket.
- Guidebook - A trail manual containing TAI, interpretive information, scenic photographs, directions to the trailhead, and other information about trails within a given recreational area.
- Computerized Visitor kiosk - An interactive accessible computer display at a visitor center providing trail selection tools, TAI, and visual and audio descriptions of images at selected destinations. Guidelines recommended for making kiosks accessible to people with mobility and vision impairments are available through the U.S. Access Board.
- Website - www.trailexplorer.org encourages TAI for trails to be posted on a comprehensive website for trails throughout the United States.





Management Plans

Risk management is the identification, assessment, and prioritization of risks followed by coordinated and economical application of resources to minimize, monitor, and control the probability and/or impact of unfortunate events or to maximize the realization of opportunities.

To effectively manage the potential risk associated with a trail, trail organizations/managers should adopt a trail management plan before a trail is opened.

A trail management plan establishes policies and procedures for:

- Trail Rules
- User Safety and Security
- Natural and Cultural Resource Management
- Physical Corridor
- Trail Events
- Conflicting and Completing Uses
- Maintenance and Operations Procedures
- Regarding the Use of Volunteers
- Trail Closures
- Use of Other Powered-Driven Mobility Devices

The management plan must be adapted to reflect the requirements of the uses being managed on your trail.

The Regional Trail Corporation (RTC), the agency that oversees the Great Allegheny Passage Trail, extending from Pittsburgh to Washington D.C., developed written policies for their shared use trails. Their policies serve as a good model and can be adapted for most types of non-motorized trails.

The RTC policies were reviewed, adapted, and are presented herein as model policies. These model policies serve as a point of reference for trail sponsors needing to develop and adopt such policies. Carefully review and adapt these policies to influence a trail's sponsor guiding policies.



Trail Rules

Generally trail rules begin with identifying the permitted uses for the trail. Often the rules also identify uses that are prohibited along the trail. The following rules have been established by the RTC for the Great Allegheny Passage:

1. No motorized vehicles except wheelchairs.
2. Bicyclists and horse riders should wear helmets. Pennsylvania State Law mandates that children under 12 years old wear safety-approved helmets.
3. Keep right, except to pass.
4. Warn before passing sound your bell or horn and announce audibly “Passing on your left.”
5. Bicyclists yield to walkers/hikers.
6. Equestrians are allowed on specified sections of the trail. Please keep horses off the improved hiker/biker surface. Stay on the grass.
7. Trail open daily, dawn to dusk. No overnight use, excepted in authorized areas. Reservations are required for groups of 10 or more.
8. Stay on trail.
9. No fires.
10. Keep pets on a short leash, under control, and pet waste off the trail.
11. Do not discharge or carry loaded firearms on or near the trail.
12. Please respect private property and stay on the trail.
13. Dumping and littering are prohibited. If you see litter, please pick it up.
14. Permits are required for any special group event of twenty or more.



Dr. J Freeride Trail Rules

Photo Credit: Pashek Associates

Policies Concerning User Safety and Security

The most important consideration along your trail is the safety of the trail users. Therefore, proactive planning, coordination, and response policies are necessary.

Cellular Phones and Cellular Service

On websites, at trailheads, and on all printed trail maps, identify whether cellular phone service is available along your trail corridor, and provide the telephone numbers for emergency responders. The RTC’s policy is as follows:

Cellular phones have become a staple for those traveling. Cell phones provide contact with emergency service personnel during times of need. However, given the remote locations of many trails, one cannot always assume cellular service will be available when needed. Therefore, to promote awareness and eliminate reliance on service that may not be available; place cell service notices at trailheads and access points. Notices should indicate whether cell service is



available, and if so what carriers can access it along a particular corridor, further, this notice should indicate where service is ‘spotty’ or where access service is unreliable.

Police, Fire, and Ambulance Coordination

Often a trail will pass through a number of municipalities each of which having jurisdiction over the corridor. As a result, potential conflicts for conflict in providing emergency services exists. The state of Pennsylvania has created the Pennsylvania Emergency Management Agency (PEMA) to facilitate the provision of emergency services in a coordinated fashion. The PEMA policies and procedures govern emergency response procedures, and responsibilities along the trail corridor.

In many jurisdictions, first response to medical emergencies on the trail is calling the designated first responder number. In many locations, dialing 911 can initiate this. It is critical to post the phone number(s) for the appropriate first responders at trailheads and trail access locations. Local volunteer trail monitors who have access to first responder radio networks, or can contact them by cell phone - assuming cellular service is available - can provide faster response.

Law Enforcement

Each state agency is empowered to enforce state conservation laws on its lands and, in certain cases, on other lands throughout Pennsylvania. In addition, state conservation agencies can sometimes enforce the state’s criminal and vehicle code on agency-owned conservation lands. Finally, state and municipal law enforcement agencies are empowered to enforce general laws of the Commonwealth of Pennsylvania on all lands.

The following is a general description of agency responsibilities. These can be divided into the following categories:

- General responses to criminal actions that require law enforcement officials
- General law enforcement actions on state lands

General Responses to Criminal Actions

All responses to criminal actions are handled by either the State Police or by a local law enforcement agency. Where a full-time local law enforcement agency is available, that agency will serve as the primary law enforcement agency and the State Police will function as backup. Where a part-time law enforcement agency exists, that agency will serve as the primary law enforcement agency during its business hours; State Police will fill in during off-hours. Where no local agency is available, the State Police will provide necessary police service on a full-time basis. Emergency law enforcement response can typically be activated by dialing 911.



Law Enforcement on State–Owned Lands

State agencies have powers to enforce conservation codes and some criminal code provisions for violations that occur on state conservation lands. Some overlap exists among agencies having powers over conservation codes and public use of state land areas. A description of each agency's powers is provided below.

- Forest officers of the Department of Conservation and Natural Resources Bureau of Forestry enforce Chapters 51 and 52 of Department rules and regulations governing state forests and state forest picnic areas. These regulate activities in Commonwealth forest lands, including boating, motorized vehicle use, fuel wood harvesting, camping, hunting and littering. In addition, state forest officers are empowered to enforce the snowmobile and all-terrain vehicle law, fish and game laws, and certain criminal and vehicle code provisions on state forest lands. The Bureau of Forestry is also charged with protecting state endangered plant species from collecting and harvesting.
- Rangers of the Department of Conservation and Natural Resources - Bureau of State Parks enforce Chapter 31 of the Department rules and regulations pertaining to state recreation areas. The chapter regulates activities within state recreation areas, including traffic and parking, trespassing, possession of firearms, littering, hunting, camping and a wide range of other uses. Park rangers are also empowered to enforce snowmobile and all-terrain vehicle laws, fish and game laws, and vehicle and criminal codes within Commonwealth recreation areas.

Emergency Medical Services (EMS)

Emergency medical service response can typically be activated by dialing 911. Post EMS response contact information at trailheads and trail access points.

Search and Rescue

Initiate all search and rescue efforts by calling the designated first responder number. In many locations, dialing 911 can initiate this. Receive immediate help by directly contacting state, county, and municipal agencies. Local and county emergency management coordination services provide search and rescue service on private lands. In case of more than one municipality being involved, the county having jurisdiction will provide coordination of search and rescue efforts.

Wildfire Control

Typically, County 911 dispatch centers have jurisdiction as the wildfire control agency, and coordinates efforts, regardless of land ownership. Most rural fire companies maintain their own trained volunteers and equipment, to fight forest and grass fires, and maintain a communication network.



Other Emergencies

With other types of emergencies that may arise on the trail, the local emergency response network manages responses.

Emergency Response Procedures

Trail managers should establish an emergency response policy that addresses whether motorized response is permitted along the trail corridor. The RTC policy is as follows:

On non-motorized trails the use of motorized vehicles is typically discouraged unless necessary for the preservation of life. The use of mountain bike response teams (MBRT) can be a suitable alternative that can be encouraged for initial response to an emergency, unless the emergency is life threatening i.e. person not breathing. Even at this, dispatch the team automatically to facilitate rapid access to the patient without putting other trail users in danger.

When the MBRT encounters the emergency, they could begin resuscitative measures, and/or locate any necessary equipment. This equipment will proceed at a low rate of speed to prevent accidents and negative impact on the trail and its users. The team shall sound warnings to other trail users about the coming ambulance or other emergency equipment as they respond to the emergency. Mountain bikes are able to reach relatively safe speeds of up to 25 to 30 mph. This would be hazardous in an ambulance due to the size of the vehicle. In addition, consider the use of ATV's as an alternative unless it is owned and operated by a local EMS agency.

Mountain bike response teams are successfully being used along the Allegheny Passage. Where there are local law enforcement agencies, their members should be trained to patrol and respond on bicycles. Bicycles can be transported to an access area in an ambulance or fire truck which contributes to the mountain bike response team's flexibility.

The common practice of fire and ambulance personnel responding to the scene of an emergency in their private vehicles must be limited to trail access points to meet up with necessary equipment. This prevents congestion on the trail, as well as further reduces the risks to trail users and the environment.

Milepost Signs for Emergency Response

Maintain milepost and 1/2 mile or 1/4 mile post markers to allow users to identify their location along the trail. For example, if a person calling in an emergency, states that the victim is near milepost 8.5, response personnel can easily identify an appropriate vehicular staging area to allow for the



quickest access to the victim. This prevents response personnel from having to cruise the trail looking for the victim.

River, Stream, and Pond Related Emergencies

Should a report of a possible drowning be received, the mountain bike response team (MBRT) should be dispatched immediately. If the location of the victim(s) is unknown, the mountain bike response team will be able to scan the stream's banks much faster and more efficiently than could be done on foot or in a motorized vehicle. When the victim is located, resuscitation can be initiated, and the necessary equipment summoned, thereby reducing wasted efforts and negative impact on the emergency and the environment.

Fire

In the event of a fire or possible fire on the trail, fire chiefs should verify the nature and extent of the fire prior to moving equipment onto the trail. Saving a few trees will not mean as much if the trail surface is left unusable by large firefighting equipment. MBRT could be used here as well, to verify the nature of the burn.

Campfires and open burning should be discouraged along the trail through local ordinances. This restriction should be visible on trail maps and brochures, and on signs at trailheads and trail access points.

Emergencies of Rainy Days

The possibility of an emergency on a rainy day is slim due to fewer users being on the trail, but nonetheless is still present. The use of the MBRT can be especially important at this time. The team must be cautious, however, because trails are typically more receptive to damage when wet and saturated.

Winter Emergencies

In the event a skier or hiker should require medical aid, it may be necessary to gain access to the victim by the use of snowmobiles. Trail conditions may prohibit the moving of the MBRT, ambulances, or other heavy equipment onto the trail. Therefore, local response personnel should identify partners who can provide snowmobiles should the need arise. Registration of these snowmobiles with trail partner organizations will help to prevent recreational use and abuse by the snowmobile owners. Snowmobiles should be staffed by EMS personnel, as either the operator or passenger. A sled for patient transport should also be on hand. Any use on the trail not associated with an emergency would be considered recreational and the appropriate penalties must be applied. Snowmobile operators must be aware of the possibility that other skiers or hikers may be present on the trail, and extreme caution should be taken when operating a snowmobile on the trail during response efforts.



Policies Concerning Natural and Cultural Resource Management

The resource is not only the trail itself but also the context in which it is used. The trail is a functional, aesthetic resource; protection measures should therefore be designed and undertaken with these considerations in mind.

Adjacent Land Use – Zoning

The issue of adjacent land use is central to protection of the trail as a resource. Since the trail is a relatively narrow strip of land, the quality of the trail users' experience is largely a function of how the land adjacent to the right of way is developed. This is one critical reason for viewing the rail trail corridor as a whole.

Overall direction and policy formulation should therefore be derived from a broad-based planning process. The resulting development guidelines should be formalized in a model ordinance, which can in turn be used by local municipalities to modify their land use plans and zoning ordinances.

Visual Management and Open Areas

Open areas and scenic vistas are desirable to maintain at certain points along the trail to ensure variety and an interesting trail experience. Visual management and open area maintenance should be directed primarily by the local municipalities having jurisdiction, and coordinated by the corresponding County Planning Department.

Timber and Vegetation Management

Timber management is carried out for safety and economic reasons. Since the trail is intended as a scenic recreational corridor, it is unlikely that timber harvesting within the trail proper will be permitted. More likely is the fact adjacent lands could be used for timber harvesting.

Weed control is another issue. All possible steps should be taken to ensure that resulting environmental and aesthetic impacts on the corridor are adequately considered and mitigated. Environmentally friendly alternatives to herbicides and pesticides must be explored and exhausted before resorting to chemical measures to control weeds and pests. Fortunately, the corridor is immediate to the resources available at the Pennsylvania State University, and, can serve as an opportunity for research and design of alternative control methods where none currently exist.



Threatened and Endangered Species

Threatened and endangered species are protected under state and federal laws. Trail development, use, and interpretation shall take into consideration these protections, and work within the state and federal regulations to protect these resources.

Wildlife

The trail corridor can provide narrow but continuous habitat for a wide variety of wildlife. This may benefit both wildlife and trail users alike, since viewing wildlife is one of the many reasons people use trails. Protection of wildlife is the responsibility of the Pennsylvania Game Commission, including game, non-game, endangered and non-endangered wildlife species. Should wildlife disturbance be a concern along your corridor, contact your regional Pennsylvania Game Commission office.

Historical and Cultural Resources

Typically, historical and cultural features are located along trail corridors. These features add to and enrich the trail experience. Strive to protect and enhance significant historic and archeological resources to the maximum extent possible.

When maintenance activities might negatively affect or disturb cultural or archeological resources, contact the Pennsylvania Historic and Museum Commission by completing a Cultural Resources Notification as described in The Permitting Process, located in Chapter 1.

When a project is planned that may disturb a historic structure or area, it is necessary to contact the PHMC to determine if the structure or area is of significant historical value. When a site is historically significant, the PHMC may require additional research, possibly including archeological surveys of the area.

Wetlands

An inventory of wetlands and jurisdictional determination is often required during the design of a trail. The U.S. Army Corps of Engineers and the Pennsylvania Department of Environmental Protection regulate impacts to jurisdictional wetlands. Wetlands should be viewed as an essential component of the trail corridor and protected accordingly. Wetland disturbance may also be necessary while performing maintenance activities. Should you anticipate needing to disturb wetland during maintenance of the trail, follow the procedures described in The Permitting Process, located in Chapter 1.



Policies Concerning The Physical Corridor

Trailhead Access and Parking

The initial design development will include the planning and development of trailheads, trail access points, and parking facilities. Other minor access points along the trail will develop as needed over time, in full coordination with local officials and property owners along the section in question.

Consider whether to publicize local access points to avoid non-local use. Finally, there may be restricted access points, which will only be available to emergency services, law enforcement, and trail maintenance personnel.

Drinking Water and Toilets

Unless marked otherwise, consider natural sources of drinking water such as springs and streams as unsafe and unreliable. Therefore, consider using water and toilet facilities within the service area of local municipal systems where they are available. Also, consider composting waterless toilets where municipal systems are not available.

Signs

Adopt a standardized sign system consistent throughout the length of the trail during the trail's initial development. An effective sign plan includes directional signs to nearby trail services, mileage markers, regulatory and interpretive signs for historic, cultural, and environmental features. Local trail groups are often responsible to maintain signs.

Bridges, Stream Crossing, and Drainage Structures

Routine maintenance of existing structures should be performed by the trail sponsor. Any larger project requiring a State or Federal waterway permit will be jointly planned and executed by the agency owning the land and the trail sponsor.

Landscape Screening

Often, during the design and development phase of the project landscape screening is considered along portions of a trails. Decisions on a case-by-case basis made in the field enhance the trail experience and ensure privacy for adjacent landowners. Consider material and maintenance requirements of screening options so they are in harmony with their surroundings and achieve the desired effect. Further, plant any landscaping along a trail with native plant materials. Maintenance and replacement of evergreen plantings will be the task of the maintenance groups.

Connecting and Side Trails

It is important to maintain the continuity of the trail experience throughout a trail's entire length. Therefore, where appropriate, plan and develop a trail



with its regional significance in mind. The addition of side trails or spurs off the trail may occur in the future to provide access to nearby parks, historic, cultural, and environmental features. Trail sponsors typically maintain side and connecting trails with the formalized written consent of the agency owning the land.

Overnight Use/Camping Areas

The development and management of overnight facilities may require more detailed policies and regulations as time passes and would depend upon the local community environment where such facilities are located.

Develop overnight campsites for trail users following the Trail Master Plan/Feasibility Study. On long distance trails campsites should occur at intervals of approximately 10 miles.

The minimum facilities typically include a water well, a picnic table, and a waterless toilet. Youth Hostels, Bed and Breakfast facilities, and groups of commercial campgrounds with a high level of facilities usually are available in or near towns with public sewer and water utilities. It is not typically necessary to provide camping when lodging is available. Provide Wayfinding signs to direct trail users from the trail to these locations.

Policies Concerning Programming and Environmental Education Activities

Special Event and Large Group Use

When the use of a trail is likely to involve special events, such as walk-a-thons, group bicycle rides, and large group use by various organizations, regulations, permit applications, and participant releases/waivers should be used. Waivers decrease impacts on other trail users as well as adjacent communities and land uses while maximizing the educational, economic, and community-building benefits of the event. Local municipalities must be involved in forming regulations and providing subsequent enforcement of them for special events and large group use. Special events and large group uses should be permitted through the agency responsible for the day-to-day management of the trail.

Sponsors of events with 25 participants or more should be required to complete a special use permit application.

The trail's managing agency will review the application and, barring any conflicts, issue the permit along with the regulations and guidelines letter. Special uses by definition should not occur without appropriate review and control. All special uses will be reviewed and allowed or disallowed by a trail's partnership organizations. Sponsors should be required to provide additional facilities as required to meet the needs of their event's participants. This

NEED PHOTO OF GROUP RIDE

Photo Credit: XXXX



many include garbage and recyclable collection; providing temporary sanitary facilities, tents and marquees, emergency and delivery routes, etc. All major activities should be centered at a trail access point capable of hosting the special event.

Public Information and Education Program Policies

The success of a trail depends on the awareness of its existence by potential users and supporters. Therefore, it is important to ensure that all appropriate means are used for informing the public of the trail. This should be accomplished through direct efforts of state, county, and local agencies and through marketing efforts by local municipalities and commercial interest groups.

Public education programs can be developed to illuminate various aspects of the trail and adjacent areas such as history, plant and wildlife biology, ecology, and recreational workshops. The trail could become a valuable educational resource for local elementary and secondary schools, as well as local colleges and universities within the region of the trail.

Caretaker and Trail Rider Programs

These programs would serve several purposes, such as information and education for users, monitoring of use patterns, facilitating emergency responses, maintenance of facilities, and providing a presence along the trail. The nature of the trail, including its proximity to a population base with dedicated volunteers, should be used to determine if such a program would be successful for a given trail. Local municipalities can be a resource to aid organizations in recruiting volunteers to increase local community involvement.

Policies Concerning Promotion of the Trail and Trail Related Events

Established trails, and its corresponding events, should be promoted within their region. A coordinated effort for promoting the trail should begin with respective County's tourist promotion agency (TPA). TPA's receive funding for tourist promotion through a bed tax. Furthermore, TPA's can help with applying for transit oriented development (TOD) signs through PennDOT, and in applying for and securing funding for promotion and tourist development opportunities throughout the trail corridor.

An often-overlooked aspect of security along the trail is the fact that vandalism, trespassing, and criminal concerns along the trail are generally less likely to occur as more people use the trail. Vandals, trespassers, and those dumping along the trail are generally fearful of discovery. Therefore, they look for isolated areas and areas of limited traffic to reduce the potential of detection. Encouraging significant use of the trail is an automatic deterrent.



Management, operation, maintenance, and security of the trail are only important if the trail is used. Therefore, a concise effort must be made by the trail sponsor, and respective County (ies), municipalities, and trail organizations to promote the use of the trail for recreation, transportation, health and wellness, social, and historical, cultural, and environmental interpretation opportunities.

Policies for Conflicting and Competing Uses

Many of the following issues typically require the enactment of local ordinances by participating municipalities.

Motorized Use

Many non-motorized trails have adopted policies that prohibit motorized use of a given trail. In these instances, the design of the trail should be completed in a manner to discourage motorized use of the trail. Cooperative enforcement efforts by municipalities, counties, and the State Police are necessary to preserve motorized use restrictions. Adopt local ordinances to prohibit motorized use of the trail corridor and establish an escalating scale of penalties for offenders.

Litter and Graffiti

Enforcement of litter and graffiti ordinances will be the responsibility of the relevant municipality or the State Police. Routine monitoring and patrolling by trail users and local support groups will help to decrease these problems. Littering and vandalism is less likely to occur repeatedly in areas where trail monitoring occurs. Therefore, remove litter and graffiti from the trail corridor.

Hunting

As the right to provide public access to the trail corridor is acquired, the trail's managing agency must establish their the policies regarding hunting along the trail corridor.

If hunting is going to be permitted, consider the trail the same as a highway, requiring hunters be at least 100 yards away and not shoot towards the trail. Post warning signs at access areas during hunting seasons to advise trail users of the activity and recommend users wear blaze orange vests when on the trail.

Horseback Riding

Where horseback riding is a permitted use of a non-motorized trail, horses have the right of way; bikers and hikers will yield to horses; bikers yield to hikers.



Road Access Control

When a trail is designated exclusively for non-motorized use, measures should be taken to ensure that easy access by motor vehicles is discouraged. Construct appropriate signage and barriers at at-grade crossings to prevent vehicular access to the trail. However, construct barriers to permit access to emergency response maintenance, and security personnel, as needed.

Road access control, as with the other enforcement issues, will largely depend upon the enactment of non-motorized vehicle ordinances by local municipalities and the subsequent cooperation of local and state law enforcement. Routine monitoring by volunteers with local police response accessed by radio is the anticipated method of enforcement.

Utilities and Communications Facilities

Use of trail corridors by utilities and communications providers is not unusual, since the nature of railroad alignments lend themselves to such uses. An example of this is a trail in which a portion of the railway right-of-way was purchased upon abandonment by a local municipal authority. Today, the right-of-way contains a sanitary sewer line and the authority has an agreement with the local municipalities to use the right-of-way for public trail use.

Where the municipalities obtain ownership of the trail corridor, licensing of utility companies for such uses is often a viable means of helping to recoup trail development costs and operational expenses. To the extent that pipelines and cables are buried, the uses generally do not conflict. Each application for such use should be reviewed carefully, however, with particular attention being paid to expected frequency of maintenance, required access by motorized maintenance vehicles, and the mitigation of any resulting impacts. The local municipality having jurisdiction shall take the lead in regulating this policy within its municipality.

Policies Regarding the Use of Volunteers

In addition to implementing policies to comply with the Pennsylvania Child Labor Law and Pennsylvania Child Protective Services Law, as recommended in Chapter 4, trail organizations should consider adopting liability waivers.

Liability Waivers

A Liability Waiver should be signed by individuals and groups who desire to perform any form of work associated with a trail. Before adopting such a waiver your organization must review it with legal counsel to ensure its applicability and durability in a court of law. The form on the following page is a typical waiver of liability.



Sample Adopt-a-Trail Volunteer's Release, Hold-Harmless and Indemnification Agreement

WHEREAS, the undersigned, _____ wishes to volunteer his/her services by participating in _____ Adopt-A-Trail Program, in order to inspect and maintain the _____ trail; and

WHEREAS, participation in the project will take place at different times over a year; and

WHEREAS, participation in this project will involve using tools and equipment; and

WHEREAS, the _____ wish to cooperate in said endeavor;

THEREFORE, in consideration for said cooperation the undersigned agrees to the following:

1. To release and hold harmless the _____, and its employees, officers, and agents for any claim or claims which might arise out of any incident connected with or in any way related to participation in the Adopt-A-Trail program. This includes claims for personal injury, property damage, and/or any other type of harm or injury.
2. To release and hold harmless the _____, and its employees, officers, and agents for any claim or claims arising out of any incident connected with or in any way related to the undersigned's participation in the Adopt-A-Trail Program, including claims for personal injury, property damage, or any other type of harm or injury, made or asserted by any other person(s) against the _____.

I HAVE READ THIS RELEASE AND UNDERSTAND ALL OF ITS TERMS. I SIGN IT VOLUNTARILY, WITH FULL KNOWLEDGE OF ITS SIGNIFICANCE, AND WITH THE INTENT TO BE BOUND BY IT.

Signature

Date

Name (printed)

Telephone Number

Address



Volunteer Trail Patrol

Photo Credit: Pashek Associates

Trail Monitoring/Patrols

Monitoring is critical for a number of reasons, but primarily for determining levels and types of use, for providing information for effectively meeting trail users' needs, and for providing a presence along the trail. Monitoring requires a cooperative effort between the trail sponsor, local municipalities, participating partner organizations and local trail support groups. A monitoring role may be assigned to an appropriate group or organization, either for the trail as a whole, or by sections as in the currently popular "Adopt a Highway" program. Written agreements should document the roles and responsibilities of the trail sponsor and the monitoring/patrol agency or group.

Monitors should obtain certification by attending a training class conducted by trail partnership organizations. Monitors should wear bright shirts with "Trail Patrol" displayed prominently in large black letters.

Monitors should keep a record of their time and distance monitored at each outing. They should report their monitoring times and distances to their groups monitoring captain, who in turn will submit monthly reports to the trails managing partner.

Monitors will include any incidents, such as helping with bike repairs or administering first aid. Any incident of an urgent nature, such as a wash out or fallen tree should be reported immediately to local authority having jurisdiction.

Policy for Trail Closures

Trail users need to be cautious of the construction and periodic maintenance of trails, as well as roads and other facilities they intersect. Forewarn trail users of trail closures and give reasonable detours, to bypass closed sections of the trail. If there is no alternative route then close the entire trail until the respective trail segment can be re-opened. Acknowledge trail closures at trailheads and access points with appropriate signage.

Closure of trails generally occurs for the following reasons:

- Seasonal Closures – i.e. hunting, snowfalls, breeding seasons, etc.
- Construction Closures – i.e. trail construction, culvert maintenance and/or replacement, bridge construction/replacement, etc.
- Closures Do to Overuse/Negative Resource Impact

Best practices for trail construction zones include standard signing at the entrance to the affected segment of trail, including but not limited to information on the dates of closure. Clearly sign trails closed to public use and gate them, or otherwise block them to prevent access to the closed segment of the trail.



Trail Patrol

Photo Credit: Montour Trail Council



Policy Concerning Wheelchairs and Other Power-Driven Mobility Devices on Trails

In July 2010, the U.S. Department of Justice issued an ADA ruling that addresses the use of wheelchairs and power driven mobility devices on trails.

The DOJ rule, which implements the ADA, applies to state and local governments and public entities. The ADA does not apply to federal agencies with the exception of wheelchair use in federally designated wilderness as detailed in ADA, Title V Section 508 (c). The federal agencies are governed by the 1968 Architectural Barriers Act (ABA) and the 1973 Section 504 of the Rehabilitation Act.

The DOJ now categorizes mobility devices for individuals with mobility related disabilities either as a Wheelchair or as an Other Power Driven Mobility Device (OPDMD).

The DOJ defines a wheelchair as:

“a manually-operated or power-driven device designed primarily for use by an individual with a mobility disability for the main purpose of indoor or of both indoor and outdoor locomotion.” This definition does not apply to federal wilderness areas; wheelchairs in such areas are defined in section 508(c)(2) of the ADA, 42 U.S.C. 12207(c)(2).

And, the DOJ defines an OPDMD as:

“Other power driven mobility device: any mobility device powered by batteries, fuel, or other engines - whether or not designed primarily for use by individuals with mobility disabilities - that is used by individuals with mobility disabilities for the purpose of locomotion, including golf cars, electronic personal assistance mobility devices (EPAMDs), such as the Segway® PT, or any mobility device designed to operate in areas without defined pedestrian routes, but that is not a wheelchair within the meaning of this section. This definition does not apply to Federal wilderness areas; wheelchairs in such areas are defined in section 508(c)(2) of the ADA, 42 U.S.C. 12207(c)(2).”

The DOJ rule requires a public entity to make reasonable modifications in its policies, practices, or procedures to permit the use of OPDMDs by individuals with mobility disabilities unless the public entity can demonstrate that the class of OPDMD cannot be operated following legitimate safety requirements.

This is based on actual risks and not speculation regarding the device or how it will be operated per § 35.130(h), or on any of the other assessment factors



Other Powered Driven Mobility Devices

Photo Credits:
Top - Americantrails.org
Mid - Actiontrackchair.com
Bottom - Triangleglides.com



DOJ has provided in § 35.137(c) and § 35.137(b)(2) . Public entities can only use these factors in determining whether a particular OPDMD can be allowed in a specific facility as a modification to its policies, practices, or procedures. The focus of the analysis must be on the appropriateness of the use of the device at a specific facility, rather than whether it is necessary for an individual to use it. These assessment factors help public entities to determine whether allowing the use of a particular OPDMD in a specific facility is reasonable.

OPDMD Assessment Factors

1. The type, size, weight, dimensions, and speed of the device;
2. The facility's volume of pedestrian traffic (which may vary at different times of the day, week, month, or year);
3. The facility's design and operational characteristics (e.g. whether its service, program, or activity is conducted indoors, its square footage, the density and placement of stationary devices, and the availability of storage for the device, if requested by the user);
4. Whether legitimate safety requirements can be established to permit the safe operation of the OPDMD in the specific facility; and
5. Whether the use of the OPDMD creates a substantial risk of serious harm to the immediate environment, natural, or cultural resources, or poses a conflict with Federal land management laws and regulations."

Under the DOJ rule, public entities must make reasonable modifications to their policies, practices, or procedures to allow the use of an OPDMD by an individual with a disability. Further, public agencies must develop and publicly post their policy stating the following:

- The procedure by which claims that the OPDMD device is being used for a mobility disability will be assessed for legitimacy (i.e., a credible assurance that the device is being used for a mobility disability, including a verbal representation by the person with a disability that is not contradicted by observable fact, or the presentation of a disability parking space placard or card, or State-issued proof of disability);
- The type or classes of OPDMDs that are permitted to be used by individuals with mobility disabilities;
- The size, weight, and dimensions of the OPDMDs that are permitted to be used by individuals with mobility disabilities;
- The speed limit for the OPDMDs that are permitted to be used by individuals with mobility disabilities;
- The places, times, or circumstances under which the use of the OPDMD is or will be restricted or prohibited;
- Safety, pedestrian, and other rules concerning the use of the OPDMD;
- Under which circumstances, storage for the OPDMD is made available; and how and where individuals with a mobility disability can obtain a copy of the OPDMD policy.



If that class of OPDMD is allowed, a person who has a disability may not be denied the opportunity to use that device. The public entity may ask a person using a power-driven mobility device if the mobility device is necessary. A public entity shall not ask a person using a mobility device questions about the nature and extent of the person's disability.

Sample OPDMD Policy

Lancaster County has developed an OPDMD policy based on the following rationale:

Based upon an evaluation of the trails in the Lancaster County Park Trail utilizing the assessment criteria provided in the DOJ ruling, the Lancaster County ADA Trail Accessibility Policy was developed to maximize trail accessibility for individuals with mobility disabilities while at the same time maintaining the safety of other hikers, bikers, and horseback riders who use our trails and being responsible stewards of the environment on and around these trails.

Bearing this in mind, the following is the rational for the requirements regarding the use of other powered driven mobility devices (OPDMDs):

- Electric-powered devices only: Internal combustion powered devices are prohibited as they emit exhaust that is detrimental to the health of other trail users. In addition, Lancaster County Rails Trails run parallel to and in close proximity of County waterways thus creating potential for significant environmental risk from a spill of fuel or oil used in a gas or similarly powered combustible- fuel device. Finally, the loud sound generated from combustion powered devices is neither consistent with peaceful enjoyment of the outdoors nor appreciation of nature and is thus inherently contradictorily to the functions of the park trails.
- Motors producing 300 watts or less: (One of the DOJ assessment parameters for regulating OPDMDs is speed. Because speed cannot be accessed from vehicle specifications, the power of the motor has been used as an equivalent, with higher power translating to higher top speed.) A fit individual using a traditional bicycle can maintain a power output of approximately 300 watts for a period of ten minutes. Allowing mobility devices equivalent to a fit bicyclist produces no safety threat to other trail users. Allowing OPDMDs with motors of 300 watts and above, and thus higher speeds, does present increased danger to other trail users.
- Vehicles are no more than 36 inches wide: The Lancaster County Rail Trails have been built to 8 feet wide specifications. With two-way traffic, that gives a four feet width for vehicles to pass. Any OPDMD wider than three feet begins to encroach on the opposite lane and



becomes a safety hazard to both the operator and other trail users. In addition, bollards and gates at trail access areas and road crossings have a limited size and thus are significant barriers to wider vehicles.

Given this rationale, Lancaster County has adopted the ADA Trail Accessibility Policy on the following page.

In order to avoid challenges to any restrictions you may impose on your trails, your agency must adopt an OPDMD policy for its trails. Otherwise, your trails are considered open to every type of OPDMD, regardless of whether they have been planned, designed, and/or constructed for those uses.

If more than one entity is responsible for a trail corridor, it is important that all parties involved collaborate in developing a uniform policy for the entire trail corridor.



Lancaster County Department of Parks and Recreation ADA Trail Accessibility Policy

A. Lancaster County Park trails are available to individuals with a mobility disability as follows:

Wheelchairs: Wheelchairs, as defined by 28 CFR § 35.104, are permitted on all Lancaster County Park trails approved for pedestrian access.

Other power-driven mobility devices (OPDMDs)

OPDMDs, as defined by 28 CFR § 35.104, are permitted on multi-use County Park rail trails including the Conewago Recreation Trail, the Lancaster Junction Recreation Trail, the Conestoga Greenway Trail and the Iron Horse Trail as follows:

1. The OPDMD must be electric-powered. Internal combustion engines are not permitted.
2. The OPDMD must have an electrical output of no more than 300watts.
3. The OPDMD must be no more than 36" in width.

Please note that electric bikes (ebikes) meeting the above criteria, and which allow the user to pedal or alternatively run on battery power, are permitted on all Lancaster County Park Trails approved for bicycle use. User discretion is advised as some approved bicycle trails may not be suitable for all types of ebikes.

B. Users of a OPDMD or wheelchair must operate the device at a safe speed considering the condition of the trail and the other users traveling on the trail.

C. The adoption of this Policy does not represent an endorsement that the Park trails or other Park properties are safe for the use of an OPDMD or wheelchair. Users must exercise reasonable caution and care while operating such devices within the Lancaster County Park System.

If you have any questions regarding this policy or the accessibility of Lancaster County Park trails, please contact the Park Office at (717) 299-8215.



Trail Maintenance

The development of a trail does not end with its construction; rather it is the beginning of a process. Proper planning and design, along with quality materials and workmanship during construction will keep the maintenance costs low. Conversely, trail degradation will occur quickly without an effective maintenance program, no matter how well you plan, design, and construct a trail. Overall, if routine maintenance does not occur then maintenance costs will dramatically increase.

Therefore, it is equally important to have a maintenance plan in place to ensure you have the work force, materials, and finances to maintain your trail properly. Ensuring proper trail maintenance is an ongoing activity once a trail is dedicated.

Good trail maintenance practices are important because they:

1. Reduce liability and provide a defense when an accident does occur
2. Reinforce that the trail is an important regional recreation and transportation resource
3. Are effective in deterring vandalism, litter, and dumping in the trail corridor
4. Promote positive relationships with adjacent landowners
5. Help to create a sense of community throughout the trail corridor, and encourage a sense of pride

To ensure the success of the trail system, local partners, agencies, and municipalities should work together to define the roles and responsibilities for maintaining the trail and to ensure good maintenance practices are adopted. This does not mean the municipalities perform all of the work. More often, it means they coordinate the efforts of volunteer organizations and provide aid when needed. State, local, and county agencies would help groups where possible in planning trail improvements and providing material and equipment for trail maintenance. Volunteer efforts must be coordinated to ensure the provision of their services in a timely manner. Document such agreements in writing, between the participating parties.

Maintain trails for their full width so as not to allow the edges of the trail surface to unravel. A “pack it in, pack it out” policy for the trails will eliminate the necessity to provide trash collection along the entire route. Locate trash receptacles and/or dumpsters at key locations including trailheads. De-icing agents should be avoided as they can damage the trail and can be harmful to the environment. Drainage must be designed in a way so it does not interfere with the trails. Should drainage problems arise after construction, address them immediately to avoid damage of the trail surface.



Maintain trail surfaces in a safe and usable condition at all times. All defects in a trail surface must be corrected as soon as feasible through the removal and replacement or corrective action that restores the trail surface to a safe and usable condition. All potential hazards must be clearly identified and marked to alert users until they can be repaired.

The operation and maintenance of trails is ongoing and a necessary activity that will ensure the continued use of safe trails.

Monitoring Trail Use

The level of use for the trail will have a direct correlation to the frequency of maintenance required. Monitoring the type and volume of use, and evaluating the effect that level of use has on the trail, provides the trail manager information he needs to schedule inspections and routine maintenance. Monitoring the type and volume of use can be done by several methods. Type of use can be determined by monitoring trailhead use and observing riders on the trail.

Trail use can be determine with:

- Mechanical counters
- Sampling of trailhead parking
- Visual counts
- Trail registers

Monitoring trail use is important to determine the impact of use on the trail and to determine the frequency of routine maintenance activities. Continued monitoring alerts the trail manager to a change in the level of use, which will require a change to the maintenance schedule.

Trail Maintenance Record

Maintain maintenance records by documenting the history and schedule of maintenance activities. The length and complexity of the trail system will determine how maintenance records are organized. For example, you may have separate records for each trail in a multiple loop trail network. For long trails, you will likely keep a separate record for various reaches of the trail. The maintenance record should include:

- Trail management objective sheets
- Detailed trail map
- Scheduled maintenance and trail improvements
- Deferred maintenance or trail improvements
- Inspection reports and checklists
- A history of maintenance



**North Country National
Scenic Trail Register**

Photo Credit: PA DCNR



- Documented type and level of use
- Trail design and construction plans

Reviewing maintenance records will help identify areas or sections of trails that require more maintenance than expected. Evaluate these records for a design or construction problem. You may need to re-design, realign, or make improvements on a trail where the level of maintenance required is unacceptable or too costly.

Trail Inspection

A complete inspection of the trail should be a routine scheduled event, performed by trained personnel familiar with the trail. Perform trail inspection in conjunction with scheduled maintenance by the maintenance crew or as a separate action to determine the need for additional work.

Inspectors should review the trail management objectives for a section of trail before inspecting the trail. It is also helpful to review the previous inspection results and the records for maintenance performed since the last section. The inspector should determine the current condition of the physical features of concern and document all deficiencies, change of conditions, and the need for corrective maintenance.

Inspection Procedures

Trail inspectors should carry a trail map, inspection checklists, and tools for minor maintenance relating to safety. Photographing the current trail conditions is an excellent way to document and monitor changes in conditions. Inspectors must identify deficiencies that create safety concerns and hazards to the trail users.

Items to look for include:

- Erosion
- Failure of water diversion features
- Trail degradation
- Vegetation or materials extending into the horizontal or vertical clear area
- Deadfalls, dead limbs, or standing dead trees that could fall within the clear area
- Loose impediments on the treadway (that exceed the level of difficulty)
- Sloughing or erosion of embankments
- Missing or damaged regulatory or warning signs and other signage
- Vandalism

After identifying safety issues and hazards, the second highest priority is repairing damage from improperly functioning drainage features. The



inspector should inspect the inlets and outlets of all culverts, ditches and swales, to ensure they are free of debris and functioning properly. Evidence of a breakdown of drainage includes erosion, ponding of water, wet areas, and rutting of the trail tread. Correction of improperly functioning drainage may be a matter of maintenance or may require trail improvements or re-routing. Allowing a trail to continue to degrade from poor drainage quickly leads to costly corrective maintenance or trail closure.

The inspector should also be attentive to evidence of heavy use and trail tread condition requiring immediate maintenance. Record the general condition of the tread throughout various sections of the trail.

The information collected should be used to schedule the frequency of normal maintenance, such as re-grading and compaction of the treadway.

Patrolling

Patrolling may be necessary for certain high use trails. Patrolling of the trail can serve many purposes to include:

- Security
- Protection from vandalism
- Promoting safe riding
- Handling conflict
- Trail safety inspection
- Emergency aid

Persons assigned to conduct patrols should be trained in trail inspection.

Washington & Old Dominion Trail Patrol Handbook

The Friends of the Washington & Old Dominion (W&OD) Trail have prepared the *W&OD Trail Patrol Handbook*. This handbook serves as a good example of a how to establish policies and procedures to train, establish, and manage volunteer trail patrols. Download the the handbook from the Rails-to-Trails Conservancy's website, <http://community.railstotrails.org/media/p/10279.aspx>.

Another good resource for developing a volunteer handbook is the U.S. Forest Service's publication *Volunteers in the Forest Services: A Coordinator's Desk Guide*, available from www.fs.fed.us/t-d/pubs/pdfpubs/.../pdf09672814dpi72pt03.pdf.

Volunteer trail patrols can be trained to address urgent, emergency, and other extraordinary situations that might arise for users along a trail. Patrol members are the "eyes and ears" of the trail. For most of the length of many trails, no other users are there specifically for the well-being of other users; the volunteer patrol is there to help in these situations.

W&OD Trail Patrol Handbook

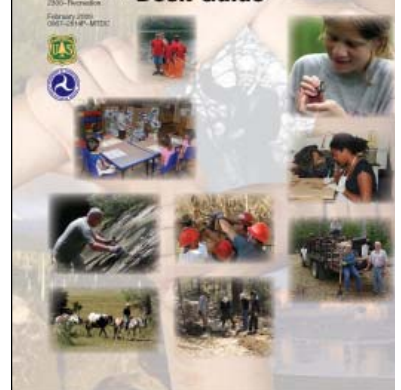


April 2009
(Version 1.0)

W&OD Trail Patrol Handbook Friends of the W&OD



Volunteers in the Forest Service: A Coordinator's Desk Guide



Volunteers in the Forest Services: A Coordinator's Desk Guide U.S. Forest Service



Although these types of situations are rare, the consequences for the individual involved can be extreme. In case of accidents, injuries, or security incidents, the primary responsibilities of the volunteer patrol are to contact the appropriate authorities to respond to the situation, and to try to improve the immediate situation with reasonable measures. Under no conditions is the volunteer patrol expected or encouraged to attempt what is beyond their authority, training, or capability.

A volunteer patrol handbook helps patrol members to respond with appropriate actions in emergencies to provide reasonable aid. This handbook provides a model for policies and procedures related to volunteer trail patrol activities and can be referenced when developing a similar document for your trail.

The *W&OD Trail Patrol Handbook* contains the following information, policies, and procedures:

- About the W&OD Trail Patrol
- Goals of the W&OD Trail Patrol
- Promote Trail Courtesy and Safety
- Provide Information to the Public
- Come to the Aid of Trail Users When Possible
- Who Makes a Good Patrol Member?
- W&OD Trail Patrol Policies
- W&OD Park Rules
- Treatment of Trail Users
- First Aid Treatment for Trail Users
- Patroller Safety Equipment
- Training
- Use of Patrol Jerseys
- Patrol Member Requirements
- Registration
- Orientation
- Training
- Patrol Member Code of Conduct
- Trail and Patrol Property or Equipment
- Patrol Notifications
- Patrol Member Location
- Patrol Reports
- Minimum Participation
- Patroller Identification
- Membership in Friends of the WOD Organization
- Patrol Concept of Operations
- Getting Started
- First Patrols
- Unsupervised Patrols
- Patrol Reports
- Patrol Management and Operations
- Advanced Training Patrol
- Mandatory Equipment
- Recommended Equipment
- Example First Aid Kit
- General Patrol Activities
- Dealing with the Public Persons
- Reporting Procedures



Trail Maintenance and Volunteer Patrol Checklist

A maintenance plan should establish a routine maintenance schedule, preventative maintenance, and risk management inspection schedule. Maintain trails in a safe and usable condition at all times. All defects in trail surface, rough edges, depressions, uneven pavement, etc., must be corrected immediately by making the required repairs to restore the trail to a safe and usable condition. Clearly identify all potential hazards and mark them to alert users. Failure to properly operate and maintain the trail may result in an unsafe trail, which has the potential to become a liability to the entities involved.

The trail management plan should specify who is responsible for conducting trail inspections and their frequency. Either paid staff or trained volunteers can conduct trail inspections. Regardless of who completes the inspection, the results must be documented and kept on file in case a liability claim is filed by a trail user. By following the policies and procedures documented in the trail management plan, the risk of liability can be reduced.

During the planning for the Bellefonte Central Rail Trail the following Volunteer Patrol Checklist, on the following pages, was developed.



Bellefonte Central Rail Trail Volunteer Patrol Checklist

The Arboretum at Penn State

Date of Walk-Through:

Name of Volunteer / Phone:

Please send completed form to **Brian Phiel, The Pennsylvania State University, 119 Physical Plant Building, University Park PA 16802**, and a copy to **Kate Reeder, Arboretum program assistant, 336 Forest Resources Building, University Park PA 16802**. Report emergency trail condition issues to the *Office of Physical Plant Work Reception Center* — **865-4731**.

TRAIL OBSERVATIONS	NO	YES	COMMENTS
Crushed Limestone Surface			
Is the trail surface level?			
Is pothole patching needed?			
Is repair needed for washouts?			
Are there plants growing on trail surface and along trail edge?			
Asphalt Surface (at intersections)			
Is there gravel or debris on the asphalt?			
Are there washouts?			
Is the coating or sealing faulty?			
Are there potholes?			
Is pavement in disrepair?			
Vegetation			
Is grass on shoulders higher than 8 inches?			
Is vegetation/brush in the additional cleared area (see attached map) growing higher than 2 to 3 feet?			
Is there evidence of vandalism, illegal cutting, or illegal vehicular access?			



TRAIL OBSERVATIONS (continued)	NO	YES	COMMENTS
Are tree limbs/leaves growing close to the trail or hanging within the corridor used by bicyclists and hikers?			
Are any trees, tree limbs, or large shrubs laying close to or on the trail?			
Is there a large quantity of leaves on the trail?			
Signs and Structures: Any apparent need to clean, repair, or replace?			
Signs			
Fences			
Gates			
Bollards			
Storm Drains			
Culvert			
Trash Removal			
Is litter pick-up needed?			
Illegal Activities			
Is there evidence of vandalism, illegal cutting, or illegal vehicular access?			

Frequency of Inspection

The frequency of inspection depends on the maintenance history of the trail, sustained level of use, and special events. Special events include acts of nature, as well as planned events such as organized club events or poker runs. Inspections should be frequent enough to correct potential problems before they become a safety issue or lead to more costly corrective maintenance.

Inspect trails immediately after significant storm events. Give emphasis to safety issues and evidence of a failure of drainage systems or structures.



Trail Mapping

Good trail maps play an important role in the maintenance program. Maps should include an inventory of the physical features of the trail. They must be of appropriate scale and detail to enable personnel to record information regarding maintenance directly on the maps or reference physical features. A scale of 1" = 500' is a reasonable scale for an overall maintenance map, generally permitting each mile of trail to be displayed on a single 8-1/2" x 11" map. Detailed maps at a scale of 1"=50' may be necessary to show details and location of specific maintenance required or performed.

Use trail maps to document any maintenance requirements identified through inspections and to record all maintenance performed.

Inspection Checklists

Use checklists to ensure that inspections are consistent and complete. Develop a checklist for normal trail inspection. Specific checklists may also be developed for trailheads, play areas, and physical features such as culverts, bridges, and retaining walls.

Structural Inspection

An experienced and licensed professional engineer must be used to inspect structural facilities such as bridges and retaining walls. Schedule the frequency of structural inspections based on the age and condition of the structure.

Types of Maintenance

There are many types of maintenance activities necessary for trails.

Types of Trail Maintenance

- Scheduled maintenance
- Seasonal closure and opening
- Winter maintenance
- Regular vs. periodic grooming
- Tread, drainage, and trail structure repair
- Brush cutting/mowing
- Small Scale reconstruction - Structure replacement and reconstruction
- Large scale reconstruction - Rerouting or trail hardening projects

Establish trail maintenance tasks in a Trail Maintenance Plan adopted by the agency/organization responsible for the trail. Maintenance requirements are dependent on the type of trail and amount of visitation it receives.

As with trail inspections, maintenance activities, regardless of type, should be documented and kept on file for risk management purposes.



Scheduled Maintenance

Scheduled maintenance is the normal maintenance needed to restore a trail to its intended standard after prolonged wear and tear of normal use and exposure to the elements.

Develop an annual trail maintenance schedule using historical maintenance and known maintenance requirements from previous inspections or deferred maintenance. This living document must then be adapted to the changing conditions of the trail.

Typically, scheduled maintenance tasks are follows. This list can be altered as needed to meet the needs of the trail, its users, and the communities that a trail passes through:

- Trimming or removing vegetation, dead limbs, or standing dead trees
- Removing debris, deadfalls, or loose impediments
- Cleaning out ditches, swales and culverts
- Repairing and revegetating minor erosion on slopes or embankments
- Grooming the tread surface
- Minor repairs such as replacing missing or broken posts or signs
- Mowing
- Trimming
- Trash removal
- Pruning
- Weeding
- Invasive Removal
- Brush hog
- Signs – inspect / repair / replace
- Fence – inspect / repair / replace
- Culverts – inspect / repair / replace
- Storm Drains – inspect / repair / replace
- Gates - inspect / repair / replace
- Bridge - inspect / repair / replace
- Maintain dips
- Grade ditches
- Trail grooming
- Trail surfacing
- Landscaping
- Storm damage
- Vandalism
- Repair washouts
- Garbage pick-up at trail access points
- Mowing of berms
- Cleaning of restrooms at trailheads
- Sweeping trails with a rotary brush to remove dirt and leaf litter
- Erosion control, repair of drain pipes and cleaning of swales
- Patching, regrading, and compacting of surface
- Inspecting, repairing, replacing signs, traffic markers, bollards, and gates
- Cleaning culverts, catch basins, and other drainage structures
- Maintaining and completing preventative maintenance on support facilities
- Inspecting trail-related structures to ensure they are in a safe condition
- Plowing trailhead parking lots in the winter



The following is a typical calendar for scheduled maintenance:

Frequency of Scheduled Maintenance												
Maint. Activity	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
EXAMPLE- Mowing			•	••••	••••	••••	••••	••••	••••	•••		
Mowing												
Trimming												
Trash												
Pruning												
Weeding												
Invasive Removal												
Brush Hog												
Signs - inspect/repair/ replace												
Fence - inspect/repair/ replace												
Culverts - inspect/repair/ replace												
Storm Drains - inspect/ repair/replace												
Gates - inspect/repair/ replace												
Bridge - inspect/repair/ replace												
Maintain Dips												
Grade Ditches												
Trail Grooming												
Trail Surfacing												
Landscaping												
Storm Damage												
Vandalism												
Repair Washouts												

Seasonal Opening and Closure

If seasonal closures occur along a trail, inspect and maintain portions of the trail that require maintenance. Sometimes this maintenance can be intensive, especially in areas prone to blow downs. Further for those trails with winter closures the tread surface may need to be graded, compacted, and or groomed in preparation for increased use during wet conditions of the spring season. Fall maintenance should focus on clearing and trimming of vegetation sufficiently to account for the next growing season. Repair and stabilize eroded areas to prevent future erosion.



Winter Maintenance

Collaborating agencies must decide and document what the winter maintenance policy will be for the given trail. Avoid plowing trails that including winter uses such as cross country skiing and snowshoeing. In urban environments, where pedestrian and bicycle use is expected to continue throughout the winter months consideration may be given to plowing the trail. Regardless of the decision made, communicate to users of trails in inclement weather to establish expectations of other users. This is an important issue and it may warrant space on a trail kiosk or rules sign.

Periodic Grooming

The frequency of periodic grooming of the tread surface is dependent on the level of trail use, and the amount of surface displacement. Simple grooming may require re-grading and compaction of the in-place tread surface materials. More intensive grooming may require placement and compaction of additional tread surface materials to address minor rutting or erosion.

Corrective Maintenance

Corrective maintenance is necessary for the restoration of areas or structures severely damaged or destroyed by overuse, inadequate scheduled maintenance, abuse, vandalism, or unexpected natural events. Corrective maintenance is usually unexpected and may require more planning or design than scheduled maintenance. Corrective maintenance includes:

- Reinforcement and replacement of trail subbase and tread surface
- Stabilization of severely eroded or sloughed embankment
- Reconstruction of grade dips, or other water control structures
- Replacement or major repair of culverts, bridges, or retaining walls

Regardless of the level of planning and effectiveness of the scheduled maintenance program, the unexpected will occur. Address unexpected emergencies by establishing a contingency item in the annual maintenance budget. Ideally, a long-term capital improvement budget will be in-place to repair/replace major structures.

Regardless, the need will arise at some time where corrective maintenance must be deferred pending availability of financial resources.

Deferred Maintenance

Deferring maintenance is a deliberate decision to delay maintenance due to a lack of labor, budget considerations, or specific conditions. Defer routine and scheduled maintenance in order to make funds, maintenance personnel, or equipment available to perform corrective maintenance. As an example, deferral of scheduled maintenance for general grading and re-compacting may occur in order to make a maintenance crew available to perform a major replacement of subbase and tread surface for a section of trail that



Clearing the Trail Corridor

Photo Credit: PA DCNR



has become severely rutted due to wet conditions and higher than expected use during the spring season. Then, reschedule deferred maintenance when resources become available. Never defer corrective maintenance that is safety related unless that section of trail can be bypassed or closed.

Structural Inspection

An experienced and licensed professional engineer must be used to inspect structural facilities such as bridges and retaining walls. Structural inspections should be scheduled based on the age and condition of the structure.

Sign Maintenance

Incorporate a regular maintenance program for signs into your trail management plan. Signs are highly visible and their maintenance or lack of maintenance leaves the visitor with a positive or negative impression about the trail and the park. Well-maintained signs convey a sense of pride and reduce vandalism while poorly maintained signs may contribute to a diminished visitor experience, including disorientation of trail users.

The following guidelines are recommended:

- Maintain a record of all signage, including location, type of sign, and photo.
- Inspect signs regularly, especially after each winter season, for weathering and visibility.
- Repair or replace damaged or missing signs as soon as possible.
- Secure loose or tilting signs in an upright position.
- Clear vegetation from around signs to maintain visibility.
- For signs mounted on living trees, loosen fasteners as necessary to accommodate growth of the tree.
- Review signage content to ensure continued relevance and accuracy.
- Obsolete, damaged, or surplus signs should be reused or recycled whenever possible.
- When signs have been weathered or otherwise damaged or destroyed, consider the reasons for the damage. If the sign was eaten by wildlife, consider less palatable materials. If weather or natural events damaged the sign, consider stronger materials, a different location, or a different system for mounting the signs. If the sign is damaged by water or decay, consider applying a sealer or preservative (assuring compatibility with color, aesthetics, and environmentally sustainable practices) or replacing the sign with a more water-resistant material.



Priorities for sign maintenance are:

1. Signs required for user safety
2. User restrictions and advisory signs
3. Destination and identification signs, blazes, and trail logos, and
4. Informative and interpretive signs

Hazard Tree Identification, Inspection, and Corrective Action

A tree is considered hazardous when structural defects are likely to cause failure of all or part of a tree within striking distance of a target. A target can be a vehicle, building, or an area where people or their equipment are likely to stop and congregate, such as park benches, campgrounds, picnic tables and environmental education areas. Identifying hazard trees is not an exact science. Many times trees will fail though they appear healthy. One cannot always accurately determine the hazard potential of every tree, but following a few basic inspection techniques can identify many hazard trees so corrective action can be taken.

Tree age, size and condition are important factors in determining the seriousness of potential hazards. Older trees have had more time for injuries and infections to develop into structural defects; and larger trees have greater structural stresses due to their size and weight. Trees in poor condition may have dead or dying branches, off-color foliage, or multiple branch shoots along the trunk, known as epicormic branching. Trees in poor condition should be inspected more closely for structural defects including cracks, cankers, decay, weak branch unions and root problems. Be sure to inspect trees carefully and systematically. Examine all parts of the tree, including the trunk flair, main stem, branches and branch unions. A pair of binoculars or a spotting scope will allow for a more complete exam of the tree canopy.

Close inspection should occur in any public overnight and day use areas, along trails and roadways, and adjacent to buildings. Again, a hazard tree is defined as having a defect and a target. Tree inspections should be conducted using consistent procedures. A widely accepted method of assessment is the USDA, USFS Risk Assessment Scale, which prioritizes hazard trees based on three criteria. The three assessment criteria are tree condition, failure potential, and probability of target impact. Tree condition considers the amount of deadwood, foliage and trunk decay. Failure potential is rated according to the severity of the defects. Probability of target impact is ranked based on frequency of use. Trees determined to have low probability of failure and target impact should be considered for their value as wildlife habitat.

Tree inspections should occur in the spring before leaf-on, in mid-summer, and after leaf-off in the fall. This inspection schedule allows the tree to be observed in various states and provides ample time for corrective action to be taken. Areas more heavily used by the public should be inspected



Removing Blow Downs

Photo Credit: PA DCNR



immediately after severe storms that cause blow downs, leaning trees, or large hanging branches.

Tree inspections should be documented. This documentation should include the date(s) of the inspections, the area(s) inspected, the person(s) conducting the inspection, and whether this inspection was a routine inspection or conducted because of a storm event. Using the USFS Risk Assessment Scale allows for easy and consistent documentation over a multiple year period. Trail maintainers can provide this document to the landowner as a recommendation for corrective action.

Following the identification and inspection of trees, if a tree is found to have a deficiency, the appropriate corrective action should be taken promptly. Use the USFS Risk Assessment Scale to prioritize hazard trees that require corrective action. These corrective actions could include moving the target, pruning the tree to remove the hazardous portion, or removing the entire tree. Landowners/managers can contact a professional arborist if they do not have sufficient equipment or properly trained people to carry out tree pruning or removal.

Additional information on hazardous trees and the USDA, USFS Risk Assessment scale can be found at: <http://www.na.fs.fed.us/spfo/pubs/uf/utrm/index.htm> or http://www.fs.fed.us/r10/spf/fhp/hazard/rating_method.htm.

Revegetation and Restoration

It is important to restabilize and restore vegetation in areas damaged by trail construction. Restoration often leads to better habitat for animals, a richer diversity of species, healthier ecosystems, and cleaner water. Furthermore, this process allows for the establishment of aesthetically pleasing natural areas rather than those marred by overuse and abuse.

Restoration work can vary from simple undertakings to multi-year efforts to revegetate and erase the impacts of human effects on the land. Areas that are in need of this type of maintenance include: bootleg trails, shortcuts across switchbacks, inappropriate campsites, permanently closed trails, logged-out areas, and barren riparian areas. In some cases, especially in heavily impacted areas no longer subject to erosion, agency personnel may determine that restoration efforts are not necessary. This is because these areas will probably not suffer any further damage regardless of future use.

The decision to restore an area is best made by informed land management personnel. This planning should insure that the work is appropriate and will be effective, as well as have a realistic understanding of the size, duration, expense, and demands of the undertaking. Restoration projects require careful planning and a commitment to achieve objectives. This work includes activities such as site and soil preparation, seeding, transplanting native vegetation and container-grown plants, as well as on-site plant propagation.



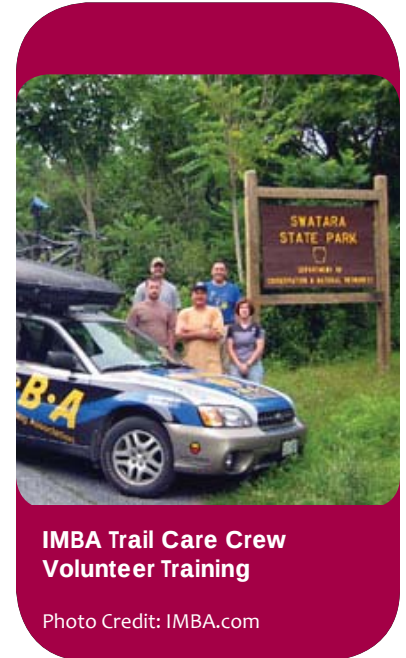
While the effectiveness of revegetation and restabilization efforts are not fully understood today, it is important to take these steps in order to reestablish the land as close to its natural state as possible. Monitoring of restoration sites will allow project managers to judge the effectiveness of environmental repairs over time. Written records, maps, and photographs will reveal these changes and allow future users and land managers to understand the negative and positive impacts an area has withstood.

Education

Education is critical to the success of your trail. You can not assume trail users, volunteers, or staff have the knowledge and skills necessary properly use the trails, and, to perform their duties and functions the manner required to maintain a safe, inviting, and well maintained trail environment.

The following table summarizes various educational components and their intended audience.

Education Component	Audience						Frequency and Location
	Designer	Land Manager	Maint. Staff	Trail Patrol	Volunteers	Trail Users	
Safety Training		•	•		•		
Trail Etiquette						•	
Proper Trail Use					•	•	
Pack-In and Pack-Out					•	•	
Trail Construction	•	•	•		•		
Trail Maintenance			•		•		
Water Crossing Construction			•		•		
Trail Inventory Procedures	•	•		•			





Trail Etiquette

DCNR's pamphlet *Trail Etiquette: Promoting User Compatibility Through Understanding and Cooperation* provides the following recommendations for Trail Etiquette based on the type of trail activity being pursued.



Hiking

Trail encounters while hiking with:

Another Hiker with a Dog

Speak to the hiker as they approach. This lets the dog owner take appropriate action based on their dog's behavior and lets the dog know that you are not a threat or prey. Do not make direct eye contact with the dog. This is perceived as a threat by most dogs and they may react defensively. Do not attempt to pet the dog unless invited to do so by the owner, and even then, only carefully. Allow the dog to first smell the back of your hand. If the dog is off the leash, stop and allow the hiker to regain control of their pet.

A Bicyclist

Step off of the trail and allow the biker to pass.

An Equestrian

Speak to the rider as they approach. This lets the horse know that the hiker is human and not a predator. Step off of the trail on the lower side to allow the rider to pass. Do not stare directly at the horse as it nears or passes — this is perceived by the horse as a threat. Predators stare before they attack. Do not attempt to touch the horse unless invited to do so by the rider.

A Hunter

Acknowledge the hunter with a silent nod and continue on your way. Do not intentionally interfere with a hunt. This is illegal in Pennsylvania and punishable by law.

Wildlife

Remain quiet to observe the wildlife. If you are uncomfortable with the encounter, do not attempt to hide (predators hide before they attack) and do not attempt to run away (many animals will chase after things when they run away). Speak to the animal; most animals are afraid of humans and will leave the area.



Bicycling

Trail encounters while bicycling with:

Another Biker Approaching from the Same Direction

Call out, “on your right” or “on your left,” if you wish to overtake or pass the biker in front of you to let them know on which side you will be passing them. Use extreme care and pass only where there is room on the trail to safely do so. If you are being overtaken by another rider and hear, “on your right” or “on your left,” maintain course on the trail or move to the side opposite the callout to permit the other rider to pass safely.

Another Biker Approaching from the Other Direction

Dismount and clear the trail if you are traveling downhill to allow the uphill biker to continue his climb without breaking stride. If you are traveling uphill, make sure the downhill rider is giving way before continuing your climb. If in doubt, give way and clear the trail.

An Equestrian

Look up occasionally while you ride because you might be approaching horses that may not hear you. Speak to the rider as they approach. This lets the horse know that you are a human and not a predator. Dismount and get off the trail on the lower side to allow the rider to pass. Do not make direct eye contact with the horse as it nears or passes. This is perceived by the horse as a threat. Predators stare before they attack. Do not attempt to touch the horse unless invited to do so by the rider.

A Hiker

Dismount to pass the hiker unless the hiker is obviously giving way to permit you to continue your ride.

A Hunter

Acknowledge the hunter with a silent nod and continue on your way. Do not intentionally interfere with a hunt. This is illegal in Pennsylvania and punishable by law.

Wildlife

Continue on your way. If you wish to observe the animal, do so safely — no sudden stops when riding with other bikers. Never pursue wildlife. This is illegal and unsafe. If you see a rattlesnake crossing the trail, simply wait for it to move off of the trail or go around it, giving it a wide berth.



Horseback Riding

Trail encounters while horseback riding with:

All Other Non-Motorized Trail Users

You have the right-of-way in nearly all situations. However, you know best how your horse will react in different situations. Take appropriate action to keep all encounters safe. Speak to others if they have not spoken at your approach. If other users move to the high side of the trail, ask them to move to the lower side, explaining that horses perceive animals above them as predators.

Wildlife

Observe the animal safely or continue on your way. If you decide to observe the animal, do so safely. Again, you know best how your horse will react to wildlife, so you should plan accordingly.

Winter Hiking, Snowshoeing, & Cross Country Skiing

In winter, hikers and snowshoers should not walk on cross-country ski trails, as this ruins the track for skiers. Walk to the side of the set ski track. Non-motorized trail users should be alert to approaching snowmobiles and give way to allow the snowmobiles to pass.

Snowmobilers using multiple use trails need to be careful around turns or other areas where you cannot see far ahead. Give yourself plenty of time to slow down or stop, should you encounter another user.



Leave No Trace Philosophy

Leave No Trace is a national and international program designed to assist outdoor enthusiasts with their decisions of how to reduce their impacts when they hike, camp, picnic, snowshoe, run, bike, hunt, paddle, ride horses, fish, ski, or climb. The program strives to educate all those who enjoy the outdoors about the nature of their recreational impacts, as well as techniques to prevent and minimize such impacts. Leave No Trace is best understood as an educational and ethical program, not as a set of rules and regulations.

1. *Plan Ahead and Prepare*

- Know the regulations and special concerns for the area you'll visit.
- Prepare for extreme weather, hazards, and emergencies.
- Schedule your trip to avoid times of high use.
- Visit in small groups when possible. Consider splitting larger groups into smaller groups.
- Repackage food to minimize waste.
- Use a map and compass to eliminate the use of marking paint, rock cairns, or flagging.



2. Travel and Camp on Durable Surfaces

- Durable surfaces include established trails and campsites, rock, gravel, dry grasses, or snow.
- Protect riparian areas by camping at least 200 feet from lakes and streams.
- Good campsites are found, not made. Altering a site is not necessary.
- In popular areas, concentrate use on existing trails and campsites.
- Walk single file in the middle of the trail, even when wet or muddy.
- Keep campsites small. Focus activity in areas where vegetation is absent.
- In pristine areas, disperse use to prevent the creation of new campsites and trails.

3. Dispose of Waste Properly

- Pack it in, pack it out. Inspect your campsite and rest areas for trash or spilled foods. Pack out all trash, leftover food, and litter.
- Deposit solid human waste in catholes dug 6 to 8 inches deep at least 200 feet from water, camp, and trails. Cover and disguise the cathole when finished.
- Pack out toilet paper and hygiene products.
- To wash yourself or your dishes, carry water 200 feet away from streams or lakes and use small amounts of biodegradable soap. Scatter strained dishwater.

4. Leave What You Find

- Preserve the past: examine, but do not touch, cultural or historic structures and artifacts.
- Leave rocks, plants, and other natural objects as you find them.
- Avoid introducing or transporting non-native species.
- Do not build structures, furniture, or dig trenches.

5. Minimize Campfire Impacts

- Campfires can cause lasting impacts to the backcountry. Use a lightweight stove for cooking and enjoy a candle lantern for light.
- Where fires are permitted, use established fire rings, fire pans, or mound fires.
- Keep fires small. Only use sticks from the ground that can be broken by hand.
- Burn all wood and coals to ash, put out campfires completely, and then scatter cool ashes.

6. Respect Wildlife

- Observe wildlife from a distance. Do not follow or approach them.
- Never feed animals. Feeding wildlife damages their health, alters



natural behaviors, and exposes them to predators and other dangers.

- Protect wildlife and your food by storing rations and trash securely.
- Control pets at all times or leave them at home.
- Avoid wildlife during sensitive times: mating, nesting, raising young, or winter.

7. **Be Considerate of Other Visitors**

- Respect other visitors and protect the quality of their experience.
- Be courteous. Yield to other users on the trail.
- Step to the downhill side of the trail when encountering pack stock.
- Take breaks and camp away from trails and other visitors.
- Let nature's sounds prevail. Avoid loud voices and noises.



Tread Lightly!

Simply stated, the TreadLightly! program is an educational program dedicated to increasing the awareness of how to enjoy the outdoors while minimizing impacts. It emphasizes responsible use of non-motorized, and motorized, travel and low impact principles related to outdoor recreational activities. It is a long-term, information program to increase the public's awareness that outdoor enthusiasts must exercise special care in the outdoors, if opportunities to recreate are to continue, and the quality of experience maintained and enhanced.

Tread Lightly! was launched in 1985 by the USDA Forest Service to help protect public and private lands. Shortly thereafter, the United States Department of the Interior's (USDI) Bureau of Land Management adopted the program. To maximize its effectiveness, program responsibility was transferred from USDI in 1990, to a not-for-profit organization. Over the years, the program has steadily added new dimensions to meet the needs of all types of outdoor enthusiasts. The program's message was expanded in 1997 to include the promotion of responsible water based recreational activities. These changes brought the USDI Bureau of Reclamation and United States Department of Defense's (USDOD) Army Corps of Engineers on board as additional federal partners. Most recently the USDI National Park Service and U.S. Fish and Wildlife Service joined in partnering with Tread Lightly!.

Tread Lightly! Inc. unites a broad spectrum of federal and state government agencies, manufacturers of recreational products, media, enthusiast groups and concerned individuals who share a common goal to care for natural resources. Through education, restoration and research, Tread Lightly! empowers generations to enjoy the outdoors responsibly, ensuring future use of the land and water.



Through the services of Tread Lightly! Inc., outdoor enthusiasts learn that opportunities to use the outdoors tomorrow, depends on how they recreate today.

Tread Lightly Pledge

- T**ravel & recreate with minimum impact.
- R**espect the environment and the rights of others.
- E**ducate yourself, plan and prepare before you go.
- A**llow for future use of the outdoors, by leaving it better than you found it.
- D**iscover the rewards of responsible recreation.

TreadLightly, Inc. offers resources to help educate the public. These resources include:

- **Master Tread Training:** Send your volunteers and / or employees to become Master Tread Trainers or Tread Trainers.

The Tread Trainer™ program is designed to train participants in innovative, practical methods of spreading outdoor ethics to the public. Training is especially relevant for those who deal with motorized or mechanized recreation on managed land or waters. To become a Master Tread Trainer, training is provided through a one-day course that will qualify the participant to become a Tread Lightly! representative—a “Master Tread Trainer.” Once a participant becomes a “Master Tread Trainer” they are qualified to train other staff members, clubs, educators, enthusiasts or community members to be “Tread Trainers.”

- **Tread Training:** If your staff is unable to attend a Master Tread Trainer course, consider requesting a Master Trainer in your area to train your staff to become Tread Trainers. Tread Trainers are provided with an overview of Tread Lightly!, and become qualified to present the Tread Lightly! message of responsible recreation to local clubs, enthusiasts, guided groups and many others.
- **Brochure Series:** The most efficient way to spread the Tread Lightly! message is to distribute TreadLightly, Inc.’s one-page tip brochures. These address the best ways to minimize impact while recreating in the great outdoors. These brochures are available at no cost to members, and government entities are considered members. Brochures include tips on ATV riding, four wheeling, off-highway motorcycling, mountain biking, personal watercraft use, snowmobiling, sand duning for kids, camping, geocaching, hiking, fishing, hunting with ATV, horseback riding, water recreation, and cross country skiing.



- **Guidebooks:** Tread Lightly!’s guidebook series includes books for four-wheel driving, ATV riding, mountain biking, snowmobiling, personal watercraft use, trail biking and “leaving a good impression.” The books are based on the TreadLightly! principles and include safety tips, checklists, and ways to minimize impact on the outdoors.
- **Public Service Announcements (PSAs):** Display Tread Lightly!’s public service announcements to educate visitors to your site. Many agencies have printed the PSAs out as posters, posted them in kiosks, inserted them into publications, and have urged local newspapers and magazines to donate ad space for them. They are available on DVD at no cost to individuals who commit to use them effectively.
- **Web Banners:** Tread Lightly!’s web banners are educational and inspirational. A variety of banner sizes and recreation types are available.
- **Promotional Items:** Tread Lightly! products act as billboards for responsible recreation by visibly reminding people to “tread lightly.” Items include: apparel, decals, educational materials, maps, advertising and promotional materials, souvenirs, and vehicle accessories.
- **Logo Use:** The Tread Lightly! logo renders a strong message. Government agencies have the opportunity to use the TreadLightly! logo on brochures, maps, newsletters, posters, kiosks, signage and any other type of printed or posted materials.
- **Youth Outreach Kits:** Tread Lightly!’s Youth Outreach Kits help teach outdoor ethics to kids. The printable kit includes pledge cards, a kids’ principles sheet, a poster/psa, and coloring book pages. These items can be printed out and given away at information booths or events and can be posted on websites and kiosks. Also available in this kit is a series of kids’ web banners.
- **Posters:** Large, 30” x 22” full color posters are a great way to display the Tread Lightly! principles to your visitors. Put it up in your info booth or on a kiosk.
- **Curriculums:** Tread Lightly! has created several curriculums for hunter education, junior high teachers, and elementary school teachers.

Agencies and organizations who participate in TreadLightly!’s programs have the opportunity to access resources not otherwise available. Through their “Restoration for Recreation” Program, partnerships are established to restore recreational areas and educate the public about responsible recreation. Tread Lightly! provides expertise and guidance to its partners



by administering the restoration and educational elements of each project. Initial funding for a project is generated through Tread Lightly!’s official sponsors, making them partners in Restoration & Education. In most cases, Tread Lightly! finds government and foundation partners willing to provide additional funding, leveraging as many investments as possible. A strong educational element incorporated into every project includes Tread Lightly! presentations in local schools, near a project, to empower the next generation to enjoy the outdoors responsibly.

Railroad Safety Education and Outreach

When a trail parallels or crosses an active railroad corridor, it is important to make trail users aware of the corridor, and the hazards associated with it. Many railroads participate in community outreach in many ways such as posting signs at trailheads and crossings, attending community events, regular monitoring of tracks, and penalties for trespassers. Most also support and participate in Operation Lifesaver.

When an active railroad line is associated with your trail corridor, contact the State’s Operation Lifesaver Coordinator to arrange for presentations on pedestrian safety and railroad trespass prevention for trail organizations and other users.

Operation Lifesaver, Inc.

Operation Lifesaver serves as a nationally recognized nonprofit association committed to educating the public about the dangers associated with railroad rights-of-way and highway-rail grade crossings. The program strives to decrease the number of injuries, collisions, and deaths at highway-rail grade intersections and on railroad property. This is sponsored cooperatively by a wide variety of partners, including national railroads, highway safety and transportation organizations, as well as federal, state, and local government agencies.

This program seeks to improve driver, bicyclist, and pedestrian compliance with crossing signs and signals at highway-rail grade crossings. Operation Lifesaver recognizes the importance of strong enforcement and engineering improvements, such as the consolidation and closure of redundant highway-rail crossings. In recent years, Operation Lifesaver has also increased its efforts to educate the public that trespassing on railroad rights-of-way, tunnels, trestles, and other railroad property remains both illegal and deadly.

Operation Lifesaver is an extremely valuable resource for both rail-with-trail managers and all public and private railroad companies. Its award-winning safety materials include videos and brochures about the dangers of rail trespassing, as well as information for pedestrian and bicycle safety at crossings.

Operation Lifesaver!
Mr. Donald Lubinsky,
Pennsylvania Coordinator
4600 Deer Path Road
Harrisburg, PA 17110
302-659-1209
Donlsr10@verizon.net
www.oliofpa.org



Identify where you can make safety information materials available on a regular basis. Consider the willingness of local bicycle sales or rental shops to distribute safety information, and consider other means for encouraging safe use of approved trails .

TIPS FOR BICYCLISTS

Hey, bike riders! Operation Lifesaver, Inc. (OLI) and its safety partners, the Federal Railroad Administration, the Federal Highway Administration and the National Highway Traffic Safety Administration want you to be alert when bicycling near and/or crossing railroad tracks.

Remember - highway-rail grade crossings are the only places where it's legal for bicyclists, pedestrians and/or vehicles to cross railroad tracks!

TO STAY SAFE, KEEP THESE LIFESAVING TIPS IN MIND:

Look out! If you see a railroad crossing, Always Expect a Train -- on any track, in any direction!

Watch for warning signs and pavement markings as you approach the crossing. At the crossing, look for crossbuck signs, stop signs, flashing lights and gate arms.

Think before you cross! LOOK in both directions. LISTEN for a train. PROCEED across the tracks only after making sure that no trains are coming and that no warning devices are activated.

Don't let your bike wheels get caught in the rails. Always try to cross at a 90-degree angle to the tracks. Never bicycle across tracks at less than a 45-degree angle -- instead, dismount and walk your bike across.

Did you know that an optical illusion makes trains seem farther away and slower moving than they actually are? Don't take chances by trying to "beat" a train across the tracks!

If you see or hear a train coming, or if warning lights start flashing and/or gates are lowering, SLOW DOWN AND STOP a safe distance (at least 15 feet) from the railroad tracks.

Stay alert at crossings with more than one track! Even after a train passes, before crossing look and listen for other trains on other tracks coming from either direction.

Wet train tracks can be slippery. Be extra careful when crossing railroad tracks if it's rainy, snowy, foggy or just plain wet. Dismount and walk your bike across the tracks if the crossing looks hazardous. Step over the rails, not on them.

Gravel service roads and green space beside railroad tracks are usually railroad property. It's illegal -- and dangerous -- to ride your bike on these areas.

Trains are wider than the tracks! Locomotives and railroad cars can extend as much as three feet beyond the rails on both sides. When a train is passing, stay at least 15 feet from the tracks, behind any gates or "stop lines" marked on the pavement.

Some railroad crossings can be rough. Slow down and be careful -- a bumpy crossing may cause you to lose control of your bike and loosen accessories or cargo.

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Risk Management

As noted earlier, risk management begins with the planning and design of the trail, continues through construction, and is implemented as the trail is placed into service. Trail managers must periodically evaluate its risk management activities and adapt them as necessary to ensure the safety of trail users.

IMBA's publication "Managing Mountain Biking: IMBA's Guide to Providing Great Mountain Biking," presents 16 considerations for risk management planning.

Risk Management Considerations to Ensure a Standard of Care

- Design and construct the trail appropriately
- Inspect and maintain the trail appropriately
- Address unreasonable hazards and post warnings
- Anticipate foreseeable activities and take reasonable care to protect users

IMBA's 16 principals summarize the components of a good trail risk management plan.

IMBA's 16 Considerations for Risk Management Planning

1. **Create a risk management team and designated leader:** Recruit a team that will be responsible for making sure the risk management program is developed, constructed, and documented. Designate a risk management coordinator. This person must be conscientious and provide attention to detail. This person will also serve as the point person for the risk management team.
2. **Write a philosophy statement:** Document your organization's commitment to safety and risk.
3. **Establish a trail design and construction policy:** Adopt policies and guidelines for trail design and construction and ensure your organization abides by them.
4. **Establish a trail inspection and maintenance policy:** Many lawsuits are related to faulty maintenance practices rather than improper design. Develop an inspection and maintenance plan and make sure it is followed. Specify the frequency of inspections, provide a detailed inspection checklist, and have a policy for reporting and addressing maintenance issues/concerns within a reasonable time frame. File inspection and maintenance records in case they need to be referenced at a later date.



5. **Maintain the trail system as specified in the policy:** Your trails should correspond to trail design and construction policies as well as inspection and maintenance policies.
6. **Eliminate unreasonable hazards:** Eliminate unreasonable hazards if they cannot be eliminated, post warning signs.
7. **Develop an effective sign program:** Signs are vital to a risk management program. Signs should indicate trail difficulty, remind visitors of trail rules, encourage responsible riding, etc.
8. **Implement a trail difficulty rating system:** Post clear signs that indicate the experience level needed and the relative level of difficulty of each trail.
9. **Institute rules and advise users:** Post rules at key locations to advise users of their responsibilities.
10. **Develop an emergency action plan:** An emergency action plan is often overlooked. Users expect reasonable medical care will be provided if they are injured and the failure to provide it may expose you to liability when an accident occurs.
11. **Purchase or review an insurance policy:** Insurance is a necessary and worthwhile nuisance. Consult a risk management expert to recommend the best way to reduce your liability.
12. **Create a record keeping system:** Instituting a risk management program is only valuable if your efforts are documented and can be located and provided to defend yourself when a lawsuit does occur.
13. **Develop an incident reporting system:** Analyzing incidents will allow you to identify and address hazards and emergency situations.
14. **Deploy a trail patrol:** A volunteer trail patrol can be the “eyes and ears” in the field. They can enhance user experiences, help land managers with inspection and maintenance, promote trail stewardship, and provide incident response.
15. **Recruit outside advisors:** Enhance your risk management program by consulting with a risk management consultant or professional recreation manager to expose potential liabilities that you may have missed.
16. **Perform periodic reviews of the program:** Keep your plan updated and revise it to evolve with your trail system.



Glossary of Terms

Abutment: Structure at either extreme end of a bridge that supports the superstructure (sill, stringers, trusses, or decks) composed of stone, concrete, brick, or timber.

Access: The opportunity to approach, enter, or make use of public lands.

Access points: Designated areas and passageways other than a trailhead that allow the public to reach a trail.

Access, public: The right of passage, established by law, over another's property. Can be created by an easement dedicated or reserved for public access. Legal public access exists on public land, public waters, public rights-of-way, and public easements.

Accessible (wheelchair accessible, handicap accessible, disabled access): A term used to describe a site, building, facility, or trail that complies with the Americans with Disabilities Act (ADA) Accessibility Guidelines and can be approached, entered, and used by physically disabled people.

Accessible route: A continuous, unobstructed path connecting all accessible elements and spaces of a facility or building that meets the requirements of ADAAG.

Alignment: The configuration of the trail in horizontal and vertical planes. The bends, curves, and ups and downs of the trail. The more the alignment varies, the more challenging the trail.

Amenities: Any element used to enhance the user's experience and comfort along a trail or during a trailhead or park visit.

Americans with Disabilities Act of 1990 (ADA): A federal law prohibiting discrimination against people with disabilities. Requires public entities and public accommodations to provide accessible accommodations for people with disabilities.

Americans with Disabilities Act Accessibility Guidelines (ADAAG): Design guidelines for providing access to a range of indoor and outdoor settings by people with disabilities.

Angle of repose: The maximum angle of a slope, measured from a horizontal plane, at which loose, cohesionless material will come to rest on a pile of similar material.

Assessment, trail or corridor: Physical evaluations undertaken to better understand a trail or corridor. Assessments include an accurate description and documentation of native elements and an inventory of built structures along the trail or corridor.

At-grade crossing: When a trail intersects and crosses directly in the path of another user.

Backcut: The vertical part of a bench cut that is blended into the backslope.

Backslope (cutslope, cutbank, backcut): The cut bank along the uphill side of the trail, extending upward from the tread, and transitioning into native hillside by varying degrees, depending on bank composition and slope stability.

Backfill: Material used to fill a void behind a trail structure.

Base course: The main load spreading layer of the constructed trail, normally constructed of crushed stone.



Ballast: Stone, cinders, gravel, or crushed rock fill material used to elevate a railroad bed above the surrounding grade. It drains water away from the ties, spreads the track load over softer subgrade, provides an even bearing for ties, holds ties in place and checks the growth of grass and weeds.

Barricade: A portable or fixed barrier having object markings, used to close all or a portion of the trail right-of-way to traffic.

Barrier: A structure installed to protect an environmentally sensitive area. A barrier can be hard (fence); live (planted); a combination of hard and live; or a terrain feature (berm). A barrier can be physical (obstructing passage) or psychological (detering access).

Barrier-free design: A trail design that promotes the elimination of physical barriers that reduce access by people with disabilities.

Base course: The layer or layers of specified material of designed thickness placed on a trailbed to support surfacing.

Base map: A map showing the important natural and built features of an area.

Batter: Leaning back from vertical towards the uphill slope.

Bed: The excavated surface of a trail upon which the tread or finished surface lies.

Bench: The excavated surface of a trail across a sidehill location; a full bench is constructed with no fill, the tread being entirely located on undisturbed mineral soil; partial bench is partially built on fill material.

Bench cut, full: The total width of the trail tread is excavated out of the slope, and the trail tread contains no compacted fill material. The most durable and recommended style of bench cut trail.

Bench cut, half: Half the width of the trail tread is excavated out of the slope and the downhill (outside) half of the trail tread contains the excavated and compacted material.

Bench cut, partial: Part of the width of the trail tread is excavated out of the slope and the rest of the trail tread is made up of fill material.

Berm: A raised outside edge of a trail.

Best management practices: Practical guidelines that can be used to reduce the environmental impacts of land uses or operations by means of careful planning, location, design, construction, management, and maintenance.

Best practices: Those that offer exemplary or model planning guidelines, design standards, development strategies, and management programs that lead to successful trails and programs.

Blaze: A painted trail marker.

Buffer (buffer zone, buffer strip): Any type of natural or constructed barrier used between the trail and adjacent lands to minimize physical or visual impacts. Buffers also provide a transition between adjacent land uses.

Capacity building: Activities that improve an organization's ability to achieve its mission or a person's ability to define and realize their goals or do their job more effectively.

Catch basin: The excavated or constructed basin at the inlet of a culvert cross drain pipe used to collect water and direct it to a culvert pipe. Catch basins slow velocity of moving water, which encourages sediments to drop out of the flow before entering the pipe.

Center Line: An imaginary line marking the mid-line of the trail tread. During construction, the center line is usually marked by placing a row of flags or stakes.



Clearing height (vertical clearance): The minimum vertical height which must be cleared of all tree branches and other obstructions that would otherwise obstruct movement along the trail.

Clearing limit: The area over and beside a trail that is cleared of trees, limbs, and other obstructions.

Climbing turn: A trail that continuously gains elevation as it changes direction, similar to a switchback but does not include a level platform at the apex of the turn. Rather, the trail follows the fall line for a short segment. On gentle grades a climbing turn is acceptable, on steeper grades it will result in erosion on the fall line segment of the turn.

Clinometer: Hand-held measuring tool used to measure the slope between two points.

Compacted: The degree of soil consolidation that is obtained by tamping with hand tools or heavy equipment.

Compaction: Tendency of soil to lose pore space and become compressed and impermeable to water when repeatedly traveled upon.

Contour: Line on a map connecting points of the same elevation. Parallel contour lines on a map represent a vertical change in grade in the landscape.

Control point(s): Places that influence where a trail goes. These features should be flagged and used to help layout a trail. The beginning and end of a trail are basic control points. Other control points include parking areas, trailheads, structures, slopes for turns or switchbacks, road or water crossings, and other trails. Also features that trail users will want to naturally head towards, or try to avoid.

Control point, negative: Places trail users should avoid.

Control point, positive: Places trail users want to visit.

Creep: Slow mass movement of soil down relatively steep slopes, primarily by gravity and water.

Cross drain: Installed or constructed structures such as culverts and pipes that move water from one side of the trail to the other.

Cross section: A drawing depicting a section of the trail sliced across the width.

Cross slope: The gradient of the hill slope as measured directly down the fall line.

Culvert: Hollow pipe for draining water beneath a trail.

Cut and fill: A method of trail construction in which the trail is built by cutting into the hillside and spreading and compacting the spoil materials downslope.

Cut slope: An earthen slope that is cut. For example, a trail built lower than the existing terrain would result in a cut slope.

Daylighting: Clearing a ditch or drain so that water can run freely, or all the way to “daylight.” Clearing of vegetation that is shading a trail or road corridor.

Deadfall: A tangled mass of fallen trees or branches that have died due to disease, insect or other damage, or decay.

Deberming: Removing the ridge of dirt that forms on the downhill side of the trail, preventing water from flowing across the trail.

Debris: Any undesirable material that encroaches on a trail and hinders the intended use

Descent: A downward incline or passage or process of descending from a higher to a lower location.

Desire line: An informal path that users prefer to take to get from one location to another rather than using the designated trail.



Dip (grade dip, drainage dip, rolling dip, rolling grade dip, broad-based dip): A reverse change in grade or gradual dip in the grade of the trail.

Displacement: The forces moving material sideways.

Disturbance: Any management activity that has the potential to accelerate erosion or mass movement. Also, any other activity that may tend to disrupt the normal movement or habits of a particular wildlife or plant species.

Disturbed area: Area where vegetation or topsoil has been removed, or where topsoil, spoil, or waste has been placed.

Downed tree: Fallen tree that blocks the trail.

Downslope: The downhill side of a trail.

Drainage ditch: Open ditches that collect water and carry it away from the site or trail. A drainage ditch is also an element of a waterbar, providing an escape route for water diverted from the trail by the bar.

Drainage structure: A water diversion structure constructed across the trail tread to remove water flowing down the trail tread or to prevent it from entering the tread.

Drop-off: Slope that falls away steeply on the outer edge of a trail.

Easement: Grants the right to use a specific portion of land for a specific purpose or purposes. Easements may be limited to a specific period of time or may be granted in perpetuity; or the termination of the easement may be predicated upon the occurrence of a specific event. An easement agreement survives transfer of land ownership and is generally binding upon future owners until it expires on its own terms.

Easement, conservation: Places permanent restrictions on property in order to protect natural resources.

Easement, construction: Temporary area or corridor needed to construct a trail or facility.

Easement, maintenance: Additional permanent area or corridor (not open to the public) needed to maintain the trail and adjacent vegetation.

Easement, scenic: Places permanent restrictions on a property in order to protect the natural view.

Elevation: The height of a place (mountain or other landmark) given in the number of feet or meters above sea level.

Embankment: An artificial deposit of material that is raised above the natural surface of the land and used to contain, divert, or store water, support roads, railways, trails, or for other similar purposes.

Equestrian: Of horses, horseback riding, riders, and horsemanship.

Equine: Hoofed mammals having slender legs and a flat coat with a narrow mane along the back of the neck.

Erosion: Natural process by which soils move downhill or downwind; may be greatly accelerated on trails due to water, wind, and user traffic.

Erosion, accelerated: Soil loss above natural levels resulting from human activities

Erosion control: Techniques intended to reduce and mitigate soil movement from water, wind, and user traffic.

Event: A single, structured, organized, consolidated, or scheduled meeting or occurrence for the purpose of recreational use. An event may be composed of several related activities.



Exotic species: A plant introduced from another country or geographic region outside its natural range.

Exposure: The relative hazards encountered when on trails; takes into consideration obstacles, alignment, grade, clearing, tread width, tread surface, sideslope, isolation, and proximity to steep slopes or cliffs.

Fall line: Straight up or down a slope; flowing water will follow the fall line; the fall line is perpendicular to the contour line.

Zone fall: The area on either side of or below a technical trail feature that provides a clear landing for a rider who has failed to negotiate the obstacle.

Fill: Material (usually mineral soil and rock) excavated from the trail or a borrow site to fill holes in trail tread, stabilize rock steps, or to pack behind retaining walls and other structures.

Fill Slope: Area of excavated material cast on the downslope side of trail cut (also called embankment).

Firmness: The degree to which a surface resists deformation by indentation when, for instance, a person walks or wheels across it. A firm surface would not compress significantly under the forces exerted as a person walks or wheels on it.

Flagging: Thin ribbon used for marking during the location, design, construction, or maintenance of a trail project.

Flagline (Flag Line): A series of flags indicating the intended route for trail construction. Ribbon usually tied in trees or pin flags placed on the centerline, inside edge, or critical edge of the proposed trail project.

Flags, pin (wire flags, wire stake): Wire wands with square plastic flags at one end for field layout and marking of new trail or relocations of trail sections.

Ford: An unimproved route across a stream usually selected for its wide, shallow character, and typically containing a stone bottom.

Gabion (gabion baskets): Rectangular containers that can be wired together, and then filled with gravel or cobble to make quick retaining walls for erosion control.

Geocaching: Involves hiding a small object in a remote location and recording its location using a GPS unit. The coordinates, along with a few helpful hints, are then posted on a website for other GPS-wielding geocachers to look up and then hunt for—a modern day treasure hunt.

Geocells: Polyethylene strips bonded together to make a three-dimensional honeycomb structure. Fill material placed within the cells stabilizes and reinforces soil by confining substrates in cells to prevent lateral movement.

Geographic Information System (GIS): A spatial database mapping system (computer and software) that contains location data for trails and other important features.

Geogrid(s): Polyethylene sheeting configured into an open grid with high tensile strength. Used for reinforcement and often placed on top of a layer of geotextiles to provide separation.

Geotextile fabric: Textile made of synthetic fibers, usually non-biodegradable, to form a blanket-like product. In trail construction these are used as moisture barriers, for separation, for reinforcement of soils, filtration, and for drainage.

Global Positioning System (GPS): Small handheld receiver or unit that reveals a user's latitude, longitude, and altitude by timing signals received from orbiting satellites maintained for navigation by government agencies.



Grade: The slope of the trail along its alignment. The slope is expressed in a percent ratio, or the ratio of the elevation compared to the distance traveled, rise over run.

Grade reversal: A spot at which a climbing trail levels out and then changes direction by dropping subtly, before rising again. Grade reversals are known by several different terms, including grade dip, grade brake, drainage dip, and rolling dip.

Grade separated crossing: When a trail crosses the path of another user independently without impeding the other user.

Guideline(s): A statement and/or illustration describing a recommendation or principle for a preferred development technique or a course of action. Guidelines are not mandatory actions.

Hardening: Process of improving or firming the trail's treadway so it can withstand traffic and erosion.

Hazard tree (danger tree, widow maker): Tree or limb that is either dead, or has some structural fault, that is hanging over, or leaning towards the trail or sites where people congregate.

Hazard(s), foreseeable: Hazards that trail builders or managers might perceive, but the average user would not see. An example is dead snags that serve as hazard trees when close to a trail.

Hazardous materials: Anything that poses a substantive present or potential hazard to human health or the environment when improperly treated, stored, transported, disposed of, or otherwise managed.

Headwall: A support structure at the entrance to a culvert or drainage structure.

Headwaters: The area in the upper reaches of a watershed typified by unconfined surface water flows. Headwaters can coalesce to form rivulets or first order streams with distinct channels. Headwaters can often be ephemeral (wet only part of the year).

Helical pier: Steel post with auger-shaped bit end that is rotated into wet soils either by hand or with specialized hydraulic tools to establish a foundation for a boardwalk.

Hub(s): Area(s) that anchor a network and provide an origin or destination for elements moving to or through it.

Impermeable material: A soil or material whose properties prevent movement of water.

Impervious surface: Hard surfaces that do not allow absorption of water into the soil and that increase runoff. Examples of such surfaces include concrete or asphalt paved trails and parking areas.

In-kind contribution(s): Labor or materials donated toward the match for a grant. Can include value of donated labor or equipment, real property, professional services, materials, etc.

Infiltration: The portion of rainfall or surface runoff that moves downward into the subsurface rock and soil.

Inlet: The opening in a drainage structure of pipe where the water first enters the structure.

Inslope: A trail cross section that is sloped towards the uphill side. Insloped trails drain towards parallel swales that collect water which is periodically conveyed under the trail to suitable discharge areas.

Interpretation: Communicating information about the natural and/or cultural resources and their associated stories and values found at a specific site or along a trail. Tours, signs, brochures, and other means can be used to interpret a particular resource.



Interpretive: Technique that assists audiences through communication media in making both emotional and intellectual connections with heritage resources.

Interpretive center: A facility where opportunities are provided for people to forge emotional and intellectual connections between their interests and the meanings that arise from learning about the resource. The facility may or may not be staffed, and can range in scale from a kiosk to a complex of buildings and natural sites, but always provides information about the natural and cultural resources.

Interpretive display (sign): An educational display usually in an interpretive center or at a trailhead that describes and explains a natural or cultural point of interest on or along the trail.

Intersection: Area where two or more trails join together.

Inventory: The systematic acquisition and analysis of resource information needed for planning and management purposes.

Invert: The bottom surface of a pipe, ditch, or culvert over which water flows.

Invasive exotic: Non-native plant or animal species that invades an area and alters the natural mix of species.

Knick: Shaved-down section of trail, about 10 feet in diameter, with an exaggerated outslope. Like a rolling grade dip, a knick is used to shed water off a trail and is a useful remedy for wet spots on relatively flat trails.

Land Ethic: The desire humans have to conserve, protect, and respect the native landscape and other natural resources because their own well being is dependent upon the proper functioning of the ecosystem.

Leadership in Energy and Environmental Design (LEED): A green building rating system for the design, construction, and operation of high-performance green buildings. LEED recognizes performance in five key areas of human and environmental health: sustainable site development, water savings, energy efficiency, materials selection, and indoor environmental quality.

Lease: The temporary grant of an interest in land upon payment of a determined fee. The fee does not have to be monetary, but some consideration must be given for the right to use the land, or the lease will not be legally binding.

Leave No Trace: Educational program designed to instill behaviors in the outdoors that leave minimum impact of human activities or occupation.

Lead off ditches: Excavations designed to divert water away from the trail or trail side swale in order to reduce water volume and velocity in the swale.

Length: Dimension of a feature measured parallel to the direction of travel.

Level spreading: A way of preventing concentrated flows and erosion by maintaining water runoff as sheet flow and dispersing it into dense vegetation for filtration and infiltration.

Low water crossing: A low water crossing is a constructed feature that creates a temporary stream crossing that is expected to wash out during high water.

Maintenance: Repair, improvements, or other work that is carried out on or near a trail to keep a trail in its originally constructed serviceable condition or to improve the safety and sustainability of the site. Usually limited to minor repair or improvements that do not significantly change the trail location, width, surface, or structures.



Maintenance, annual: Involves four tasks done annually or more often as needed: cleaning drainage, clearing blowdowns, brushing, and marking.

Maintenance, cyclic: Preventive maintenance activities that recur on a periodic and scheduled cycle.

Maintenance, deferred: Road/trail maintenance that was not performed when it should have been or when it was scheduled and which, therefore, was put off or delayed for a future period. When allowed to accumulate without limits or consideration of useful life, deferred maintenance leads to deterioration of performance, increased costs to repair, and decrease in asset value. Trail repair, rehabilitation, replacement and/or decommission can reduce or eliminate deferred maintenance. Used to be referred to as “backlog.”

Maintenance, heavy: Work usually done to repair damage normally expected from seasonal and occasionally unusual natural conditions or occurrences.

Maintenance, routine: Work that is planned to be accomplished on a continuing basis, generally annually or more frequently.

Marker, trail: An appropriate and distinctive symbol with the name of the trail, used to mark a trail route.

Marker, reassurance (confidence marker): Reconfirms the identity, location, or route of the trail.

Memorandum of Understanding/Agreement (MOU/MOA): A signed, written agreement entered into by various governmental agencies and nonprofit groups to facilitate the planning, coordination, development, and maintenance of a trail or trails system.

Mineral soil: Soil free of organic matter used for trail tread construction.

Minimum tool rule: Use the right tool for the job. A principle that guides the use of tools, and especially equipment, that are efficient, cost effective, and minimize the likelihood of environmental damage.

Minor field adjustments: Deviations of the trail alignment made during the course of normal construction or maintenance as determined by the supervisor or crew leader, and not part of an original survey.

Mulch: Organic matter spread over recently disturbed areas to help stabilize soils and protect them from erosion.

Multimodal: Facilities serving more than one transportation mode, or a transportation network comprised of a variety of modes.

Multiuse trails (also referred to as shared use paths): Refers to trails designated for pedestrian, equestrian and mountain bicycle or other non-motorized wheeled use, such as wheelchairs (with or without motors). Multiuse trails can also include accessible trails.

National Scenic Trail: a trail designated by the Secretary of the Interior or Secretary of Agriculture as part of the national system of trails authorized under the National Trails System Act of 1968, www.nps.gov/legacy/nomenclature.html

Nonmotorized: Trail recreation by modes such as bicycle, pedestrian, equestrian, skate, ski, etc.

Noxious weeds (noxious plant): Plant species designated by federal or state law as generally possessing one or more of the following characteristics: aggressive and difficult to manage; parasitic; a carrier or host of serious insects or disease; or nonnative, new, or not common to the United States.



Organic soil: Top layer of soil which contains roots, leaf litter, and decomposing branches and other vegetative materials.

Other power-driven mobility device (OPDMD): Any of a large range of devices powered by batteries, fuel, or other engines - whether or not designed solely for use by individuals with mobility impairments - that are used by individuals with mobility impairments for the purpose of locomotion, including golf cars, bicycles, electronic personal assistance mobility devices (EPAMDs), or any mobility aid designed to operate in areas without defined pedestrian routes.

Outfall (outlet): The drainage channel of a dip, waterbar, or switchback drainage structure where the water exits the structure.

Outflow: Water and its sediment load once it leaves the trail and the area it flows through.

Outlet protection: Devices or material placed at the outlet of pipes or drainage structures to dissipate energy of flowing water, reduces its flow velocity, and prevent erosion.

Outslope: A trail cross section that is sloped towards the downhill side of the trail to facilitate efficient movement of water runoff across and off the trail.

Overlooks and viewing areas: Trail features that are specifically designed and constructed to provide unobstructed observation of a vista or a specific point of interest.

Passing space: A section of trail wide enough to allow two users to pass one another or travel abreast.

Passing space interval: The distance between passing spaces.

Pavement: Any hard covering of stones, bricks, tiles, asphalt or concrete formed over a solid base to create a smooth surface. That part of a trail having a constructed hard paved surface for the facilitation of wheeled trail traffic.

Pavement, porous: A special type of pavement that allows rain and snowmelt to pass through it, thereby reducing the runoff from a site and surrounding areas.

Percent slope: Number of feet rise, vertical, divided by feet of run, horizontal, multiplied by 100 to get percent slope.

Permit (system): Use-authorization forms issued by agencies to control the amount of use along trails or in wilderness areas.

Pitch: A short section of trail that is steeper than the maximum design grade.

Porous Concrete: A concrete mix that has few fines, resulting in many small voids through which water can pass.

Prescription: Management practices which are selected and scheduled for application in a specific area in order to attain goals and objectives.

Quadrangle, 7.5-minute: A USGS paper map at 1:24,000 scale covering 7.5 minutes of latitude and 7.5 minutes of longitude. Therefore one map covers approximately nine miles in a north-south direction and approximately 6.5 miles east-west. Features shown include elevation contours, roads, railroads, trails, water bodies, buildings, and wetlands. An automated version of the 7.5-minute quadrangle is called the digital raster graphic or DRG. It is informally known as 7.5-minute quad.

Quiet Title: An action brought in state court to establish legal rights to property. In the legal proceeding called “action to quiet title,” the word “quiet” means to pacify; to render secure or unassailable by the removal of unsettling causes or disputes. Under this proceeding, the plaintiffs title to land is established by bringing into court an adverse claimant and there compelling them to either establish their claim or be forever after stopped from asserting it.



Quitclaim deed: Document that transfers ownership of real estate, but contains no guarantees that the seller has a valid right to do so, or that others do not have rights to the land.

Rail-with-Trail: A rail trail constructed within or adjacent to an active railway right-of-way.

Railbank(ing): Retaining a rail corridor for future railroad uses after service has been discontinued. The National Trails System Act provides for interim public use of the corridor, allowing the establishment of recreational trails.

Railroad bed: The earth bank or bed supporting railroad tracks.

Rail trail: Conversion of a railway easement into a shared use path.

Ravel: Constant movement of loose or coarse material on a slope. Also a process where the coarse material on a trail surface comes loose and separated from the trail tread because of lack of binder or poor gradation of tread material.

Rebar: Steel reinforcing rod, available in a variety of diameters, useful for manufacturing pins or other trail anchors.

Relocation: New and different location for the trail.

Register, trail: Located at overnight stops or trailheads to allow users to make comments to those behind them, and read comments from those ahead. Registers can be an important safety measure to pinpoint the location of trail users.

Rest area: A level portion of a trail that is wide enough to provide wheelchair users and others a place to rest and gain relief from the prevailing grade and cross slope demands of a path.

Rest area interval: The distance between rest areas.

Restore: To bring back to a former, normal, or productive condition by repairing or rebuilding.

Restrictions, road or trail: Limitations placed on the use of a road or trail. Sample codes: S-seasonal closure, Y-closed yearlong to motorized vehicles, R-restriction on types of traffic allowed on road or trail, L-limitations on vehicle dimensions, weight or speed, N-no restrictions applied, B-no bicycles, E-no equestrians, M-no motorized vehicle, P-permit required for use.

Restroom (comfort station, pit toilet, vault toilet, composting toilet, chemical toilet, latrine, bathhouse): Facility for human waste disposal.

Revegetation: Process of planting or transplanting vegetation on bare soil to develop plant cover.

Reversionary interest (reversion): The right of a property owner to the future enjoyment of property presently in the possession or occupancy of another.

Ridgeline: A line connecting the highest points along a ridge and separating drainage basins or small-scale drainage systems from one another.

Right-of-Way: A linear corridor of land held in fee simple title, or as an easement over another's land, for use as a public utility (highway, road, railroad, trail, utilities, etc.) for a public purpose. Usually includes a designated amount of land on either side that serves as a buffer for adjacent land uses.

Rills: The result of concentrated flow of water resulting in erosion, typically up to one inch in depth. Gullies are considered to have developed when rills exceed one inch of depth.

Right of way: A legal easement that grants the one person right to pass over the land of another.

Riparian (riparian zone, habitat zone): A habitat that is strongly influenced by water and that occurs adjacent to streams, shorelines, wetlands, or other water bodies, dominated by high soil moisture content and influenced by adjacent upland vegetation.



Riparian vegetation: Plant species growing adjacent to a wetland area, including a perennial or intermittent stream, lake, river, pond, spring, marsh, bog, meadow, etc.

Riprap: Well graded, durable, large rock, ideally with angular surfaces, sized to resist scour or movement by water and installed to prevent erosion of the soil.

Risk, assumption of: The legal concept that users assumed and perceived an activity was potentially dangerous and the users willingly assumed those risks and participated anyway.

Risk, perception of: Subjective judgment that people make about the characteristics and severity of a risk. For example, perceiving a steep drop off on a trail as being dangerous and therefore not getting too close to the edge.

Risk management: An element of safety management that evaluates the effects of potential hazards on safety by considering acceptance, control, or elimination of such hazards with respect to expenditure of resources.

Road, designated: Specific roads identified by the land management agency where some type of use is appropriate and allowed either seasonally or yearlong and which have been inventoried and mapped and are appropriately signed on the ground.

Road, forest: A temporary or permanent road connecting parts of the forest to existing public roads. They provide access to forestland for timber management, fish and wildlife habitat improvements, fire control, and a variety of recreational activities.

Road-crossing: Intersection of trail and road traffic.

Rolling dip: Water flowing over the surface of an unpaved trail always carries sediment with it, no matter how carefully the trail is designed and constructed. A rolling dip functions by collecting runoff and sediment at a low point in the trail and routing them off the trail to daylight. The frequency and locations of dips are based on trail width, grade, tread material, and soil erodability.

Ruts (rutting, entrenchment): Sunken tracks or grooves in the tread surface cut in the direction of travel by the passage of trail users or water.

Saddle: A topographical term for a “sagging” ridge that connects two summits.

Scour (scouring): Soil erosion through the force of moving water.

Sediment: Soil particles that have been transported away from their natural location by wind or water action and re-deposited in a different area down-slope or down-stream.

Sediment deposition: The accumulation of soil particles on the trail tread and banks.

Sedimentation: Deposition of soil particles or other material carried in water; usually the result of a reduction in water velocity below the point at which the material remains in suspension.

Segment: A portion of a trail.

Sensitive species: A species that is listed under the Endangered Species Act as threatened, endangered, candidate or proposed by the Pennsylvania Natural Heritage Program or is a species of concern for listing.

Shared use path: A bikeway physically separated from motorized vehicular traffic by an open space or barrier and either within the highway right-of-way or within an independent right-of-way. Shared use paths may also be used by pedestrians, skaters, wheelchair users, joggers and other non-motorized users. Rail trails are typically classified as shared use paths.



Shared use trail (multi-use trail): A combination of harmonious recreation uses sharing the trail resource.

Sheet flow: Desirable condition in which water flows in smooth sheets rather than concentrated flow; slower flow (velocity) and less concentration (volume) results in less erosion.

Shoulder: The surfaced or unsurfaced width of a trail next to the tread, or travelway.

Shy distance: The distance between the trail's edge and any fixed object capable of injuring someone using the trail.

Sidehill: A trail location that cuts across a slope, following the contour, as opposed to a fall line trail that goes straight up and down the slope, or a ridgeline trail that follows the top of slope.

Sidehilling: Process of excavating or cutting a bench across a slope.

Sideslope: Angle on inclination of land along the fall line, usually measured in percent slope or in degrees from horizontal.

Sight distance: How far a person can see along an unobstructed line of sight.

Sight stopping distance: Sight distance while taking into consideration the time it takes a traveler to see something, react to it, and to stop safely before the object.

Sign (signage): A board, post, or placard that displays written, symbolic, tactile, or pictorial information about the trail or surrounding area. Signage increases safety and comfort on trails. There are five basic types of signs: Cautionary, Directional, Interpretive, Objective, and Regulatory.

Sign, cautionary: Warns of upcoming roadway crossings, steep grades, blind curves, and other potential trail hazards.

Sign, directional: Gives street names, trail names, direction arrows, mileage to points of interest, and other navigational information.

Sign, interpretive: Offers educational information that describes and explains a natural or cultural point of interest on or along the trail.

Sign, objective: Provides information about the actual trail conditions, including grade, cross slope, surface, clear trail width, and obstacle height. This allows users to make more informed decisions about which trails best meet their trail needs and abilities.

Sign, regulatory: Tells the “rules of the trail” by prohibiting certain uses or controlling direction of travel.

Slope: Angle of the ground from a level position measured as a percent of rise over run, or in degrees. Generally refers to the hillside (land) and not the trail, as trail “slope” is called the grade.

Slope, cross: The slope or gradient of the undisturbed hillside; the amount or grade of the pre-existing sloped that is perpendicular to the direction of the trail.

Slope, running: The average slope of a contiguous section that is in the same direction as the trail; measured by averaging the values of slope measurements taken periodically at different points along the trail.

Slope, running cross: The average cross slope of a contiguous section of a trail; measured by averaging the values of cross slope measurements taken periodically at different points along a given section of trail.

Slope, percent: Number of feet rise (vertical) divided by feet of run (horizontal) times 100 to get percent slope.

Slope, toe: When soil and rock move downslope and come to rest, they form a toe slope, named for its similarity to the human form.



Slope stability: The resistance of a natural or artificial slope or other inclined surface to failure by mass movement.

Social/renege trails: Informal, unofficial paths or shortcuts that have been created over the years by consistent human use. While these may appear no different than other trails to users, social trails tend to impact natural, cultural and historic resources more than other routes that were designed and constructed as trails. Social trails need to be evaluated to determine whether they are appropriate and sustainable, or whether they should be renovated or closed as part of proposed trail system improvements.

Soils: The surface material (mineral materials, organic matter, water, and air) of the continents, produced by disintegration of rocks, plants, and animals and the biological action of bacteria, earthworms, and other decomposers. The four fundamental groups of soils are: gravels, sands, loams, and clays.

Soil, erodible: Soils that are likely to have high soil loss when exposed to water runoff.

Soil, heavy: A soil with a high percentage of clay.

Soil, Hydric: Soil that is saturated or flooded during a sufficient portion of the growing season with anaerobic conditions in the upper soil layers. Examples include bogs, marshland, and swampland soils.

Soil, Inorganic: Mostly sandy soils containing little or no plant and animal remains.

Soil, Light: A coarse-textured soil with a high percentage of sand and a low percentage of clay.

Soil, mineral: A soil comprised of rock fragments, sand, and smaller sized particles, and relatively free of organic matter. Mineral soil is typically buried under layers of surface organic matter. It is relatively stable and is the preferred material upon which to build a trail tread. When compacted, it also provides good support to rock walls and other trail structures.

Soil, organic: Soil that is made up of leaves, needles, plants, roots, bark, and other organic material in various stages of decay, and has a large water/mass absorption ratio. Generally the first (outermost) layer of soil.

Soil, residual: A soil formed from and resting on consolidated rock from which it was formed.

Soil, saturated: A condition in which all of the pore space in the soil is filled with water. This may be a temporary condition associated with large amounts of rainfall or snowmelt, a semi-permanent, or permanent condition.

Soil, surface: The uppermost part of the soil profile (solum).

Soil cement (Cement-Treated Base): A mixture of pulverized soil combined with measured amounts of portland cement and water and compacted to a high density. As the cementing action occurs through hydration, a hard, durable semi-rigid material is formed. It must have a seal coat to keep out moisture and a surface that will withstand wear.

Soil compaction: A decrease in the volume of soil as a result of compression stress.

Soil map: A map showing the kinds of soil types and their boundaries in all the detail significant to soil use and management.

Soil profile: Site-specific arrangement of soil layers from surface to bedrock.



Soil stabilization: Measures that protect soil from the erosive forces of raindrop impact and flowing water. They include, but are not limited to, vegetative establishment, mulching, and the application of soil stabilizers to the trail tread.

Soil stabilizer: Material, either natural or manufactured, used to hold soil in place and prevent erosion due to water, gravity, or trail users. Stabilizers include soil cement, geogrid, etc.

Species, invasive or exotic (alien, introduced, nonindigenous): Non-native plant or animal species that invades an area and alters the natural mix of species by aggressively out-competing native species.

Species, sensitive: Any plant or animal species for which population viability is a concern as evidenced by significant current or predicted downward trends in population numbers or density, or habitat capability that would reduce a species' existing distribution.

Species, threatened or endangered: Any plant or animal species that is in danger of extinction throughout all or a significant portion of its range, and has been officially listed as endangered by the Secretary of Interior or Commerce under the provisions of the Endangered Species Act. A final rule for the listing has been published in the Federal Register.

Specifications: Written provisions and requirements (standards) for the performance of work and type of materials to which trails (tread, clearing, grade) and trail structures (bridge, culvert, puncheon) are built and maintained according to type of use.

Sponsor: Organization or government agency that will sign agreements and contracts and be responsible for a trail or greenway project.

Stakeholders: Group or individual who can affect, or is affected by, the achievement of the organization's mission; examples include managers, employees, policy makers, citizens, and community groups.

Stakes, line: Temporary stakes set by the trail locator to establish the centerline of the trail.

Standards, design: The specific values selected and documented from the design criteria become the standards for a given trail or greenway project. These standards will be identified and documented by the designer. Standards will related to the physical characteristics, users, location, and environmental factors of a project.

Statewide Comprehensive Outdoor Recreation Plan (SCORP): Recreation management plan developed periodically (about every 10 years) by each State to help federal, state, and local agencies assess recreational use trends and the needs for future management and facilities.

Station: One hundred feet or other set measurement along the centerline of the trail or road; used in surveying and construction.

Steward: The person taking responsibility for the well-being of land and water resources and doing something to restore or protect that well-being.

Stewardship: Taking responsibility for the well-being of land and water resources and doing something to restore or protect that well-being. It usually involves cooperation among people with different interests and sharing of decision-making. It is generally voluntary. It is oriented towards assessment, protection, and rehabilitation of trails and greenways as well as sustainable use of renewable resources.

Subgrade: The surface of trail tread upon which the base course is constructed.

Spur trail: A trail which terminates at a destination, i.e. vista, campground, water source, etc.



Tamping: Using a narrow machine compactor, a tamping bar, or the handle of a shovel or other tool to compact earth backfill around a post, pole, or pile.

Terminus: Either the beginning or end of a trail.

Title search: A legal review of deeds of record in the chain of title to a piece of property analyzing all encumbrances or prior sales of the property, to make sure that a piece of real estate can be sold without anyone else claiming rights to it.

Toe: The break in slope at the foot of a bank (trail or stream) where the bank meets the bed.

Top bank (top of bank): The break in slope between the bank and the surrounding terrain.

Topographic (topo, USGS topographic, contour) map: Maps that indicate built and natural features (buildings, roads, ravines, rivers, etc.) as well as elevation changes and land cover. United States Geological Survey maps are available from many government offices, outdoor shops, and map stores; or from digitized versions on the Internet.

Topographic: Of or having to do with topography or the physical features of a place.

Topography: The elevation and slope of the land as it exists or is proposed. It is represented on drawings by lines connecting points at the same elevation. Typically illustrated by dashed lines for existing topography and solid lines for proposed.

Topsoil: The top layer of earth, dark and rich in organic matter.

Trail: Travel routes that are designed, designated or constructed for recreational hiking use or provided as a pedestrian alternative to vehicular routes within a transportation system.

Trail, access: Any trail that connects the main trail to a town, road, or another trail or trail system.

Trail bed: The tread plus base materials.

Trail clearing limit: The area beside and above a trail tread that is cleared of trees, limbs, and other obstructions; often the edges of the trail corridor.

Trail clearing width: The space to each side of the trail tread that is cleared for trail users. Typically, there is an uphill and downhill clearing width prescribed.

Trail corridor: The strip of land in which trails and other facilities are built. This includes the tread itself, the shoulder, and the cleared area adjacent to the trail. Multiple tread corridors include the tread, clearing, and vertical clearing width for all treads.

Trail, contour: A trail constructed or existing such that it follows a contour, with its elevation remaining constant. Constructed at full bench with outslope, grade reversals, and sustained grades.

Trail, fall line: Trail constructed on the fall line (direction water flows down a hill) which encourages water to run down the trail.

Trail, natural surface: A tread made from clearing and grading the native soil, and with no added surfacing materials.

Trail, regional: An extended or longer trail that may cross one or more land management agency jurisdictions and connects diverse trail systems.

Trail, rolling contour: A trail characterized by gentle grade, grade reversals, and outsloped tread.

Trail, sustainable natural surface: A trail that supports currently planned and potential future uses with minimal impact and negligible soil loss while allowing the naturally occurring plant systems to inhabit the area, recognizing required pruning and eventual removal of certain plants over time. The sustainable trail will require little rerouting and minimal maintenance over extended periods of time.



Trail access information: Objective information reported to trail users through signage, about the grade, cross slope, tread width, and surface of a trail.

Trail design: Designing and layout of trails requires special training, knowledge, experience, and skill. When designing trails, many different factors are taken into account including hydrology, topography, soils, flora, fauna, management objectives, user expectations and characteristics, and trail design standards. The designer will utilize data collected from area site analysis, environmental assessments, public meetings, and area trail and management plans.

Trail management objective (TMO): The goals and objectives pertaining to a specific trail management project (new construction, maintenance, reroute, bridge, etc.). Usually intended to provide guidance for a trail crew leader.

Trail system: A collection of individual trails that may or may not be connected to one another, whereby each retains its distinctiveness, and yet belongs to the system by association with a federal, state, local, or bioregional context.

Trailbed: The finished surface on which base course or surfacing may be constructed. For trails without surfacing, the trailbed is the tread.

Trail vertical clearance: The space over the trail that is clear of obstructions.

Trailhead: Points of access to trails intended for public use. Sometimes provides parking, information signs, shelter, etc.

Trailway: The portion of the trail within the limits of the excavation and embankment.

Trailway clearance: The trailbed plus the area to either side that is needed to accommodate the construction of cuts and fills to accommodate the trail.

Travel management (access and): A continuous process of analyzing, controlling, and regulating travel to accomplish management objectives. It is that portion of the planning and implementation process that develops clear specific direction on the appropriate levels of access and travel opportunities to be made available. It takes into account long-term social, biological, economic, and physical considerations; it combines a variety of design considerations that are commensurate with how access will be provided and travel will be managed; and it also involves sharing this information with the public.

Travel management areas: Polygons or delineated areas where a rational approach has been taken to classify areas open, closed, or limited, and that have identified or designated networks of roads and trails that provide for public access and travel across the planning area.

Travelway: The trail as a whole, including the trail tread and the cleared areas on either side of the trail.

Traverse: To cross a slope horizontally going gradually up and across in lieu of the more direct up-and-over (up the fall line) approach.

Tread (treadway): The travel surface of the trail.

Tread, trail (treadway): The surface portion of a trail upon which users travel, excluding backslope, ditch, and shoulder. Common tread surfaces are native material, gravel, soil cement, asphalt, concrete, or shredded recycled tires.

Tread creep: When the loose soil of the trail tread moves (sags or slides) downhill because of erosion or use. Specific causes include bushes or trees protruding into the trail from above, exposure of roots from an uphill tree, an improper bench cut, or poor trail flow.



Tread Lightly!: Educational program designed to instill outdoor ethics of responsible behavior when participating in outdoor activities, www.treadlightly.org.

Trench (trenched, trenching): Badly eroded trail in which the user travels in a ditch down the center that may be knee deep or deeper. Any long furrow or ditch cut in the ground.

Trespass: The use of private or public land without authority, resulting from an innocent, willful, or negligent act.

Trespasser: Person who uses property without the owner's implied or stated permission and not for the benefit of the property owner.

Tributary: A river or stream feeding into a larger waterway or lake.

Turn: Where a trail or river changes course or direction.

Turn, climbing: A turn on a hill to reverse direction that doesn't have a constructed turning platform or landing. The upper and lower legs of a climbing turn are generally joined by a short section of trail (the apex of the turn) that lies directly in the fall line. As a result, climbing turns located on hillsides with a grade of more than 7% are erosion prone and should be replaced with well-built switchbacks.

Turn, inside: On a trail traversing a hillside, concave, or naturally banked turns in which the sideslope helps direct trail users around the turn.

Turn, outside: Convex or off-camber turn (usually on trails that traverse hillsides) that are more difficult to navigate, as centrifugal force pulls trail users to the outside of the turn. A turn in which the ground slopes toward the outside, making it harder to keep (wheeled) traction as speed increases.

Turnout: A place where the trail is widened to permit trail traffic traveling in opposite directions to pass.

Underdrain: A buried trench filled with coarse aggregate, coarse sand, or gravel typically placed in a swale line along the trail. An underdrain functions by diverting subsurface water away from trail subgrade and conveying it to a suitable outlet point. Underdrains use uniform sized rock, may be wrapped in geotextile fabric, and may have a perforated drain pipe in the bottom of the trench.

Underpass: An underground tunnel or passage enabling trail users to cross under a road or railway.

Understory: All forest vegetation growing under the canopy or upper layers of forest vegetation.

Universal design: Access without limitation or exception.

Universal Trail Assessment Process (UTAP): An inventory process that can be used by trail managers to assess a trail to determine compliance with design guidelines and to provide objective information to trail users regarding grade, cross slope, tread width, surface, and obstacles.

Unravel: To lose material from the edges of a retaining wall (revetment, cribbing).

Upland: Land at a higher elevation than the alluvial plain or low stream terrace; all lands outside the riparian-wetland and aquatic zones

Vegetation, aquatic: The vegetation that grows as either submerged (below the surface of the water) or emergent (some portion of the plant floating on or rising above the surface of the water) in an aquatic habitat.

Vegetation, native: Indigenous species that are normally found as part of a particular ecosystem; a species that was present in a defined area prior to European settlement.



Walks: Exterior pathways (including sidewalks) with a prepared surface intended for pedestrian use. Although federal regulations make a distinction between trails and walks, portions of some trails at developed sites are classified as walks and are included under the general classification of trails.

Wall, retaining (Revetment, Cribwall, Cribbing, Mono-wall, Multi-Tier Wall): A structure used to prevent soil from slumping, sliding, or falling; usually made of log, stone, bags, block, or pavement. Often used to provide stability and strength to the edge of a trail or stream bank.

Wall, rubble: A roughly built wall or structure of irregular or greatly differing sizes of stone usually laid at or nearly at the same angle as the cross slope; frequently used to discourage shortcutting corners and to armor native slopes to prevent erosion.

Wall, wing (wingwall): A structural component of a retaining wall, which is interlocked with the facer or front of the wall. The wing generally intersects with the facer at a 45° angle, but may be at an angle between 1° and 90°. This component is anchored by tie logs and both assists the facer in retaining the fill material and helps prevent flanking.

Wash rack: Area set aside for cleaning or cooling animals after a ride or washing off bicycles.

Washout: Erosion of a relatively soft surface, such as a trail, by a sudden gush of water, as from a downpour or floods; a channel produced by such erosion.

Water course (watercourse): Any natural or built channel through which water naturally flows or will collect and flow during spring runoff, rainstorms, etc.

Water pollution: The presence in water of enough harmful or objectionable material to damage the water quality.

Water trough (stock tank): Used to provide drinking water for animals such as cattle or horses. Troughs or tanks can range in size from 30 to over 1500 gallons and typically are made of galvanized steel. Many riders prefer watering their stock in clean, freshly filled water troughs.

Water quality: The chemical, physical, and biological characteristics of water with respect to its suitability for a particular use.

Waterbar: A drainage structure (for turning water) composed of an outsloped segment of tread leading to a barrier (log, stone, or timber) placed at a 45° angle to the trail. Water flowing down the trail will be diverted by the outslope or, as a last resort, by the barrier. This type of drainage structure is no longer recommended for construction or use on trails. Grade dips are preferred instead.

Waypoint: A point between major points on a route, as along a track.

Waysides: The side or edge of a trail, road, way, path, or highway. Sites along a trail that allow users a place to stop to sit, rest, eat, enjoy a view, or read an informational display. They can be located where there are noteworthy natural or cultural resources, attractive views, or a lack of other nearby facilities.

Weathering: The physical and chemical disintegration and decomposition of rocks and minerals.

Wetland: Areas that are inundated by surface or ground water with a frequency sufficient to support, and that under normal circumstances do or would support, a prevalence of vegetation or aquatic life that requires saturated or seasonally saturated soil conditions for growth and reproduction.

Wetland, jurisdictional: Area subject to the regulations of the Clean Water Act of 1977; generally concave or low-lying topographical forms that collect, store, or flow water frequently enough to favor a majority of plants that are adapted to saturated soil conditions.



Wheelchair: Mobility aid, designed for and used by individuals with mobility impairments; may be manually operated or motorized.

Width, clearing: The outer edges of clearing areas (cleared of trees, limbs, and other obstructions) as specified by trail use.

Widening: The instance when the main trail has expanded beyond its original width.

Width, design: The width specification that a trail was designed to meet, generally considered part of the trail.

Width, minimum clear: The narrowest point on a trail; created when significant obstacles, such as utility poles or tree roots, protrude into and reduce the design width.

Width, tread: The width of the surface portion of the trail used for travel.

Wildlife: Any undomesticated animal species living in its natural habitat including birds (raptors, songbirds, upland game birds), mammals (fur bearers, big game, non game mammals), reptiles, amphibians, and fish.

Windfall (windthrow, downfall): Anything (trees, limbs, brush, etc.) blown down on the trail by the wind.

Windrow: A ridge of loose soil that is produced by the spill from a grader or dozer blade.

Wire flags: Wire wands with square plastic flags at on end for field layout and marking locations of new or relocated trails.

Zero-mile mark: The point at which a measured trail starts.